data flow at a time when it might be highly desirable for forecasters tracking the hurricane's path. NOAA has taken various steps to reduce this risk, including:

1. Installing "hurricane rated" 16.5 m GOES antennas at the WCDAS that can operate through a Category 3 hurricane (130 mph winds) and can survive a Category 5 hurricane (155+ mph winds).
2. Placing a back-up GOES antenna further inland at the Goddard Space Flight Center in Greenbelt, MD. This back-up would be close-enough to the WCDAS (~150 miles distant) to deploy antenna operations personnel prior to the hurricane tracking up the east coast to be ready if needed.
3. Having a back-up GOES antenna located at the NOAA Fairbanks, AK CDAS (FCDAS) that can support the GOES-West spacecraft if needed. At 64.5° N latitude, Fairbanks is not the optimal location for a geostationary satellite ground terminal, but, NOAA operates the Fairbanks site primarily to support their Polar Orbiting Environmental Satellite (POES) spacecraft operations. In 2003, the Japanese government asked if a GOES spacecraft could be temporarily drifted over the Pacific Ocean to provide weather imagery of the area around Japan (a Japanese weather satellite had failed and they needed interim support until they could launch a replacement). Since the WCDAS would not be able to communicate with a GOES satellite that was moved this far out over the Pacific, a GOES antenna was put in place at the FCDAS to provide this link. Although support to Japan ended in 2005, this GOES antenna remains as a back-up for the GOES-West spacecraft.

A few things stand out in this example:

- When using dedicated antenna assets it is important to think through contingency situations. What happens if that single dedicated antenna fails? How much risk is acceptable if mission data can not be recovered for some period of time while the antenna is being repaired or is stowed during high winds? Putting in back-up dedicated antennas may lower risk, but can be a high cost solution. An alternative can be to have an agreement with a commercial service provider to provide "on call" service when a mission dedicated antenna is non-operational. The challenge with this approach is that the spacecraft and ground system architecture must be compatible with the commercial provider's network architecture, service offerings and network connectivity. In addition, these commercial assets have many customers, so requests to be inserted into their schedule on short notice may be difficult for them to accommodate.
- Dedicated antennas may provide outstanding support during nominal on-orbit operations, but, will they support non-nominal operations such as contingencies where the spacecraft may be in a non-nominal attitude and may have failed over from a primary on-board HGA to an emergency LGA? How about during launch and early orbit (LEOP) operations when the orbital parameters ground track and navigation requirements may be very different from the on-orbit operational scenario? In the case of GOES, L&EO operations are supported by a combination of commercial services and NASA Deep Space and Near Earth Network antennas that provide the geographic distribution required to support the frequent contacts required to support orbit raising maneuvers as well as the northern/southern hemisphere coverage required to collect accurate tracking data to support orbit raising maneuver planning. The NOAA Office of Satellite Operations [2011] provides additional information.

LEO: Far Ultraviolet Spectroscopic Explorer (FUSE) Puerto Rico Ground Station

The Far Ultraviolet Spectroscopic Explorer (FUSE) spacecraft was built and operated for NASA by the Johns Hopkins University (JHU) [2011] in Baltimore, MD. The FUSE observatory was in a 775 km circular orbit at a 25 deg inclination with an orbital period of ~100 min. Operationally, participating universities from around the world would submit observing requests which would be scheduled by the JHU operations team based on predicted viewing intervals, spacecraft positioning constraints, and the needs of each science program. These detailed observing schedules were then uplinked to the satellite which would store and execute the schedule without the need for direct interaction by ground controllers.

To support this equatorial LEO mission, a single ground antenna was placed at the University of Puerto Rico, Mayaguez. Contacts between FUSE and the Puerto Rico antenna lasted ~10 min/contact and there were ~6 orbits in a row of contacts followed by ~8 orbits without contacts. During contacts, an observing schedule was uploaded and captured scientific data, which had been stored while the spacecraft was out of contact, was downlinked for processing. The on-board recorder was sized to ensure that far ultraviolet observations as well as spacecraft state-of-health data could be collected and stored during the daily ~13 hour out-of-contact periods. In addition, the communication link was sized to ensure that data rates could be supported that would enable downlink of this stored data within the limited 10 min contact periods once the Puerto Rico antenna was again in view.

Although this appeared to be a low-cost solution for supporting FUSE operations, the Puerto Rico-based antenna was a single-point-failure within the communication architecture. In 1998, after the antenna was installed but prior to FUSE launch, hurricane George came across Puerto Rico and severely damaged this unprotected asset (luckily this happened prior to launch). After repairs, and the addition of a hurricane rated radome, the antenna was once again ready for operations just prior to launch. After

launch, the FUSE project signed an agreement with Universal Space Network (USN) to use a USN 13 m antenna at Hawaii as an "on call" back-up to the Puerto Rico antenna, reducing the risk of mission data loss while only incurring this additional support cost when the Puerto Rico antenna was unable to support the mission requirements.

SmallSat: Alaska Research CubeSat (ARC)

Like many universities, the University of Alaska, Fairbanks (UAF) is designing a CubeSat mission, the Alaskan Research CubeSat (ARC). For most of these CubeSat development teams, the limited budget available for development and operations drives them to creative low-cost ground system solutions. The UAF ARC CubeSat solution described below is used to illustrate one approach of the many being developed by this community.

Among a number of objectives, the ARC spacecraft will collect Earth images and relay them to the ground. To receive these images, the ARC ground station is being designed to support the 3 frequencies most commonly used by amateur satellites (VHF (144–146 MHz), UHF (420–450 MHz), S-Band (2.0 GHz)) (Note: Since these are heavily used transmission bands, it is important to try to locate the ground station in "radio quiet" environment as possible). Although the ARC satellite will only operate at the UHF and S-Band frequencies, the ground station was designed to cover all three frequencies so that they could offer support in the future to other CubeSat missions built by UAF or other universities. The single ground station consists of two circularly polarized 6 m long yagi antennas, one for the VHF band and one for the UHF band, as well as a dish antenna to support the S-band downlink (a ~2 m television satellite dish is adequate for these type of applications). A common elevation/azimuth steering system allows the antennas to be pivoted for tracking. Ground software, driven by orbit predictions, points the antennas prior to each pass.

A downconverter will take the S-band signal down to VHF so that a VHF/UHF radio can process all incoming signals. Downlink rates are limited to 1,200 and 9,600 symbols/second due to receiver design constraints (the addition of filters to allow the S-band downlink to use a rate of 38,400 symbols/second is being considered). A low data rate beacon on the ARC spacecraft will be turned on and remain on after launch vehicle deployment. The beacon signal contains limited information regarding the power level available on the spacecraft. Whenever the beacon is captured at the ground station, a determination can be made as to whether or not there is sufficient power to operate the transmitter at full power to begin a higher rate UHF/S-Band transmission session. After each high rate transmission is complete, the spacecraft will go back into beacon transmission mode.

Another nuance of the design, since the system is being deployed in the Alaskan interior, is that it must be able to operate down to temperatures as low as –60° C. Validation of the ground station operations will be performed by the use of weather balloon flights which will carry engineering models of the ARC communication equipment for end-to-end testing.

Additional information on NASA's CubeSat [2011a] program can be found at their website.

28.1.2 Space Relay Systems

For some missions in LEO, the constraint of only being able to communicate with the ground for ~20% of each orbit is not compatible with the mission's operations concept. For example, for Space Shuttle crews circling the globe every ~90 min, being in contact with their Earth-based Mission Control Center for ~18 min/orbit would severely restrict the amount of data that could be transmitted to and from the Shuttle during each mission. In addition, these short and infrequent contact periods would severely limit opportunities for ground crews to support the Shuttle crew in troubleshooting anomalies or handling emergencies that may occur during a mission.

Another example is the Swift Gamma-Ray Burst mission. Swift is a LEO mission that is searching the sky for gamma ray bursts coming from the most powerful explosions in the Universe. To study these bursts, which can be seen from Earth for periods of only seconds or minutes before they die out, the Swift spacecraft needs to be able to send a message to the ground-based Gamma-Ray Coordination Network (GCN) within a few seconds of identifying a new burst. These messages include coordinates for the burst so that Earth-based and space-based gamma ray telescope facilities around the world can be quickly notified that a burst has occurred so that they can point their telescopes and begin their own data collection. After announcing that a burst has occurred, Swift then trains its own scientific instruments on the burst to begin collecting detailed data that can be downlinked later for analysis.

In both cases, and many more, latency requirements, and data volume requirements force designers to consider the use of a space-based relay system for support. Space-based relays, placed in geosynchronous orbit, can provide LEO spacecraft with a significant improvement in the percentage of each orbit during which time they can be communicating with the ground. For each GEO relay satellite, LEO satellite coverage of ~45% can be achieved (Note: specific coverage periods are based on the specific orbit altitude and inclination of the LEO satellite). With a constellation of 3 GEO relay satellites, spaced evenly every ~120 deg in longitude, continuous coverage for satellites in LEO can be achieved.

In addition to providing expanded coverage over ground-based antennas, GEO relay satellites have a few different link budget considerations when compared to direct space-to-ground communication. First, the free-space loss term of the link Eq. (21-4) is significantly greater when using space-based relays since the LEO-GEO link distance is much greater than for LEO-to-ground communication links. On the other hand, space relay systems can make use of higher frequency communication links, such as K-band and optical, where space-to-ground link signal attenuation, especially during rainy and cloudy weather, may make these shorter wavelength frequencies impractical for establishing reliable communication links. Using

Table 28-2◀, Fig. 28-5◀, Eq. 28-2◀

these shorter wavelength links, LEO spacecraft can communicate at data rates above 1 Gbps through GEO relays (although the link between the GEO relay and its ground terminal may need to rate buffer this data as it transmits it through the atmosphere).

Space-based relay satellites are typically placed in equatorial orbits where east-west station-keeping maneuvers are key to keeping them at their proper operational longitudinal position. Lunar-solar perturbations and the ellipticity of the Earth's equator also tug on the GEO orbiting satellites, requiring north-south station-keeping to maintain an equatorial orbit. Since north-south station-keeping requires a significant amount of propellant, some GEO satellites are allowed to drift in inclination. This drift continues until the inclination has peaked at 15 deg after 27.5 years of drifting from the equator (see Sec. 9.5.1). At this point, the forces mentioned above begin to bring the inclination back down to 0 deg over the next 27.5 years. This drift in inclination can impact visibility periods between GEO relay satellites and the user spacecraft that they support. As the inclination increases, these satellites spend more of the day at latitudes where it is possible for them to communicate with satellites orbiting near the Earth's North and South poles. For some countries, such as Russia, where much of the country is at high latitude, GEO relay satellites at higher inclinations can also be valuable for sustaining communication links to high latitude ground terminals.

Although numerous countries have either experimented with, or developed operational space-based relay satellites, NASA's Tracking and Data Relay Satellite System (TDRSS) is used here as an example of what type of support can be provided through a space-based relay system. With an almost 30 year operational history, the TDRSS constellation provides support for LEO missions from various nations.

28.1.2.1 Tracking and Data Relay Satellite System (TDRSS)

In September of 1983, Shuttle mission STS-8 was able to download more data to Earth than all 7 previous Shuttle missions combined. This was made possible by the introduction of the first of NASA's new geosynchronous communications relay satellites, the Tracking and Data Relay Satellites (TDRS). Previously, ground-based antennas located around the world provided limited coverage for all NASA missions since LEO spacecraft were generally only in view of ground antennas for ~20% of each orbit. The TDRS System (TDRSS) provides constant coverage for LEO satellites using an architecture that includes

- A constellation of three geosynchronous TDRS that serve as bent-pipe delivery systems between user satellites and their ground based control centers.
- Ground terminals used for receiving/sending user spacecraft data to/from the TDRS and for monitoring and maintaining the health of the TDRS constellation.

In 1989, with the launch of a third TDRS, low-Earth orbiting satellites in orbits above 1,050 km (and below ~9,000 km) were now able to communicate with Earth at any time in their orbit through the TDRSS White Sands Ground Terminal (WSGT) located at the White Sands Complex (WSC) in New Mexico. During this period only two TDRS were used operationally while the third was an on-orbit spare. The two operational TDRS were placed in orbits that allowed them both to view the single ground terminal at White Sands while being spaced as far apart in longitude as possible to maximize the overall coverage the constellation could provide to LEO satellites. With a two satellite constellation, LEO spacecraft orbiting below 1,050 km encountered a *Zone of Exclusion* (ZOE) during each orbit where they were out-of-view of both operational TDRS and therefore unable to communicate. To address this ZOE, in 1993, a re-configuration of the TDRS constellation placed a third TDRS into operational service. These three operational satellites were now evenly spaced around the world in longitude (every ~120 deg). Since this spacing in longitude meant that one of the operational TDRS would not be able to view White Sands, a second ground terminal in Guam (the Guam Remote Ground Terminal (GRGT)) was added. The three satellite constellation, allowed for constant communications with Earth orbiting satellites down to 72 km orbit altitude. In 1994, as the number of orbiting TDRS was increased to 5 (including both operational spacecraft and on-orbit spares), a Second TDRS Ground Terminal (STGT) was added at WSC to support this growing constellation. The Space Network (SN) space/ground elements are shown in Fig. 28-6.

Each ground terminal establishes links to individual TDRS through Space-Ground Link Terminal (SGLT) equipment chains that are connected to 19 m antennas used to uplink/downlink both user spacecraft data as well as TDRS state-of-health data (shown in Fig. 28-7). SGLT connectivity to the NASA Integrated Services Network (NISN) provides the user spacecraft control center interface into the ground terminals.

The TDRSS is designed to provide the following user services:

- Forward Service: A communication path that originates at the user control center and passes through the WSC or GRGT on its way to the user spacecraft through a TDRS. This user spacecraft command path can be supported in S-band, K_u-band and K_a-band. A separate ranging channel can be established as part of this service as well.
- Return Service: A communication path that originates at the user spacecraft and passes through a TDRS to either WSC or GRGT-to-WSC on its way to the user control center. This path allows the flow of science data as well as user spacecraft state-of-health data to be returned to the user control center. S-band, K_u-band and K_a-band Return Service options are available.

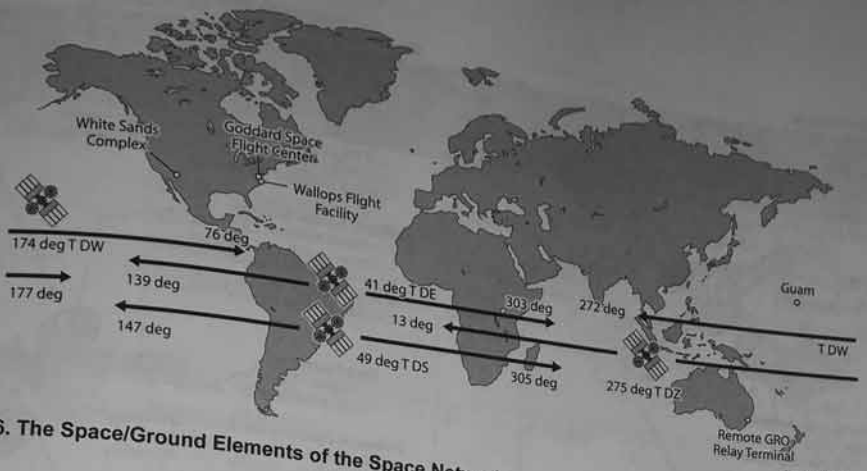


Fig. 28-6. The Space/Ground Elements of the Space Network.

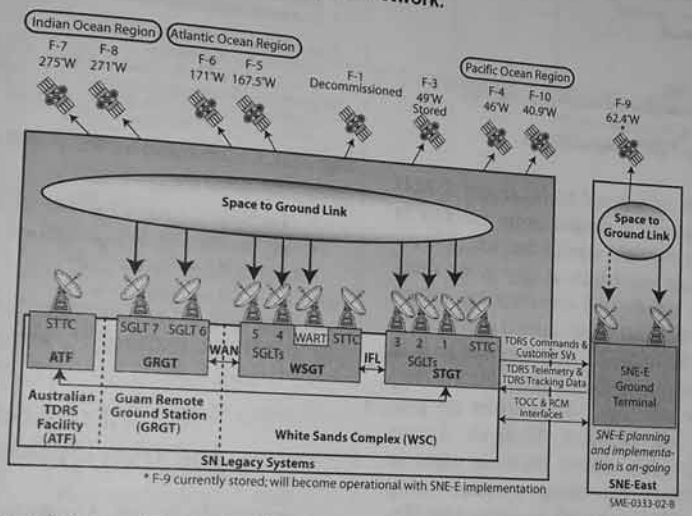


Fig. 28-7. TDRSS Ground/Space Architecture.

From a user perspective, understanding the capabilities of the TDRS spacecraft is key to designing a communication system that will properly interface with the TDRSS. The TDRS constellation currently consists of 6 operational spacecraft, each capable of supporting over 20 user spacecraft at any given time. The TDRS constellation has evolved from:

- A first generation series of 7 satellites (see Fig. 28-8 for capabilities)
  - 5 of which are still operational in 2011
  - 1 was destroyed in the 1986 Challenger accident
  - The first (TDRS-1) was retired in late 2009
- To a second generation of 3 satellites (see Fig. 28-9 for capabilities)
  - All of which are still operational (as of 2011)
  - With two updated second generation satellites, TDRS-K & TDRS-L, scheduled for launch in 2012 and 2013

Some TDRS are maintained in a storage mode where they are not available to support user spacecraft but they are available to be placed into operations quickly if any issues arise with an operational TDRS.

All TDRS spacecraft are maintained in their operational location by east-west stationkeeping maneuvers while, to preserve propellant, north-south stationkeeping maneuvers are not performed. The resulting inclination drift has made it possible for some TDRS spacecraft to have a clear line of site to the North and South Poles for a part of each day. Starting in 1998, TDRS-1 was the first satellite to achieve an inclination where it could be used as a South Pole TDRSS Relay (SPTR). Currently, other first generation TDRS have inclinations allowing them to support SPTR requirements. SPTR provides a critical communication relay for scientists working at the pole. At a 15 deg inclination, a TDRS spacecraft can provide up to 12 hours of communication access for forward/return services to/from antennas based at the Earth's poles. Basic TDRS capabilities include (apply to first and second generation TDRS except where noted):

Table 28-2◀, Fig. 28-7, Eq. 28-2◀

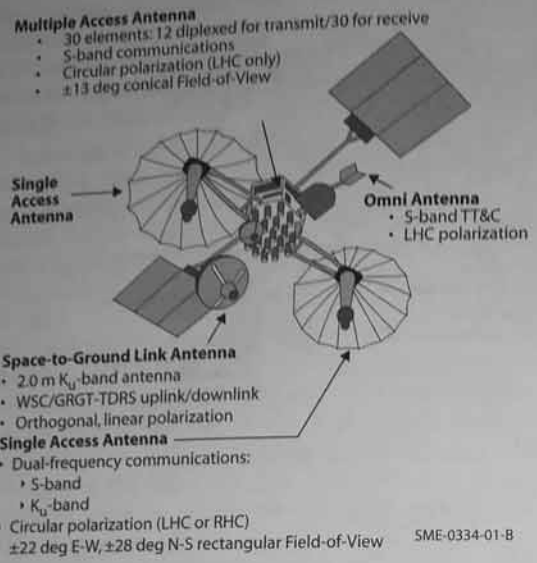


Fig. 28-8. First Generation TDRS Capabilities.

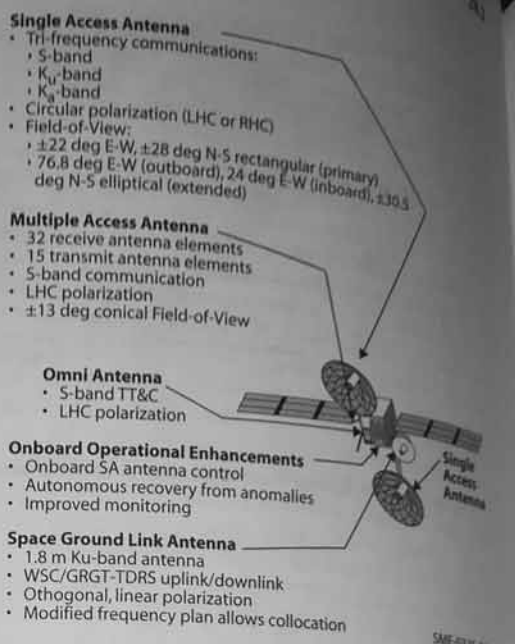


Fig. 28-9. Second Generation TDRS Capabilities.

- *Multi-Access (MA) or S-Band Multi-Access (SMA) Service*: A multi-element phased array antenna is used to support one forward service and multiple return services to user spacecraft at any given time. Limitation on the number of simultaneous return service customers that can be supported is driven by limits on the amount of SGLT user services ground processing equipment and the level of self interference between user satellites. MA services are provided using a Code Division Multiple Access (CDMA) technique that allows multiple users to communicate through the same system, using spread-spectrum technology with a unique identifying Pseudo Noise (PN) Code assigned by the NASA Spectrum Manager to each user.

- *Single Access (SA) Service*: Two 4.8 m, steerable dish antennas are used to each support a link to a single user spacecraft at a time.

- *Cross-Support Service*: In this service, user spacecraft use both the S-band SA (SSA) service and MA simultaneously, one for the forward service, one for the return service.

- *Tracking and Clock Calibration Services:* Available to all users except K_a-band SA (KaSA) users, ranging, two-way Doppler measurements and data on the offset between the user spacecraft clock and Universal Time Coordinated (UTC) time are available.

- *Demand Access System (DAS) Service*: DAS equipment at the ground terminals can be configured to allow for MA users to establish an extended duration link that is handed-off from TDRS to TDRS as the user satellite orbits the Earth. DAS Services can only be supported by the first generation TDRS.

- Technical specifications of TDRS antennas operating in the modes described above can be found in Table 28-3.

Additional information on NASA's Space Network [2011d] can be found on their website.

28.2 Data Accounting and Distribution Services

Connectivity between antenna ground stations and user facilities is a critical link for ensuring captured telemetry can get to the end-user within the required latency, and that commands can get from the user facility to the spacecraft during ground station contacts. Although this section will not cover the activities that occur at the user facilities (such as satellite image processing and command data generation), it will cover the tools, processes and protocols used to transfer information to and from the ground stations that interface with these user activities.

28.2.1 Data Capture/Transmission Verification

Early space missions used a *Time Division Multiplexing* (TDM) formatting scheme where data was continuously multiplexed in defined fixed length frames. Each mission would define their own multiplexing scheme as well as their own method for verifying that commands/telemetry were reaching their destination. For most missions during this period, command reception verification was accomplished by *end item* verification where the resulting command action, for example, seeing a spacecraft heater turn on after being commanded to do so, was typically the only way that the flight operations

Table 28-2◀, Fig. 28-9, Eq. 28-2◀

Table 28-3. TDRS Technical Capabilities Summary.

Service Type	Forward Max Data Rate	Forward Modulation Schemes	Forward Minimum EIRP	User Return Links / TDRS	Return Max Data Rate[10]	Return Modulation Schemes[10]	TDRS G/T (nadir minimum)
MA	300 kbps	QPSK, BPSK	PFOV[2]: 1st Series: 34 dBW 2nd Series: 40 dBW LEOFOV[3]: 1st Series: 34 dBw 2nd Series: 42 dBW	5/TDRS up to 20/WSC 2/TDRS through GRGT	1st Series: 300 kbps 2nd Series: 3 Mbps[8]	1st Series: Rate ½ Convolutional 2nd Series: Rate ½, 1/3 Convolutional	Formed Beam: PFOV 1st Series: 22.0 dB/K F8(cold)–F10: 3.2 dB/K F8(hot): –0.2 dB/K LEOFOV 1st Series: 3.1 dB/K F8(cold)–F10: 4.5 dB/K F8(hot): 1.2 dB/K
SSA	7 Mbps	QPSK, BPSK, PCM/PM	PFOV & EEOV[4]: Normal: 43.6 dBW High[5]: 1st Series: 46.3 dBw 2nd Series: 48.5 dBW	2	6 Mbps[8]	Rate ½, 1/3 Convolutional	PFOV: 9.5 dB/K EEOV (2nd Series only): 9.5 dB/K
KuSA	25 Mbps[1]	QPSK, BPSK	PFOV & EEOV[4]: Normal autotrack[6]: 46.5 dBW High autotrack[6]: 48.5 dBW Normal program track: 40.5 dBW High program track: 42.5 dBW LEOFOV: Normal autotrack: 46.5 dBW High autotrack: 48.5 dBW Normal program track: 44.0 dBW High program track: 46.0 dBW	2	300 Mbps[9]	Rate ½, 1/3 Convolutional or uncoded	PFOV (1st Series) and EEOV (2nd Series): Autotrack: 24.4 dB/K Program Track: 18.4 dB/K LEOFOV Autotrack: 24.4 dB/K Program Track: 21.9 dB/K
KaSA[7]	25 Mbps[1]	QPSK, QPSK	PFOV: Autotrack: 63.0 dBW Program track: 56.2 dBW LEOFOV: Autotrack: 63.0 dBW Program track: 59.5 dBW	2/TDRS up to 8/WSC	300 Mbps	Rate ½, 1/3 Convolutional or uncoded	PFOV and EEOV: Autotrack: 26.5 dB/K Program Track: 19.1 dB/K (PFOV only) LEOFOV Autotrack: 26.5 dB/K Program Track: 23.0 dB/K

Notes:
1. Current ground terminal equipment limits forward K_u/K_a-band forward link to 7 Mbps but plans are to increase this capability to 25 Mbps
2. PFOV (Primary Field of View)
3. LEOFOV (LEO Field of View)
4. EEFOV (Extended Elliptical Field of View)
5. Use of High Power mode is restricted and must be coordinated with the Space Network prior to use
6. The KuSA forward autotrack performance EEFOV is a goal and will be supported on a best-effort basis
7. KaSA return service only available on 2nd generation TDRS
8. Available when using rate ½ Convolutional Encoding
9. GRGT support of KuSA return data rates in excess of 150Mbps should be discussed with the SN
10. Maximum data rates may increase, and new modulation schemes may be added as the SN evolves

team knew that a command had successfully reached the spacecraft. Telemetry reception verification during this period was primarily accomplished through the manual inspection of downlinked data that had been recorded on magnetic tape days to months earlier at a ground station. If missing data was discovered, it was most likely already erased from the spacecraft recorders so there was little recourse and the data would be considered lost. With advances in ground transmission capability and bandwidth in the 1970s and 1980s, it became possible for at least some spacecraft data to be sent in real-time from the ground station to the user facility. At this same time, development of Internet Protocol (IP) for terrestrial applications in the 1970s and 1980s, was establishing protocols for high Quality of Service (QoS) (QoS refers to a data transmission mechanism that ensures a specific bit-error-rate or probability of dropped packets on a transmission link) transmission protocols using the concept of standard format "packetized" data. The convergence of the development of IP protocols and the establishment of sufficient bandwidth terrestrial networks to allow spacecraft real-time transmission directly to the user facilities, led to the development of high QoS spacecraft telemetry and command protocols/systems. The goal was to establish a standard packet length along with the appropriate metadata and protocols to allow for the receiver to auton-

Table 28-3, Fig. 28-9◀, Eq. 28-2◀

omously verify reception, and for the sender to autonomously retransmit data that does not reach the receiver.

28.2.1.1 Command Verification Protocol

One of the first packet transmission protocols developed for space application in the early 1980s was the Consultative Committee on Space Data Services [CCSDS, 2011a] Communications Operation Procedure 1 (COP-1), and command transmission protocol. COP-1 protocol is focused on autonomously ensuring that commands are received by the spacecraft in the order that they were transmitted and that all commands get to the spacecraft successfully. Since the command link is typically transmitting a lower data rate (Kbps for command links vs. Mbps for telemetry links) and since high QoS command reception can be critical to spacecraft safety and mission success, it made sense for an acknowledgement base command protocol to be developed first. COP-1 protocol is a closed-loop, QoS scheme that uses a Frame Operation Procedure (FOP) on the command transmission end and a Frame Acceptance and Reporting Mechanism (FARM) on the receiving end. The FOP, operating at the user control center, takes the command virtual channel frames that are ready for uplink and inserts a sequential Frame Sequence Number (FSN) in the frame header prior to transmission to the antenna ground station for uplink. After transmission of the command frame to the spacecraft the on-board FARM checks to ensure that the FSN of arriving commands is sequential and it also validates that the incoming command frames successfully passed through any command error detection/correction software. Once successful command reception is verified, the FARM generates an updated Command Link Control Word (CLCW) for downlink. The CLCW is the part of each virtual channel telemetry frame that contains information on the last command FSN that has been verified as well as information on any out of sequence commands that have been received with a request for retransmission of "missing" command frames. The FOP interrogates the CLCW and retransmits command frames as requested. The FOP also has a timeout function that initiates command retransmission for any command frames that are not verified in the CLCW within a specified amount of time (the timeout function needs to be adjusted as spacecraft get further and further from Earth to take into account light-time travel delays that will lengthen the time between command uplink and receipt of an updated CLCW). COP-1 protocol is now a standard used by most spacecraft. There are some spacecraft where COP-1 is less valuable, specifically, for missions headed deep into the solar system where the light-time delay between command transmission and reception is so great that commands may not be verified during a single ground station contact. For these missions, a more manual method for verifying spacecraft command reception is sometimes used. On other, low-cost missions, command reception verification is still accomplished by command *end item verification*. End item verification is the process of identifying that a spacecraft has received a command by verifying that the action associated with the command has occurred. This was the primary method of command verification prior to the development of COP-1 protocol. For example, a command to turn a heater on could be end-item-verified by examining telemetry to ensure that the heater circuit has been turned to the "on" state and/or that nearby thermistors are registering a rise in temperature. For example, end item verification was used for command reception verification on the NASA/DoD Fast, Affordable, Science and Technology Satellite (FASTSAT) mission developed by Marshall Space Flight Center and launched in 2010. Although this approach saves money on flight software it also makes command verification difficult, especially during periods where telemetry is intermittent and requires the flight operations team to manually determine when command retransmissions are required.

28.2.1.2 Telemetry Verification Protocol

Capturing a high percentage of the data stored in spacecraft mass memory is typically critical to mission success. Currently, for most spacecraft, ensuring a high data capture percentage is a manually intensive process that relies on mission control center personnel verifying the receipt/non-receipt of data and then either releasing successfully captured spacecraft memory segments for overwriting or commanding the spacecraft to retransmit memory segments that were not successfully returned to the control center. Although some missions have developed mission-unique software to support the data recovery process, an international standard for automating this function has not existed until recently. The goal is to provide an FTP-like file transfer protocol that is more highly tolerant of the existing space communications infrastructure, long time delay, higher error rate, environment that space communication links pose compared to the terrestrial communication environment that the FTP protocol was designed to support. CCSDS File Delivery Protocol (CFDP) automates the data recovery process, ensuring as high a data capture percentage as possible while freeing spacecraft memory segments for overwriting as quickly as possible to ensure the maximum quantity of storage is available to the spacecraft at all times. CFDP operates over a wide range of underlying protocols (see Fig. 28-10) [CCSDS, 2011b].

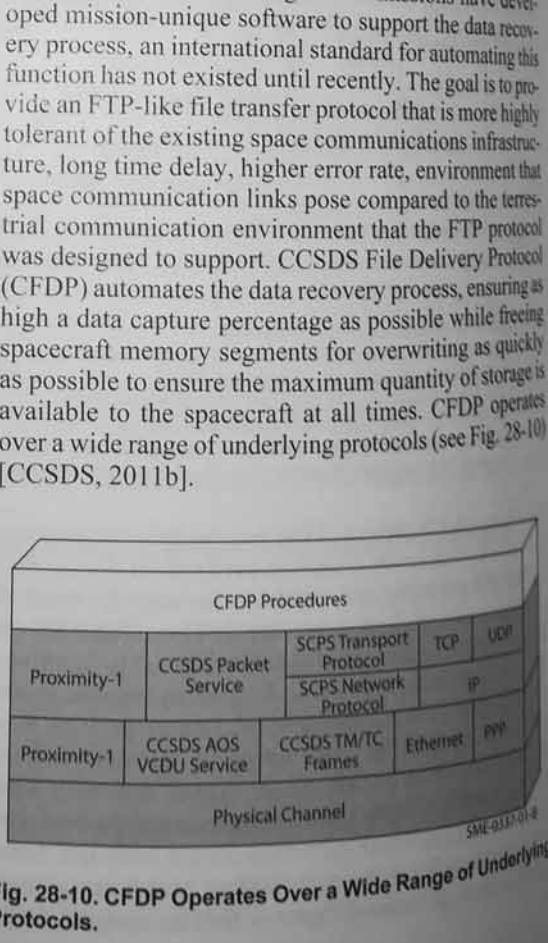


Fig. 28-10. CFDP Operates Over a Wide Range of Underlying Protocols.

Table 28-3◀, Fig. 28-10, Eq. 28-2◀

Although CFDP works on the transmission/reception of *files*, the user can define the file size to fit their operational scheme. As CFDP files are transmitted to the ground via CFDP Protocol Data Units (PDU's), the ground system CFDP software (running either at the antenna ground station or back at the mission control center) logs incoming PDU's. Some PDU's contain CFDP *File Delivery Units (CFDUs)* containing the contents of the file being transferred while others contain metadata used to track the progress of the transmission. For PDU's not received at the ground, two approaches for recovering this data are available. In the first, as soon as a PDU is received out of sequence, indicating that there is missing data, a *non-acknowledgement*, called a *nack*, is sent to the spacecraft to request retransmission of the missing PDU's. The other option is to wait until an *end-of-file* PDU is received and then send the spacecraft all *pending nack* requests for that CFDP file. Either way, once receipt of the entire CFDP file has been acknowledged by the ground, the spacecraft can now free this memory space for overwriting. Another facet of CFDP implementation to consider is that the ground based CFDP software can either be run at the mission control center or at the antenna ground station, both implementation options are permissible—depending on the operational requirements and time available for retransmission. CFDP software running at the antenna ground station shortens the time to respond to missing data and can operate even when transmission links to the mission control center are not available, but, since this mode requires the antenna ground station to send "command data" back to the spacecraft some spacecraft operators would prefer to close the CFDP-loop at the mission control center. If the CFDP-loop is closed at the mission control center, an option is to have the ground station store the downlinked recorder data and generate *nacks* and send them to the mission control center where they can be turned around and sent back to the ground station for uplink. This hybrid model allows the mission control center to be the source of all spacecraft commands and to be responsible for the data management process, while minimizing the real-time bandwidth requirement between the ground station and mission control by not requiring PDUs to complete the journey to the mission control center to be acknowledged. CFDP can allow for data management across ground station contacts and across multiple ground stations although the ground implementation (mission control center vs. antenna ground station) will drive how CFDP metadata will need to be exchanged to support these scenarios. A number of commercial products are now available that implement this standard and for spacecraft with significant data recovery requirements and missions where lights-out operations are being considered, implementation of CFDP offers distinct advantages. For example, the LRO mission successfully applied CFDP to ensure they could achieve a high percentage of science and spacecraft housekeeping data recovery using a K_a-band telemetry link that could be adversely impacted by weather in the transmission path.

28.2.2 Standard Transmission Protocols

Section 28.2.1 discussed a number of protocols used to ensure the successful transfer of data to and from spacecraft. These *Space-Link* protocols address only one leg of the journey that data transits on its way to and from a spacecraft. The other leg of the journey is the terrestrial link between ground elements. One of the most popular standard protocols being used to provide high QoS links between ground stations and user facilities is the CCSDS Space-Link Extension (SLE) protocol. Formed in 1982, the CCSDS, established numerous international standards for the link between spacecraft and ground stations. More recent efforts, [CCSDS, 2011d], have focused on establishing a similar set of inter-Agency "Cross Support" standards that address connectivity between ground segment elements to extend the space link directly to the end user (Fig. 28-11).

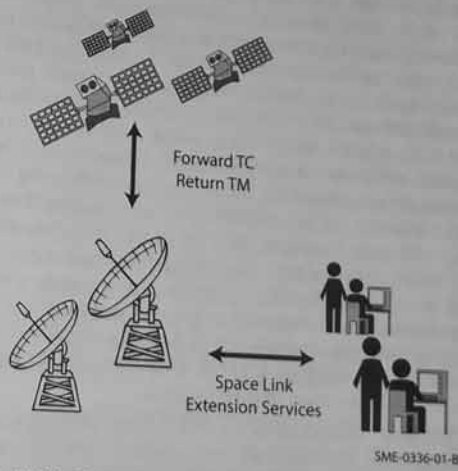


Fig. 28-11. Extending the Space Link Back to the Control Center.

The SLE protocol is a layered architecture that uses existing communications infrastructure to enable antenna ground stations and mission control centers to be connected seamlessly. In a world where both national Space Agencies and private companies have developed a large amount of ground-based spacecraft support infrastructure, these standard protocols enable interoperability between diverse ground elements. SLE protocols address both a set of TCP/IP-based Data Transfer services, such as, telemetry distribution based on Virtual Channel or Applications ID (apid), command data transfer as communications link transfer units (CLTU). All services moving through this layered protocol pass through an Authentication Layer that provides security for the transmission path, authenticating users before "binding" the service path to allow data transfer.

The SLE Cross Support Service Management protocols enable the transfer of information between mission users (Utilization Management) and service providers

Table 28-3◀, Fig. 28-11, Eq. 28-2◀

(Complex Management) required to: establish a Service Agreement on Data Transfer services to be provided, define Configuration Profiles for these services, schedule service instances, and/or transfer information needed to support service execution (e.g., trajectory predictions, equipment requirements). These standard protocols can be used from the early definition phase when negotiating a Service Agreement through the service execution phase.

28.2.3 Bandwidth/Connectivity Between Ground Stations and User Facilities

One of the limiting factors in getting data to and from the spacecraft is the bandwidth available between the ground station and the user control center facilities. As spacecraft telemetry data rates rise toward 1+ Gbps, terrestrial communication links to send all this data back to the user facilities in real-time can become cost-prohibitive. In addition, since many missions now require multicast data distribution to numerous user facilities, bandwidth requirements can be increased even further to send copies of the data to each user. To evaluate bandwidth requirements, latency and data quantity/unit time requirements are taken into consideration. For missions with long latency requirements (days, weeks, months), physical media such as DVDs or USB flash drives can be used to collect data at the ground station for shipping to user facilities. For medium latency requirements (~6 – ~48 hours), transfer from the ground station to the user facility can be done using internet file transfers using FTP or other file transfer protocols. For missions with short latency requirements (few minutes), dedicated transmission lines, for example, T1 (1.544 Mbps/1.536 Mbps available to the user) or OC3 (155.52 Mbps/148.608 Mbps available to users) line may be required.

The other consideration is data volume and multicast requirements. For missions with moderate data volumes where the available bandwidth back to the user can deliver data to the end user within the required latency, rate-buffered transmission directly from the ground station to the users may be the best option. The ground station can also save a copy of all transmitted data to respond to requests for retransmission from the user although this retransmission capability will have a lower priority for bandwidth than any real-time data that may be coming from the ground station at the same time. For missions where data volume or multicast requirements are high and bandwidth to the ground station is limited, ground stations may transfer their data to a network storage/distribution facility located at a site with high bandwidth connectivity to the users. This central storage facility can provide longterm storage for user data, can retransmit data to users on request, and can multicast data as required.

A final consideration is whether or not an alternative path is required to deliver data to/from ground stations. Alternate paths can be used to increase the probability that latency requirements can be met. Alternate paths can protect against network outages resulting from cable cuts or weather related outages between the ground station and the user. For many missions, the cost of alternate paths for data delivery may be prohibitive. For remote locations, alternate paths for data delivery may all lead to a similar *last-mile* path. For example, there may be only one set of utility poles, or a single bridge leading to the ground station, or a single network interface card at the ground station resulting in a single-point failure in the transmission path.

28.3 Ground System Driving Requirements and Sizing

There are many aspects of mission design that will drive the design of the supporting ground system. The following information provides a summary of some of the primary mission design drivers and their impact on ground system design.

Orbit/Trajectory

1. LEO, High Inclination, Low Eccentricity: Ground stations at high latitude locations provide coverage for almost every orbit each day. Stations near both poles provide half the orbit coverage; near one pole only provides once an orbit coverage with increased latency.
2. LEO Mid-Latitude, Low Eccentricity: GEO Space-relay systems can provide constant coverage. Ground stations each provide single contact opportunities for less than ~40% of orbits each day (specific percentage depends on satellite inclination and station latitude).
3. High Eccentricity: Based on the orbital period, it may be possible to locate ground stations where the ground track location of the satellite's apogee repeats over time.
4. GEO: A single antenna and data processing, storage, distribution center located in constant view of the GEO satellite provides an optimal ground architecture solution.
5. Interplanetary Trajectory: Networks with 3 nodes separated from each other by ~120 deg of longitude (e.g., NASA's Deep Space Network) provide greatest duration of daily view periods. A combination of stations above and below the equator is optimal for collecting tracking data that can be used to establish orbit/trajectory with high accuracy.
6. Launch Trajectory: Ground Stations at locations to view key events (e.g., upper stage shutdown/ignition, separation, array deployment) (stations must be capable of tracking at high rates during these low altitude contacts). Space-relay systems can be used as long as their antennas can slew at the high rates necessary to track these low-orbit/high velocity events.

Table 28-3◀, Fig. 28-11◀, Eq. 28-2◀

Latency (from collection of data on board the spacecraft to arrival of processed products at user facility)

- **Minutes:** Requires dedicated (or close to dedicated) antenna services to support. Also, requires significant redundancy and robustness in the ground architecture to ensure that low latency can be maintained when specific contacts are missed due to equipment or operational issues. GEO space-relay systems are optimal for supporting low latency LEO requirements since they can provide coverage over a high percentage of each orbit. A ground-based antenna service would need to have a large number of nodes spread evenly throughout the satellite's groundtrack to provide multiple partial-orbit opportunities for downlinking data. For GEO satellites, a single dedicated ground station can provide constant contact to support low latency, and for lunar and planetary missions, a 3-node ground station configuration equally distributed in longitude can provide this constant availability (although transmission time delays for spacecraft beyond the Moon make "minutes" latency difficult or impossible to achieve). Once on the ground, verification of data and requesting re-transmissions must also be completed quickly using advanced automated protocols such as CFDP (Sec. 28.2.1.2). Data processing and transmission to users directly from the ground station is optimal for achieving low latency (unless integration of data between multiple ground stations is required as part of the data processing function —although this integration would be difficult to perform while still meeting this low latency requirement). To perform this function, adequate bandwidth from the ground station to the user must be available to minimize the amount of rate-buffering/delay that occurs in the data transport to the user.
- **Hours:** Reduced amount of required contact opportunities makes a shared, distributed ground station architecture a cost effective solution to meet this requirement. If a distributed ground station architecture is used, integration of data at a single location may be valuable for performing data accounting, storage, and distribution. This architecture can include rate-buffering of data at the ground station, reducing ground station bandwidth requirements/cost while still achieving required latency.
- **Days/Months:** Infrequent scheduled contacts on a shared, distributed ground station architecture provides a cost effective method for supporting this requirement. Also, once data is on the ground, low cost options for transporting data to a central processing center and to the final user can be employed such as using standard internet FTP (where bandwidth availability can vary unpredictably) and even the shipping of data tapes/flash-drives from the ground station can be used instead of paying for expensive communication data lines.

On-board Storage Capability

- **No On-board Storage:** GEO space-relay satellites can provide a constant orbital coverage capability for most LEO satellites. Experience on NASA's Compton Gamma Ray Observatory (CGRO), where the on-board recorders failed during the mission, showed that GEO relay satellites could effectively collect/transmit user spacecraft data as long as the GEO satellites could dedicate the required assets to support the mission. In another case, LEO satellites with ops concepts that only require the collection data over specific ground locations can operate with a dedicated ground based antenna at these locations to capture data in real-time. An example of this would be an Earth imaging satellite designed to image major cities with a series of ground antennas located at each city. Lastly, GEO satellites with dedicated ground based antennas can operate without on-board storage as well since there is constant view between the ground and spacecraft antennas. NASA's geosynchronous Solar Dynamics Observatory (SDO), for example, has two dedicated ground antennas in New Mexico that provide primary/back-up data receptors for a constant stream of SDO telemetry. In all of these cases, downlink data rates are driven by spacecraft data collection rates. In addition, co-location of telemetry processing/distribution at ground stations is often advantageous in these scenarios since received data can be processed and shipped immediately (no need to wait for "redumps" of missing data when there isn't any on-board recorder).
- **Minimal On-board Storage:** Ground architecture needs to focus on quickly evaluating recorder data dumps and initiating re-dumps as required to fill gaps in initially recovered dataset and free on-board memory for overwriting. Antenna services need to provide frequent opportunities for recovering recorded data. Also, advanced protocols, such as the CFDP (Sec. 28.2.1.2) can assist in quickly evaluating recovered data, freeing on-board memory space as recovery is verified and automatically initiating re-dumps as required to fill gaps. Other advanced protocols, such as DTN (Sec. 28.5.3.1) can also provide a method for passing data from the spacecraft to an intermediate location (could be a space-based relay satellite) which would then take responsibility for the data transmission, allowing the spacecraft on-board memory to be released for overwriting.
- **Significant On-board Storage:** Possibility of going multiple orbits without downlinking data greatly reduces the number of ground stations required to provide support. Also, simplifies data recovery process, allowing for more time for evaluation of captured data by humans or machines, prior to re-

Table 28-3◀, Fig. 28-11◀, Eq. 28-2◀

leasing on-board memory for overwriting. Note, if re-transmits are spread over multiple ground stations, transfer of all data to a common location for level-zero processing/data accounting to build final datasets may be required.

Required Data Recovery Percentage

- **<~95%:** There can be evaluation of setting more lenient requirements throughout the ground system for fault tolerance, robustness, and redundancy to match with this higher tolerance for missing data and to keep down ground system costs. Reductions in re-transmissions can also be included to save time. This reduction in re-transmissions can be tied with a simple and highly automated ops concept that can be more tolerant to missed contacts and does not need to include as many fault identification and recovery schemes.
- **>~95%:** Ground systems supporting missions with data recovery percentages approaching 100% have traditionally focused on having large operations staffing who focus on "chasing" every bit of data. Improvements in automation have started to reach levels where even these strict data recovery requirements can be met by automated systems. New data capture protocols, such as CFDP, provide a highly accurate method for maximizing data recovery. The biggest challenge for automated systems is handling anomalous situations, such as, a missed contact, where rapid rescheduling of additional contacts to ensure full data recovery can be challenging. Dedicated ground stations for providing antenna services is optimal for these missions since rapid rescheduling in a multi-user ground station network can be difficult based on other user priorities and the ability to interact (particularly when automation is being used) with the antenna service provider in real-time to reschedule contacts quickly to minimize data loss.

Level of Automation

- **Fully Staffed Operations:** Highly human intensive operations allows for simplification of the ground system architecture. All nominal and non-nominal situations are handled by the operations staff who are also responsible for monitoring and identifying non-nominal situations. Operations staffing is typically comprised of discipline specific team members.
- **Day Staff Only:** Automation is used to handle all nominal operations while staff are "on-call" to analyze and react to non-nominal situations that the automation has identified. In this *lights dim* concept, the biggest challenge is balancing data capture requirements against the role of automation to ensure that human response time will not result in unacceptable data loss. In addition, in many of these systems commanding the spacecraft is a task retained by the operations staff and not entrusted to the automation. Operations staffing also shifts to a team where almost all members are multi-skilled.
- **Full Automation:** *Lights out* operations with minimal human intervention in ground system operations involves development of automation systems with significant levels of fault tree analyses programmed to allow for the ground system to identify, troubleshoot and react to non-nominal situations. In addition, use of advanced data capture protocols such as CFDP can provide high data capture percentage performance without the need for human involvement in recorder management.

28.4 Mission Examples

The following mission examples, outline how requirements drove specific ground system designs:

Advanced Composition Explorer (ACE)
Mission Summary: NASA, Explorer Class Mission to provide up-to-date information on solar wind speed and particle/energy flux
Launched: 1997
Orbit: Lissajous at Earth-Sun L1 (1.5×10^9 m from Earth)
Ground System Design: 34 m DSN antennas, at sites around the world, are used to support single daily S-band 3-hour duration playbacks of stored data (76 kbps playback rate) (Note: ACE spacecraft is available for contact as long as the Sun is above the horizon at the ground station). In addition, smaller aperture antennas around the world are used to capture 464 bps Real-Time Solar Wind (RTSW) downlink which is transmitted 24/7 (except during single daily stored data playback). RTSW data contains critical information on variations in the speed and intensity of the solar wind. RTSW data is shipped directly to NOAA's Space Weather Prediction center where it is processed and made available as quickly as possible to organizations, such as spacecraft operators and power utility companies, who are concerned about potential impacts from increased solar activity on their systems.

High-rate playback and real-time data from DSN sites is transferred to GSFC where a lights-out automated multi-mission control center ingests ACE data and performs level-0 processing on all captured data and analysis and trending of spacecraft state-of-health data. Flight Operations team members are automatically alerted if there are anomalous spacecraft health readings or problems with the ground system. Completed level-0 products are shipped to the ACE Science Center at the California Institute of Technology where level-1 and other products are generated for distribution to the science community.

Polar Operational Environmental Satellite (POES)
Mission Summary: NOAA's LEO weather monitoring satellites. Provide visible and infrared images and soundings in addition to ozone, space weather and other readings
Launched: Series of satellites with launches dating back to the 1970s

Orbit: A two-satellite constellation with morning and afternoon ascending-node crossing times. POES are in Sun Synchronous 833 km/870 km (morning/afternoon) circular orbits

Ground System Design: Dedicated Command and Data Acquisition (CDA) stations located at Wallops Island, VA and Fairbanks, AK use 13 m antennas to contact each POES spacecraft ~11 times/day for ~12 min/contact. Since each POES spacecraft orbits the Earth ~14.1 times/day, there are ~3 orbits/day where the spacecraft orbit ground track path does not intersect the *mask* of any of the CDA antennas. During CDA contacts, 2.66 Mbps playback are performed to download data stored during the previous orbit (previous 3 orbits when the spacecraft is coming out of the non-contact period described above). During these contacts, real-time data on the spacecraft state-of-health and the status of the recorder playbacks is transferred to the POES Satellite Operations Center (SOC) in Suitland, MD where a 24/7 staff of operations personnel assess the spacecraft health and ensure recovery of playback data is completed. Playback data is collected at the CDA during the contact and is then rate buffered and transmitted via domestic commercial communication relay satellites (*DOMSAT*) back to the Central Environmental Satellite Computer System (CEMSCS) facility located in Suitland, MD (the goal is to complete these transmission within 25 minutes of the completion of the CDA contact). Automated systems at CEMSCS then process this data as quickly as possible, producing weather products that are available to weather forecasting, and environmental science organizations around the world.

DSN 34 m antennas and the Air Force Satellite Control Network (AFSCN) provide contingency support as required.

Ramaty High Energy Solar Spectroscopic Imager (RHESSI)

Mission Summary: NASA Small Explorer mission to explore the basic physics of particle acceleration and explosive energy released in solar flares

Launched: 2002

Orbit: 600 km circular, 38 deg inclination

Ground System Design: Primary data recovery is through the automated, multi-mission 11 m Berkeley Ground Station (BGS) antenna located at the University of California, Berkeley. The BGS antenna support to RHESSI consists of ~6 contacts/day at ~9 min/contact. 11 Gbits/day of data are recovered using a downlink rate of 3.5 Mbps during BGS contacts. Data on the spacecraft state-of-health and recorder playback status are transferred in real-time from the BGS to the Berkeley Mission and Science Operations Center (MSOC) where a multi-mission, automated ground system assesses the incoming data and alerts flight operations personnel of any detected spacecraft or ground system anomalies. Recorder playback data is captured at the BGS and is transferred to the MSOC post-pass using a secure FTP push. The MSOC performs level-0 and above processing and generates science products for the RHESSI science team.

If high levels of solar activity generate larger than normal volumes of data or if high winds or other issues make the Berkeley Ground Station antenna unavailable, the spacecraft can communicate through NASA's NEN ground stations at Wallops Island, VA and Santiago, Chile are used as back-up. When using Wallops, RHESSI data is transferred post-pass to a NEN central data storage facility at GSFC where an email is then sent to the MSOC alerting that a file is ready for transfer. The MSOC then initiates an FTP transfer of this file over the open internet.

Radio Aurora Explorer (RAX)

Mission Overview: University of Michigan / SRI International joint venture 3U CubeSat designed to study large plasma formations in the ionosphere that can disrupt communications between Earth and orbiting spacecraft.

Launched: 2010

Orbit: 650 km circular, 72 deg inclination

Ground System Design: Existing ground radars are used to transmit high power signals into the ionosphere where they are scattered. This scattered signal is detected and analyzed by the RAX spacecraft as it orbits. The resulting data is then downlinked to over 20 existing amateur radio operator sites around the world that have volunteered to collect and process RAX data. By operating at 437.5 MHz (FM) at 9600 baud, amateur radio operators can capture the signal while RAX is overhead. The University of Michigan provides interested radio operators with Beacon Decoding software that they can use to process RAX data and then ship it back to the RAX Science Operations Center

28.5 Technology Trends

The future of spacecraft ground systems is focused on increasing the volume of data that spacecraft can transmit to the ground over their operational lives, reducing the time it takes to get the data from the spacecraft to the end user, and increasing the interoperability between ground systems from different countries and manufactures.

28.5.1 Shorter Wavelength Communications

As modern spacecraft collect ever increasing quantities of data, transferring this information to the ground fast enough to free up on-board memory before it fills to capacity becomes increasingly challenging. This challenge is particularly severe for Earth orbiting spacecraft that are trying to send their information directly to ground stations where contact periods can be short and contacts infrequent. To address this challenge, many spacecraft are moving toward higher frequency communication bands combined with advanced modulation and encoding schemes to increase their maximum downlink data rates. In addition, increasing use of lower frequency bands for a variety of Earth-based communication applications provides another incentive for spacecraft developers to consider using higher frequencies for communication links.

Table 28-3◀, Fig. 28-11◀, Eq. 28-2◀

For many science missions, short wavelength K_a-band communication is a consideration. K_a-band has the potential for downloading data from LEO satellites at rates above 3 Gbps. Moving up into K_a-band for direct-to-Earth communication links does come with the challenge of overcoming atmospheric attenuation, particularly when there is rain in the transmission path. For this reason, some ground stations have been situated in low rainfall areas (For example, NASA's LRO downlinks K_a-band data to White Sands, New Mexico) or downlink periods have been limited to higher spacecraft elevations above the antenna mask to minimize the path distance, and associated attenuation, through the atmosphere. Another approach to K_a-band direct-to-ground transmission links in the presence of varying atmospheric attenuation is to use adaptive link techniques. Adaptive systems either vary the data rate, modulation scheme, or encoding scheme to maintain a positive link margin. Adaptive systems use constant feedback from the receiving to the transmitting antenna to establish the link margin. Based on this feedback, the transmitting system adjusts specific link parameters to maintain a positive link margin. K_a-band is also gaining popularity for space-to-space communication where atmospheric attenuation isn't a concern (for example, NASA's TDRSS constellation supports both K_u-band and K_a-band communication with user spacecraft).

Moving beyond K_a-band, research into communication using lasers is gaining popularity. Optical Communication can potentially return 10 times to 100 times more data with 1% of the antenna area of current RF systems, while requiring less communication system mass and power on the spacecraft side. Optical communication can be used for LEO, planetary and deep space applications and can be used to support direct-to-ground as well as space-to-space links. Since optical communication wavelengths are 3 to 5 orders-of-magnitude shorter than typical RF wavelengths, the beamwidths are much narrower than RF systems with a large fraction of the power concentrated in a narrow beam. Narrow beamwidths impose the need for high precision pointing between sending/receiving antennas to maintain a positive link during spacecraft contacts.

28.5.2 Reduced Latency

The operational applications for satellite data has grown significantly over the years. Today's satellites perform a variety of missions including serving as communications relays, monitoring the Earth's weather and climate, and collecting data related to all sorts of natural and man-made disasters (tsunamis, earthquakes, forest fires, nuclear reactor accidents). As we have become increasingly dependent on satellites to support decision making on, for example, the deployment of troops in a combat theater or the deployment of search and rescue crews in a disaster zone, reduction in latency requirements has been a consistent theme.

For some applications, such as hurricane tracking, satellites in geosynchronous orbits provide a platform for continuous monitoring over some large portion of the globe. In these cases, strategically placed ground stations can provide a continuous, almost instantaneous, link between satellites and the customers who require their data. NASA's Solar Dynamics Observatory (SDO) launched in 2010 would be a good example of this type of satellite. The SDO ground system architecture (shown in Fig. 28-12) includes two dedicated 18 m K_a-Band S-Band antennas located in White Sands, New Mexico. These antennas both receive the same signal and ground system software is used to select the best source for processing and sending forward to the data customers. This architecture provides a fault tolerant system with a very high data capture percentage and low latency since solar observation data is constantly streaming in from the satellite, the best source is selected quickly, and data is shipped out to customers with minimal latency. Since solar events, such as flares and coronal mass ejections, can have a significant impact on the operation of the world's satellite fleet and on power grids and communication systems on Earth, rapid knowledge of evolving solar conditions can be critical to minimizing the damage caused by major solar events.

Satellites orbiting below geosynchronous orbit pose a greater latency reduction challenge since their groundtracks cover a larger swath of the Earth's surface over time, requiring either numerous ground stations or a geosynchronous space-based relay system to allow for low latency operations. Human Spacecraft such as the International Space Station and the Space Shuttle utilize the TDRSS space-based relay network to provide low latency operations. Since each TDRS has an >120 deg of longitude view, a constellation of these satellites can provide, as long as they have adequate capacity, continuous links between numerous LEO/MEO satellites and the ground. These systems have a number of challenges in supporting low latency operations, such as limited space-based equipment/capability limiting the number of active satellite links they can support at any given time and the high cost of operating a space-based relay network. As an alternative, the majority of LEO polar orbiting satellites use a store-and-forward architecture which relies on ground stations spread along their groundtrack to meet latency requirements. A good example of a ground-based architecture being designed for low-latency for polar orbiting spacecraft would be the Receptor Net architecture originally designed as part of the National Polar Orbiting Environmental Satellite System (NPOESS) Program. Even after the decision was made to dissolve the NPOESS program and to return to a mode where civilian and military LEO weather satellites would be developed independently (similar to former POES and DMSP programs), the Receptor Net ground system remained as a common component to support the low latency requirements of both programs. The Receptor Net architecture included 15 dedicated ground stations (called receptor sites) spread out on all 7 continents. Working in K_a-Band to reduce the possibility of RFI when downlinking around the world, civilian and mili-

Table 28-3◀, Fig. 28-11◀, Eq. 28-2◀

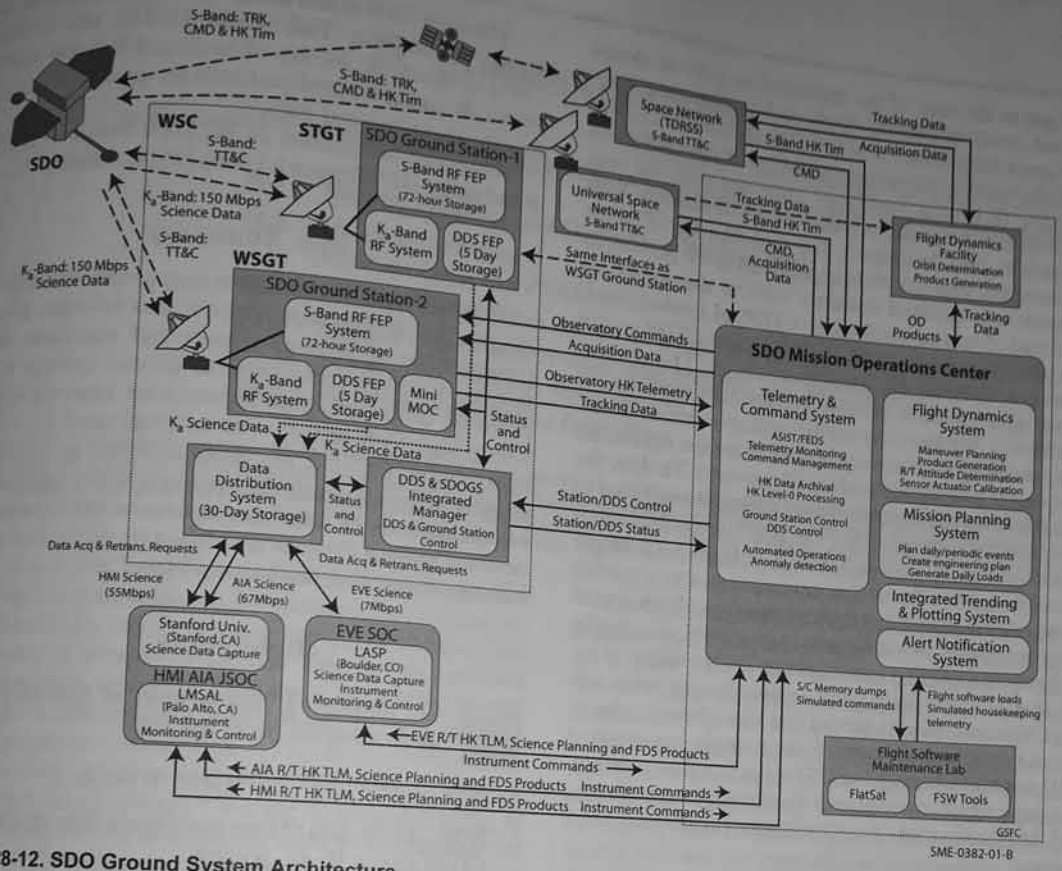


Fig. 28-12. SDO Ground System Architecture.

tary LEO weather satellites dump stored data at each receptor site (with a second dump of this data as well as newer data being performed at the following site to increase data capture probability). With this architecture, latency of between 15 and 30 minutes from the time a weather image is collected until it is in the hands of the end-user are achievable. Although this architecture requires a significant investment in receptor sites and interconnecting infrastructure, mission requirements for rapid collection and distribution of weather data from polar orbiting satellites could not be achieved by using the existing NOAA CDAs and the AFSCN networks that had previously supported LEO weather satellite contacts.

28.5.3 New International Standards—Increasing Interoperability and Data Capture Percentage

As an ever increasing number of satellites are placed in orbit by an ever increasing number of countries, international standards in the ground system interfaces and space/Earth communication protocols have been evolving to support these mission needs. In addition, standards are being developed and deployed to move the space/Earth communication link toward an architecture more similar to ground-based internet protocol where data accountability is more automated and transparent to the end user to ensure high data capture percentages. As discussed in Sec. 28.2, SLE and CFDP are two such international standard protocols that have been implemented in support of a number of missions. CCSDS continues to work on developing new standards that can benefit spacecraft operations. A primary example of this work is DTN.

28.5.3.1 Delay (or Disruption) Tolerant Networking (DTN)

Unlike most Earth-based communications links where establishing and maintaining connectivity throughout a data transfer session is straightforward, communication with spacecraft over great distances provides a challenging environment where connectivity may be lost or recovered multiple times while data is being transferred to the ground. Originally conceived as the "Interplanetary Internet", DTN makes use of store-and-forward techniques within the network in order to compensate for intermittent link connectivity. Compared to the internet which is a "connected" network where low latency (milliseconds) and low bit-error-rates (BER) allow for the end-to-end path to be established/verified/maintained throughout a data transfer session, DTN operates as a "disconnected" network that must perform in a high latency (for space applications, for example, 1.7 sec one way path time to the Moon, 8 min to Mars), high BER (caused by solar radiation for space applications) environment with intermittent connectivity (caused by blockage between spacecraft and Earth during some periods when planets, moons, and the

Table 28-3◀, Fig. 28-12, Eq. 28-2◀

Sun get in the way of the transmission path or shared ground station resources). Evolved from the retransmission procedures of the CCSDS CFDP protocol, DTN uses custody transfer, where responsibility of data segments (bundles or bundle fragments), migrates with the data segment as it progresses across a series of network hops to reach its final destination. The goal, like for CFDP, is to ensure that data sent from a spacecraft to the ground, or vice-versa, is received correctly without the additional overhead of manual verification. CFDP, as implemented, is a point-to-point protocol. A DTN-based CFDP would provide a multihop file transfer capability. The convergence layer protocols provide the standard methods for transferring bundles over various communication paths. The bundle agent discovery protocols are the equivalent of dynamic routing protocols in IP networks. To date, the location of bundle agents has been managed analogous to static routing in IP networks. The goal is to establish end-to-end data transfer through a series of hops. An example would be a Mars Rover that communicates to a surface based relay terminal, that communicates to a Mars orbital relay spacecraft, to a DSN antenna on Earth, and finally to the rover's mission control center. DTN could also apply to a Medium Earth Orbit communication relay network that uses various ground and space based nodes to transfer data between users. Even a single spacecraft in LEO can take advantage of DTN to reduce the operational complexities involved with handovers, rate buffering, and data distribution. The advantages of employing DTN include:

- Reduction in manual data delivery verification
- The ability to rapidly "erase" data from spacecraft memory by providing a reliable store-and-forward custody transfer protocol
- Application of a prioritization scheme to ensure that high value data can take precedence as bandwidth becomes available
- Delay tolerance where bundles can be produced without concern for when they will be transmitted (includes a "time to live" field that allows old data to be erased from memory if it is not able to be transmitted within a defined timeframe)
- Long haul retransmission of non-file data (CFDP is based on file transfer)
- More rapid retransmission request since each custody transfer can result in a request for retransmission (for example, data not arriving at a DSN ground station can be requested for retransmission instead of waiting until the mission control center identifies that the data is missing)
- A standardized method for store and forward and rate buffering data terrestrially

Although DTN is still being developed towards an Internet Engineering Task Force (IETF) and CCSDS [2011c] standard, the set of store-and-forward bundle protocols and associated protocols are quickly advancing toward an operational state.

Additional information on the DTN Protocols can be found at the DTN Research Group [2011] website.

28.6 Summary

Ground system architectures vary greatly based on the mission link-budget requirements, data volumes, latency requirements, QoS requirements and numerous other factors. A constant evolution of ground systems is focused on enabling higher transmission rates while providing high QoS service with minimal need for human intervention. CCSDS protocols continue to evolve to support these advanced requirements while maintaining standardization of interfaces to ensure interoperability between ground systems developed by individual countries or commercial service providers.

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29 Mission Operations

with support from members of the AIAA Space Operations and Support Technical Committee*

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29.1 Mission Planning and Operations Development
29.2 Mission Execution
Mission Operations Processes
29.3 Mission Termination and Post-Mission Activities
29.4 Mission Operations Process Improvement and Best Practices
Process Improvement; Best Practices
29.5 The Future of Mission Operations

* I would like to acknowledge the support of Joachim Kehr (DLR, retired), J. Jeffrey Devine, and Richard Campion (Honeywell Technology Solutions, Inc.), and the following members of the AIAA Space Operations and Support Technical Committee: Franz Newland (Com Dev International), Patricia Klein (Naval Research Laboratory), David LaVallee (Johns Hopkins University Applied Physics Laboratory), Carrie Olsen (NASA Marshall Space Flight Center), and David Welch (LASP/Honeywell Technology Solutions, Inc.).

It is possible to design and build the best possible spacecraft and even launch it into space, but it is useless unless there is a way for it to accomplish its mission. This is the role of *mission operations*. Contrary to a popular misconception, mission operations is not limited to what happens in the Mission Control Room or even the Mission Operations Center (MOC). That is really only the tip of the iceberg. Mission operations includes what happens on the spacecraft or launch vehicle, at ground stations, in engineering offices and science labs, to accomplish the goals of the mission, from design and development through mission execution. It is an integrated system of people, hardware, software, and activities that have to work together to ensure the successful execution of the mission. All organizations that fly space missions, from government agencies to industry and academia, perform mission operations activities. They nonetheless have different requirements for mission operations and often have their own philosophies and methodologies for developing and implementing mission operations. What is presented in this chapter are the methods that have worked for myself and for others from all types of missions—manned, robotic, Earth orbit, deep space, small, large, single and multiple satellites.

Mission operations can be categorized into four basic functions: spacecraft or launch vehicle operations, payload operations, ground operations, and mission management. Mission operations also vary by the phase of the mission life cycle and are divided into two fundamentally different modes separated by the launch. Mission operations design, development, and testing occurs during the study (Phase A), design (Phases B and C), and the assembly, integration, and testing (Phase D) phases of the mission. Mission operations execution occurs during the flight phase (Phase E) and termination phase (Phase F).

The following definitions of the four basic mission operations functions are based on those by Kehr [2007]:

Spacecraft Operations/Launch Operations covers the preparation and implementation of all activities to operate a space vehicle (manned and unmanned) or launch vehicle under nominal, non-nominal and emergency conditions. This includes the specification, design, production and qualification of all means (tools, procedures, and trained personnel) to perform the task of spacecraft/launch operations. It also involves designing operability into the space segment. The main challenges in this area are the cost-efficient combination of tools, degree of automation (for both space and ground segments), and staffing to provide secure and reliable operations. A very prominent role is played by the mission database containing all pertinent spacecraft and ground operations data parameters to be maintained throughout the mission. It is initially created by the spacecraft designers and handed over to the operations personnel during Phases C–D to be augmented by the specific ground operations parameters.

Payload operations cover the preparation and implementation of all activities related to the payload,† which is generally the primary reason for the mission. Details of typical spacecraft payloads are covered in Chap. 15 and not repeated here. Payload operations differ from spacecraft operations in that unique mission-specific expertise may be required to make decisions with regard to tasking the payload and retrieving and interpreting its data. That being said, for small spacecraft and relatively simple payloads, the payload operations are often included as another subsystem within spacecraft operations. In this case, the spacecraft operations team may retrieve and distribute the payload data to clients directly, without the intervention of a dedicated payload operations team. Large spacecraft, on the other hand, often have complex

† Although the payload is usually contained within the spacecraft bus, sometimes it might be separate, such as a separable probe or inspector vehicle.

and independent payloads, such as science instruments, requiring particular mission-specific expertise and warrant having their operations treated separately from that of the spacecraft bus operations. These missions would have separate spacecraft and payload operations centers with separate, but coordinated, operations teams. Payload operations may also include dedicated receiving stations for payload data. It is also possible to have simple payloads that are operated independently of the hosting spacecraft for organizational reasons.

Ground operations covers the design, implementation, qualification, operations and management (including configuration management) of a mission-dedicated ground segment, including infrastructure components such as communications networks, antennas, control centers, simulation test beds, and databases. The main challenge in this area is the implementation and operations of high-quality, cost-efficient, and secure space-to-ground, space-to-space and ground-to-ground communications respecting a multitude of different standards and protocols across a suite of related but distinct hardware and software components. This also includes the procurement of new equipment which might turn out to be a major task if complex, new subsystems or systems have to be installed respecting internally/externally regulated procurement rules and interfaces.

Mission management covers all tasks for preparing and operating a particular mission. The main talent expected is an expert level of technical understanding of spacecraft and ground operations requirements resulting in the planning and execution of all defined activities within a given schedule and within the available budget. Managers also should have a good understanding of the organizational influences on spacecraft and ground operations to minimize risks of mishaps and accidents and the behavioral aspects of human error in the conduct of the mission operations. For cooperative international missions, managerial skills in dealing with international partners and agencies and intimate knowledge of the "culture" and policy of a particular organization or agency are considered to be mandatory.

Although mission operations uses hardware, software (both space and ground), and facilities (often collectively referred to as the *operations infrastructure*) to carry out its functions, this chapter will examine the operational aspects (e.g., techniques and methods, procedures, management, use of tools) and not the infrastructure, which is covered either under the chapters on the spacecraft bus subsystems (especially flight software) or the ground system.

Although this chapter deals primarily with the operations supporting unmanned missions, most of the principles and techniques discussed are valid for both manned and unmanned missions. Some differences between the two are mentioned in the text when they are of particular significance.

The misconception that mission operations is only what happens in the MOC and ground stations after launch has been an institutional misconception since the early years of the space age. Many space mission managers consider mission operations only late in the development cycle and sometimes do not get serious about it until approaching the launch date. After all, if the engineers could control the spacecraft during integration and testing, it must be even easier to control it during flight when all the bugs have been eliminated. However, operational failures occur during flight even to the point of mission failures. This points to a difference in philosophy between the spacecraft designers and the operators. The former, are required to demonstrate it will work, while the latter, needs to prove they can control it, should it fail. Many flight operations teams discovered after launch that the spacecraft was not designed to facilitate ease of operation or recover from failures, thus reducing efficiency and increasing their workloads with the resultant increase in recurring operations and maintenance costs. Sometimes the necessary operation tools had not been developed prior to launch and had to be developed afterwards.

Often, spacecraft design decisions impact significantly how operations are conducted. Project managers during spacecraft development quite often, for cost pressure reasons, will go with less operable design choices (saving a few millions of dollars or months of schedule) but these decisions can drive up the cost of operations (which is usually not their responsibility). It is sometimes better to have a slightly more expensive design that will allow for cheaper operations. This means looking at the total life cycle costs that include both the non-recurring development costs and the recurring costs (mostly operations).

To mitigate the problems caused by ignoring or minimizing consideration of mission operations until late in the development cycle, the discipline of mission or spacecraft *operations engineering* has become more widespread in recent years. It is now generally recognized that operations need to be closely involved with the conceptual and detailed design of the spacecraft as well as the ground systems [Ryan and Fatig, 1993].

Operations engineering is a part of systems engineering focused on incorporating operational experience, knowledge, and insight into the development process to manage life-cycle costs (LCC) and operability. The goals of operations engineering are to reduce the LCC throughout development, testing, and operations of a system and to improve system performance and operability. Operations engineering provides a systems level view of operations throughout the system life-cycle [Alcott and Peters, 1996].

Operations engineers participate during each phase of the development cycle, and help the spacecraft or ground system engineers to design a spacecraft and supporting system that facilitates ease of operation and failure management after launch. Operational and contingency modes are defined, and guidelines for commandability and observability of on-board systems are used. It is not good enough to make a spacecraft that works—it has to be easy to operate under nominal conditions so that the chance of human error is minimized. The more complex

Table 29-1. Operations Characteristics for Different Mission Types. These ratings are subjective. Ops responsiveness means how flexible and timely operations have to be to handle changing situations (but not anomalies). Mod = Moderate

Mission Type	Ops Complexity	Staffing Levels	Autonomy S: Space G: Ground	Acceptable Risk Level	Data Volume	Data Importance	Ops Responsiveness
Manned	Very High	High: 24/7	S: High G: Mod	Very Low	Mod to High	Mod to High	High
LEO Science/RS	Mod to High	Low to Mod	S: Mod to High G: Mod	Mod	Low to High	High	Mod to High
GEO Comm	Low	Low	S: Low G: High	Low to Mod	High (PL)	High	Low
Deep Space Probes	Low to High	Low to Mod	S: High G: Low to Mod	Low	Very Low to Moderate	High	Low to High
Landers/ rovers	Very High	Mod to High	S: Mod to High G: Mod to High	Low to Mod	Very Low to Mod	High	High
Small Sats	Low to Mod	Low	S: Low G: Low to High	High	Low to Mod	Low to High	Mod to High
Launch Vehicles	Low to Mod	High	S: Low to Mod G: High	Low to Mod	Low	High	High

or "time dense" the operating procedures required, the greater the chance of errors. Also, as autonomy is used more often in the spacecraft operational design and the MOC uses automated processes and procedures, the impact of these complex procedures can be minimized. As long as the operations are well-defined or characterized so that they can be automated, then human error is mitigated or significantly reduced (but not eliminated).

The mission operations paradigm is not a case of "one size fits all." There are substantial differences in the mission operations required for manned versus robotic missions, geostationary communication satellites versus science or remote sensing satellites, LEO satellites versus deep space probes, satellite constellations versus single satellites, large versus small satellites, and in their own unique categories, launch vehicles and planetary landers/rovers. Table 29-1 shows the typical operations characteristics of these various mission types.

Despite the wide range of vehicles and missions, there are still some functions that are common to all their mission operations. You have to communicate with the spacecraft (or payload), to monitor and analyze its condition and success in accomplishing its mission, and to send commands to it, whether in real-time or as time-delayed commands. To be able to accomplish this, there are many additional functions occurring behind the scenes to support the operations. The mission operations functions can be broadly separated into two categories: the discrete *operations functions*, and the *support functions* as shown in Fig. 29-1. This figure shows the eight functions that are common to most space missions. Although there are many opinions on what constitutes a function, the common functions identified in the following sections are in fundamental agreement with other operations functional descriptions [Wertz, 1999; Welch, 1996; Squibb et al., 2006]. To these can be added additional functions that are required by specialized missions. Three examples are shown in the figure: Launch Operations (which is sometimes considered separate from mission operations if it involves the actual control of the launch vehicle), Crew Operations for manned space flights, and Rover Operations for planetary rover missions. Other examples include rendezvous and docking operations, and entry/re-entry and landing operations.

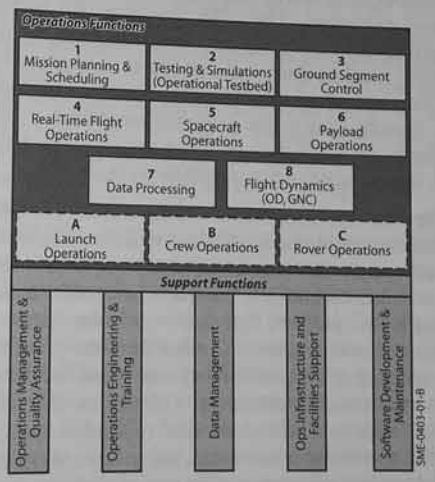


Fig. 29-1. Mission Operations Functions. There are eight typical mission operations functions, which perform distinct tasks during space missions. Although these functions are common to most missions they may be allocated together in some cases (e.g., 4–6), or alternatively, there may be other specialized functions (e.g., A–C) needed for some missions. These discrete operations functions are all supported by five major support functions, which are services typically provided to all of the operations functions.

The *Mission Planning and Scheduling (MPS)* function consists of the facilities, equipment, software, and personnel used to plan, schedule, and program the activities of the space and ground segments to ensure that the space mission and its objectives are met in the most efficient and effective way possible, given the budget and schedule constraints. The MPS function provides the master schedules for the spacecraft and ground network activities and deconflicts those schedules based on mission priorities, status of the spacecraft and ground network, and ground station availability. The MPS function generates the pass/contact plans that are passed to flight operations for configuring and performing real-time operations.† This function also creates the command loads for the spacecraft activities, although the command loads for payload activities may be provided to the MPS function by the payload customers, which is typical for many missions. The MPS function integrates the submitted payload command loads with the generated spacecraft command loads into a final command load set that may be verified using the Testing and Simulations function and then submitted to the Flight Operations function for upload to the spacecraft and subsequent execution. The MPS functions are normally performed in advance and not during real-time, although dynamic rescheduling and command load generation may be required, especially during checkout operations after launch. Due to the non-realtime nature of these functions, the MPS personnel can usually work regular work hours and are not required to work on shift.

The *Test and Simulations (T&S)* function consists of the facilities, equipment, software, and personnel used to support operations training and rehearsals, test command scripts before uplinking to the spacecraft, perform anomaly resolution, perform flight software maintenance, and test system modifications and enhancements. All of these tasks are done off-line so the personnel involved can usually work standard work hours, with the possible exception of mission simulations and rehearsals that may exercise operations teams over a day or days of typical or critical operations phases. The testing and simulations function often uses an *operational test bed (OTB)*, which integrates actual spacecraft or ground station hardware (e.g., brassboards, the flight segment engineering model) and flight software with environmental software models. The models may drive the space dynamics (trajectory and attitude) or thermal and power dynamics needed to run simulations, and often also include models of subsystems or components for which the hardware is not available. Software-only simulations are also common. Some programs have both (hardware-and-software test beds, and software-only simulation), others have one or the other. The decision of which to use depends on the availability and cost of hardware, number of instances required/desired (more instances typically drives towards a software-only solution), training requirements, acceptable levels of mission risk and technical considerations. The T&S function is used during spacecraft and ground system integration and test (I&T), and then operations rehearsals before and sometimes after launch. For this task, the T&S function typically interfaces with the MOC at command and telemetry frame or packet level, such that the T&S functions look like an actual spacecraft or ground station telemetry stream and responds to commands in some realistic or representative fashion. The hardware and software simulators can also be used after launch to test command loads prior to execution on board the spacecraft, especially for first-time activities. The T&S function is also used during flight to help with anomaly resolution and for flight software maintenance activities.

The *Ground Segment/Station Operations (GSO)* function consists of the facilities, equipment, software and personnel used to monitor and control the mission's ground network, including both the ground stations and the connecting communications network. The GSO function ensures that the ground segment assets can reliably support spacecraft passes, including tracking the spacecraft (antenna control) and collecting, transferring, and storing its data stream. The GSO function monitors (and as the situation requires and authority allows, modifies) the ground network contact schedule, and ensures that the network is properly configured in time to support each scheduled contact. As network anomalies occur, the GSO function is also responsible for real-time troubleshooting and implementing work around procedures to maximize the chances of contact success. Finally, the GSO function maintains a log of all activities for each contact and notifies maintenance or network engineering support function of system outages and problems that may require maintenance or repair. During nominal operations the GSO function may be authorized to configure, reconfigure or control the ground network. In these situations, other functions that are monitoring the network (e.g., real-time flight operations) may request configuration alterations. However, all these requests go through the GSO function, which may be solely responsible for implementing these changes. The GSO function is required whenever there are contacts with the spacecraft and is thus part of the real-time operations. Nearly all of the GSO functions supporting real-time operations can be automated, with human involvement only in monitoring the GSO performance and to intervene when certain anomalous situations occur. In many cases, the ground stations are also fully automated and do not even require real-time monitoring as long as they were configured beforehand to execute the pass. In environments with many concurrent mission operations activities, the GSO function is often shared between missions. While the ground segment configurations may be tied to a particular mission, the GSO *activities* are typically not mission specific.

† In Wertz [1999] the MPS function was broken up into two functions: *Mission Planning*, which was at a higher level more orientated to the payload (also performed the Ops Engineering support function), and *Activity Planning & Development*, which developed the activity plans and command loads/sequences. Either method is equally valid and is a matter of preference or experience.

The *Real-time Flight Operations (RTFO)* function consists of the facilities, equipment, software, and personnel that provide the real-time monitoring, command and control of the spacecraft in the MOC during ground contacts. This function monitors the spacecraft during the real-time contact, ensures that commands and uploads are successfully received and executed by the spacecraft, and that both state-of-health (SOH) and payload telemetry are downlinked as planned. This function is active during ground contacts with the spacecraft, and thus, is often a shift job if human monitoring and possible intervention is required or expected.† Recently, it has become common for smaller or less-complicated (e.g., communications satellite) missions to employ unmanned or "lights out" flight operations during routine spacecraft contacts. In these cases there is usually an automated SOH monitoring function that will alert operations personnel if an anomaly is detected.

The *Spacecraft Operations (SO)* function consists of the facilities, equipment, software and personnel that provide the engineering support for the spacecraft bus during the mission. The SO personnel are often either spacecraft engineers, or are members of the RTFO with support from spacecraft engineers that might only be on call or spend a small amount of their time in the SO function (for instance, if they are already working on developing the next mission). The SO personnel monitor and analyze the spacecraft SOH data contained in the downlinked telemetry to determine and ensure spacecraft safety and mission effectiveness. They use processed spacecraft SOH telemetry data and ground network data for analysis to detect any anomalous spacecraft subsystem or ground network performance, and to assess long-term trends. Trending analysis can help prevent malfunctions by noting and correcting potential problems before they reach a critical stage. The SO function also reports spacecraft status, and coordinates with spacecraft component manufacturers and integrators as required to solve anomalies. If the spacecraft requires some action to correct or maintain the welfare of its systems, or to optimize performance, the SO function may provide the data to the Payload Operations functions for them to assist in the determination of the action required. The SO function may also have partial responsibility for monitoring the health and status of the payload (if this is not done by the payload customer). The SO personnel form the core of the anomaly resolution team, and help the MPS function by specifying tasks required by the spacecraft bus. They are also available to monitor testing of any command loads if required. One of the tasks of the SO function is to generate regular status reports on the health and condition of the spacecraft, and possibly update such data on a website.

† In some operations centers, the RTFO may include the pre-contact (e.g., ground station preparation) and post-contact activities (e.g., console log reports), in others only activities actually occurring during the contact come under RTFO while all other related functions come under Spacecraft Operations (SO).

The *Payload Operations (PO)* function consists of the facilities, equipment, software, and personnel used to monitor and maintain the health and welfare of the payload instruments (or other equipment), to monitor the performance of the instruments, and sometimes any ground data processing and ground-based data quality measurements, in accomplishing the mission objectives. It may make changes in the payload and payload processing configuration as required. The PO function interacts directly with the MPS function to schedule and provide commanding and uploading of control data (e.g., calibration tables). The PO function also interacts closely with the SO function to help ensure the performance of the payload is within specifications. Depending on the nature of the observations, and the maturity and stability of the payload system, this function may or may not require participation in the RTFO, and thus possibly use shift work to match the contacts schedule. However, most of the PO function is usually non-shift work. In some cases, much of this function can be automated or may be combined with the SO function as mentioned earlier.

The *Flight Dynamics (FD)* function consists of the facilities, equipment, software, and personnel used to determine and predict the spacecraft's orbit and position in space, and also to determine or calculate its orientation (this function is sometimes done on board the spacecraft). After determining the orbit of the spacecraft using tracking data received from the ground stations (or other means), the FD function produces an ephemeris or state vector and distributes it to the rest of the operations team as required. If the spacecraft has a propulsion system, then the FD function plans and analyzes trajectory maneuvers. This includes deorbit maneuvers at the end of life, determining relative position of more than one space asset, and for rendezvous or constellation operations activities. Attitude maneuvers are also often planned and tested by the FD function. A growing task for FD deals with space debris conjunction analysis and avoidance. The FD function needs to compute likely conjunctions with other spacecraft or debris and determine trajectories/burns to ensure sufficient separation between spacecraft and debris while minimizing mission perturbation. Flight dynamics is usually a non-realtime operation, so the FD staff will generally be able to work regular work hours except possibly during maneuver execution activities.

For some projects, most of the SO, PO, and FD functions may all be combined together under a single "Mission Analysis" function, but these have been broken separately in this discussion to show how each of these three functions can be distributed yet complement each other.

The *Data Processing (DP)* function consists of the facilities, equipment, software, and personnel used to accept the flow of data (spacecraft telemetry or ground network status) received at the MOC or some other data processing facility, such as the SOC, process it to a usable or specified form (if required), archive the data, and dis-

Table 29-2. Description of Science Data Levels. These levels are used mostly for processing imaging and science data. Although they come from NASA, they are now internationally recognized.

NASA Level	Description
Packet Data	Telemetry data stream as received at the ground station with science and engineering data embedded.
Level 0	Instrument science data (e.g., raw voltages, counts) at full resolution, time ordered, with duplicates and transmission errors removed.
Level 1A	Level 0 data that have been located in space and may have been transformed (e.g., calibrated, rearranged) in a reversible manner and packaged with needed ancillary and auxiliary data (e.g., radiances with the calibration equations applied). Also referred to as Experimental Data Records (EDRs).
Level 1B	Irreversibly transformed (e.g., resampled, remapped, calibrated) values of the instrument measurements (e.g., radiances, magnetic field strength). Also referred to as Calibrated Data Records (CDRs).
Level 1C	Level 1A or 1B data that have been resampled and mapped onto uniform space-time grids. The data are calibrated (i.e., radiometrically corrected) and may have additional corrections applied (e.g., terrain correction).
Level 2	Geophysical parameters, generally derived from Level 1 data, and located in space and time commensurate with instrument location, pointing, and sampling.
Level 3	Geophysical parameters mapped onto uniform space-time grids.

tribute the data to the next function in the process. For the low-rate SOH data, the DP function typically converts the telemetry into engineering units that can be displayed by the real-time system or delivered to the engineering trending and analysis function (although with increased computing power on board the spacecraft, this function is sometimes done before the telemetry is downlinked). For the high-rate payload data, the DP function can perform pre-processing, typically to Level 0 (see Table 29-2) and deliver the data to the payload customer for further processing and analysis. The data processing function is not the same as data management, that is part of the support functions,† although in some architectures it can be part of the data management function (thus becoming the Data Processing and Management function). The DP function performs specific tasks in accepting, modifying, and outputting telemetry data. The data management function facilitates the transfer and manipulation of all data between the other functions and provides the master storage or archives used by the whole system. The DP function can be almost entirely automated, including the collection and reporting of data processing metrics, with alerts sent for anomalous conditions, although interpretation and use of the data usually requires domain expert involvement.

The *Launch Operations (LO)* function consists of the facilities, equipment, software and personnel that provide the engineering support for testing the spacecraft and ground network during pre-launch and post-launch tests and activities. It can also be defined to include the actual monitoring and control of the launch system itself, which generally uses a separate Launch (Operations) Team. Although this is a discrete function performed by the operations team for most missions, it is classified here as a specialized function because it is only performed for a limited time at the start of the mission and is not an ongoing function throughout the mission. In some operations plan, this is just a specialized phase of the operations for the other functions (such as SO, PO, DP) and not a separate function. If this is treated as a separate function, then it usually has limited automation and is performed by the operations personnel as required.

The following are the support functions, which provide their services to support the previous operations functions. They provide the five pillars of support for mission operations as shown in Fig. 29-1.

The *Operations Management & Quality Assurance (OM&QA)* function consists of the facilities, equipment, software and personnel used to provide overall management of the mission operations functions. The OM&QA function ensures that both the entire system and the individual subsystems are operating within specifications and are helping to accomplish the mission goals. Included in the system management function may be any hardware and software used to monitor and optimize the performance of the various interfaces between the MOC subsystems. The quality assurance function gathers performance metrics from key points within the system and the operational process, processes the metrics, analyzes the resulting data, and assesses mission risk. The results of the analyses are included in reports to the mission management, but recommendations for process improvement can also be a product of this function. The OM&QA functions can be highly automated with little human oversight or intervention, other than to implement process changes and make administrative decisions.

The *Operations Engineering & Training (OE)* function consists of the facilities, equipment, software and personnel used to perform the operations engineering functions for the mission such as procedure development, validation, activity plan development, and training of personnel.‡ Prior to launch this is the function that is mainly responsible for preparing operations for the mission. After launch, operations engineering maintains and improves the spacecraft performance, and continues to develop operations procedures as needed to support

† Telemetry and payload data processing functions may be separated, depending on the division of responsibilities between the spacecraft and payload operations groups.

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changes in the mission. In this operational phase, the OE function is sometimes known as the Maintenance and Sustaining Engineering function. This function also has the expertise to develop additional ground segment capabilities in response to the evolving payload requirements and manage ground segment obsolescence issues for long-term mission activities. Should a major upgrade be needed, the operations engineering function performs an analysis to determine the most cost-effective approach for the upgrade. The OE function also generates reliability, maintainability, and availability (RMA) statistics on the primary operational applications, and investigates process improvement potential of other software solutions.

The *Data Management (DM)* function, includes the data management server and the data archive. The DM function is not the same as the Data Processing function described earlier, although it supports that function. The DM function does not simply deal with spacecraft or ground network telemetry (SOH and payload data)—it also includes mission plans, command and communication histories, and records of communication between the mission team and the customers. It provides the data interfaces between the various functional elements of the operations system and handles the distribution and manipulation of the data. The DM function allows easy access to all mission data as well as any additional information about the origin, history, and use of the data. The customer interface may be as simple as a website where the payload SOH data, reports and communications from the mission team are made available to the customer. An interface tool (or website) may include a function for communication from the customer to the mission team; this includes requests (formal and informal) and constraints.

The *Operations Infrastructure and Facilities Support (OIFS)* function consists of the facilities, equipment, software and personnel used to monitor, control, and maintain the MOC Local Area Networks (LANs), ground network including ground stations, and the computers and processing equipment used to support mission operations, as well as, the physical facilities. This includes the communication interfaces with external elements including the Internet, any security, and information assurance constraints (e.g., firewalls) required to ensure the integrity of the MOC system. The OIFS function monitors the performance of the LANs and external communications networks, and makes adjustments to maintain data flow within the operational specifications. The OIFS function typically coordinates all vendor maintenance activities, and maintains the inventory of all applications and their licenses. Monitoring and routine operation of the OIFS function is often automated and does not require human intervention. However, after the acceptance test phase, the mission management usually maintains a staff to provide sustaining engineering and on-call maintenance of the system hardware and software and to make enhancements as necessary.

The *Software Development and Maintenance (SDM)* function consists of the facilities, equipment, software and personnel used to develop and maintain the software used by mission operations, whether writing new applications for the mission, or integrating commercial off-the-shelf (COTS), government off-the-shelf (GOTS), or legacy software into the system. The SDM function is responsible for the underlying software infrastructure that: (1) provides an open and accessible set of building blocks for software relating to spacecraft construction and operations; and (2) supports the higher level aspects of spacecraft construction and operations specific to the mission operations system itself. To accomplish this, the SDM function is usually an integrated system of software components, protocol definitions, and code management procedures. The SDM software usually includes a suite of functional libraries, on top of which are built various software engines and user commands. These can then be assembled, along with off-the-shelf or legacy components, to make various processes and graphical tools. The SDM function software development can also work on optimizing existing algorithms, and developing new algorithms. It also provides the procedures and protocols to ensure quality assurance and software uniformity and compatibility throughout the mission operations system. Once the operations software has been deployed and operational, the SDM function is responsible for maintaining and upgrading it as needed. It can also provide support for the flight software development and testing.

29.1 Mission Planning and Operations Development

Mission operations development nominally covers the period from mission concept until launch and the completion of the launch and early operations phase, although in practice, operations development continues throughout the mission. During the operations development phase, the mission operations development team (MDT) consists of the personnel from the operations engineering or various functions described earlier (SO, PO, GSO) and is partly or fully responsible for the following:

- Operations concept and plans
- Space segment operability specification
- Mission operations architecture definition, operations requirements definition and assignment to architecture elements
- Staff planning (numbers, roles, training requirements)
- Operational tools/facility development
- Operational procedures development

‡ In some organizations the Operations Engineering function is an umbrella function that contains the SO, PO, and GSO functions, each of which performs the tasks of OE described here for their own area of responsibility.

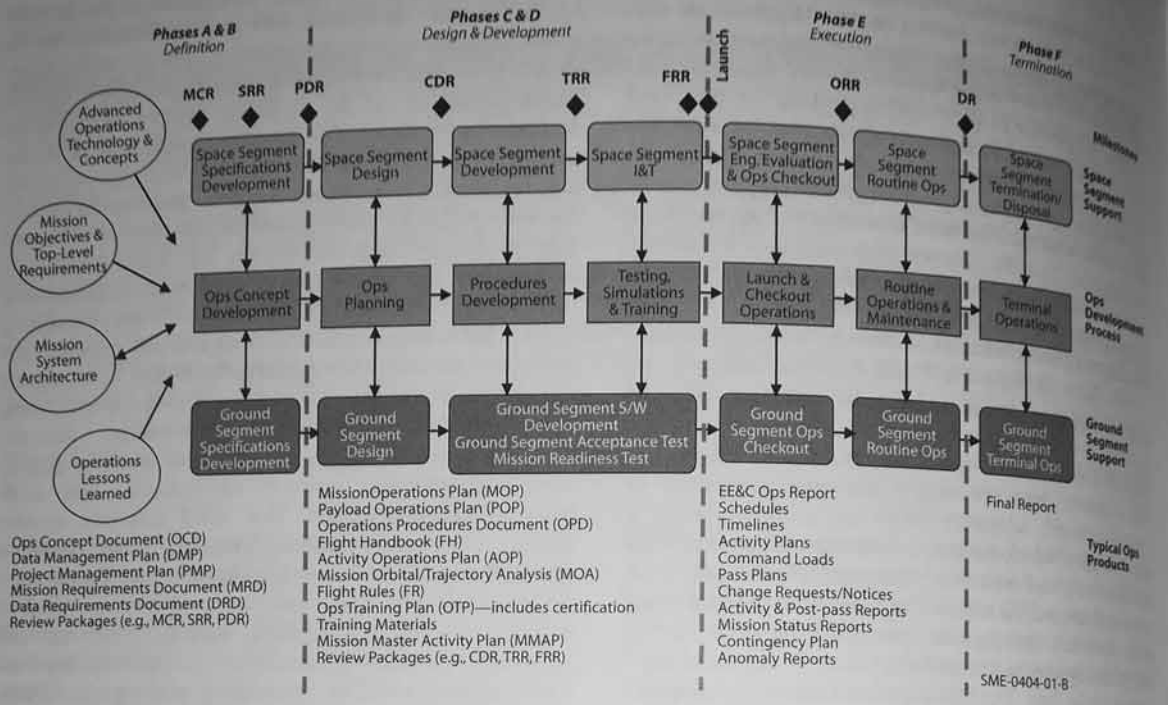


Fig. 29-2. Mission Operations Life Cycle. Mission Operations is an important part of the life cycle of any space mission. Inputs from various sources are integrated into an operations concept during the Definition Phase, then the design and implementation is completed during the Development Phase and is used to conduct the mission during the Execution Phase. The Termination Phase incorporates the mission-ending activities. This figure shows the interaction between mission operations and the space and ground segments. Also shown are the typical mission milestones and operations products during each phase.

- Spacecraft and ground system knowledge (understanding) capture
- Training of all operations personnel for launch and on-orbit operations
- Development of command procedures and telemetry displays
- Validation of mission database content (collectively operations products)
- Verification and validation of developed procedures and displays
- Validation of ground system capabilities to support the mission
- Support of appropriate ground system and spacecraft tests (hardware and software)
- Support of all mission and system operations reviews

Key phases of the mission operations development process and how these relate to project/mission phases is shown in Fig. 29-2. The four pre-launch operations phases (middle row) are: (1) concept development; (2) operations planning, (3) procedures development, and (4) operations test, simulation, and training. The three post-launch operations phases are: (1) launch and checkout; (2) routine operations and maintenance; and (3) terminal operations.

Table 29-2◀, Fig. 29-2, Eq. 29-0

The operational procedures and processes are tested, modified and supplemented as a result of actual operational experience during the Testing, Simulations and Training sub-phase. At the end of the Launch and Checkout Operations sub-phase (also known as Launch and Early Orbit—L&EO, or Engineering Evaluation and Checkout—EE&C), the operations development is considered complete and nominal routine operations started. The transition point into nominal operations is typically called the Initial Operational Capability (IOC).† However, operations development activities may continue, in order to optimize operational processes and procedures, develop or augment automation of the processes, to meet new or evolving mission needs including extended mission operations, apply lessons learned, and deal with failures or anomalies.

Mission development includes processes for influencing the spacecraft and payload development. This includes participating in space segment, ground segment, and operations trade studies early, providing inputs to requirements documents, and making recommendations

† For constellation ops, this can have a different meaning. For some constellations, mission objectives only starts to be achieved after a few assets (satellites) are deployed, at which point IOC is declared. Nominal operations on the first asset may have started some time previously, requiring other assets achieving nominal operations to achieve the constellation IOC.

regarding designs, especially with regard to operability (typically decomposed into commandability, observability, and autonomy). To ensure that the operations are efficient and cost-effective, it is important that the engineers from the operations team participate with the spacecraft and instrument development engineers throughout the life cycle, and especially to be active participants in the integration and test process. This will ensure knowledge capture for the operations team and also help the development of procedures that will enhance the operability of the spacecraft and ground system. This is a fundamental application of operations engineering discussed earlier. During this phase, processes and databases can be developed that will help with automating operations.

Another approach commonly used is a red-team review with external experienced reviewers, where the new team develops its concepts, approaches, and procedures, but pulls in agency, industry, and other experts, under suitable non-disclosure agreements and for a cost, to evaluate the team and its approach. With a new organization, such a red team can participate at gates throughout the process development. Often, agencies are ready to support such activities for cost-reimbursement only.

Mission operations prelaunch support is subdivided into the two basic processes that revolve around the two major system elements: the space segment and the ground segment. In general, during operations preparation, the spacecraft integrators have the critical role of preparing the space segment while the mission operations personnel act as subordinates to support these activities in their efforts to learn how to operate each subsystem. Conversely, it is the mission operations personnel who have the critical role in preparing the ground segment for the mission and the spacecraft integrators who are more peripheral. They provide the space segment components of the mission database (typically comprising such information as command, telemetry, and parameter definitions including parameter calibration information, space segment reference frames, mass, inertias, and sensor and actuator alignments), and they usually are a user of this system during mission rehearsals and early mission support (so there is a benefit to their knowledge and understanding of the ground segment). Spacecraft integrators also potentially deal with software updates or hardware troubleshooting on an as-needed basis. It may be an appropriate management strategy to ensure that the spacecraft/payload integration team members are provided roles to play during the routine operations phase, providing them the necessary expertise in an emergency situation. The size and tasking for this "standing army" should be considered against mission complexity, planned lifetime, cost, and risk tolerance at the mission management level. Similarly, during the mission definition and development phases of a project, the foundation for effective and efficient routine operations (the ultimate goal) may be laid by maximizing the OE team's exposure to the spacecraft and all its subtleties at the earliest opportunity. It is crucial that the OE team is formed with ready accessibility to the development facilities, personnel and processes starting no later than the space segment Preliminary Design Review. As the development of the mission transitions into the I&T phase, this OE team may increase its participation so that each position identified for ultimate inclusion in the post launch operations team has a chance to participate and fulfill their training requirements. During preparation, these operations personnel are often subordinate or observers to the space segment integrators and engineers while shadowing their activities and absorbing their expertise. Ultimately, the OE team will support spacecraft integration and test, procedure development, real-time team training and readiness activities as they ready themselves to be the primary operators of the spacecraft following launch and early orbit checkout. Note that in small organizations and missions, it may be that the integration team eventually becomes the *Mission Operations Team* (MOT), absorbing the spacecraft expertise by default. However, the core principle of considering operations from the start of the project, as illustrated in Fig. 29-2—should remain the same regardless. **Much frustration and disappointment can be avoided—and mission success better assured—if the team has experienced operations personnel involved from the beginning of the project.**

As shown in Fig. 29-2, mission operations typically develop various documentation products during each phase of the mission. Some of the more common products are listed in Table 29-3 along with their definitions. Although most of these documents are under the ownership of mission operations, some important products are listed which have operations as a contributor rather than the owner (e.g., Mission Requirements Document, Project Management Plan, Risk and Configuration Management Plans, review packages). It should be emphasized that large and complex missions will usually have many more documentation products than those shown, while for small, low-cost missions, several of the products shown can be eliminated or combined.

The milestones shown in Fig. 29-2 are some of the more common ones used in most space projects and are based on the NASA and European Cooperation for Space Standardization (ECSS) conventions. Phases A & B are the Mission Concept Review (MCR) and System Requirements Review (SRR). The transition between Phase B and Phase C is the Preliminary Design Review (PDR), which can be considered as either the end of Phase B or the beginning of Phase C. The transition from Phase C to Phase D is the Critical Design Review (CDR). A Test Readiness Review (TRR) is held during Phase D to ensure that the project is ready to enter the system integration and testing.

Once the testing is completed, a final Flight Readiness Review (or Mission Readiness Review, Pre-Shipment Review, or Launch Review) is held before launch. At the end of the checkout period of launch an Operations Readiness Review (ORR) or Commissioning Review (CR) is

Table 29-3. Typical Documentation Products for Mission Operations.

Document Title	Description
	Phases A & B (Definition)
Mission Requirements Document (MRD)	The mission operations top-level requirements and constraints, and the derived requirement (with traceability) are listed as a section in this document.
Operations Concept Document (OCD)	Provides a high level description of how the mission will be operated to support and accomplish the mission objectives. It includes operations in both space and ground segments.
Project Management Plan (PMP) *Project Implementation Plan (PIP)*	Ops provides an important section in these documents that are used by the project management to describe the project and how it will be managed. Some projects incorporate the PIP into the PMP. These documents also cover procurement procedures for hardware and software. If not included in the PMP/PIP, a post-launch Phase E/F Management Plan should be developed.
Configuration Management Plan (CMP)	This document provides the plan for managing the configuration of the project software, hardware, databases, and operations staff, and procedures.
Risk Management Plan (RMP)	The RMP focuses on identifying, assessing, planning for, and dealing with areas or events having a potential for causing unwanted results (i.e., risks). This may be included in the PMP.
Data Requirements Document (DRD)	Document describes the requirements of the data collection, processing, and distribution. For simple missions (from a data standpoint) this may be incorporated into the MRD.
Mission Software Plan (MSP)	Describes the design process, including staffing, schedule, and tasking, to develop and test the flight and ground software needed to accomplish the mission
Documentation Plan (DP)	A plan for the creation of the project documentation. The plan helps to ensure that this documentation is planned early in the project, all work involved in creating the documentation is identified, and ownership of both technical writing and development team input tasks is established. The DP includes a list and description of all the project documentation and provides an identification nomenclature and ownership of each document.
Project Review Packages	Ops presentation materials given at project reviews such as the MCR, SRR, and PDR (which is the transition between Phases B & C).
	Phases C & D (Design and Development)
Mission Operations Plan (MOP)	Addresses in detail how the mission operations will be developed, tested, and then used to conduct the mission including routine, L&EO, and contingency support operations. It includes a system description, operations overview (mission ops, ground ops), ops staffing & training, technology development plan, phase ops (prelaunch, on-orbit, mission planning & scheduling, pass execution, analysis & reporting, anomaly resolution), and data management. Includes ops management and improvement process.
Data Management Plan (DMP)	Document describes the design, process, and procedures to handle data (including SOH and payload) within the system, with emphasis on data received from the satellite and ground network.
Payload Operations Plan (POP)	This document is similar in scope to the MOP, but addresses the payload and payload data. It may be produced by the payload customer or may be incorporated into the MOP depending on the mission.
Operations Procedures Document (OPD)	Captures the operations procedures required to support (1) launch operations; and (2) mission operations. Contains the specific procedures to be followed for different activities, such as rebooting computers (flight or ground), anomaly identification and resolution.
Flight Operations Handbook (FOH)	Provides a general reference for operations personnel that summarizes the space and ground systems including schematics and specifications; mission timeline summary; standard and contingency operations definitions; contact lists; and other useful information as needed.
Activity Operations Plan (AOP)	Provides a detailed plan for specific planned mission activities (usually payload related) such as an experiment, systematic mapping, observation of a target event, flyby observation of a target, or deployment of a payload. The AOP provides background, analyses, timelines, instrument settings, procedures, data management plans.
Mission Orbital/Trajectory Analysis (MOA or MTA)	Provides orbital and/or trajectory analyses for the mission, including predictions of orbit changes, lifetime, altitude maintenance requirements, maneuver specifications, and lighting analysis.
Flight Rules (FR)	Lists the rules to be followed by the mission managers for various nominal and off-nominal conditions that could be encountered during the mission. Provides conditions under which the mission or activities should be aborted or modified. For small missions, the FR is not a separate documents but is usually incorporated in the FOH.
Technology Development Plan (TDP)	Describes the methods planned to develop the technology needed to complete the mission.
Operations Staffing and Training Plan (OSTP)	Describes the plan to staff, train and certify mission operators to support the mission. May be included in the MOP.

Table 29-3. Typical Documentation Products for Mission Operations. (Continued)

Document Title	Description
Contingency Plan (CP)	Describes the contingencies for which recovery operations have been designed and developed (such as safehold recovery, how to swap redundant units, how to implement workaround procedures to bypass failures.)
Mission Master Activity Plan (MMAP)	Describes all the major discrete spacecraft and payload activities that are expected to occur during the mission and a timeline of when they are planned to occur, or windows of when they could be scheduled. Each activity has information concerning its goals, the responsible person (e.g., PI), any special power, communication, or other requirements or constraints to aid the mission planners.
Training Materials	Any handbooks, workbooks, or presentations used to train the ops team. Development and updating of these materials can continue into Phase E.
Project Review Packages	Ops presentation materials given at the CDR (which is the transition between Phases C & D) and other reviews such as the TRR and FRR.
	Phase E (Execution)
EE&C Ops Report	Documents the performance of the spacecraft and supporting systems during the engineering evaluation and checkout period after launch.
Schedules	Long-term and short-term schedules to show upcoming events for the mission including communication passes with ground stations.
Timelines	Show the spacecraft and/or ground station activities as a function of time (typically one orbit or one day depending on mission). Often shows other information such as orbit position or lighting condition (umbra periods).
Activity [Operations] Plans (AP or AOP)	Detailed descriptions of planned events and activities for the spacecraft with information concerning resources required, tasking, timelines, commands, data handling, and instrument settings.
Command Loads	The actual command sequence uploaded to the spacecraft for execution. These are traditionally time-delayed commands that are triggered by specified execution times.
Pass Plans (PP)	Lists the communication pass objectives coupled with specific files, keystrokes, and verifications that the operator (or system) must complete within the pass. May take the form of electronic files, paper printouts, and/or handwritten directives.
Change Requests (CR) Change Notices (CN)	A formal record of the operational procedure changes maintained by each flight controller. All CRs & CNs will be stored in a database and used to tracking the evolution of mission operations and to capture lessons learned.
Activity Reports (AR)	Reports the results of mission activities (such as payload experiments/observations). This is usually an engineering report rather than a science report, i.e., does not analyze the results of the experiment, just the ability to execute it.
Post-Pass Reports (PPR)	Describes what happened during a communication pass including listing any files transferred, S/C or GS anomalies detected, time of AOS and LOS.
Mission Status Reports (MSR)	Periodic (usually daily) reports on the status of the mission, including spacecraft bus, payload, and ground network SOH, activities and results, anomalies, and progress towards meeting objectives.
Anomaly Reports (AR)	The written explanation that follows an anomaly investigation detailing how and why an anomaly occurred, and the plan adopted to resolve or minimize the effects of the anomaly.
	Phase F (Termination)
End-of-Mission/ Decommissioning Plan	Describes the plan and procedures to terminate the mission.
Final Report	Final comprehensive report on the mission and its operations prepared for the customer or for the historic record.

held to ensure that the mission system (space segment, ground segment, customers) are ready for routine or primary operations. At the end of the life of the mission, if the spacecraft has to be actively terminated (decommissioned), a Decommission Readiness Review may be held.

Developing the Mission Operations Plan

The Mission Operations Plan (MOP) is the defining document for mission operations development, which describes what mission operations will entail, what are its objectives, how it will be accomplished, and lays out a detailed plan for the development, testing, and application of mission operations. The MOP describes, in operator and user terms, the operational attributes of the flight and ground-based elements of the mission. However, the MOP is not just for the mission operators and users. It is used by the mission management and organization as a tool to understand and thus effect changes in mission operations or other aspects of the mission.

Although Fig. 29-2 shows the MOP as being a product of the Development Phase (Phases C & D), it really begins with the development of the Operations Concept Document (OCD), the data and software plans, and the

Table 29-4. Process for Developing a Mission Operations Plan. These steps are summarized in the text and explained in more detail on the book website.

Step	Notes
1. Identify top-level requirements and constraints that affect mission operations, ops lessons learned, and advanced technologies & techniques for operations	• Determine constraints (e.g., cost, schedule, teaming) • Ops requirements come from various mission areas: • Orbit • Communications • Payload (including data latency and tolerance for data losses) • Ground network
2. Develop mission operations concept and supporting architecture	• May have multiple mission ops concepts during Phase A or basic changes in concept during Phase B
3. Develop ops scenarios and techniques that accomplish mission concept	• For each scenario develop sample timelines to ensure feasibility of scenario
4. Determine ops functional activities required in Step 3, allocate to functional areas, and develop functional flow block diagram	• To satisfy mission concept identified in Step 2 • Basic functional areas are: Mission Planning and Analysis, Real-time Operations (Contact Execution), Data Management, Trending and Analysis
5. Identify ways to implement functions identified in Step 4	• Assesses state-of-the art and legacy solutions that are available with reference to previous similar missions
6. Determine level of automation for both space and ground segments	• Determine how much of the processing and control will be done autonomously on the spacecraft, how much at the ground stations, and how much in the MOC • Determine if the MOC and ground station functions can be automated or require personnel, given the mission cost and schedule constraints
7. Determine whether capabilities identified in Step 5 and 6 exist, are obtainable or must be developed.	• Are these resources attainable or do they need to be developed?
8. Develop staffing plan and identify other resources required	• Determine the operations and support organization including roles (positions) required and level of experience needed • Determine if shifts are needed and work out the shift plan • Determine number of personnel required to support operations (MOC and ground segment) • Determine the staffing profile (rate at which personnel are brought on-board) • Determine what hardware and software resources are required
9. Perform trades to determine best solution from Steps 5–8	• Compares availability, performance, and cost (non-recurring and recurring) of various options identified
10. Determine the operations Work Breakdown Structure (WBS) and ROM mission operations cost	• This is an important element in determining the viability of the current mission concept. Cost is typically based on: • WBS • Staffing requirements • Hardware & software procurement • Hardware & software development • Facility/infrastructure development or sustainment for existing facilities
11. Repeat steps 4–10 for development phase (Mission Phases A–D as appropriate)	• If the initial Phase E analysis was just completed, then the developmental phases also need to be analyzed
12. Assess mission utility, complexity, risk, and cost drivers	• This is to help select the best mission concept for the baseline by determining which best meets the mission objectives
13. Repeat for alternate mission concepts (starting with Step 2) if required	• This is skipped if the baseline mission concept has been selected
14. Identify derived requirements	• This is done to the selected baseline mission operations concept
15. Develop training plan	• Plan to train and certify flight ops personnel and rehearsal plan
16. Generate MOP that includes technology development plan, personnel staffing and training plan, and documentation plan	• MOP includes the processes governing operations development and validation • Technology development plan identifies the ground segment components that need to be developed, such as databases, or planning software. • Personnel staffing and training plan identifies the positions anticipated during operations and the certification plan for those positions. • Documentation plan identifies the documents that are needed (e.g., Flight Handbook) and how to develop them
17. Document and iterate/refine as needed	• Publish the MOP and revisions as needed

Mission Requirements Document development in the Definition Phase (Phases A & B). The MOP is built upon the foundation formed by these earlier documents (see Table 29-3 for a more detailed description of these documents). Table 29-4 outlines a 17-step process to develop the MOP starting with the top-level requirements and the OCD. This is an expanded and modified version of the methodology that was presented in [Wertz, 1999]. Note that the development of the MOP is an iterative process. The plan will necessarily be modified as the project matures and the nature of the space and ground elements is better understood, and thus how to operate them to best fulfill the mission objectives.

Step 1. Identify Top-Level Requirements and Constraints that Affect Mission Operations, Ops Lessons Learned, and Advanced Technologies & Techniques for Operations

This first step is required to formulate the operations concept (Step 2). A new mission initially starts with a mission statement that defines the purpose of the mission. This statement is then parsed into achievable primary and secondary objectives. The mission objectives define what the mission elements (such as spacecraft, crew, ground segment, and payload users) must do to successfully accomplish the mission. After the objectives have been defined, they are normally assigned success criteria to provide a means to measure the relative success of the mission. (See Sec. 5.4.) These success criteria help in the overall mission design to identify levels for which the system can be designed to achieve steps of partial success on the way to full mission success.

The mission objectives lead to an initial mission concept or description that presents in broad terms an overview of the mission, including some aspects as trajectory profile, launch windows, payload type, mission phases and duration. The mission definition process then determines the top-level requirements and constraints (see Chap. 14) to help define the mission so that its objectives can be achieved within the project constraints. Note that this step is initially iterative with Step 2, since only the top-level requirements derived directly from the mission objectives will be available. These are not enough to define all the information needed to proceed on to Step 3—first you have to have a mission concept to determine the top-level requirements within the various system and subsystem areas. There will be various mission concepts considered, and these may have different requirements and constraints on the system. When information is not available, you have to make assumptions until the development matures to a level where the assumptions can be replaced with actual information.

Step 2. Develop Mission Operations Concept and Supporting Architecture

One of the most important documents developed during the Definition Phases is the Operations Concept Document (OCD). The OCD is the documentation product that describes how the mission will be executed to accomplish its objectives, and attempts to define the required software, interfaces, timelines, procedures, and various spacecraft modes. The ops concept usually covers the operations of both the space and ground segments, and looks at operational scenarios. The resultant OCD is important in helping to define the mission system architecture and also the requirements for the various system elements and subsystems.

As shown in Fig. 29-2, there are several external inputs that contribute to the development of the ops concept. These include the mission objectives, and top-level requirements and constraints that were determined in Step 1. Another important factor in the development of the ops concept is the mission architecture. The development of the system architecture is usually iterative with the ops concept—an ops concept is needed to develop an architecture, and the architecture defines many of the elements described in the ops concept.

One problem with current practices in mission operations is the lack of common standards. Various organizations, such as AIAA, IEEE, SAE, CCSDS, and ECSS, have and are attempting to set standards for space missions, but it is difficult to come to an agreement and to implement changes in legacy systems. When the ops concept is defined, some decisions in standards, especially with interface formats, will have to be made. However, most of the decisions on standards can wait until the development of the Mission Operations Plan.

Step 3. Develop Operations Scenarios and Techniques that Accomplish the Mission Concept

Once you have developed the general concept for the mission operations, you need to flesh it out and test it, both to check its feasibility, and to help determine the elements and tasks that are needed to support it. The best way to do this is to put together some operational scenarios that represent various events or phases in the mission, such as deployment, nominal payload data takes or events, communication contacts, eclipse periods. Early in the development cycle you may not be able to determine a precise power profile to support the scenario, but you should be able to determine the sequence of major events to support the scenario. With the help of the power engineers, you should be able to calculate the associated power profiles, at least to a rough order of magnitude.

Step 4. Determine Ops Functional Activities Required in Step 3, Allocate to Functional Areas, and Develop the Functional Flow Block Diagram

When you have finished Step 3 you should have a good understanding of what mission operations will entail and which tasks or functions are required to fulfill the mission. You can now list the specific tasks or functions that need to be performed during the execution of the mission. Each function identified should have data input and data output and perform a specific function in a process flow. Once you have listed the functions, you can now produce one of the most important products in the design process to understand what needs to happen to accomplish the oper-

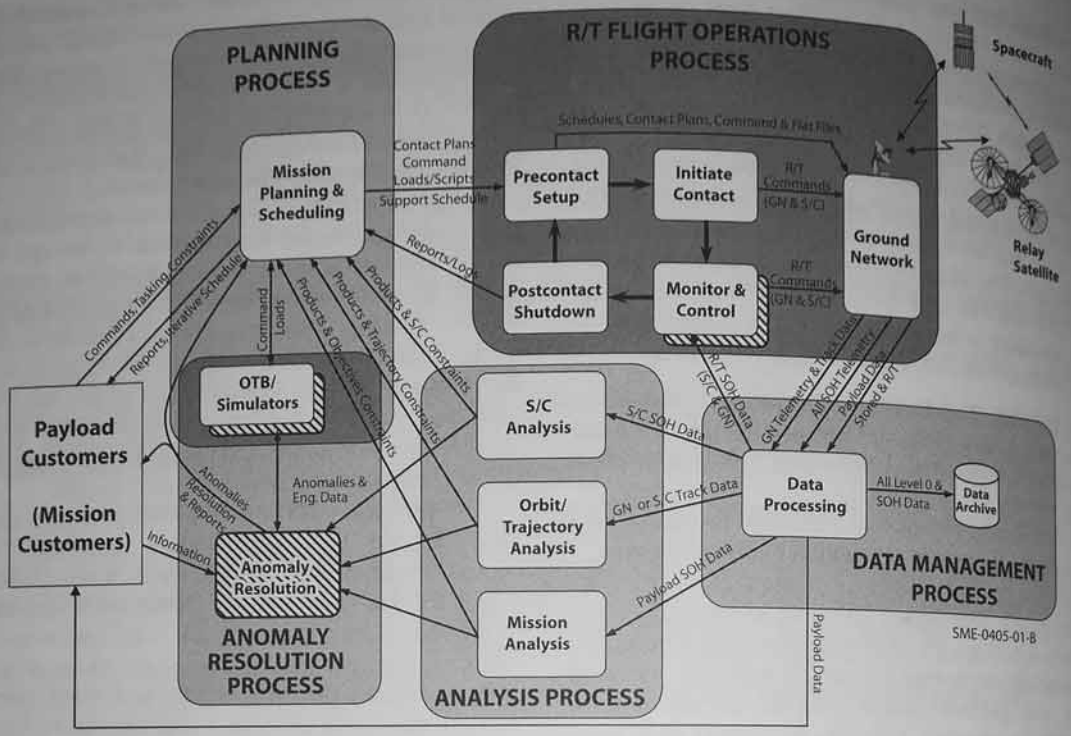


Fig. 29-3. Mission Operations Functional Flow Block Diagram. Operations performed during a space mission usually consist of the following basic processes performed on a cyclic basis: Planning, Real-Time Flight Operations, Data (Processing &) Management, Analysis, and Anomaly Resolution. The actual processes used by a particular mission may vary from this model, but usually the same functions are performed. Some specialized missions (e.g., manned or surface ops) may add additional processes.

ations—the Mission Operations Functional Flow Block Diagram (FFBD) or the Functional Block Diagram (FBD). The FFBD shows the functions in the operations process and the flow of process or data between them. The FBD is often the same as the FFBD, although it sometimes just shows connections between the functions without identifying the data flows between them. An example of a typical FFBD showing the execution of the operations cycle for a space mission is in Fig. 29-3.

This FFBD shows the major functional areas in the operations process (Planning Process, Real-time Flight Ops Process, Data Management Process, Analysis Process, and Anomaly Resolution Process) and the major functions within each of these processes. The arrows show the flow between the functions and are labeled to show the nature of the data. The FFBD provides a visual way to understand what happens during operation of a spacecraft and thus, what you need to include in your design. Note that the example given in the figure is typical of operations for small autonomous satellites in Earth orbit and are basic functions required by almost any space mission. However, when you have a more complicated mission or a mission of a different type, such as manned missions, or deep space missions, the operations process may differ. Communications satellites in geosynchronous orbits also have a more steady state operational process rather than the cyclic process shown. We will be looking at the operations functions in this process in more detail in the next section.

Table 29-4◀, Fig. 29-3, Eq. 29-0

Step 5. Identify Ways to Implement Functions Required in Step 4

If you have worked in mission operations previously, then you have a head start in accomplishing this step. In that case you will probably start with your previous operations system and then determine where changes need to be made due to the nature of the mission, or where improvements can be made based on lessons learned or advances in technology (identified in Steps 1 and 3).

However, we will assume that this is your first time developing mission operations. Each of the operations functions identified in Step 4 is different and requires different methods and tools to accomplish. It is helpful to set up a spreadsheet table that lists the major processes with their associated functions listed in the leftmost column, and then place existing tools (generally software) listed across the top. These tools are mostly COTS applications (products) that are sold by many vendors or available from various government agencies. You can identify the candidates to occupy this trade space by using web searches, looking at advertisements in trade periodicals, review recent technical papers from conferences (where many of these applications or tools are presented), visiting exhibitor booths at the major conferences (see the list in Step 2), and contacting NASA or other government agencies that operate spacecraft (they are generally happy to share technical information and sometimes can even provide legacy systems or modules).†

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Table 29-5. Definitions of Autonomy Levels. Table (a) shows the basic levels of autonomy for space segment health management, while (b) shows the basic levels of autonomy for mission execution. These are based on the ECSS standards regarding space segment autonomy, but are fairly standard globally.

(a) Space Segment Health Maintenance Autonomy Levels		
Level	Description	Functions
1	**Closed-loop:** Pre-planned automatic response to detection of a particular state or condition that does not require ground input	For example, thermal control (automatic powering of heaters when temperature drops)
2	**Closed-loop/Open-loop:** Establish safe space segment configuration following an on-board failure and wait for ground response	Includes actions that can help prevent potential anomalies that would require safing of spacecraft (e.g., power management to prevent critical power condition)
		Identify anomalies and report to ground segment
3	**Closed-loop**: Re-establish nominal mission operations following an on-board failure	Reconfigure on-board systems to isolate failed equipment or functions
		Place space segment in a safe state
		As Step 2, plus configure to a nominal operational configuration
		Resume execution of nominal operations
		Resume generation of mission products

(b) Mission Execution Autonomy Levels		
Level	Description	Functions
1	**Open-loop**: Mission control under ground control; limited on-board capability for safety issues	Real-time control from ground for nominal operations
		Execution of time-tagged commands for safety issues
2	**Open-loop/Closed-loop**: Execution of pre-planned, ground-defined, mission operations on-board	Capability to store time-based commands in an on-board scheduler
		Limited capability to react to execution of commands (e.g., retry if unsuccessful)
3	**Closed-loop**: Execution of adaptive mission operations on-board	Event-based autonomous operations
		Execution of on-board operations control procedures
4	**Closed-loop**: Execution of goal-oriented mission operations on-board	Goal-oriented mission re-planning

Step 6. Determine Level of Automation for Both Space and Ground Segments

Before you can size mission operations for the number of personnel needed, and cost, you need to determine how much of the operations will be automated and how much requires human operators. This includes not only the spacecraft, but also the ground segment [Calzolari, 2007]. As mentioned in Chap. 20, the advances in miniaturization and computing power have enabled the placing of more tasking responsibilities and automation on the spacecraft rather than on the ground [Sherwood and Chien, 2007]. Mission operations recognize the importance of automation as a productivity enhancer, useful for performing repetitive tasks and responding to well documented contingencies. However, there can be a performance downside to extensive automation. Validating complex automation can itself be extremely complex and costly. Over-dependence on automation results in lower operator technical proficiency and only experienced, technically current and knowledgeable personnel can effectively respond to non-standard situations.

Looking at the space segment, there are three basic levels of automation for space segment health management and four basic levels of mission execution automation as shown in Table 29-5. These levels are based on the ECSS standards regarding space segment autonomy [ECSS, 2008]. The reference also lists two levels of autonomy for mission data management.

Step 7. Determine Whether Capabilities Identified in Steps 5 and 6 Exist, are Obtainable, or Must be Developed

Once you have determined the operations functions required by your project, and the expected level of automation in both the space and ground segments, you need to determine which of these can be addressed by assets already possessed by or available to your team, and which ones you will have to obtain from external sources (e.g., vendors or government sources) or have developed (either by the team or contracted out). The answers can have a large effect on both the schedule and cost of the project. Some of this step may have been done during the accomplishment of steps 5 and 6 just by gathering the information on the various options.

Step 8. Develop Staffing Plan and Identify Other Resources Required

One of the most important factors in determining the cost and performance of mission operations is the staff-

Table 29-5, Fig. 29-3◀, Eq. 29-0

† For example, all operators in ESA member countries can use MICONYS (SCOS2000 spacecraft monitoring and control tools) and SIMULUS (Simsat software-based spacecraft simulation tools) free of charge for a license with ESA, and in the USA, tools such as ITOS and AMMOS are free from NASA.

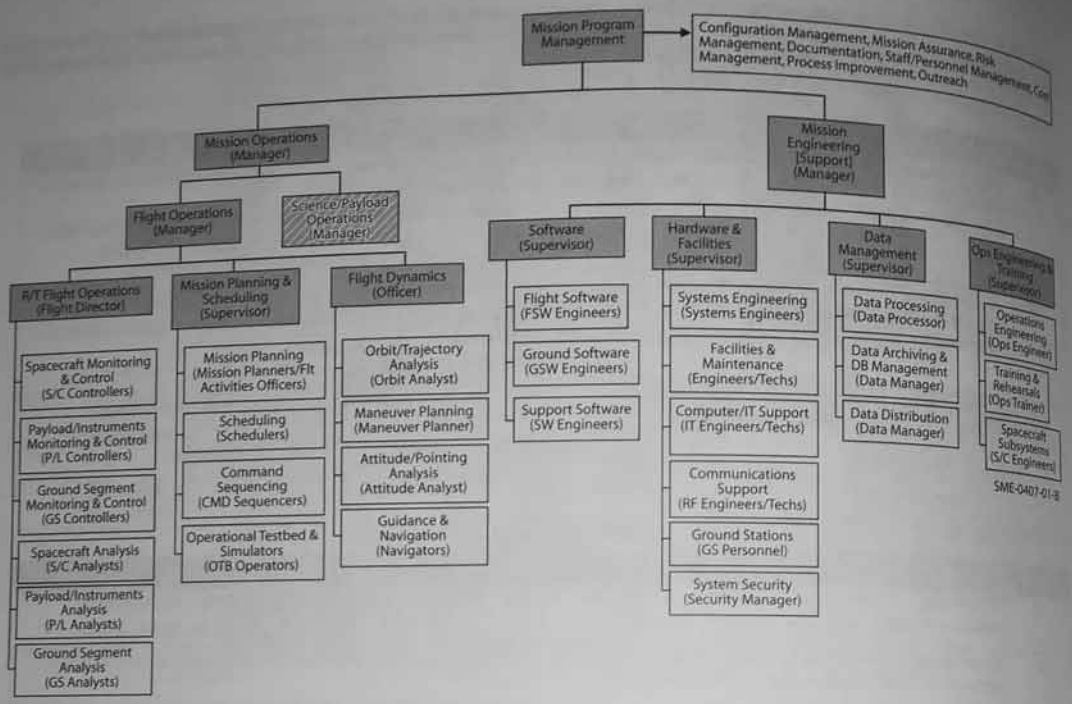


Fig. 29-4. Sample Mission Operations Organization. This organization chart shows typical groupings of operations functions, which may or may not map directly one on one to personnel. Common staff positions that fulfill each function is shown in parentheses below the function name. This is just an example and actual operations organizations can vary significantly from this—smaller, larger, or using different terminology for the functions and/or positions. The Science/Payload function is often a separate organization.

ing—both the level of staffing throughout the project life cycle and the personnel qualifications required. In Steps 4–6 of this operations development process, you determined the functions required to do the operations for your mission, the means to accomplish the functions, and the level of automation that you could use. Now you should look at those functions again to see which require the support of human operators and to what extent. Are the operators just needed for supervision of an automated process, required to make decisions at critical points in the process, conduct the function entirely manually, or just be on standby to intervene when something anomalous occurs? The frequency of the human participation needs to also be considered. In manned space missions, like for the Space Shuttle or ISS, human operators are present continually—24/7. For many autonomous spacecraft, such as nanosatellites orbiting the Earth or probes deep in space with long communication lags or infrequent contacts, little time may be needed to support operations, although some non-realtime functions, such as mission planning or data analysis, do not depend on having contact with the spacecraft but will be staffed during office hours.

One of the products to come out of this step is an operations organization chart or tree, an example of which is shown in Fig. 29-4. At this point it is useful to define the typical staff positions used for operations. Note that these are just some of the more common ones, some of which will be found in nearly all operations teams, but they may have different names in a particular organization. There are also some other positions that may be used based on the nature of the mission or the organization's operations philosophy. For smaller missions, some of these functional positions may not be in the functional organization and the responsibilities of those positions will be assumed by other positions. For example, the Mission Operations and Flight Operations functions, or the Flight Operations Director and the Real-time Flight Operations Supervisor functions may be combined.

The mission staffing plan accounts for the criticality of an operator in the loop that is supplemented by, but not supplanted with, automation tools. Basically, the mission operational staffing concept is primarily dependent on several variables including the duty cycle of mission payload events; the complexity of the operations; the cost; and the degree of automation that has evolved. Mission operations staffing can be divided between management, science, flight operations, and engineering support elements. Together they comprise the Mission Operations staff. The organization is first developed to be functional rather than actual. This means that one actual person might perform more than one function shown in the organization or a function may not have a person dedicated to it. Note that the organization shown does not include ancillary positions such as administrative assistants, for example. It should also be noted that some positions may be shared between more than one mission, so the costs would also be shared.

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Table 29-6. Example of a Staffing for Highly-Automated Small Satellite Mission Operations. This example is for a small imaging LEO satellite that can perform unmanned routine "lights-out" ground contacts for transfer of files. More personnel are required during EE&CO, but the number required can be decreased as the ops mature and become repetitive.

Position	EE&CO FTE	Nominal FTE	Mature FTE	Shift hr/day	Comment
Operations Manager	1	0.5	0.1	8/5	
Spacecraft Controller	2	0.9	0.1	8/5	8/7 for EE&CO
Ground Segment Controller	1	0.1	0.1	8/5	
Mission Planner	2	1	1	8/5	
Data Manager	1	0.5	0.1	8/5	
S/C Analyst	4	0.5	0.1	8/5	8/7 for EE&CO
Software Analyst	2	0.5	0.1	8/5	8/7 for EE&CO
Orbit Analyst	1	0.5	0.1	8/5	
TOTAL	14	4.5	1.7		

Besides automating the maximum number of repetitive and predictive tasks, another key way to keep operations costs down is to reduce staff size to the minimum necessary to meet all requirements. If the mission staffing approach is based on a minimization concept, such a cost savings objective is attainable. However, reduced staff, while effective at reducing costs, might introduce risk that must be mitigated if quality service is to be provided with a high degree of reliability, maintainability, and availability (RMA). The primary risk of minimized staff is lack of depth at key positions. For some small missions, many of these functions might be performed by one person or several functions might be shared by a few people thereby providing backups. On some big missions one function might be done by several people.

Once the staffing positions have been defined, the next step is to determine which functions the operations personnel need to perform (maybe not all the identified staff positions are required), and when they are needed. A very important factor in this determination is the level of operations automation (space and ground), which is why this step is included in the repetitive trade study segment of the mission operations development process. Table 29-6 shows an example for the staffing levels in Full-Time Equivalents (FTEs) for a highly automated small imaging satellite in LEO that is capable of routine "light out" ground contacts to transfer data files, with the major human activity being required to plan and implement the imaging of various targets. This provides a minimal staffing plan especially suited for university-class nano- or micro-satellites. The number of personnel required to perform and monitor mission operations decreases as confidence in the system increases and the operations become mature and repetitive. A high level of staffing is required to check out the satellite until IOC.

The tool that is used for helping to determine the number of personnel needed, and to later determine the staffing costs, is the Work Breakdown Structure (WBS). This basically defines the tasks that have to be performed to successfully develop or conduct operations, and when applied to workload (FTE personnel) estimates and a schedule, becomes the basis for the staffing plan. The WBS and how to develop one are described in Sec. 11.1. An example WBS for mission operations is shown in Table 29web-5.

Once the WBS has been developed and it has been fitted to a schedule to identify when in the project life-cycle its tasks need to be performed (and thus staffed), the workload for each task can be estimated along with identifying the staff position to accomplish it. There are several good COTS software packages available to perform this project management function such as Microsoft Project® and dotProject, which is an open source project management tool.

For Phase E missions operations, you need to determine whether or not 24/7 operations are required by some of the team, or whether a 40 hour work week (or less) is sufficient. If the real-time team is required 24/7, then you must determine the number of teams and shift schedule for the teams. Usually, there are four or five teams with shift lengths of 8 or 12 hours. Some popular shift plans can be found on the Business Management Systems website. As an example, the Clementine mission [Horan, Sorensen, Tran, and Hoffman, 1995] used a four team (eight persons per team) scheme for real-time operations on a rotating, day-night-day, 12 hour shift with four days on (10 AM to 10 PM) followed by four days off, then four nights on (10 PM to 10 AM) followed by four nights off. This shift schedule combined a high degree of predictability with an avoidance of commuter rush hour. However, this scheme does require some overtime work and has long shifts.

Once the positions required for mission operations have been identified and their expected workload and utility have been determined, as well as when they are required in the project life cycle, a staffing profile can be built. The first step is to list all positions on a spreadsheet that shows the schedule of the project at an appropriate time resolution (e.g., monthly). Note on the spreadsheet

when each operations team member is brought into the project and at what level as Full Time Equivalents (FTEs). The monthly staff level in FTEs can then be determined by adding the values per month. From these totals, you can produce a graph showing the staffing profile for the mission.

Although we have concentrated on the operations staffing so far in this step, the other part of the step is to identify other resources that will be required by the mission. These could include ground station coverage (utilization), computers, communications equipment, consoles and workstations, or software applications that are needed to support operations. Some of these are covered by other areas, such as the Ground Segment, but may come under the responsibility of Mission Operations or under joint responsibility. Once you have the staffing requirements done, as well as the communications and other mission requirements as designed into the current system architecture, you will be able to make estimates for these other resources. However, the resources include not only hardware and software, but also services, such as precision orbit determination, that might be required, and for which a cost will have to be determined.

Multi-mission operations can have different loading in both staff and resources. For example, a 0.2 FTE orbital engineer may be normal as that function can be multi-mission. Likewise, a ground station is nearly always multi-mission, while a monitor and control tool is probably mission specific (although within that mission it may control multiple satellites).

Step 9. Perform Trades to Determine Best Solution from Steps 5–8

At this point in the MOP development process, you will have identified the functions and level of automation required to complete your mission. You probably have identified several tools or packages (whether existing or not) that may work to accomplish these functions. The next step is to perform a trade study between the various candidate options to determine which would be best for your particular application. If this is being done early in the development cycle, then optimization is not expected, but rather obtaining a solution that is acceptable and will accomplish the mission. Later in the development cycle (i.e., a later iteration), then a more detailed analysis and comparison may be required to optimize your solution.

You must select the criteria that are best for distinguishing between the candidates in the trade studies. Typically these fall into the following four categories: performance, schedule, cost, and risk. These will be examined in turn.

Performance is probably the most important attribute in your trade study. This is a measure of how well the object being evaluated fulfills the functions for which it is being considered. Your selection must be able to function sufficiently well to accomplish the tasks needed to fulfill the mission objectives. If it will not do the job required, then it should be discarded. However, the selection cannot be made based on performance alone, but must be considered in light of the other selection parameters. When you score the attributes of the candidate solutions, performance is generally weighted the highest.

Schedule is important because no matter how well the candidate fulfills the other trade criteria, it is worthless if it cannot be available on time, especially if there is no flexibility in the project schedule. Schedule is also closely related to cost for elements that have to be developed for the project, because of the increase in labor cost as the time required increases.

Cost is critical for some projects, especially for universities and small programs, but inflated costs can kill even the high-end missions as well. For projects with tight budgets, cost becomes the dominant constraint and can even out-weigh performance as the primary selection parameter. In this case, "good enough" might be the selection mantra, even though it might cause reduction in mission objectives, lifetime, or increase in risk. Cost includes both the non-recurring costs for the pre-launch phases, and the recurring costs during the conduct of the mission. This is where automation can be a deciding factor—higher non-recurring costs for developing and testing automation may lead to less overall project costs due to reduced recurring costs (e.g., fewer personnel required to operate the mission). When including cost as a selection parameter, it might be worthwhile dividing it into non-recurring and recurring costs, with possibly different weighting. This difference in weighting might not just be related to the total project costs, but could instead be tied to a funding profile for the project—plenty of funding during development but little funding for operations would put the weighting in favor of the non-recurring costs, while vice versa would put it in favor of the recurring costs.

Risk can encompass many areas, such as safety, probability of successfully completing the mission, political risk (e.g., high visibility projects may have high political risk—failure could have far-reaching consequences) and technology readiness level. Risks are identified using a process of risk management, and the willingness to accept the risks has to be determined. When using this as a trade selection parameter, you might want to differentiate between the different types of risk, because each may have different weighting for your project.

The simplest way to do the trade study may be to take the table you used to determine the functionality of the candidates in your trade space (e.g., Table 29web-5) and assign numeric values to each of them for each of the four trade parameters instead of the simple "X" as shown in Table 29web-5. However, a more rigorous methodology for performing trades is presented in Sec. 5.3 and is applicable for use in operations as well.

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Step 10. Determine ROM Mission Operations Cost

By this step you have determined the factors that are needed to develop a rough order of magnitude (ROM)

cost estimate for mission operations. These major factors include the tasks (WBS), staffing profile, hardware, software, facilities, and communications costs as well as maintenance/refurbishing and replacement cost for long-term missions. Detailed cost analysis methodologies are described in Chap. 11. See Cost Model (NICM) in Chap. 11.2.5.

The cost of operating a mission depends most on the number of personnel involved (people are very expensive as a recurring cost) and operations complexity. Both of these depend on the amount of automation employed both in the spacecraft and in the ground segment. Adding more automation probably increases the non-recurring costs during Phases A–D, but may allow for lower recurring operations costs during Phase E. The complexity of the operations depends on the mission objectives, the design of the spacecraft and payload, communications and ground segment design, and operational risk policies.

Step 11. Repeat Steps 4–10 for Development Phase (Mission Phases A–D as Appropriate)

The core of mission operations for any project is what is required to execute the mission after launch. This is the phase of the project that the mission operations are designed to fulfill—this is what accomplishes the mission goals. The purpose of the operations development done prior to launch (Phases A to D) is to design mission operations that will run smoothly to accomplish the mission. That is why we look at the Phase E operations first to develop our Mission Operations Plan, and to size and cost these operations. However, to complete the MOP, we must go back and determine what needs to be done to evolve the operations into the efficiently functioning system required for flight operations. Now is the appropriate time to go back and repeat the steps of the MOP development to account for the operations development in Phases A to D. In some of your previous steps, you may have already included some of this early work (e.g., staffing profile), so now is a good time to go back and revise it if necessary based on your later determinations. At the end of this step you should have a revised ROM cost estimate and be ready to press on to the final steps of the mission operations plan development.

Step 12. Assess Mission Utility, Complexity, Risk, and Cost Drivers

The mission utility of the operations (i.e., determining how well the operations help us meet the mission objectives) is calculated using the method outlined in Sec. 5.4. It is helpful if you have simulations that you can use for determining the performance and characteristics of the system and for which you can input a range of values for the design parameters to determine their effect on the cost and performance. Operational simulations such as these have been developed and used extensively [Rainey, 2004], and are available from various sources such as NASA. The development of the Mission Operations Plan is iterative, and as the plan becomes more mature, the differences in the mission utility parameters between iterations should decrease, and also helps provide confidence in the viability of your operations system design.

Mission operations cost depends largely on the complexity of the operations. To determine how the complexity affects the cost drivers, we can use a complexity model which expresses operational parameters in terms of FTE operations personnel. Each of the operational activities is graded as low, medium, or high complexity, using the analogous method (i.e., comparing with the same function on similar class missions that have been flown). This complexity factor is included in the algorithms of the model that can be used to evaluate the operations and to reduce operations cost. This model and method is described in detail in Squibb, et al. [2006].

Step 13. Repeat for Alternate Mission Concepts (Starting with Step 2) if Required

This step is used early in the life cycle of the mission design when trade studies are still being done to determine alternate mission concepts, but is skipped if the baseline mission concept has been selected. The purpose of doing operations concepts for alternate mission concepts is to determine how operations affects the performance, schedule, cost, and risk of each mission concept being evaluated. This becomes an important factor in deciding the baseline mission concept to be used.

Step 14. Identify Derived Requirements

In Step 1, the top-level requirements, both for the mission and for operations, were defined. However, to implement a design, requirements need to be specified that go to a more detailed level in each system element. Although this is particularly obvious with spacecraft subsystems, it is also important for operations as well. You need to know how you are going to meet the top-level requirements and there will be constraints or design factors that have to be considered as you get deeper into the design. These are captured in what are called derived requirements (see Sec. 6.1). Each *derived requirement*, as its name suggests, is derived, or allocated, from a higher level requirement. This means it has traceability all the way up to the mission objectives. Requirements also need some way to be tested or verified. It is important that requirements be numbered and tracked in the Mission Requirements Document, System Specification Document, or a similar document. In Table 29-7 is a simple example of top-level requirements and their first generation derived requirements.

Step 15. Develop Training Plan

When you have identified the operations staff positions required by your mission and have determined the staffing profile, you need to develop a Training Plan (sometimes called a Training and Certification Plan). All operational positions require training but not all require certification. Although different organizations may have different policies and techniques for training and certification, the basic training plan and methodology described here is based on NASA-sanctioned Instructional Design Processes, and should work for most missions with little

Table 29-7. Example of Derived Requirements (Child Requirements) Obtained from Top-Level Operations Requirements (Parent Requirements).

Top Level Operations Requirements (Parent)	Derived Operations Requirements (Child)
The Mission Operations function (Mission Ops) shall provide management and control of the spacecraft and Ground Segment to help ensure the successful completion of the mission.	• Mission Ops shall perform real-time contact support (monitoring and commanding). • Mission Ops shall perform mission planning. • Mission Ops shall perform timeline and command script generation. • Mission Ops shall perform orbit analysis and provide orbit ephemeris to mission users. • Mission Ops shall perform spacecraft engineering trending and analysis. • Mission Ops shall perform anomaly resolution and contingency operations. • Mission Ops shall determine operations statistics. • Mission Ops shall perform data management. • Mission Ops shall provide operations report generation functions for the mission. • Mission Ops shall be involved with the design of the spacecraft throughout the project life cycle. • Mission Ops shall support spacecraft integration and test activities.
Mission operations shall be designed for human control or autonomous "lights out" nominal contact support.	• Mission Ops shall be automated to allow the option of "lights out" nominal contact operations after IOC.

modification. However, it should be noted that some organizations do not want their satellite operators to "think" for themselves, but rather just to follow prescribed procedures. This is more common in satellite operations within organizations where the operators are there only as a temporary assignment and may be operating assets worth billions of dollars. There have been instances where an operator has jeopardized a mission by trying to solve a problem that was relatively benign to start with (i.e., the fix was worse than the problem).

The most efficient method of training, especially for smaller organizations, is to use a "right level" on-the-job training (OJT) style approach to training its operators where the emphasis is on position proficiency and not universal knowledge. Hands-on experience is emphasized far more than classroom instruction. This approach requires position cross-training, especially with limited staffing resources, while up-training is a privilege that must be earned. Personnel should be encouraged to learn the basics of performing higher level functions and expanding their technical worth to the program. In addition, those exhibiting high degrees of initiative and motivation can be up-trained as a career enhancing opportunity. However, positional up-training as a function of an operator's employment is not necessary and is not usually formally pursued unless the situation (e.g., imminent departure of ops team member) drives such a requirement. Within the mission operations team, the training and certification authority and administrator will vary depending on position. Basically training is divided into two distinct parts, new-hire basic training and mission specific training.

To effectively implement the classroom/hands-on and field training phases of the program, a variety of methods of instructional delivery should be used that yields qualified satellite operations personnel, cost effective implementation expenditures, and allow for a follow-on training program. A formal certification program is included to evaluate and ensure the proficiency of operations personnel thus mitigating risk and increasing the overall level of confidence in mission success. Training objectives are linked with skills required for certification to ensure operations personnel competence in all required mission related tasks.

The training process is designed to develop a spacecraft flight operations training and certification hierarchy that contains generic, as well as program or skill category specific training. At each stage, there are certification procedures that ensure knowledge bases and skill sets before a trainee is allowed to progress to the next stage of training.

> NOTE: Not every engineer or technician is suitable for the pressure of real-time operations. There have been cases where persons are outstanding in planning and implementing time-delayed spacecraft operations (with no ground contact), but when faced with the pressure of real-time operations, they "freeze" and cannot act as required. Sometimes repeated training will take care of this, but not always. Some people are not suited for real-time mission operations and should be utilized in other areas such as mission planning and scheduling, mission analysis, or flight dynamics. It is important that your training and certification process is able to differentiate between the different capabilities of the trainees and does not certify people into an area for which they are not suited. Probably the best way to do this is to put your trainees into as realistic high-stress simulated scenarios as possible and see how they perform. This should be done several times with different scenarios to see if their performance improves with experience.

The training plan is a project document under configuration control where the methodology, processes, course topics, personnel, schedule, and certification process and criteria are captured. For some organizations, especially the smaller ones, the

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Training Plan may not be a separate document, but incorporated instead into the Mission Operations Plan or a similar document. It is important that the Training Plan is kept up-to-date and actually used by the operations team. It helps to have someone designated with this responsibility.

Step 16. Generate MOP That Includes Technology Development Plan, Personnel Staffing & Training Plan, and Documentation Plan

If this is the first time through this process, then you now have all the pieces to put together a basic Mission Operations Plan. You should include in this document the background description of the mission, the mission statement, mission objectives, and the top-level requirements and constraints†, the mission architecture, the operations concept (description), operations architecture and functional flow block diagrams, operations organization, WBS, staffing plan, schedule, training and certification plan, technology development plan, and the documentation plan. For large and complex projects, some of these parts of the MOP are standalone documents (e.g., Training and Certification Plan or Technology Development Plan), but they would be summarized in the MOP. For smaller projects, these parts would appear in the entirety in the MOP as separate sections or maybe appendices.

The Documentation Plan lists and defines the various operations documents to be used for the project (see Table 29-3), who will be responsible for producing the document, when and how it should be developed, and its intended usage. This includes all the procedures to be used to conduct and support operations.

If this is not the first iteration of this MOP process, then this step is where you review, revise, and improve the MOP based on new information obtained in the previous steps of this iteration. This is the primary document of mission operations development and is very important. Remember that the MOP is the blueprint to develop and successfully conduct operations for your mission.

Step 17. Document and Iterate/Refine as Needed

This is the final step of the mission operations development process, where you publish the base (or revised) MOP that you have developed in the previous steps. If this is early in the life-cycle of the project, then the base MOP will probably not be complete or at least not mature, and one or more iterations through the steps will be needed to finish the MOP sufficiently for extensive use and reference by the project. The various iterations of the MOP should be made accessible and reviewed by other members of the project so that their input and feedback can be used to improve the plan.

† For large and complex mission you may want to just include the top-level mission operations and constraints and refer to the Mission Requirements Document (or equivalent) for the remainder.

29.2 Mission Execution

So far, we have examined the functions of mission operations and discussed how to put together a Mission Operations Plan and develop the products and elements needed to operate the mission. During Phases C and D of the mission project (see Fig. 29-2) is where the procedures and documents for operations are developed and tested, and the operations personnel are trained. The readiness of the team to support operations is examined in a final review before launch, such as the Mission Readiness Review (MRR), Operations Readiness Review (ORR), or Flight Readiness Review (FRR). Once the spacecraft is delivered and mated to the launch vehicle, the project moves into the Execution Phase (Phase E). This is where all the hard work and preparation for the mission pays off. This phase is divided into the following distinct segments of activity for operations:

Launch and Checkout Operations—The operations team may be involved with checking the functionality of the spacecraft after it has been integrated with the launch vehicle. The involvement of the MOT in the launch itself depends on the state of the spacecraft. Some spacecraft are launched "cold," which means it remains unpowered and inert during launch and is usually activated by a microswitch during deployment, by the launch vehicle avionics, or by transmitted signal just prior to deployment. Other spacecraft are launched in a "warm" state, which means that they have a minimal level of subsystems powered on during ascent, and then fully powered on after or just prior to deployment. Spacecraft may also be launched in a "hot" state, which means the basic subsytems are fully powered on. This is the state used for spacecraft with biological payloads (e.g., human) or where the spacecraft may need to take measurements during the launch. Launch operations usually last from launch until deployment of the spacecraft from the launch vehicle, although that may depend on whether or not a rocket is being used to boost the spacecraft into a different orbit or trajectory. These booster rockets are sometimes called the Inertial Upper Stage (IUS), Payload Assist Module/Motor (PAM) or Apogee Kick Motor (AKM). Of course, if the firing of this motor depends on the participation of the spacecraft, then it must be activated. Depending on the trajectory desired, there may be several minutes, hours, or even days between orbit insertion and the firing of the booster.

Following launch of the spacecraft, there is usually a phase of operations called Engineering Evaluation & Checkout (EE&C), Launch and Early Operations (LEOP), On-orbit Checkout (OOC), or something similar. This is a transition period bridging integration and test (I&T) and nominal operations. It is the phase at which the mission operations development gradually ends and the nominal mission operations begin. In some ways, the checkout phase of operations is different from nominal operations and in other ways, it is the same. This phase may include deployment of the spacecraft from the launch vehicle/booster, initial configuration and check-

out of the spacecraft, and if applicable, insertion into the transfer orbit or cruise to the final trajectory or target point, and insertion into the final orbit/trajectory. After establishing its final operational orbit, the mission begins a period of checking out the spacecraft's subsystems, operational procedures, and the entire mission system, including the entire ground segment. This checkout period normally lasts for several weeks.

The checkout phase is characterized by a highly dynamic environment where subsystems are initiated and checked out, and processes and procedures are thoroughly tested, and if necessary, redefined and rewritten. It is a period where the operations personnel (including spacecraft engineers) can expect long hours, few days off, maximum vigilance and complete flexibility. If applicable, it is also when responsibility for defining the mission operations transitions from the Mission Development Team (MDT) to the Mission Operations Team (MOT). During this phase, all subsystems are checked out for nominal performance and the spacecraft is configured toward nominal operations. The checkout phase may require spacecraft manufacturers and integrators to be physically present at the MOC supporting the planning, execution and analysis efforts in accordance with the Early Orbit Checkout Plan. The operations processes used during checkout are the same as during the nominal mission. However, the level of effort is usually more intensive, and the application of "one time only" operations more prevalent. Once the checkout phase is completed, the spacecraft/mission/project achieves what is commonly referred to as Nominal Operational Status (NOS) or Initial Operational Capability (IOC).

Nominal Operations—Once IOC has been achieved, the mission operations moves into what is normally called nominal operations phase. Depending on the mission type, the nominal operations can consist of vastly different operational activities based upon the progress of the mission. For instance, deep space probes (such as Mars or asteroid/comet missions) will have a long cruise period with very little activity other than maintenance of the spacecraft health and possibly occasional trajectory adjustment burns or ancillary payload activities, followed by intense operational activities once the target has been reached. This can also occur on a smaller scale for geosynchronous satellites as they are being placed into their correct final orbit. However, many LEO or spacecraft orbiting other planetary bodies may have near continuous high levels of activity (e.g., Hubble Space Telescope or Mars Reconnaissance Orbiter).

The nominal operations phase can also contain non-nominal operations, such as contingency operations that are required to resolve an anomalous condition. The process of handling anomalies will be discussed later in this section. Finally, for science missions, the nominal operations phase can evolve due to changing requirements and scope due to being granted an extended mission, which generally comes with new science goals and a budget cut.

29.2.1 Mission Operations Processes

The mission operations process consists of four major subprocesses (see Fig. 29-3) that are performed on a continually cycling basis: planning process, execution (R/T flight operations) process, data management process, and the analysis process. There is a fifth process that is not part of the normal operations cycle and is only performed as needed, and that is the anomaly resolution process. In order to achieve low-cost nominal operations, each of these processes should be automated as much as possible without compromising mission success. Each of these processes will be examined in turn.

The *Mission Planning and Scheduling (MPS) process* (Fig. 29-5) is the process of determining a spacecraft's state (position, velocity and attitude/rate), selecting operational objectives, generating the products used to implement these objectives and coordinating with the real-time team to ensure these objectives are realized. It produces the master schedules, timelines, pass plans, and command loads for the space and ground segment activities. Inputs to MPS usually include the Mission Operations Plan, flight rules, mission requirements, status of the spacecraft and ground network with any resultant constraints, as well as task requests from the customers (usually payload customers) and from the spacecraft operations team if needed. When these inputs have been factored in along with products from the Flight Dynamics function (the spacecraft orbit or trajectory ephemeris and sometimes attitude/pointing information), a baseline plan and schedule are created.

As part of the Flight Dynamics function, guidance, navigation, and control (GNC) functions are provided for mission planning. Based on orbit/trajectory analysis and the desired mission plan, trajectory maneuvers and pointing targets can be provided as part of their standard products along with the orbit ephemeris and attitude information. After the Flight Dynamics products are received, variously sorted visibility products and orbit events are output and published for Mission Planners to review. These products provide all the information that is required by the MP to identify operations opportunities, select passes and determine maneuver requirements. Spacecraft operations planning primarily concerns space segment assets but not ground segment scheduling. However, there may be occasions when on-board activities and passes may be concurrently scheduled. Usually, these would be considered special events with a high priority, and will require that the spacecraft analyst or even a customer technical representative be in the MOC to oversee the support.

Pass scheduling is performed to produce a schedule of ground station contact support required to support the mission during the planning period. This usually starts with a schedule of available support times and assets provided by the Ground Segment schedulers. For high priority missions or missions with dedicated ground station assets, the planners will tell the GS schedulers which passes are required instead of choosing just from those

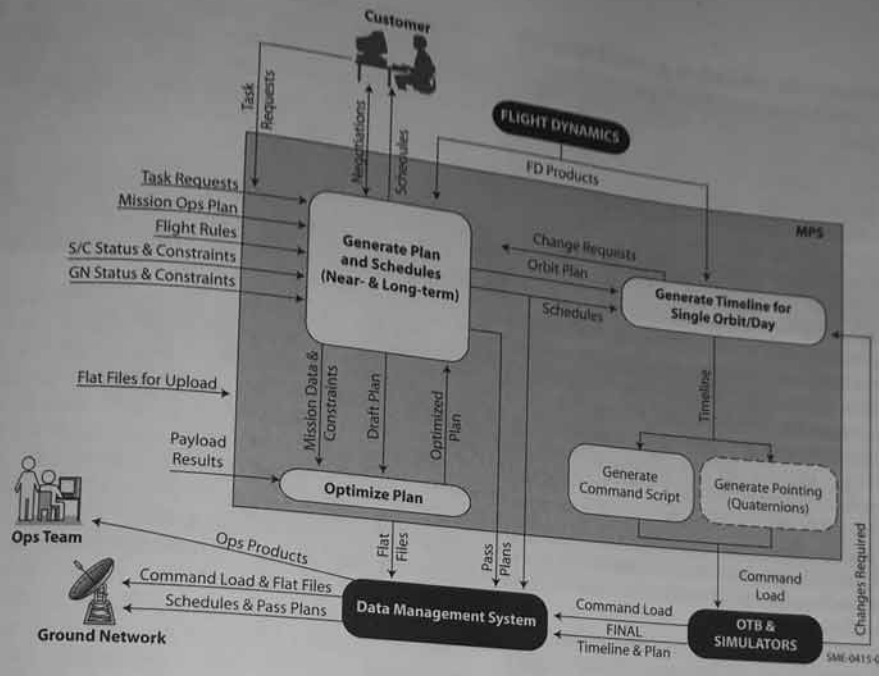


Fig. 29-5. Mission Operations Planning and Scheduling Process. The Mission Planning and Scheduling (MPS) process takes various inputs, including the mission plan, payload/customer requests, status of the spacecraft and ground network, orbit ephemeris data and using mission constraints, defines a plan, schedule, and command loads to be used to perform the mission during the next execution cycle. Most missions test the commands on an operations test bed or simulators for verification before uploading. The MPS process shown is loosely based on that used for the Clementine lunar mission [Sorensen et al., 1995], but is typical for many space missions.

that are still available. Some missions know that certain events are critical for mission success but are unable to specify a date and time for these events. Examples include pre-launch simulations, one time encounter opportunities, opportunistic observations, launches and GS scheduled maintenance. The MPS function should allow these users to reserve a block of time based on these events. In some cases, event scheduling could involve higher prices and cancellation penalties. However, it also ensures that the customers are provided with a pass priority for the critical events of their missions.

The MPS process is responsible for being cognizant of operations constraints and coordinating with the customer when objectives and constraints conflict. These constraints should be both documented in the MP's operation procedures document and programmed into the command generation and verification process. Possible constraints that would affect either party may include the following:

- **Environmental:** Cloud cover, smoke, sun glint, solar panel jitter, or solar exclusion zones. All mission specific and found in the ICD and other coordinated documents. Use tools created to reduce the impact of such constraints.
- **Positional:** Angle, elevation, or ground obstructions relative to a target.
- **Data Requirements:** Recorder capacity, amount of data planned for collection, amount of data per pass that can be downlinked, compression scheme, and latency of data (i.e., how quickly needed).
- **Single vs Multiple Payloads on One Spacecraft** (single mission) Operational constraints such as not being able to run payload A while payload B is powered up.
- **Spacecraft Performance Capability:** Electrical power capacity, thermal management, attitude determination, pointing and slewing, and maneuvering for example. Information that is provided from the Spacecraft Analyst or other subsystem experts that identify spacecraft threatening performance shortcomings.
- **Single vs Multiple Spacecraft on One Mission:** Primarily a coordination constraint.
- **Alteration Lead Time:** If the customer is selecting operations but the MPS process is producing the commands, there must be a constraint against the customer trying to change the operation after a certain time.

There might also have to be some negotiations with the customer if the baseline plan cannot fully meet the tasking requests of the customer. During this planning

process all conflicts in the schedule (e.g., conflicting task requests, or availability of ground stations) are resolved. The baseline plan must also be internally conflict free and not violate any spacecraft or payload constraints, such as those listed above. This is called the *deconfliction* process. The baseline plan and schedule may then go through an optimization process based on payload priorities, mission constraints, and status of data collections. In some systems, this optimization process is integrated with the process that generates the baseline plan and schedule and not a separate process as shown.

There are generally two levels of planning. The output of the *long-term planning* is a sequence of spacecraft and payload activities that may cover from days to months in the future that ensure the mission objectives will be met and to inform the team of when activities are planned and what assets (e.g., ground station coverage) will be required. *Short-term planning* includes detailed targeting using the current orbit, the current state of the spacecraft and payload, and any absolute time of timed commands. Depending on the mission, short term planning can occur daily and within a few days prior to the execution or may cover longer periods.†

Part of the short-term planning is the generation of *pass plans* (also called *contact plans*) to provide details to the ops team for real-time contacts with the spacecraft. The pass plan contains the essential information for executing the real-time pass, such as times of AOS, LOS, maximum elevation, duration of contact, which assets (e.g., antenna) will be used, the settings of the ground equipment, the objectives of the pass, which files are to be uploaded or downloaded, any commands or special activities that the ops personnel may have to perform. For spacecraft that are in continuous or long-term communications with the ground, then the pass plan as defined here does not have much meaning. In these cases there may be a daily activity or special activity plan.

The optimized plan and schedule can then be used to create detailed timelines for spacecraft and support activities. The timeline is a human readable form of the activities to be done by the spacecraft and ground network as a function of time. Timelines are usually generated for an orbit or for a day, depending on the type of mission, and combines the orbital events provided by the orbit ephemeris (one of the FD products) along with the commands to be performed by the spacecraft or ground. This provides the detail needed to execute what was laid out in the short-term plan. In the case of manned missions, the timeline also includes references or links to detailed crew procedures for each individual task. Development of the timeline traditionally has been done manually by modifying previous similar timelines ("cut and paste"), although software tools are now available to streamline the timeline generation and reduce the chance of errors introduced by the manual method. Note that if the time between the development of the timeline and its execution on the spacecraft is large (as defined for the mission), then the "final" timeline might have to be tweaked with an updated ephemeris. If the difference between the ephemeris during planning and the ephemeris close to execution is no more than a second or two, then it might not matter, while if the difference is many seconds or tens of seconds, it may matter and thus have to be reduced.

Once the timeline has been completed, reviewed, and approved by the appropriate authority in the operations organization, it must be converted into a command sequence that can be uploaded and executed by the spacecraft. Although this can also be done using manual methods, it is becoming more common to use a software tool called a Command Script Generator (CSG) to produce the command sequence in a form readable by the spacecraft computers. Typically, the timeline is first parsed by the CSG to produce software code in the language used to program the spacecraft's on-board computer (called the *command script*), and then compiled to produce the binary machine language code that is actually uploaded and executed by the spacecraft (called the *command load*). The actual details of this process depend on the spacecraft's software and command architecture and may differ from what is described here, although the end objective, a sequence of commands to be read and executed by the spacecraft, is the same.

In Fig. 29-5, there is a parallel function to the command script generation shown which generates pointing for the spacecraft. This function is not required by all missions. Even if pointing has to be determined for the spacecraft to execute the commands, with increased computing power that is now available to spacecraft, the pointing may be calculated on the spacecraft and not on the ground as was traditional. Many spacecraft use quaternions to define pointing vectors and attitudes, but not all, which is why this is shown in parentheses. If the pointing tables, angles, or quaternions, are calculated by the MPS function, then they are combined with the command script and subsequent command load derived from the timeline.

The command load is usually tested on an Operations Test Bed (OTB) or simulators before leaving the MPS function. If any problems are found, then changes can be made in the plan, timeline, and command script to eliminate them. Once the command load has been verified as working as expected, the final version is passed along to the mission database (part of the Data Management function) where it is combined with the pass plans and flat files. The flat files are data files that need to be uploaded to the spacecraft or to the ground stations, but are not part of the command sequences. Examples of flat files include software patches, instrument or calibration settings. The command load is sent to the ground network for uplinking to the spacecraft, while the schedules, pass

† Deep space missions often upload timelines covering much longer than a day. For example, Cassini went from a 45 day schedule during primary mission to a 10-week schedule during its extended mission. Commands for the entire schedule were uploaded at the beginning of the 10-week period. Other missions that rely on the DSN must provide schedules longer than a day.

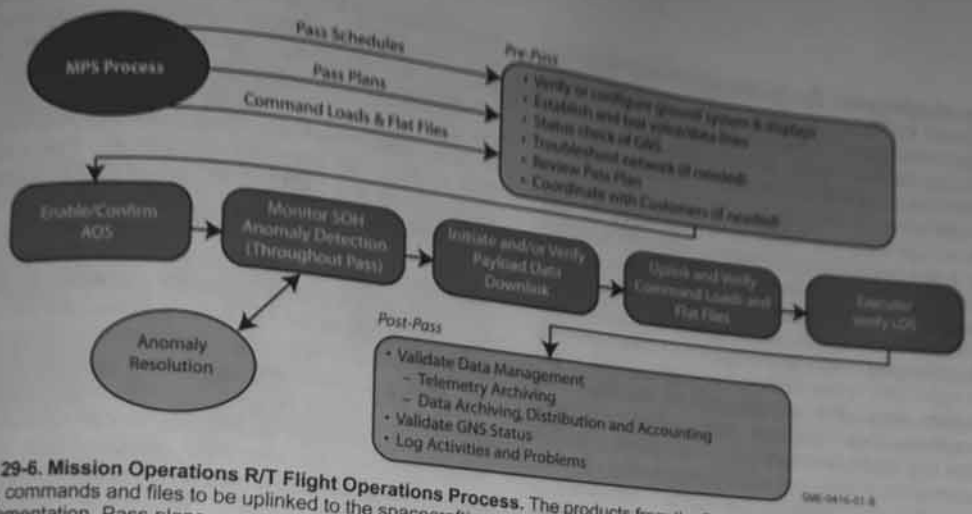


Fig. 29-6. Mission Operations R/T Flight Operations Process. The products from the Planning Process (pass plans, schedules, commands and files to be uplinked to the spacecraft) are usually delivered to the Flight Operations Team in the MOC for implementation. Pass plans are reviewed and status of the ground network checked before AOS to ensure that everything is ready. After AOS, the spacecraft SOH is checked, and if any anomalies are detected, they are resolved. During the pass, any payload data are downloaded and files are uploaded as required. Pass termination procedures are executed before LOS.

plans, timelines, command scripts are the operations products sent to the operations team while the schedules and pass plans are sent to the ground network to support the spacecraft contacts.

The MPS functions are essential to the responsiveness and flexibility of the entire mission operational system to the requests of the payload customers. For many missions, it is thus important to implement automation and process improvement methodology in the MPS functions to ensure the best possible performance.

The *Real-time Flight Operations or Execution process* (Fig. 29-6) performs the activities associated with real-time contact with the spacecraft.† The process we describe here is that associated with LEO satellites that have to pass over ground stations for communication. This process is modified somewhat for missions where the spacecraft is in communication with the ground either continuously (e.g., geosynchronous satellites or spacecraft using TDRS such as the ISS) or very lengthy communication periods (e.g., deep space spacecraft). In these cases the definition of a pass is either moot or arbitrary. One advantage that these latter cases have is that they can usually schedule contacts with the spacecraft at convenient times (dependent of availability of ground station assets) such as during regular working hours. For LEO missions with just a few ground stations for communication, the passes may occur at any time of the day depending on the orbit.

The MPS function delivers its products (e.g., schedules, timelines, pass plans, command loads, flat files) to the mission data system where they are available to the operations team or sent to the ground stations network ready for their support of the upcoming passes or uploading to the spacecraft as appropriate. The process shown here assumes that the operations center and ground stations will have human operators, although many of the functions will still be performed even if this process is partially or even fully automated.

At a specified time before the upcoming pass (i.e., contact with the spacecraft), the pre-pass activities are started to prepare everything for the pass. These activities include configuring computers and consoles, configuring and testing the ground station equipment (e.g., receivers, transmitters, antennas), establishing and testing data links and voice lines (if needed), reviewing the Pass Plan with personnel supporting the pass to ensure that everyone understands what to expect and what they need to do. If the customers are supporting the pass (e.g., a payload event is planned during the pass), then coordination with them will be verified.

Often, during the pre-pass activities, the MOC automatically sends test commands to the ground site and receives command echoes back verifying that the commands were properly received and interpreted. The ground station can also send test data to the MOC. The test data are spacecraft telemetry data stored at the site and injected into the front end of the ground station processed through the entire system and sent to the MOC for validation. If the ground station has a high rate link to the spacecraft, this may also include a sample of the payload data. The ground stations may also track a collimation transponder located nearby so the antenna pointing, ranging, and Doppler functions can be tested and verified.

The ground station is responsible for detecting and establishing a communications link with the spacecraft. The ground station uses a profile of antenna elevation and azimuth angles as a function of time that is based on the

† With interplanetary missions the real-time contact may actually be several minutes or hours behind real-time on the spacecraft. Even with LEO missions, there is a small time lag, so "real-time" is a relative term.

spacecraft ephemeris. This profile might be produced at the ground station or calculated at the operations center and forwarded as part of the pre-pass products.

If a communication link with the spacecraft does not occur within the allowed window of time, then contingency activities for locating and establishing communication will be initiated. These activities might involve doing a pre-defined search pattern with the antenna around the expected location of the spacecraft or switching receivers or other equipment in case of performance problems.

Once the spacecraft's carrier signal has been detected, a firm signal lock is established to provide a stable communication link. The ground station receives the spacecraft downlink signal, successfully de-commutates the data and sends the telemetry to the MOC. At this time, commands can be sent to the spacecraft as needed. All during the contact with the spacecraft, the ground station is sending to the MOC the spacecraft telemetry; possibly tracking data in the form of angles, range, and Doppler; and ground station status data, although often some of these data are provided post-pass.

The spacecraft, instruments, and ground network state-of-health (SOH) data are monitored and controlled when necessary by the MOC. This includes monitoring of the payload instruments and spacecraft telemetry received by each ground station and sent to the MOC in real-time. The MOC command and telemetry system converts the telemetry to engineering units and performs limit checks on all telemetry values and reports automatically any telemetry parameters that are out of bounds or approaching an out of bound limit. On many missions the SOH telemetry monitoring is done automatically.

The controllers (spacecraft, instrument, and ground network) are responsible for implementing the plans provided by the MPS function. The human controllers, normally, just monitor the smooth running of the execution process, but are ready to intervene or make decisions as necessary. The spacecraft monitoring and control process verifies the spacecraft response to uploaded commands, monitors "tactical" spacecraft performance, detects spacecraft anomalies, notifies the spacecraft analysts of new anomalies, and logs the details of each contact. At times, the controllers may implement certain contingency plans, and will routinely implement alternative operations as required. However, the monitoring and control function does not usually investigate or resolve undocumented anomalies (unless critical to the safety of the spacecraft), it merely detects and reports them. The reason for this approach is that the function's primary purpose is to ensure the safety of the spacecraft and system in real-time, which could be compromised if resources are diverted to anomaly hunting. The MOC may also be responsible for monitoring and controlling the ground network, although some systems may have a separate Ground Segment Control function.

Routine dumping of the archived SOH or payload data that are stored on-board the spacecraft's memory is initiated by ground command (either manual or automatic) since this can only be done when the spacecraft is in communication with the ground station (or relay satellite). The MOC or ground segment control initiates the payload data dump when the correct link has been established and the data capture and processing equipment at the receiving site are properly configured. If the ground station is automated, the configuration and reporting of station status are done via an electronic link from the ground site to the MOC. If the ground station is manned, the status of the ground station may be relayed by voice to the MOC where the MOC Spacecraft Controller would then initiate the spacecraft commands to dump the recorded data. To ensure that the payload data set was correctly received on the ground before the spacecraft payload memory is reused by subsequent observations, data protocols, such as the CCSDS File Delivery Protocol (CFDP) [CCSDS, 2007], may be used for transporting and verifying the data. This protocol provides for automatic retransmission requests for corrupted or missing packets. In addition, the ground segment network function can perform several tests depending on how the data communication and processing are implemented. The ground segment control function can report on the number of valid frames and the number of correctable and non-correctable errors. If the payload data are also sent to the Data Management System in real or near real-time, then additional verification can be performed such as format verification, internal frame sequencing verification, and even initial Level 0 processing.

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As the end of the pass approaches, the payload data dump is stopped and the spacecraft is put into the configuration required after the pass. This is usually done automatically by the spacecraft, but in some cases may have to be commanded by the ground. One thing that must be avoided is to reach loss of signal (LOS) with the spacecraft in a high power or attitude configuration that could jeopardize its welfare without autonomy on the spacecraft that recognizes and can rectify this condition. This has happened on actual space missions with the loss or near loss of the satellite.

Once LOS has been verified, the ground station is restored to its non-active state and the post-pass activities are performed. These may include validating the management of the SOH and payload data received. Both the SOH telemetry and the payload data are typically archived at the ground station until it has been confirmed that they have been safely received at their respective destinations. The data are distributed with proper accounting. The status of the GNS is verified, and finally the pass log is generated with any problems or comments by the spacecraft and ground controllers being included.

The *Anomaly Resolution process* (Fig. 29-7) is performed as a result of detecting anomalies in the spacecraft (or ground segment) SOH data while monitoring the real-time data during passes, or when analyzing the stored data later.

In the figure, the process for the two basic cases is shown. During the pass the realtime data are monitored continuously and checked for out-of-limit or other anom-



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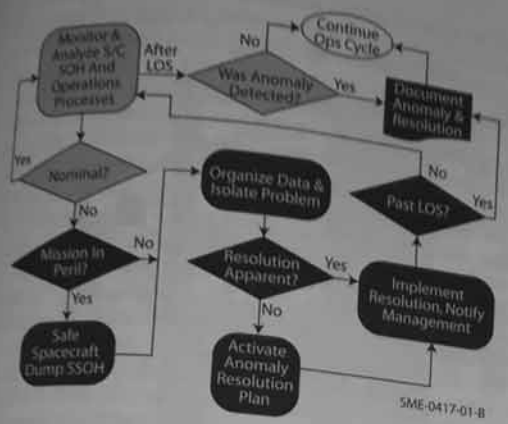


Fig. 29-7. Spacecraft Anomaly Detection Process During Flight Operations. During real-time flight operations the downlinked SOH telemetry is monitored for any anomalies. If any are detected the first thing to determine is if it threatens the spacecraft or mission. If so, then the spacecraft must be put into a safe condition while the anomaly is resolved. Otherwise, if the anomaly is minor or a fix has already been approved, it can be implemented. If not, then the anomaly resolution plan, which usually involves calling in appropriate engineers, must be activated. The anomaly and resolution should be documented, but if the pass is time critical, then this could be deferred until after LOS (as shown in figure).

alous values. The first thing to do is determine if the anomaly could jeopardize the mission. If so (and there is not a predetermined fix available), then it is best to set the spacecraft into a safe (lifeboat) state and dump all the archived SOH data which may be important in determining the cause of the anomaly. Obviously, if the spacecraft is manned, the first concern in the face of an anomaly is preservation of life, even before preventing loss of mission. Anomalies that endanger human life, while rare, initiate well thought out, well-practiced emergency procedures. To address the case of such incidents occurring during LOS, both the ground and on-board crews are able to both recognize and react to such situations. If human life is not at stake, on-console spacecraft operators should not attempt to solve unknown anomalies. They should only do as they have been trained to do and leave the anomaly identification and resolution to the spacecraft engineering experts. Safing the spacecraft (and the human crew) is the preferred response in this situation. After safing the spacecraft or if the anomaly was not critical, then the data should be organized if needed and the problem identified and isolated (highlighted) in the data. If there is a known resolution, then it can be implemented and management notified. If the resolution for the anomaly is not apparent, then the anomaly resolution process should be activated, and if resolved in time (before LOS), implemented as described before. After the pass, the anomaly and its resolution should be documented.

If the anomaly was detected during post-pass analysis, then the anomaly resolution plan should be activated, a resolution determined (possibly with the help of an Anomaly Resolution Team which may include spacecraft or payload engineers as required). Once the resolution has been determined, it should be tested on the OTB/simulators, if possible. Once verified, the tasks or commands can be sent to the mission planners or directly to the spacecraft controllers as appropriate (see Fig. 29-3). Figure 29-3 shows screened shadow boxes below the OTB/simulators and the Monitor & Control boxes to signify the part they play in anomaly resolution.

When the anomaly and resolution are documented, they should also become part of the flight handbook so that the resolution is readily available in case the anomaly occurs again.

The *Data Processing and Management process* (sometimes shortened to just *Data Management process*) is the collection, logging, processing, storage, retrieval, transfer, distribution, and reporting of all the spacecraft and mission-related data. All the data, both SOH and payload, are downlinked from the spacecraft and received at the ground station where they are archived and then forwarded to the MOC (see Fig. 29-8). The real-time SOH data are sent directly from the ground station to the operators in the MOC.

The MOC begins the process of data processing by performing the Level 0 processing on the stored (and possibly real-time) SOH and payload data, i.e., removing the spacecraft and ground communication artifacts from the data such as headers, trailers, compression, special protocol characters; and correcting correctable transmission errors. The Level 0 data are then archived at the MOC. The Level 0 stored SOH data join the real-time SOH data at the front-end processor in the MOC where they are converted into engineering units (if required) and put into the form needed by the operators.

The Level 0 science/payload data are sent to the SOC (or payload customers) for further processing, archiving, and analysis. For some missions, raw science data are sent to the SOC where the Level 0 processing is performed. The SOC also has access to the archived SOH data that may be required during the analysis. The archived SOH data (and processed science data) are then available for mission data analysis with results going to the mission planners and others. Stored SOH data are used extensively during the engineering checkout phase of the mission.

The real-time data are monitored in real-time and also stored as both raw and engineering data for the life of the mission. The data management process also collects ancillary data to be combined with the payload data such as orbit and pointing information. Non-spacecraft telemetry data such as tracking data and ground station status data are also managed by the MOC. All the data received and processed by the MOC are stored in the MOC Data Archive.

The SOC applies or attaches to the payload data any calibration values for the sensors or optics, and append any ancillary data such as orbit and pointing data, sensor telemetry, and possibly the associated command sequences used for the observation. Some of this infor-

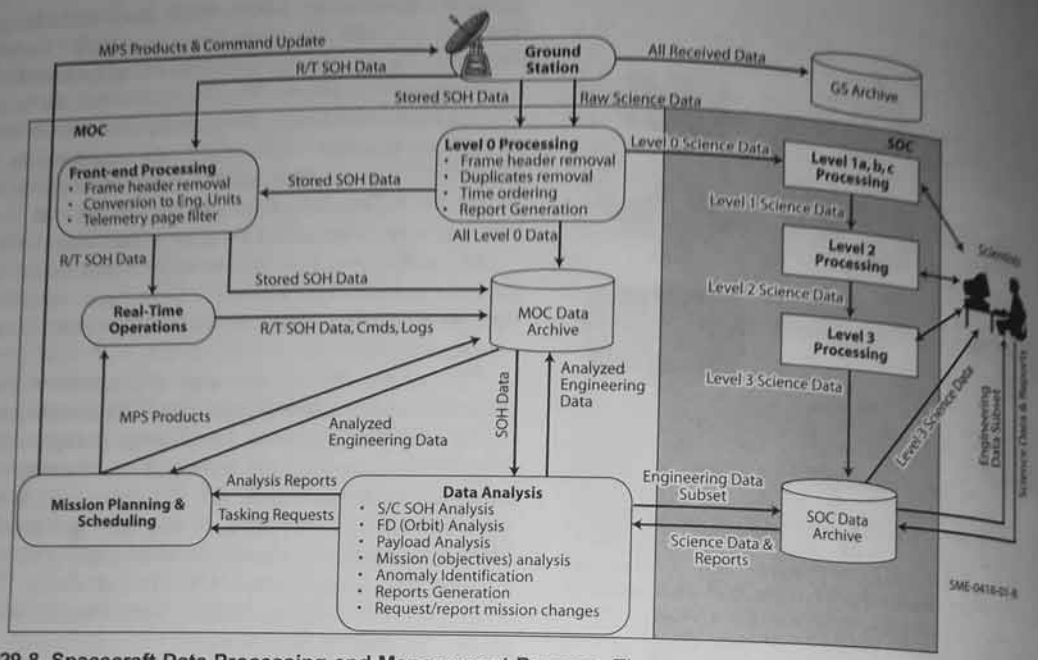


Fig. 29-8. Spacecraft Data Processing and Management Process. The process shown here is typical for a large science mission with a separate SOC, but different schemes can be used. The data downlinked from the spacecraft are received at the ground station where they are archived and then forwarded to the MOC. The real-time SOH data are sent directly from the ground station to the operators in the MOC. The stored SOH and payload data undergo Level 0 processing and archiving at the MOC. The Level 0 science data are then sent to the SOC for further processing, archiving, and analysis. The SOC also has access to the archived SOH data which may be required during the analysis. The archived SOH data (and processed science data) are then available for mission data analysis with results going to the mission planners and others. In some missions, the Level 0 processing for science data is performed in the SOC.

mation may be added later, such as the orbit and pointing data, which may need further processing to provide precise values.

Besides handling the data downlinked from the spacecraft and ground station status data, the data management process also handles other data transfer throughout the operations system, such as transfer of command loads (consisting of command sequences and flat files) to the ground stations for uplinking to the spacecraft. Other data products, such as schedules pass plans, pass logs, reports, may be handled by the data management system, which is responsible for both the transport and archiving of the data.

The *Analysis process* interprets the data obtained by the mission. Closely related to spacecraft and payload command and monitoring are spacecraft and payload analysis. The primary difference is that the monitor and control process is used with the near real-time telemetry, while the analysis process deals with the stored data that are received and processed by the system elements. Spacecraft subsystem and payload analysts review telemetry data and track the long term health and characterization of each subsystem and component. Although much of the data trending and analysis may be automated, usually there will still be human analysts required to interpret the results and make appropriate decisions based on them, although they should be able to work just normal working hours instead of a schedule based on the spacecraft contacts.

The *Spacecraft Analysis process* monitors the health and long-term trends for each of the spacecraft subsystems by examining power, thermal, pointing performance, radio links, memory storage, solar cell and battery degradation, and mechanical subsystems such as solar arrays, antennas, and the optical/payload bench. The spacecraft analysis process has the overall responsibility for determining and ensuring spacecraft safety and effectiveness. The primary functions are monitoring the health and status of the spacecraft bus and its resources, including engineering data such as temperatures, voltages and currents. This process is responsible for the trending and analysis of critical spacecraft SOH parameters and detecting anomalies or potential problems based on both short- and long-term performance trends. Spacecraft anomalies that are detected during this analysis or reported by either the spacecraft controllers or other sources are identified, documented, and resolved. If the anomaly cannot be resolved by the spacecraft analysts, then an Anomaly Response Team (ART) can be assembled and employed in the anomaly resolution process. The spacecraft analysis process also reports spacecraft status to spacecraft management and mission planners, and coordinates with spacecraft component manufactures and integrators as required.

The *Orbit and Trajectory Analysis process* is performed by the Flight Dynamics (FD) function of mission operations. On a regular basis, FD analyzes the tracking

Table 29-7◀, Fig. 29-8, Eq. 29-0

data. The analysis is the validation and calibration of the measured data used for orbit determination. The tracking data are validated automatically by comparing the measurements with the predicted orbit location, generating observation residuals and using statistical methods to identify poor quality data. Calibration using these tracking data is difficult with orbits very far from the Earth, so the use of collimation tracking data and periodically tracking other near-Earth spacecraft and analyzing the tracking residuals can be used to calibrate the angle and range data. Daily, or every few days, the spacecraft orbit is determined using the validated tracking data. This "actual" orbit is then used to produce the needed orbit products, such as a predicted ephemeris for mission and science planning, acquisition (pointing) data for the ground stations, orbit information to be sent to the spacecraft for attitude control, and a definitive orbit ephemeris for science and image analysis.

Using the definitive spacecraft orbit, FD propagates the orbit into the future and predicts when the orbit requires adjustment. A LEO satellite typically obtains an orbit ephemeris update on a daily basis. The date and parameters of the orbit maneuver are updated and made available to the spacecraft planning system. The planning system then constrains the science and other spacecraft activities around the time of the maneuver. If the exact time and location of the maneuver are not extremely critical, the planning system has some leeway as to when the spacecraft is unavailable due to the maneuver. The planned maneuver, however, is incorporated into the predicted ephemeris and used for long and short term planning. Flight Dynamics maintains the fuel consumption estimates and the propulsion performance and calibrations values.

Mission Analysis process has the overall responsibility for determining and ensuring the spacecraft payload safety and mission effectiveness. For some missions, the instrument analysis is a separate process performed by the SOC or payload engineers, while for other missions it may be performed in the MOC. The mission/payload analysis process is similar to that of spacecraft analysis, except with responsibility for the science instruments/payload instead of the spacecraft bus. This process is also responsible for determining the success of the spacecraft in accomplishing the science mission objectives of each observation, although not the science results. The mission analysis process is responsible for identifying, documenting, and resolving payload anomalies that are reported by either the spacecraft or payload controllers or other sources. The resolution process is similar to that of the spacecraft analysis function.

29.3 Mission Termination and Post-Mission Activities

Every mission comes to an end eventually, although in same cases that can be a very long time from launch. At the time of publication of this book, NASA's Voyager 1 and 2 missions are ongoing, more than 33 years after their launches in 1977. Both are still providing data about the heliosphere and interstellar space to NASA long after their primary mission to visit the outer planets had been successfully completed. Some missions complete their primary mission (or are unable to perform it) and may be funded for an extended mission, which may or may not have the same objectives. Examples of extended missions with the same overall objectives are the NASA Mars Exploration Rover missions, which have continued years past their nominal 90-day missions. An example of an extended mission with different objectives is the NASA Wide-field Infrared Explorer (WIRE) mission, which had to abandon its scientific mission of exploration in the infrared when the coolant required for the infrared sensor was exhausted early in the mission (due to a failure). However, the bus was still healthy, so the WIRE satellite was used to train spacecraft controllers and give students a "hands-on" learning experience.

There are generally four main reasons for mission termination:

1. Completion of mission (e.g., Stardust returned samples from a comet tail to Earth or LCROSS impacted the lunar surface—by design)
2. Lack of funding to continue operations—this includes obsolescence
3. Spacecraft (or payload) malfunction, including loss of contact—also includes catastrophic operations errors
4. Orbit decay and reentry or other destructive events (e.g., collision with debris)

For the first two reasons and sometimes the fourth reason (e.g., orbital decay to reentry), the mission management is aware of the mission termination in advance and can plan for it. The planning and implementation of mission termination is the responsibility of mission operations.

Since the dawn of the space age with the launch of Sputnik I in 1957, orbital debris has become an increasing problem and is recognized by the various agencies that launch satellites into Earth orbit. In order to reduce the amount of human-produced orbital debris (which includes derelict satellites as well as spent boosters, and pieces of space hardware), which can provide a real hazard to operating spacecraft (including the ISS and other manned vehicles), most Earth-orbiting space missions are designed according to criteria and standards such as outlined in the NASA document, NASA-STD-8719.14 (see also IADC [2011]). This contains a deorbit requirement of 30 years from launch or 25 years after the end of mission. To adhere to this rule, an orbital lifetime analysis must be performed. For small satellites with no on-board propulsion that can provide a deorbit capability, this means that the mission should be designed with an orbital altitude low enough that reentry will naturally occur within this time period. This threshold altitude is typically below 1,000 km, although orbital lifetime is very sensitive to changes in mass, cross-sec-

Table 29-7◀, Fig. 29-8◀, Eq. 29-0

tional area, and solar activity. If the mission has to be performed at an altitude that drives the orbital lifetime beyond the specified limits, then some sort of reentry aid needs to be included, such as a deorbit propulsive system or device to increase the cross-sectional area and thus the atmospheric drag (e.g., "speed brake" panels or an inflated balloon). See Sec. 30.2.

For satellites in a geosynchronous orbit, there is a different solution since the amount of propellant required to perform deorbit may be prohibitive. Communication satellites, for instance, occupy a particular "box" in the orbital geostationary equatorial belt and if they need to be replaced by a new satellite, then they must vacate the box without becoming a hazard. In this case, they are boosted into a higher "super geosynchronous orbit" which leaves them in orbit for centuries, but above the operating satellites and not between the Earth and geosynchronous orbit where they could still provide a hazard for collision with satellites transferring up to geosynchronous orbit. See Sec. 30.3.

Termination operations should be planned well in advance. If the spacecraft is large enough that parts of it may survive the reentry environment, then detailed analyses must be performed and the reentry burn carefully planned to ensure the surviving debris will impact in the ocean and the event should be coordinated with appropriate government and international agencies.

Before a satellite's mission is terminated, it should be put into as quiescent a state as possible. Typically, the batteries are drained, the EPS is disabled so that the solar arrays cannot recharge the batteries or provide power to the bus, the on-board memory is scrubbed, pressurized systems vented, mechanized systems put into a safe state, and the radios turned off (usually by cutting power). See TDAC [2007] for more information on safing spacecraft to reduce debris. Tracking of the satellite can be continued using the Two-Line Elements (TLEs) provided by the USAF.

For science or remote sensing missions in particular, the mission can continue long after the spacecraft's operation has ceased as the data are analyzed, reported, and used (e.g., in journal papers). There are typically some tasks that have to be performed to complete the project, such as writing final reports about the project along with other required documentation, technical papers, and documenting and sharing lessons learned (e.g., through technical conferences).

29.4 Mission Operations Process Improvement and Best Practices

29.4.1 Process Improvement

Even the most carefully planned mission operations or support plan may not survive first contact with reality. If the mission operations system has not been designed with the flexibility and processes to recognize problems or anomalies, analyze them, and provide a feedback loop to introduce improvements back into the mission operations process, then it will be very difficult and costly for the system to adapt. It is far better and cost effective to design these process improvement features into the system than to try re-engineering the system after launch. Several missions have attempted to introduce automation and process improvements into the system after launch, and have had a very difficult time doing so. It is difficult to do this without disrupting or risking the ongoing operations, and when it is possible, it is usually very costly. The golden rule of process improvement is: design the process of process improvement into the system from the very beginning so that it will appear in the design requirements.

This section, which is an abridged version of what appears in SOSTC [2003] and Sorensen, Campion, and Probert [2011], looks at some means that can be used for helping with the mission operations process improvement, from the determination of suitable metrics, methods to collect and analyze them, determination of solutions, and then feedback of the solutions into the system. Although the actual metrics and methods that are best suited for a particular mission might be different, the general principles stated herein are the results of experience obtained on several missions, including Low-Power Atmospheric Compensation Experiment (LACE), Clementine, MSTI-3, and others.

A major factor in the cost of spacecraft ground support is the effectiveness of the mission operations process. An ineffective, error-prone and labor intensive process will most likely result in increased cost, risk, and reduced customer satisfaction. In order to determine the effectiveness of how mission operations are performed and to determine areas of improvement, measures of effectiveness (MoEs) should be identified. The metrics obtained through these measures of effectiveness can then be empirically and subjectively analyzed to determine the areas of the operation that should be improved or automated to increase efficiency.

For a science mission, effectiveness factors for the mission operations include:

- Percentage completion of science objectives (e.g., number of science experiments successfully executed, coverage obtained by imaging, quality of data, quality and quantity of calibration data obtained)
- Cost of operations (comparison of actual versus projected costs)
- Response time and flexibility of the mission planning and operations process
- Efficiency (cost/data collected)

Some metrics that can help measure the effectiveness of science mission operations include: error tracking, exceptions (complexity) factor, rush factor, effort factor, response factor, fatigue factor, and morale factor. These MoEs and metrics were first identified in post-mission analysis of the Clementine lunar mission and were very useful in determining where the mission operations pro-

cess was successful and where it needed improvement. They were subsequently used in the analysis of other historical missions before being designed into an operational commercial mission operations system (Honeywell's DataLynx). MoEs and their metrics can help to determine when and where to add automation to mission operations. For example, when command script generation was automated during the MSTI-3 operations, the overall error rate for the spacecraft dramatically declined, which was obvious by tracking the error metrics [Sorensen, Campion, and Probert, 2011].

Error Tracking. This MoE tracks all the ground source errors that reach the spacecraft during the mission (although we are using the spacecraft as the end "victim" system, this MoE could be equally applied to other systems that receive external data that could cause errors in its execution). Most of the errors that reach the spacecraft are generated by the mission operations process or allowed to pass through it to the spacecraft. Spacecraft commanding and operations errors that affect accomplishment of mission goals may include:

- Planning and timeline/schedule errors
- Command script/sequence errors
- Instrument or spacecraft pointing errors
- Commands/script testing errors
- Ground system errors
- Real-time operations errors
- Spacecraft hardware errors
- Software errors (ground and flight)
- Miscellaneous errors (communication links, ground segment hardware)—this is a catchall category of unlikely or rare sources of errors. If any of these elements become a significant source of errors (e.g., communication link), then it should probably tracked as a separate error. These errors can be either human or machine

Complexity/Exceptions Factor. This is a measure of the complexity of a mission "event" (e.g., pass, observation, or experiment). If there is a "standard" sequence for spacecraft operations, then this is the number of "exceptional" events being added to that sequence (e.g., special operations added to mapping). Metrics for this MoE are chosen to meaningfully reflect complexity (e.g., number of commands or activities required).

Rush Factor. This is a measurement of time between timeline or script completion and script execution on spacecraft. The Rush Factor MoE is inversely proportional to the time, i.e., the less time, the higher the Rush Factor. Elements involved in the mission operations process that may affect the Rush Factor include time required for testing of scripts on simulator/testbeds and time required for queuing and upload to spacecraft. The Rush Factor should be low (days, not hours). However, in order to be responsive to the science team or customer in a dynamic mission, the Rush Factor may by necessity, remain high, i.e., the higher the Rush Factor, the more responsive the operations team is, although it is at a cost of putting strain on the team and processes.

Effort Factor. This is a measurement of the number of man-hours expended per mission event. It can be a measure of complexity, but it is complicated by the efficiency of the process as well as by the level of automation. The Effort Factor is desired to be low to reduce costs and possible sources of errors. Automation can reduce the Effort Factor for operations personnel, but increase it for software engineers and programmers.

Response Factor. This is an inverse function of the measurement of time between the customer's (e.g., science team) request for a mission event and its execution. The Response Factor should be weighted to account for complexity of the requested event. This factor should be maximized (i.e., the time needed between requests and execution minimized) for effective operations.

Fatigue Factor. This is a measurement of the tiredness of the operations team (e.g., hours worked). The short-term Fatigue Factor is based on shift length, while the long-term Fatigue Factor is measured over weeks or months. Other factors (e.g., complexity, rush, effort, and response) can affect the Fatigue Factor. It may be determined by subjective data (e.g., questionnaires) and the number of errors generated.

Morale Factor. This is a measurement of the satisfaction and optimism level of the operations team, but is difficult to quantify. It is mostly subjective, but some metrics can be collected to help in its determination. The possible metrics includes the turnover rate of personnel and the number of operations personnel complaints received by the operations management. It might also be possible to use routine surveys of operations personnel, but it has to be determined subjectively as to how accurately these surveys reflect the true morale of the personnel.

In order to effectively generate, track, and use these MoE metrics, they should be incorporated into the mission operation process. Due to the limited record keeping typical in many of today's missions with limited budgets, it is often difficult if not impossible to reconstruct these metrics accurately, either to generate historical test cases, or to determine retroactively how the MoE factors have changed over the life cycle of a current operations process. However, steps can be taken in the design of a new operations process to implement changes in an existing system to collect these metrics.

At each step in the process, two logs should be generated and kept. An automatic on-line log should record the time that each event starts and stops in a sub-process (e.g., recording the time that a timeline enters the script generation step and the time that generated script leaves this step to be sent on to the next step in the process, usually testing). This automatic log should also record errors detected by the computer system, especially of errors that were detected in the input data, as well as any significant

Table 29-7◀, Fig. 29-8◀, Eq. 29-0

decisions or substeps. A manual on-line electronic log should also be kept. This log is to record any errors found and corrected or changes made by the operator, along with the decision rationale if appropriate. Both logs should be archived with the files for that particular pass or event and sent automatically to the operations director or analyst for review and analysis.

Metrics Analysis Process—The operations director or mission operations analyst should regularly collect and review metrics to identify problem areas. Trending software is of particular use to see how the factors change over time. The most useful plot is the cumulative errors plot, which shows on the same chart the cumulative total errors and each of the separate errors over the life of the mission or other designated time period. The cumulative number of errors is not so important, but the slope of the line is (i.e., the derivative of the cumulative errors with respect to time). By correlating the slopes of the line (steep slopes are bad, while flat or gentle slopes are good) to the seven MoE metric tracking charts, causes of the change in errors occurring on the spacecraft can often be identified by type. Steps can then be taken to analyze the details of particular MoEs to determine the root cause of the problem (or conversely, the lack of problems).

Feedback Implementation—Once a sub-process has been identified as needing improvement, total quality methods should be used to involve operations personnel in the solution. They can help in both the identification of the root cause of the problem as well as to help determine how to rectify it and work out a way to implement the solution into the operations process. Methods and metrics to determine the success of the implementation should also be identified. In some cases, it might be necessary to include mission or program managers, and customers (e.g., principal investigators or chief scientists).

Discrepancy Tracking, Archive and Reporting—As is true for other aspects of mission operations, all discrepancy tracking, metrics collection and analysis, problem resolution and decisions should be archived. Any feedback implementations that have been decided upon should be put into the formal discrepancy tracking system and followed by the operations director until the implementation has been fully completed and tested. Any meetings involved in the operations process or improvement process should be documented to leave a documentation trail of decisions made with rationale. Any reports or minutes of these meetings and decisions should be put into the operations archive and a copy sent to the appropriate mission operations management.

29.4.2 Best Practices

Probably the major theme of this chapter, and indeed this book, is that there is no need to re-invent the wheel. For most of you planning to operate a space mission, someone has operated a similar mission before (at least in many respects). We have tried to provide you with the basics for designing and implementing an adequate operations plan based on many years of experience. It is helpful when you have access to best practices and lessons learned from previous missions. Many such documents, especially in the commercial world, are either proprietary or very difficult to locate. However, reviewing and using such lists could be very beneficial in the four basic areas of your space project: performance, cost, schedule, and risk. There are some documents available from NASA and other agencies, such as in GSFC [1999], some of which are reported in the form of technical papers at the various professional conferences. However, the Best Practices Working Group of the AIAA Space Operations and Support Technical Committee (SOSTC) started in 2001 to compile mission operations best practices into an online document, [SOSTC, 2003] the first version of which was released in 2003 and continues to be expanded and updated. This has contributions from many different authors from government, industry, and academia. We highly recommend this document as a source of best practices and lessons learned for space mission operations. Also recommended is the Annual Improving Space Operations Workshop sponsored by the SOSTC, the details of which can be found on their website [SOSTC, 2011].

29.5 The Future of Mission Operations

It is difficult to predict the future, but based on trends in the area of space mission operations, it is possible to reasonably extrapolate these trends into the near future, the next 10–20 years, or the lifetime of this book. Although the computing power and miniaturization of computer processors used in spacecraft lags behind those used in terrestrial applications, due to the need for radiation hardening and improvements in reliability, the computing power of today's generation of spacecraft far exceeds that of spacecraft in the past, and it is continuing to increase. Even a simple CubeSat has more computer capability than billion-dollar spacecraft of a couple decades ago. The result is that more of the functions that were performed on the ground as part of mission operations can now be performed on board the spacecraft. Functions such as processing payload data, orbit determination (thanks to GPS and similar systems), spacecraft attitude determination and control, and ground contact execution can now be managed by the spacecraft rather than the ground. Clementine was the first mission to perform an autonomous spacecraft operations experiment in 1994 [Sorensen, Oswald, Shook, Van Gaasbeck, 1995], but this type of autonomous operation is becoming more common. The effect of this is what was in the 1990s an innovative form of mission operations, known as "lights out" operations, is also becoming more common. This means that the spacecraft monitors its own condition and is able to execute ground station contact with no humans in the loop, even for monitoring. This happens when both the spacecraft and the ground station have been designed for autonomous or automatic operations. Of course, if there are any anomalies detected that are not within the portfolio of the automatic systems to handle, then human

operators need to be notified and brought in to resolve them. So far, lights-out operations are typically only done with standard and relatively simple operations. Humans are still needed to do long-term or short-term planning for the mission (although this is also changing), and to examine and analyze trends in the engineering SOH data, and act upon it when needed. However, as the computers (both hardware and software) become increasingly capable, the roles of humans (and the resulting high recurring costs) will decrease.

Despite the decreasing role of humans in mission operations, most spacecraft today and for the forseeable future, especially in the areas of science and exploration, are so complex and expensive that the need for humans in the operations process will remain. There is an ongoing trend, which goes back decades, to consolidate and reuse mission operations systems. It is desirable for schedule, cost, and risk considerations to reuse software and hardware from one mission to the next. It is also desirable, for organizations that operate several spacecraft, to be able to operate multiple spacecraft using the same mission operations centers and software applications. Although this has been a goal for many years, it has been difficult to implement in a cost-effective manner, especially when the spacecraft and missions are dissimilar. However, as discussed in Sec. 20.4, there are some promising efforts underway, internationally, that are using a "plug'n'play" paradigm to achieve the goal of providing mission operations cheaply and quickly for new missions and mutiple missions, simultaneously. JPL has the Advanced Multi-Mission Operations System (AMMOS), which is designed to provide the tools and services to support NASA's deep space and astrophysics missions, but is designed to be flexible enough for other missions and is available for non-NASA missions and universities [AMMOS, 2011]. Goddard Space Flight Center has its GSFC Mission Services Evolution Center (GMSEC) reference architecture, which provides a scalable, extensible ground and flight system approach for space missions. GMSEC is particularly suited for the quick and easy integration of functional components obtained by the mission project that enable a unique mission operations solution to be tailor-made [GMSEC, 2011]. Another mission operations system especially designed for end-to-end operations of small satellites is the Comprehensive Open-architecture Space Mission Operations System (COSMOS) from the Hawaii Space Flight Laboratory at the University of Hawaii at Manoa in collaboration with NASA Ames Research Center. COSMOS is a system of interconnected tools (both internal and externally provided) that are designed to be easily adaptable to new spacecraft and for installation in new MOCs. The source code is available to users for special modifications and enhancements that are not handled by the COSMOS editors and system itself. COSMOS is especially suited for nano- and micro-satellites, although it can handle larger ones as well [COSMOS, 2011].

Of course the ultimate goal of human spaceflight operations is similar to that of the aircraft operations, where the humans on board the spacecraft with the help of the on-board systems, are able to monitor and control the spacecraft with no intervention from the ground, except in emergencies. A Space Traffic Control (STC) would monitor the space lanes, the launches, and the descent of the spacecraft. The STC would hand over control to the Air Traffic Control system when the spacecraft entered the atmosphere approaching aricraft altitudes, where the ATC would integrate the spacecraft into the air traffic patterns down to a successful landing. This vision could be a reality within our lifetimes and may be sooner that you think.

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30 End of Mission Considerations

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30.1 Inter-Agency Space Debris Coordination Committee (IADC) End of Mission Guidelines

30.2 Low Earth Orbit (LEO) Disposal Options

30.3 Non-LEO Disposal Options

30.4 Passivation

Propulsion System Passivation; Power System Passivation; Passivation of Other Systems; Passivation Challenges

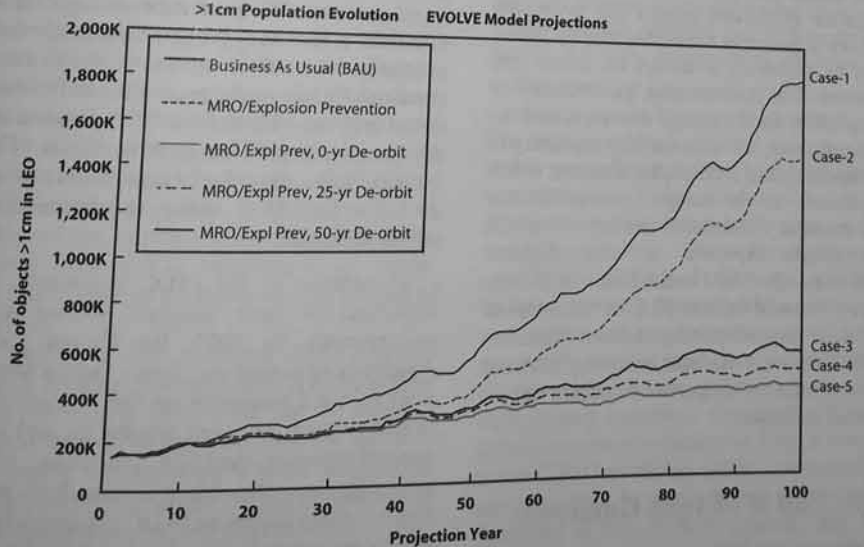
30.5 Disposal Planning

30.6 FireSat II and SCS Examples

Disposal of the FireSat II Spacecraft; Disposal of the SCS Spacecraft

While a great deal of effort goes into planning and executing successful mission operations, it is also important to consider the end of the mission during the planning, design, and operations phases of any mission. Spacecraft and launch vehicles must be disposed of properly in order to limit the generation of orbital debris (Sec. 7.5), and better preserve the orbital environment for all future missions. Figure 30-1 shows a 1990s projected growth of debris with and without the use of responsible disposal techniques. This requires early selection of a responsible disposal scenario, so that the necessary capabilities can be incorporated into the hardware designs. The mission operations must be conducted in such a way as to preserve, and then actually perform the planned, appropriate end of mission disposal.

Computer simulations have shown that the orbital debris population already present on orbit is self-propagating; that is, the orbital debris density will continue to increase through random collisions alone, unless reduced by outside efforts [Liou and Johnson, 2008]. This may well result in a cascade effect that eventually renders some orbits impractical for space operations. Since it is



Case-1: Business as Usual (No mitigation measures are taken.)
Case-2: Mission Related Objects are refrained from being released, and explosions are prevented.
Case-3: In addition to Case-2, systems are removed within 50 years after mission termination.
Case-4: In addition to Case-2, systems are removed within 25 years after mission termination.
Case-5: In addition to Case-2, systems are removed immediately after mission termination. SME-0270-01-B

Fig. 30-1. Debris Growth with Various Mitigation Approaches. [IADC, 2004a]

Table 30-0, Fig. 30-1, Eq. 30-0

not yet economically practical to remove a significant amount of existing debris from orbit, it is critical that responsible end of mission disposal be practiced for all current and future missions, in order to help control the rate of increase of mission-lethal debris objects in commonly used orbits. Had such methods been employed throughout the history of space operations, the cascade effect might have been prevented, or at least substantially delayed.

End of Mission disposal (also known as *End of Life disposal*, *Decommissioning*, or simply *Disposal*) has been addressed primarily at the international level in discussions by the Inter-Agency Space Debris Coordination Committee (IADC). The IADC is an international forum of national space agencies and the European Space Agency (ESA) for the coordination of activities related to the issues of man-made and natural debris in space [IADC, 2002]. In 2002, the IADC issued a set of guidelines (IADC-02-01) addressing, among other things, prevention of post-mission explosions, and acceptable disposal options. These guidelines, described in Sec. 30.1, were slightly refined in 2007.

Spacecraft mission designers need to consider disposal early in the design process (see Sec. 3.5), in order to incorporate the necessary hardware and procedures to ensure a safe disposal. The first step is to select a baseline disposal method, as described in Secs. 30.2 and 30.3. This disposal method will determine the key design factors which will need to be considered throughout the remainder of the design process. It may, for example, be necessary to size the propulsion system and navigation hardware for an orbit change or controlled reentry. Alternatively, it may be necessary to design the power and propulsion systems for post-mission passivation, as described in Sec. 30.4. As the design develops, and the reentry risk is determined, it is occasionally necessary to change the baseline disposal method, but at a cost which increases dramatically as the design matures. In any event, it will be necessary to develop and test spacecraft operations procedures specific to the disposal (Sec. 30.5). The disposal of the FireSat II and SCS sample missions are discussed in Sec. 30.6 as examples of the application of the disposal principles. Early consideration of the end of mission disposal is among the most effective ways to minimize the growth of orbital debris, to the benefit of all missions.

30.1 IADC End of Mission Guidelines

While they do represent agreements among the leading space agencies of the world, it is important to note that the IADC guidelines are not currently legally binding. They are generally reflected in numerous national space policies, however, and are followed at least in part for the majority of scientific and military space missions. Despite whether commercial and other missions are legally bound to comply with the IADC guidelines, meeting the standards has been shown to be crucial for limiting the growth of orbital debris, which is in the interests of all space users. The guidelines are summarized here, and the specific text is readily available on the internet [IADC, 2004b].

The IADC guidelines provide guidance for limiting the generation of orbital debris both during and after space operations. They begin by describing the need for limiting the growth of orbital debris, and defining the relevant terms used throughout the document. The guidelines refer to direct creation of debris through operational debris (e.g., lens caps), and potential breakage of tethers. They also consider on-orbit breakups caused by explosions during and after the mission, as well as intentional destruction by internal or external sources. The guidelines also define accepted disposal orbits (Secs. 30.2 and 30.3) and other conditions such as the timeline for abandoning commonly used orbits, and controlling the risk to people and property on the Earth. Finally, they address limiting the potential for damage by collisions with other space objects and with small orbital debris that could prevent the ability to successfully execute end of mission disposal. The IADC guidelines are written to apply throughout the mission lifetime, from design through operations and decommissioning.

It is worth noting that while the IADC guidelines lay the foundation for general agreements on the limitation of orbital debris, with few exceptions, they do not provide specific quantitative requirements. In fact, the stated purpose of the guidelines is to "demonstrate the international consensus on space debris mitigation activities and constitute a baseline that can support agencies and organizations when they establish their own mitigation standards." Only in the case of the definitions of the protected orbit regions, and the GEO disposal conditions, do the guidelines provide specific limits. The remaining limitations are described as qualitative measures, which are left to individual agency requirements documents to define in detail.

In addition to the IADC guidelines, various other organizations have adopted similar guidelines and requirements. In 2007, the United Nations General Assembly endorsed the "Space Debris Mitigation Guidelines of the Committee on the Peaceful Uses of Outer Space," which are very similar to, and cover the same general topics as, the IADC guidelines, while being generally less specific. Most nations look to their national space agency (often an IADC member) for orbital debris policy and control. The US has issued not only the "US Government Orbital Debris Mitigation Standard Practices," but also has a number of individual agency documents that address orbital debris limitation to varying degrees. In general, the requirements documents issued by the IADC member agencies themselves are the most specific and restrictive. Examples of national orbital debris limitation documents are shown in Table 30-1.

Table 30-1. International Orbital Debris Limitation Documents.

Domain	Document
IADC	IADC-02-01, Rev 1
US Government	US Government Orbital Debris Mitigation Standard Practices
NASA	NPR 8715.6A, NASA-STD-8719.14
US DoD	DoD Space Policy Directive, 3100.10, AFI 91-217
US FAA	Title 14, CFR Part 415.39
Japan	JAXA JMR-003
France	CNES MPM-50-00-12
Europe (ESA)	European Code of Conduct for Space Debris Mitigation
Russia	Space Technology Items. General Requirements on Mitigation of Space Debris Population

30.2 LEO Disposal Options*

Responsible exit from Low-Earth Orbit (LEO) is one of the most important steps that can be taken to limit the growth of debris in that region of space. An IADC working group has examined the long-term effects of various guideline options [IADC, 2002], and shown that by limiting the amount of time that each vehicle remains in LEO, growth of the orbital debris environment is greatly reduced. Thus, it is desirable and recommended for space objects (both launch vehicle stages and spacecraft) to be removed from the LEO region as soon as practical. The minimum orbital lifetime possible for an individual vehicle might be determined by the remaining maneuvering capability of the vehicle at the end of the mission or by the initial orbit. There are 3 basic approaches to LEO disposal. The most desirable is to perform a controlled (or "targeted") reentry into an unpopulated region of ocean soon after the end of the mission, either using onboard propulsion, or by external retrieval. If this is not possible, it may be possible to boost into a storage orbit between LEO and GEO, safely removed from both regions. Finally, a spacecraft can be allowed to reduce its orbit by atmospheric drag, (Sec. 9.4.4), resulting in an uncontrolled reentry, and impact on an unpredictable portion of the Earth's surface.

Controlled reentry is the preferred LEO disposal method for several reasons. Not only does it provide positive removal of the vehicle from orbit, but the removal also tends to occur as soon as possible after the mission. By selecting a reentry location over a large unoccupied area of ocean, the reentry risk to the ground population is minimized. This, along with the lack of post-mission passivation, can result in greater flexibility for the mission hardware designers. Controlled reentry is not without significant challenges, however. Reentry maneuvers typically require extensive planning, and notification to the relevant air and maritime traffic authorities prior to performing the maneuvers. In practice, controlled reentry is best performed using at least three separate maneuvers in order to better control and refine the orbit, with a final perigee of less than 50 km, to prevent atmospheric skip. In order to accomplish this, the spacecraft design must incorporate sufficiently large thrusters to ensure adequate control authority at low altitudes. Controlled reentry also requires that the vehicle reserve sufficient fuel to reliably perform the reentry maneuvers at the end of the mission, which will result in a larger fuel mass at launch. Section 9.6 discusses the delta V needed to accomplish controlled reentry, and provides examples. With the advent of robotic servicing and retrieval capabilities, it also may become a practical option for missions to further extend mission lifetime, and still perform a controlled reentry, but at the expense of an additional launch.

Disposal into a storage, or "graveyard," orbit may be a practical option for some high-altitude LEO missions. The storage region is located between LEO and GEO, and is generally considered to extend from 2,000 km to approximately 35,586 km (GEO – 200 km) altitude. Within this region, care must be taken to also avoid commonly used orbits, such as the circular 12 hour orbits used by navigation and other satellites. Both the apogee and perigee of the disposal orbit must be within the storage orbit region. As with controlled reentry, the spacecraft design must incorporate sufficiently large maneuvering thrusters, and fuel must be reserved for the orbit raising operation. Figure 30-2 shows the typical delta V required to maneuver from a circular LEO orbit to the 2,000 km storage orbit. Any vehicle left in a storage orbit must be passivated at the end of a mission, as described in Sec. 30.4. In general, only those missions operating above about 1,400 km altitude can reach the storage orbit region with less delta V than reentering within the recommended time frame.

If neither controlled reentry nor storage orbit disposal are practical for a LEO mission, then disposal will eventually occur by uncontrolled reentry. If possible, the final altitude and area-to-mass ratio need to be tailored to ensure that a reentry by atmospheric drag (Sec. 9.4.4) is

*Applicable standards: ISO 17875 (Space Systems). See website for reference.

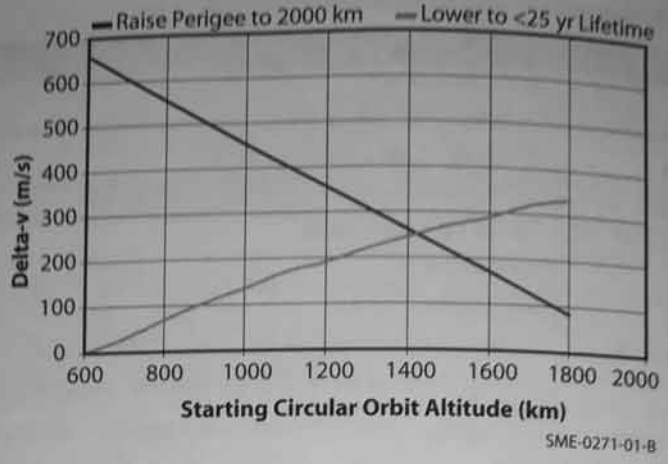


Fig. 30-2. Disposal Delta-V Requirements.

predicted to occur within 25 years after the end of the mission. The IADC study mentioned in Sec. 30.1 concluded that this orbit duration is a reasonable compromise between unlimited orbital lifetimes and immediate de-orbit at the end of the mission. For spacecraft without a propulsion system, the maximum orbit altitude is limited to about 600 to 700 km, depending on the area-to-mass ratio and launch year. The typical delta V required to maneuver from a circular LEO orbit to an orbit that will reenter the atmosphere within 25 years is shown on Fig. 30-2. Note that the solar flux is an important component of this prediction, and varies throughout the solar cycle (Sec. 7.1.1), complicating the reentry date prediction considerably. Current predictions of future solar activity are available for download from NASA, NOAA, and other sources, and are typically updated frequently. Earlier orbit decay will further reduce the likelihood of collisions, and is therefore recommended when possible.

While at first glance, uncontrolled reentry may appear to be a preferred disposal method (it is surely the simplest and lowest mass approach), there are considerable challenges to doing so responsibly. It is necessary for any non-operational object left in orbit to be passivated during the orbit-decay period, as discussed in Sec. 30.4, in order to prevent inadvertent explosion or breakup during the potentially long orbit decay period. The risk to the ground population may also be controlled by requirements of the launching or operating organization. For example, several space agencies require a detailed assessment to show that the spacecraft hardware will burn up sufficiently during atmospheric reentry to pose less than a 1 in 10,000 risk of causing serious injury to even one human. In some cases, the risk to the public from an uncontrolled reentry is sufficient to dictate that a controlled reentry is the only acceptable method of disposal from LEO.

Meeting a requirement to limit reentry risk can be extremely difficult or impossible for some large spacecraft, and usually necessitates specialized design techniques and material selection. The risk is determined by how much of the spacecraft survives reentry, and by the ground population over which the reentering debris might land. In the case of an uncontrolled reentry, the surviving debris might land anywhere in the latitude band covered by the orbit inclination, so an average population density over this band is used for the risk estimate. Object survivability is largely driven by the thermal properties of the primary construction material for the object in question, expressed as a *heat of ablation*. Heat of ablation, H_A, is typically expressed in terms of mass, but it can also be useful to express it in terms of volume (multiplying by the material density), for comparing two material options for the same size part.

$$H_A = H_S \times \Delta T + H_F \qquad (30\text{-}1)$$

where H_S is the specific heat, ΔT is the temperature change, and H_F is the latent heat of fusion.

The heats of ablation for several typical spacecraft materials are shown in Table 30-2. Notice that materials such as aluminum and graphite/epoxy composite are readily demiseable, whereas titanium, glass, and beryllium, all have high heats of ablation, and should therefore be avoided when possible, if reentry risk is a concern. In general, objects made from materials with melting temperatures greater than 1,000 K, heat of ablation greater than 1,000 kJ/kg, or 2,500 kJ/m^3 are more

Table 30-2. Heats of Ablation for Several Common Spacecraft Materials.

Material	Melting/Softening Temperature (K)	Heat of Ablation (kJ/kg)	Heat of Ablation (kJ/m^3)
Graphite/Epoxy	700	350	550
Aluminum	850	900	2,400
Stainless Steel	1,700	900	7,250
Titanium	1,940	1,600	7,050
Zerodur Glass	2,000	1,400	3,550
Beryllium	1,557	4,100	7,550

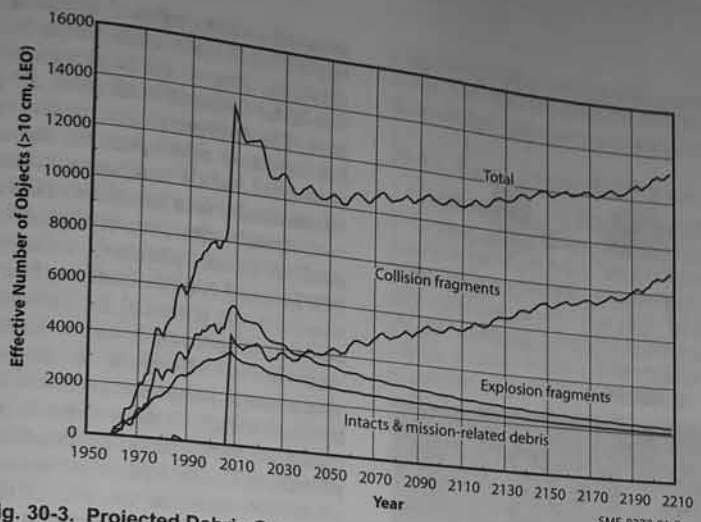


Fig. 30-3. Projected Debris Generation by Mechanism (with no new launches after 2009). [Liou, 2010]

likely to survive atmospheric reentry. *Oxidation heating* (essentially burning) on reentry also contributes to the demisability of aluminum and graphite/epoxy.

Object survivability is also influenced by the object's ballistic coefficient as shown on Fig. 9-22, which is a function of the shape, mass, and dimensions of the object, and determines its velocity (Sec. 9.4). In general, faster moving objects accumulate more heat, and are more likely to demise. Reentry risk has been successfully reduced on some flight missions by modifying a component's shape, size, or material, when possible. Another approach that can be used to reduce the reentry risk is to ensure that several high survivability objects are bound together, since multiple objects are more likely to cause injury than a single object. For example, if several surviving battery cells are contained within a robust battery box, then the single surviving box presents less risk than the multiple unbounded cells. Because the survivability of spacecraft components depends on so many factors, it is necessary to combine very specialized assessment software with detailed knowledge of the spacecraft construction in order to determine and limit the reentry risk. Early avoidance of high survivability materials in component designs can help to prevent difficult and expensive redesigns late in the design phase.

For all LEO mission disposal options, it is desirable to minimize the total time that a vehicle spends in orbit. This is done to reduce the likelihood that the vehicle will experience collisions with resident objects, which generally result in the creation of additional orbital debris. Debris generation potential increases rapidly with projectile size, and even existing objects as small as 1 cm can create additional debris on impact. Simulations predict that on-orbit collisions will be the primary source of new orbital debris, so minimizing orbital lifetime is the most important step toward limiting future debris generation. Figure 30-3 shows that even with no new launches, collisions among existing on-orbit objects will eventually cause the debris population to rise. Any fuel remaining at the end of the mission should be used to lower perigee as much as possible during the passivation process, resulting in the earliest possible reentry. One caveat to this general rule is that the last burn before an uncontrolled reentry should leave the spacecraft high enough so that its orbit remains stable long enough for ground tracking to get an accurate fix. This will allow monitoring of the vehicle for conjunction assessment and collision avoidance purposes as its apogee descends past other operating spacecraft.

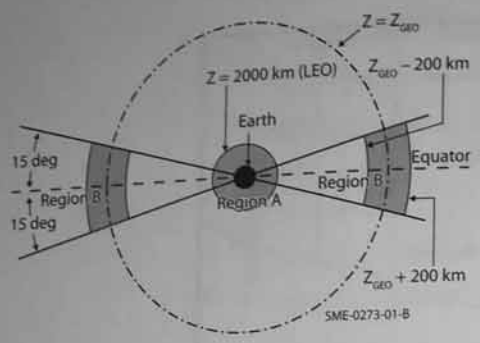
30.3 Non-LEO Disposal Options*

The IADC guidelines define two protected regions of space: LEO and GEO, as shown in Fig. 30-4. No space vehicles should be left in either of these regions on a long-term basis. The *LEO region* is simply defined as the "spherical region that extends from the Earth's surface up to an altitude (Z) of 2,000 km." Removal from the LEO region within 25 years is described in Sec. 30.2. The *GEO protected region* is defined as 35,786 km ± 200 km, with latitude of 0 deg ± 15 deg. Any mission that passes through that protected region must be maneuvered at the end of the mission to remain clear. Removal from GEO should occur as soon as practical after the end of a mission.

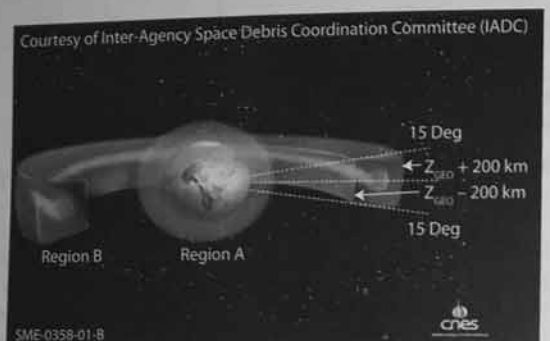
Disposal from GEO is performed by increasing the orbital radius sufficiently to remain well in excess of 200 km above the GEO altitude (35,786 km + 200 km = 35,986 km minimum altitude) for a minimum of 100 years. Due to the effects of solar radiation (Sec. 7.1),

Get Stnd Applicable standards: ISO 26872 (Space Systems). See website for reference.

Table 30-2◀, Fig. 30-3, Eq. 30-1◀



A. Definition



B. Artist's Rendition

Fig. 30-4. The IADC-defined Protected Regions of Space. [IADC, 2004a]

as well as lunar-solar and geopotential perturbations, the recommended minimum increase above GEO is defined as:

$$\Delta H = 200 \text{ km} + 35 \text{ km} + (1{,}000 \text{ km} \times C_R \times A/m) \quad (30\text{-}2)$$

where 35 km is incorporated due to effects from lunar-solar and geopotential perturbations. C_R is the solar radiation pressure coefficient (typically 1–2 kg/m^2). A/m is the spacecraft cross-sectional area to dry mass ratio (m^2/kg).

The altitude is generally increased/decreased, to prevent an accumulation of debris that future GEO missions would need to pass through to get to GEO, and to prevent potential signal interference. In addition to increasing the orbital radius, the final orbit should be circularized to an eccentricity of no more than 0.003 (see Sec. 9.4.2), and the spacecraft needs to be passivated as described in Sec. 30.4. It is estimated that the fuel required for responsible disposal of GEO missions is equivalent to that used for about 3 months of station-keeping (see Sec. 14.3) for most spacecraft [IADC, 2004b]. The "Support to the IADC Space Debris Mitigation Guidelines" document contains further details on disposal from GEO.

Disposal from Earth orbits other than LEO and GEO (e.g., high eccentricity science orbits) should minimize the orbit lifetime and avoid any highly used regions. While the IADC guidelines are not specific, some national requirements documents include a protected region for 12 hour orbits, commonly used for navigation satellites. Due to the wide variety of unusual orbits, it would be impossible to cover all possibilities in detail here. The first priority of the end of mission planning is minimization of the potential for on-orbit collisions. As with LEO and GEO missions, vehicles left in orbit must be passivated (see Sec. 30.4) after the final maneuvers.

Currently, there are few guidelines available for disposal of Lunar, planetary, or Lagrange orbit missions. The primary consideration, as it is for Earth orbits, is to minimize the potential for collisions with other current and future spacecraft, either directly or through generated debris. Therefore, it is best to avoid leaving spacecraft in long-term orbits (>10 years) at the end of a mission, except in those cases where the spacecraft continues to serve as a communications relay after its primary mission. Likewise, it is best to passivate any orbital hardware at the end of the mission to prevent explosions and other breakups, which would generate additional debris that might interfere with future missions. If disposal is to include Lunar or planetary impact, care should be taken to avoid sites of scientific or historic value, as well as preventing organic contamination whenever possible.

30.4 Passivation

At the end of a mission, the IADC guidelines call for all on board sources of stored energy to be "depleted or safed when they are no longer required for mission operations or post-mission disposal," also known as *passivation*. The main concern is that stored residual energy has in the past resulted in explosions, which have been a major source of orbital debris. Propulsion systems, batteries, and reaction wheels all contain stored energy, and are the most common components identified for passivation. Since a spacecraft or a launch vehicle could be in orbit and unattended for many years (even centuries) after the mission, it is important to passivate such components to prevent generation of debris that would increase the likelihood of collisions for other missions.

30.4.1 Propulsion System Passivation

Propulsion system explosions have resulted in some of the largest debris-producing events on orbit. In 1996, a Hydrazine Auxiliary Propulsion System (HAPS) upper stage spontaneously exploded 2 years after delivering its payload, generating over 700 pieces of debris large enough to be tracked from the ground. Due to a launch anomaly, this vehicle contained both propellant and high pressured gas, which were not able to be depleted. The most likely cause of the explosion was a long-term regulator failure, which allowed the propulsion tank to become over pressurized [Johnson, 1999].

A propulsion system (Sec. 18.1) is designed to produce a controlled explosion, resulting in the desired thrust. Unexpected events, such as small debris impact to the tank, mixing fuel and oxidizer, can produce extremely

destructive explosions. Even residual pressurant gas (despite being chemically inert) can cause a propulsion tank to rupture catastrophically if, for example, the tank is struck by small debris. Propellants should be depleted by either burning or venting, and pressurant should be vented during disposal to prevent future breakups and debris generation. Residual propellant should be used to lower the perigee of LEO vehicles to ensure that they deorbit as soon as possible. Unfortunately, some propulsion system designs do not allow for complete passivation, **unless this capability is specifically taken into account early in the design phase.**

There can be a number of challenges to depleting the fuel left on board a spacecraft at the end of a mission. Because sensors can become inaccurate, particularly near the end of a mission, when the residual propellant levels are low, it can be difficult to judge exactly how much propellant remains to be depleted. It is recommended that, if possible, the propulsion system be monitored until some positive sign of completion is observed (e.g., a rapid pressure drop from pressurant venting through a thruster). If a significant amount of propellant remains to be expelled, it must be done in a manner that will not cause excessive disruption to the spacecraft attitude, or cause the spacecraft to spin beyond its design limits. Since even directly opposed thrusters are not perfectly matched, this can be more difficult than initially imagined. Propulsion systems have also been observed to perform very differently as the propellant nears exhaustion [Hughes et al., 2006].

A unique challenge was the passivation of the TDRS-1 spacecraft after a very long and successful 26 year career. Due to a modification in its mission, when the payload finally failed, there was still a large excess of fuel on board, which needed to be expended. Since lowering perigee was not an option for a GEO mission, and using all remaining fuel to increase the altitude would cause the spacecraft to drift out of communications range during the maneuver, an alternative solution was developed. The spacecraft was first raised to the disposal orbit, then a flat spin was induced to exhaust the remaining fuel, while not disturbing the orbit or exceeding the spacecraft mechanical design. In this way, the fuel was expended and the spacecraft was passivated before it drifted out of communications range. This is an example of the creativity sometimes necessary to perform responsible passivation when it was not necessarily considered early in the design.

The IADC guidelines and most space agency regulations call for depletion of pressurant gas at the end of the mission because residual pressure exerts stress on the tanks and other hardware. The method of passivation varies based on the propulsion system design. In the case of a propellant tank with no diaphragm, it is usually possible to expel the pressurant gas by latching opposing thruster valves open at the end of the mission. Pressurant gas that is trapped behind a diaphragm, or in a separate tank, needs to be exhausted through a dedicated vent line. Such vent lines should be designed so that they will not impart a significant thrust or spin to the spacecraft during passivation. This can be accomplished by controlling the gas flow, and providing an omnidirectional outlet.

30.4.2 Power System Passivation

To date, batteries have exploded on orbit on at least 8 occasions, accounting for a portion of the debris currently in orbit [NASA, 2008]. Most battery explosions result from excess gas pressure generated inside the battery case due to overcharging. Nickel-hydrogen batteries, for example, typically operate at up to 1,000 psi (or more), directly related to the charge state. It is therefore important to provide the hardware capability to physically disconnect at least the charging capacity from the batteries at the end of a mission. The best approach is to physically disconnect the solar arrays from the power bus completely at the end of a mission. Simply minimizing the battery charging rate is not sufficient, since the setting may change due to radiation damage, part failure, or other mechanisms during the long unattended orbit-decay period. Remember that after spacecraft are decommissioned from GEO, they will be on orbit for centuries. Some designers are traditionally reluctant to include any capability to completely disconnect the solar array or battery, since this creates the possibility for premature mission failure. Section 30.4.4 provides more information on passivation methods that are designed to retain mission reliability.

Lithium-ion batteries present special concerns, since they not only store electrical energy, but also contain potentially dangerous materials. As with NiH_2 batteries, overcharge produces heat and gas buildup which can result in cell rupture. In addition, when discharged, the cells can plate out highly reactive lithium metal. Since the charge conditions cannot necessarily be guaranteed throughout the post-mission period, it is very important to prevent any possibility of discharging and recharging lithium-ion batteries. Many lithium-ion battery assemblies have built-in ground safety disconnect relays to prevent this from occurring during the integration phase. By designing in the ability to actuate these relays on orbit, the batteries can be completely isolated from the power bus after disposal, and can neither charge nor discharge. Of course, this concern is negated if the solar arrays are completely disconnected, and there is no way to re-charge the batteries. More on power systems and batteries can be found in Sec. 21.2.

30.4.3 Passivation of Other Systems

The IADC guidelines also recognize the potential for post-mission damage from other systems. Pressure vessels, self-destruct devices, and momentum wheels are all examples of components which can contain mechanical and chemical energy at the end of a mission. Provisions need to be made during the design of such components to allow for them to be vented, safed, spun down at the conclusion of a mission. Some of these passivation steps can be simple. For example, disabling power to momentum

Table 30-2◀, Fig. 30-4◀, Eq. 30-2◀

wheels is sufficient to allow them to spin down by internal friction. The IADC [2004b] document contains guidance on systems that require passivation, and safe final conditions for them. One additional system to consider for passivation is communications. Whenever possible, the transmitting capability of the spacecraft should be disabled at the end of a mission, to prevent interference with future missions. If the power system is properly passivated, both the momentum wheels and transmitter capability are passivated by default. It is best to listen for any transmissions a few days after passivation to ensure that the spacecraft has not recovered despite the intended shutdown.

30.4.4 Passivation Challenges

One difficulty in designing for adequate passivation is the potential creation of single points of failure for the mission. For example, if a single command could erroneously be sent to vent the propellant or pressurant before or during operations, the entire mission could be lost. Designers have long recognized this potential, and have employed techniques to positively prevent any such opportunities (e.g., preventing vent lines from being operable on orbit). Such hardware designs often prevent adequate passivation of the spacecraft at the end of the mission as well. In many ways, the inclusion of necessary passivation systems requires hardware designers to adopt a new approach to mission reliability. Through the use of redundant hardware and end of mission software modifications, passivation can usually be designed into the system without presenting a single point of failure.

One example of a way to provide for passivation, while preventing a single point failure mechanism is the battery passivation of the Solar Dynamics Observer (SDO). Designers have included the capability of actuating the built-in lithium-ion battery isolation relays on orbit, but omitted the commands to do so from the command database for the duration of the mission. Even if the commands were erroneously sent, they would not be recognized by the spacecraft. At the end of the mission, the command options will be uploaded into a new software patch, allowing the relay actuation commands to be recognized and executed by the onboard hardware. In addition, there is a pair of commands (arm and fire) necessary to actuate each relay. This design has allowed SDO to design for responsible passivation at the end of the mission, yet retain reliability throughout the operational phase of the mission.

Most spacecraft are also designed with autonomous fault detection and correction capabilities, which respond to abnormal spacecraft conditions in a way that keeps the spacecraft operating (or returns it to low level operation after a major anomaly). Such capabilities are contrary to the whole notion of post-mission passivation, and may work to counteract any passivation procedures. Passivation is, after all, an anomalous condition from the perspective of successful mission operations. Autonomous recovery systems, then, need to be disabled before engaging in post-mission passivation. Therefore, it is important to consider this phase of the mission when designing the operations modes and command structure. Ideally, there should be a disposal mode of operations planned from the beginning, that would allow a complete and safe disposal after the mission—with proper safeguards to prevent inadvertently entering or exiting this mode of course.

Passivating missions that were designed and launched without explicit consideration of post-mission passivation can present unique difficulties. Often, it may not be possible to fully comply with the IADC guidelines, or other requirements. Typically, it may be impossible to disable the autonomous recovery hardware built into the spacecraft design, so that the spacecraft constantly returns to a safe mode. Another example is the inability to physically disconnect the battery or solar arrays from the charging circuit. In such a case, it is important to consider the intent of passivation and debris limitation guidelines, so as to minimize the potential for creating additional debris. For example, by reducing the NiH_2 battery charging current to its lowest possible level, and leaving some benign loads enabled, it is often possible to keep the battery in a perpetual discharge condition, thus greatly reducing the risk of battery explosion due to overcharging. While compliance with the guidelines or other requirements is the most reliable approach, it may be necessary to perform less than complete passivation on some operational missions. Future missions must be designed for complete passivation at the end of a mission, unless performing a controlled reentry disposal.

When planning passivation procedures, it is necessary to consider the order in which steps must occur. For example, the transmitter should typically be disabled last so that all previous commands can be verified. There may also be attitude control factors that need to be considered in order to ensure adequate communications or power needs. Generally, passivation begins with disabling recovery mechanisms, followed by passivating the propulsion, attitude control, power, and communications systems. Bear in mind that as each system is disabled, its contribution to the mission is lost, and the remaining steps need to be completed quickly; (e.g., before battery power is lost). Each spacecraft needs to be considered on its own merits, including any hardware degradation that may have occurred during the mission. The specific passivation procedures for an individual vehicle should be reviewed and tested as thoroughly as possible before execution.

30.5 Disposal Planning*

Depending on the method of disposal employed, the process may require anything from relatively simple to highly complex preparation. For instance, a small, low-altitude science mission with no propulsion system,



* Applicable standards: ISO 24113 (Space Systems). See website for reference.

reentering within 25 years, may need only to passivate the power system at the end of the mission. A large spacecraft in LEO which requires a controlled reentry will require much more extensive planning in terms of maneuvers, communications, and staffing. Planning for disposal often requires the coordination of a number of different disciplines, and can take months of preparation and testing. Detailed written plans are often required to be submitted well in advance of the disposal.

If the disposal involves maneuvers, care must be taken to prevent collisions with other orbital assets and debris. It is important to report in advance any significant maneuvers which will change the altitude of the spacecraft more than 1 km to ensure that the maneuver will not cause a collision. In the US, the primary tracking and conjunction assessment authority is USSTRATCOM. This service is also available to some commercial and foreign entities as well, through special arrangement. In addition to checking planned maneuvers for conjunctions, it is important to plan any rapid orbit decay periods so that they take place slow enough that the decaying orbit can be adequately characterized for continuous conjunction assessment activities. This is especially important due to the potential threat to manned missions at low altitudes.

As the end of the mission approaches, there can be a temptation to continue beyond the originally planned termination point. The need for ending the mission might be driven by approaching the minimum fuel for de-orbit, declining reliability due to hardware degradation, or other factors which dictate that there may be a diminished or ineffective disposal, if not executed in the near-term. In the case of a science mission, there is a desire to continue data collection, or there may be increased profits available by continuing a commercial mission. All implications of such a decision need to be considered before continuing the mission, including increased risk of collision, orbital slot availability, RF interference, and reentry risk to the public. Only when the overall risk of continuing is truly negligible, should continuing beyond a pre-planned mission completion be considered.

30.6 FireSat II and SCS Examples

Since both the FireSat II and Supplemental Communications System (SCS) missions are assumed to be US Government-owned assets, they would need to meet the US Government Orbital Debris Mitigation Standard Practices, which is in some ways more comprehensive than the IADC guidelines. Not only the spacecraft, but also the launch vehicles for each mission must be considered for safe disposal, in terms of both orbital debris generation and reentry risk. In most cases, the individual spacecraft from each mission would likely be retired one at a time, due to equipment failure or fuel reserves. Since it is extremely rare for a mission to intentionally shed any debris during disposal, both of our example missions will be assumed to meet the operational debris requirement.

30.6.1 Disposal of the FireSat II Spacecraft

The FireSat II mission operational orbit is circular, at a 700 km altitude, and at a 55 deg inclination. At the end of the 3–5 year mission, it will be necessary to ensure that each spacecraft is removed from the LEO protected region within 25 years. Orbital lifetimes can vary based on spacecraft mass and the area presented in the velocity direction, as well as the solar flux effects on the Earth's atmosphere (Sec. 9.4). With body-mounted solar arrays, the FireSat II spacecraft is likely to tumble slowly after the mission, so an average cross-sectional area would be assumed for orbital lifetime predictions. With deployed solar arrays, the spacecraft may be more likely to adopt a gravity-gradient stabilized orientation, presenting a different (likely higher) drag area. Orbital lifetime predictions from the 700 km circular altitude can vary from just under 25 years to over 60 years, based on the launch year and spacecraft area. It will most likely be necessary, then, to intentionally reduce the perigee of each FireSat II spacecraft as a part of the end of mission disposal using all residual propellant, including the portion reserved for disposal operations.

As discussed in Sec. 30.4, it will be necessary to passivate the FireSat II systems in order to prevent fragmentation during the up to 25 year orbital decay period. Ideally, the mission operations concept would include a separate operations mode for disposal, intended to disable automatic recovery of the spacecraft after it has been shutdown. Passivation steps include venting the propulsion system (including pressurant), and disconnecting the solar arrays from the power bus described in Sec. 30.4. By permanently disconnecting the arrays, the remaining loads will deplete the battery and keep it discharged. With the recovery systems disabled, the spacecraft should remain in an inert state until it reenters.

Several of the other drivers for the FireSat II mission have led to a design which is relatively small, with insufficient fuel or control authority to perform a controlled reentry. Survival of even 15 individual pieces after reentry would cause FireSat II to exceed the required 1 in 10,000 reentry risk requirement. It will therefore be necessary to consider reentry risk throughout the mission design phase, and to minimize the use of highly survivable materials whenever possible. The spacecraft structure, for example, should be limited to graphite/epoxy composite and aluminum components. Note, fasteners are generally too small to inflict significant injuries, and need not be considered for reentry risk purposes. Solar cells are fragile enough to break up into relatively benign pieces, but any solar panel hinges and deployment hardware should avoid the use of titanium and stainless steel as much as possible. Some high survivability materials are inevitable in the payload optics, but could possibly be bundled together to minimize their reentry risk.

30.6.2 Disposal of the SCS Spacecraft

The SCS mission is intended to operate in a 21,000 km altitude, circular orbit with a 0 deg inclination, greatly simplifying the disposal procedure. In fact, this orbit is

perhaps the simplest of the commonly used Earth orbits, since all that is necessary is to passivate the spacecraft in place. The operational orbit meets the US Government requirement for Medium Earth Orbit (MEO) disposal, so there is no need for end of mission maneuvers. It is still necessary, however, to expend all remaining fuel, and perform all other passivation tasks described for FireSat II, to minimize any possibility of fragmentation during the very long post-mission storage period. It is not necessary to consider reentry risk for SCS, since the spacecraft will not reenter for centuries, by which time more permanent solutions should be available if necessary.

Conclusion

The need to control the growth of orbital debris, particularly in LEO, is clear. Toward this goal, the United Nations and individual space agencies have created guidelines and requirements that call for, among other things, responsible disposal of space hardware at the end of its useful mission life. In order to perform responsible disposal, it **must be considered during the mission planning in the earliest stages of the mission**. The chosen disposal method often drives the hardware design, including propulsion system sizing, passivation hardware, materials selection, and even potentially the orbit selection. Methods exist which will allow for passivation, while preserving mission reliability. Designing each individual mission with responsible disposal in mind is a critical step toward minimizing the risk of future collisions for all missions.

Applicable Standards:
ISO 17875: Space Systems
See website for reference.

Applicable Standards:
ISO 26872: Space Systems
See website for reference.

Applicable Standards:
ISO 24111: Space Systems
See website for reference.

References



For annotated bibliography of end of mission considerations, see the book website.

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Appendix A
Mass and Power Distribution for Spacecraft

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Table A-1 Average Mass as a Percentage of Dry Mass for Subsystems by Spacecraft Type
Table A-2 Orbit Average Power as a Percentage of Total for Subsystems by Spacecraft Type
Table A-3 List of Spacecraft Used for Mass and Power Analysis
Table A-4 Spacecraft Without Propulsion Used for Mass Analysis (Percentage of Dry Mass)
Table A-5 Spacecraft Without Propulsion Used for Power Analysis (Percentage of Total Power)
Table A-6 LEO Spacecraft With Propulsion Used for Mass Analysis (Percentage of Dry Mass)
Table A-7 LEO Spacecraft With Propulsion Used for Power Analysis (Percentage of Total Power)
Table A-8 High-Earth Spacecraft Used for Mass Analysis (Percentage of Dry Mass)
Table A-9 High-Earth Spacecraft Used for Power Analysis (Percentage of Total Power)
Table A-10 Planetary Spacecraft Used for Mass Analysis (Percentage of Dry Mass)
Table A-11 Planetary Spacecraft Used for Power Analysis (Percentage of Total Power)

Table A-1 and A-2 contain average mass and orbit average power by subsystem for different categories of spacecraft (spacecraft subsystem definitions are in Sec. 14.4). Table A-3 lists the specific spacecraft that went into the averages. Tables A-4 through A-11 show individual spacecraft data. The spacecraft were categorized by the target orbits, with the idea that the mission destination will tend to drive some aspects of the resource distributions. For instance, Telemetry, Tracking, and Command (TT&C) subsystem mass and power are significantly greater for planetary missions than they are for low Earth orbit missions, with high Earth orbit missions falling in between. The greater distance requires larger antennas and more power from the spacecraft transmitter in order to return data. The categories are:

- **No Propulsion.** This category includes spacecraft with no onboard propulsion system or kick motor. These missions are generally low Earth orbiting (LEO) missions. With the exception of the Hubble Space Telescope, they are all Explorer missions. All of the missions are astrophysics or space science missions.
- **LEO with Propulsion.** This category includes missions with an onboard propulsion system or kick motor and an orbit perigee of less than 5,000 km. Equator-S and ST-5 are the only missions in this category with apogee above 900 km. These missions are mostly Earth observing missions with station-keeping requirements.
- **High Earth.** The high Earth missions include those with perigee above 5,000 km, including geosychronous (GEO) missions, and it includes missions to Earth libration points L1 and L2 and Earth fly away missions. The energy to achieve mission orbit is significantly greater than the energy for LEO, but significantly less than the energy for a planetary mission.
- **Planetary.** This category contains missions with destinations outside of the vicinity of Earth orbit. LRO was included in this category because significant energy is required to capture into Lunar orbit, making the mission look more planetary than high Earth.

In addition to the TT&C subsystem trend noted above, several other trends stand out. Most obviously, the mass and power resources available for payload decrease significantly for missions that target more distant destinations. Launch vehicles can deliver spacecraft to a LEO orbit, but they generally can only put the spacecraft on a trajectory toward higher destinations. The spacecraft must carry much more fuel for high Earth or planetary missions, for the final orbit insertion, with a corresponding increase in the propulsion system dry mass. Out from LEO, thermal control mass increases, since the comfortable 300 K Earth no longer fills half the sky.

This spacecraft information was collected from two different NASA databases as well as from the records of systems engineers who worked on some of the missions. One database had the TT&C subsystem and the onboard processing combined for some of the missions. In cases where this occurred, the TT&C mass shows up as "0%," and I did not include either the onboard processing or TT&C mass for these missions in the overall averages. The "other" category includes balance mass and mission-unique mass that stayed with the launch vehicle.

Power breakdown for the spacecraft was not available for some missions, so the averages include only those values available. Due to the nature of averaging averages, I needed to make slight adjustments to ensure the percentages total to 100% in Tables A-1 and A-2; the other 9 tables have the raw data. I included a wide variety of missions, from the 25 kg ST-5 to the 16,221 kg CGRO. These outliers tend to skew the results slightly, but the variety should be helpful for different applications. The overall averages make sense.

All mass is shown as a percentage of dry mass. If you are starting with payload mass, you can simply take the payload mass and divide by the appropriate payload percentage from Table A-1 to get a starting point for dry mass. As you refine your design and get estimates of each subsystem mass from more detailed analysis, you can look back at these tables to check the reasonableness of your estimates. Similarly, Table A-2 will help with power estimates.

These tables are intended as a first cut at mass and power estimates as described in Sec. 14.5. There is quite a bit of variation in the actual value for each spacecraft because of unique aspects of the mission requirements. When considering a particular mission design, you may find it helpful to look more closely at the missions that most closely align with your particular mission requirements. As technology changes and new ideas come along, you can expect these averages to change.

Table A-1. Average Mass as a Percentage of Dry Mass for Subsystems by Spacecraft Type. Types include spacecraft without propulsion, low Earth orbit missions with propulsion, high Earth orbit (> 5,000 km perigee, GEO, L1, L2, and fly away), and planetary missions.

Subsystem (% of Dry Mass)	No Prop (%)	LEO Prop (%)	High Earth (%)	Planetary (%)
Payload	41%	31%	32%	15%
Structure and Mechanisms	20%	27%	24%	25%
Thermal Control	2%	2%	4%	6%
Power (including harness)	19%	21%	17%	21%
TT&C	2%	2%	4%	7%
On-Board Processing	5%	5%	3%	4%
Attitude Determination and Control	8%	6%	6%	6%
Propulsion	0%	3%	7%	13%
Other (balance + launch)	3%	3%	3%	3%
Total	100%	100%	100%	100%
Average Dry Mass (kg)	1497	2344	1258	888

Table A-2. Orbit Average Power as a Percentage of Total for Subsystems by Spacecraft Type. Types include spacecraft without propulsion, low-Earth orbit missions with propulsion, high-Earth orbit (> 5,000 km perigee, GEO, L1, L2, and fly-away), and planetary missions. Slight adjustments were made to ensure columns sum to 100%.

Subsystem (% of Total Power)	No Prop (%)	LEO Prop (%)	High Earth (%)	Planetary (%)
Payload	43%	46%	35%	22%
Structure and Mechanisms	0%	1%	0%	1%
Thermal Control	5%	10%	14%	15%
Power (including harness)	10%	9%	7%	10%
TT&C	11%	12%	16%	18%
On-Board Processing	13%	12%	10%	11%
Attitude Determination and Control	18%	10%	16%	12%
Propulsion	0%	0%	2%	11%
Average Power (W)	299	794	691	749

Table A-3. List of Spacecraft Used for Mass and Power Analysis.

Name	Launch Date	Altitude (km)	Inclination (deg)	ACS Type	Pointing (arc sec)	Downlink Data Rate (kbps)	Data Storage (Mbytes)	Battery Capacity (A-hr)	PDR to Launch (months)
SPACECRAFT WITHOUT PROPULSION									
AIM	4/25/07	750 × 750	98	3-Axis, Bias	115	2,000	8,000	5	40
COBE	11/18/89	885 × 896	103	Spin	3,600	4,096	1,000	40	75
FAST	8/21/96	348 × 4,200	83	Spin	3,600	2,250	1,000	9	59
FUSE	6/24/99	758 × 774	25	3-Axis	0.36	1,024	240	80	44
GALEX	4/28/03	697 × 697	29	3-Axis	300	24,000	64	16	55
HESSI	11/19/01	600 × 600	38	Spin	720	3,500	4,000	15	—
HST	4/24/90	613 × 613	28.5	3-Axis	0.0072	1,024	3,600	360	—
IMAGE	3/25/00	780 × 45,000	90	Spin	360	2,200	4,100	22	38
RXTE	12/30/95	500 × 500	23	3-Axis	72	256	544	100	38
SAMPEX	7/3/92	550 × 675	82	3-Axis, Bias	3,600	900	26.6	9	36
SWAS	12/5/98	638 × 651	70	3-Axis	36	1,800	110	21	72
SWIFT	11/20/04	580 × 600	20.6	3-Axis	—	2,250	—	—	—
TERRIERS	5/18/99	550 × 550	97.8	Spin	900	250	40	4.8	—
TRACE	4/1/98	600 × 650	98	3-Axis	288	2,250	300	9	30
WIRE	3/4/99	540 × 590	97.6	3-Axis	72	2,250	300	9	35
LEO SPACECRAFT WITH PROPULSION									
Astro-E	2/9/00	550 × 550	31	3-Axis	—	4,000	6,000	—	—
Aqua	3/24/02	705 × 708	98	3-Axis	35	150,000	92,000	160	61
CGRO	4/5/91	450 × 450	28.5	3-Axis	108	512	900	150	84
CALIPSO	4/28/06	705 × 705	98.2	3-Axis	126	80,000	8,000	80	69
CloudSat	4/28/06	705 × 705	98.2	3-Axis	67	1,000	275	40	68
EO-1	11/21/00	705 × 705	98.2	3-Axis	122	105,000	6	—	51
Equator-S	12/2/97	500 × 70,000	7	Spin	—	262	1,500	4	—
Fermi GLAST	6/11/08	56 × 565,000	25.6	3-Axis	7,200	40,000	20,000	125	61
GRACE	3/17/02	300 × 500	89	3-Axis	—	—	—	16	—
Lewis	6/10/97	523 × 523	97.4	3-Axis	180	2048	4,000	—	—
ST-5	3/22/06	301 × 4,600	105.6	Spin	—	100	20	8	—
Terra	12/3/99	705 × 708	98	3-Axis	151.2	150,000	140,000	100	76
TOMS/EP	7/2/96	955 × 955	99.3	3-Axis	1,800	180	16	9	58
TRMM	11/27/97	350 × 350	35	3-Axis	720	2,048	2,000	100	66
UARS	9/12/91	576 × 579	57	3-Axis	36	512	900	150	54
LandSat 7	4/15/99	705 × 705	98.2	3-Axis	180	150,000	399,635	100	—
NOAA 16	9/21/00	850 × 865	99.2	3-Axis	720	2,661.6	4,500	141	0
NOAA-N prime	2/6/09	870 × 870	98.7	3-Axis	720	2,600	7,200	120	0
HIGH EARTH SPACECRAFT									
ACE	8/25/97	L1	—	Spin	900	76.384	1,073	12	46
Chandra	7/23/99	8,700 × 140,000	67	3-Axis	0.252	512	1,800	40	—
MAP	6/30/01	L2	—	3-Axis	2,160	1,000	—	23	55
SOHO	12/2/95	L1	—	3-Axis	—	245.7	3,000	40	56
STEREO	10/25/06	Earth Fly Away	—	3-Axis	180,00	720	8,000	23	60
Triana	not launched	L1	—	3-Axis	180	250	320	9	—
WIND	11/1/94	510,000 × 1,600,000	Varies	Spin	3,600	64	2,400	79.5	61

Table A-3. List of Spacecraft Used for Mass and Power Analysis. (Continued)

Name	Launch Date	Altitude (km)	Inclination (deg)	ACS Type	Pointing (arc sec)	Downlink Data Rate (kbps)	Data Storage (Mbytes)	Battery Capacity (A-hr)	POR to Launch (months)
HIGH EARTH SPACECRAFT (continued)									
GOES 12	7/23/01	GEO	0	3-Axis	900	2,620.8	0	24	—
TDRS 7	7/13/95	GEO	0	3-Axis	108	64	—	120	66
Spitzer	8/25/03	Earth Trailing	—	3-Axis	5	2,200	1,024	16	72
IBEX	10/19/08	7,000 × 300,000	11	Spin	4,428	320,000	250	21	34
GOES-N	5/24/06	GEO	0	3-Axis	1.2	2	—	123	95
SDO	2/11/10	GEO	23	3-Axis	—	130,000	0	—	—
PLANETARY SPACECRAFT									
Cassini	10/15/97	Saturn	—	3-Axis	0.36	249	900	none	—
Clementine	1/25/94	Moon	—	3-Axis	108	128	2,000	15	—
Galileo	10/18/89	Jupiter	—	Spin/despun	702	134.4	900	none	—
LRO	6/18/09	Moon	90	3-Axis	45	100,000	50,000	80	41
Magellan	5/4/89	Venus	—	3-Axis	—	268.8	1,800	60	—
Mars Odyssey	4/7/01	Mars	—	3-Axis	—	—	8,000	16	—
NEAR	2/17/96	Eros	—	3-Axis	10.8	26.5	1,770	9	—
MESSENGER	8/3/04	Mercury	—	3-Axis	360	104	2,000	23	39
MRO	8/1/05	Mars 255 × 320	93	3-Axis	322	10,000	20,000	50	37
New Horizons	1/19/06	Pluto flyby	—	3-Axis, Bias	177.7998	0.7	16,000	none	39
Stardust	2/1/99	Deep Space	—	3-Axis	1,800	4	225.6025	—	29

Table A-4. Spacecraft Without Propulsion Used for Mass Analysis (Percentage of Dry Mass).

Name	Total Mass (kg)	Payload Mass (kg)	Payload (%)	Struc. & Mech. (%)	Thermal (%)	Power (Inc harness) (%)	TT&C (%)	On-Board Proc (%)	ADCS (%)	Prop. (%)	Other* (%)
AIM	200	64	32%	27%	2%	22%	2%	5%	10%	0%	0%
COBE	2,245	1147	51%	15%	2%	20%	1%	3%	8%	0%	0%
FAST	191	65	34%	14%	2%	31%	3%	8%	5%	0%	4%
FUSE	1,332	782	59%	19%	1%	13%	1%	4%	4%	0%	0%
GALEX	287	133	46%	14%	1%	17%	2%	7%	12%	0%	0%
HESSI	284	131	46%	24%	1%	20%	1%	5%	2%	0%	0%
HST	11,866	5,610	47%	25%	2%	9%	1%	1%	4%	0%	10%
IMAGE	492	178	36%	28%	5%	17%	2%	2%	5%	0%	4%
RXTE	3,040	995	33%	38%	2%	20%	1%	2%	5%	0%	0%
SAMPEX	162	53	32%	16%	3%	25%	3%	8%	6%	0%	7%
SWAS	293	95	33%	18%	1%	25%	2%	6%	12%	0%	4%
SWIFT	1,465	854	58%	16%	1%	15%	1%	2%	7%	0%	0%
TERRIERS	125	50	40%	6%	0%	12%	7%	9%	7%	0%	20%
TRACE	211	60	28%	23%	2%	22%	2%	4%	17%	0%	2%
WIRE	260	107	41%	12%	4%	19%	2%	3%	17%	0%	2%
mean	1,497	688	41%	20%	2%	19%	2%	5%	8%	0%	3%
median	287	131	40%	18%	2%	20%	2%	4%	7%	0%	2%
std dev	2,900	1,369	9.4%	7.6%	1.1%	5.4%	1.4%	2.3%	4.5%	0.0%	5.2%
min	125	50	28%	6%	0%	9%	1%	1%	2%	0%	0%
max	11,866	5,610	59%	38%	5%	31%	7%	9%	17%	0%	20%

* Balance and launch hardware

Table A-5. Spacecraft Without Propulsion Used for Power Analysis (Percentage of Total Power).

Name	Power Total (W)	Payload Total	Payload (%)	Struc. & Mech. (%)	Thermal	Power (Inc. harness) (%)	TT&C (%)	On-Board Proc. (%)	ADCS (%)	Prop. (%)
AIM	216	83	38%							
FAST	52	19	37%	0%	12%	11%	22%	11%	8%	0%
FUSE	520	300	58%							
GALEX	362	131	36%	0%	4%	7%	21%	16%	16%	0%
HESSI	252	174	69%	0%	6%	10%	4%	10%	2%	0%
IMAGE	238	143	60%	0%	4%	8%	16%	6%	6%	0%
SAMPEX	73	21	29%	0%	4%	13%	10%	28%	15%	0%
SWAS	182	60	33%	0%	5%	8%	6%	14%	34%	0%
SWIFT	966	593	61%	0%	3%	13%	2%	6%	14%	0%
WIRE	132	27	20%	0%	0%	11%	7%	15%	47%	0%
mean	299	155	44%	0%	5%	10%	11%	13%	18%	0%
median	227	107	37%	0%	4%	10%	9%	12%	15%	0%
std dev	258	168	15.6%	0.0%	3.2%	2.1%	7.2%	6.7%	14.2%	0.0%
min	52	19	20%	0%	0%	7%	2%	6%	2%	0%
max	966	593	69%	0%	12%	13%	22%	28%	47%	0%

Table A-6. LEO Spacecraft With Propulsion Used for Mass Analysis (Percentage of Dry Mass).

Name	Total Mass (kg)	Wet Mass (kg) (total less kick)	Dry Total Mass (kg)	Payload Mass (kg)	Payload (%)	Struc. & Mech. (%)	Thermal Control (%)	Power (inc harness) (%)	TT&C (%)	On-Board Proc. (%)	ADCS (%)	Prop. (%)	Other* (%)	Propellant (%)	Kick Motor, inc prop. (%of wet)
Astro-E	1,678	1,678	1,578	977	62%	11%	4%	11%	1%	2%	6%	3%	0%	14%	0%
Aqua	2,941	2,941	2,799	1,022	37%	26%	2%	27%	1%	2%	4%	2%	0%	11%	0%
CGRO	16,221	16,221	14,279	5,999	42%	35%	1%	11%	1%	2%	3%	2%	4%	30%	0%
CALIPSO	577	577	548	283	52%	12%	2%	19%	1%	5%	8%	1%	0%	5%	0%
CloudSat	847	847	771	255	33%	25%	3%	24%	2%	3%	7%	3%	0%	10%	0%
EO-1	579	579	556	185	33%	26%	2%	28%	1%	3%	6%	2%	0%	4%	0%
Equator-S	230	189	189	46	25%	42%	1%	20%	0%	4%	6%	0%	2%	0%	18%
Fermi GLAST	4,307	4,307	3,947	2,889	73%	11%	1%	11%	1%	1%	2%	1%	0%	9%	0%
GRACE	471	471	458	70	15%	35%	6%	15%	1%	3%	17%	8%	0%	3%	0%
Lewis	385	385	342	97	28%	20%	1%	27%	0%	7%	10%	5%	1%	27%	0%
ST-5†	25	25	24	1	3%	32%	3%	31%	8%	9%	5%	6%	0%	2%	0%
Terra	4,771	4,771	4,432	1,096	25%	33%	5%	21%	4%	7%	4%	1%	1%	17%	0%
TOMS/EP	289	289	215	34	16%	29%	2%	23%	0%	8%	12%	8%	3%	75%	0%
TRMM	3,512	3,512	2,611	640	24%	38%	4%	18%	1%	3%	6%	5%	0%	76%	0%
UARS	7,483	7,483	6,607	2,098	32%	28%	2%	13%	0%	4%	4%	2%	15%	29%	0%
LandSat 7	2,105	2,105	1,983	404	20%	27%	1%	25%	3%	9%	4%	1%	10%	14%	0%
NOAA 16	2,246	1,453	1,595	3,62	23%	20%	2%	24%	2%	6%	3%	2%	7%	5%	35%
NOAA-N prime	1,391	1,391	1,382	312	23%	34%	3%	28%	3%	4%	5%	1%	0%	1%	0%
mean	2,735	2,344	893	31%	27%	2%	21%	2%	5%	6%	3%	2%	18%	27%	
median	1,422	1,382	312	27%	27%	2%	22%	1%	4%	5%	2%	0%	11%	27%	
std dev	3,800	3,289	1,411	16.6%	9.0%	1.3%	6.6%	1.8%	2.5%	3.5%	2.3%	4.1%	22.1%	8.7%	
min	25	24	1	3%	11%	1%	11%	1%	1%	2%	0%	0%	0%	18%	
max	16,221	14,279	5,999	73%	42%	6%	31%	8%	9%	17%	8%	15%	76%	35%	

* Balance and launch hardware

†Totals for 1 of 3 spacecraft, total launch mass was 3 × 25 kg plus another 65 kg for support structure that stayed with the launch vehicle

Table A-7. LEO Spacecraft With Propulsion Used for Power Analysis (Percentage of Total Power).

Name	Power Total (W)	Payload Total	Payload (%)	Struc. & Mech. (%)	Thermal (%)	Power inc harness (%)	TT&C (%)	On-Board Proc. (%)	ACDS (%)	Prop. (%)
Astro-E	858	500	58%	—	14%	—	—	—	—	—
Aqua	1,609	987	61%	0%	3%	3%	—	20%	10%	1%
CALIPSO	508	256	50%	—	—	—	—	—	—	0%
CloudSat	598	346	58%	0%	10%	11%	3%	5%	14%	0%
EO-1	298	118	40%	4%	4%	9%	5%	12%	26%	0%
Equator-S	89	45	51%	0%	0%	16%	—	—	34%	0%
Fermi GLAST	1,646	775	47%	0%	14%	15%	2%	9%	10%	2%
GRACE	92	50	55%	1%	0%	6%	6%	32%	0%	0%
ST-5	40	8.08	20%	0%	0%	14%	52%	11%	2%	1%
Terra	2,159	992	46%	0%	5%	8%	—	34%	8%	0%
TRMM	1,019	416	41%	4%	18%	11%	5%	5%	15%	0%
NOAA 16	778	396	51%	0%	18%	3%	9%	11%	9%	0%
NOAA-N prime	858	450	52%	0%	23%	3%	8%	5%	9%	0%
mean	812	372	48%	1%	9%	9%	11%	14%	12%	0%
median	778	346	51%	0%	7%	9%	6%	11%	10%	0%
std dev	635	318	10.2%	1.6%	7.8%	4.5%	15.5%	10.0%	9.4%	0.7%
min	40	8	20%	0%	0%	3%	2%	5%	0%	0%
max	2,159	992	61%	4%	23%	16%	52%	34%	34%	2%

Table A-8. High-Earth Spacecraft Used for Mass Analysis (Percentage of Dry Mass).

Name	Total Mass (kg)	Wet Mass (total less kick) (kg)	Dry Total Mass (kg)	Payload Mass (kg)	Payload (%)	Struc. & M ech. (%)	Thermal (%)	Power (inc. harness) (%)	TT&C (%)	On-Board Proc. (%)	ADCS (%)	Prop (%)	Other* (%)	Propellant (%)	Kick motor, incl prop (%of wet)
ACE	759	759	561	168	30%	23%	7%	20%	4%	5%	1%	7%	3%	77%	0%
Chandra	5,698	5,698	4,699	2,918	62%	15%	2%	9%	0%	2%	4%	4%	2%	47%	0%
MAP	814	814	741	305	41%	23%	6%	13%	2%	3%	8%	2%	2%	22%	0%
SOHO	1,854	1,854	1,602	899	56%	12%	2%	13%	1%	4%	5%	3%	3%	35%	0%
STEREO	619	619	558	138	25%	28%	4%	24%	6%	1%	6%	6%	0%	11%	0%
Triana	2,995	855	577	104	18%	11%	2%	5%	2%	1%	8%	30%	23%	106%	71%
WIND	1,226	1,226	856	182	21%	23%	3%	31%	2%	7%	2%	5%	5%	95%	0%
GOES 12	2,275	2,275	1,058	343	32%	22%	6%	16%	4%	3%	8%	8%	0%	253%	0%
TDRS 7	2,201	2,201	1,649	311	19%	28%	3%	28%	0%	3%	6%	5%	7%	74%	0%
Spitzer	849	849	833	453	54%	17%	1%	13%	3%	2%	7%	3%	0%	2%	0%
IBEX	458	143	117	26	22%	44%	3%	13%	4%	2%	2%	6%	4%	23%	69%
GOES-N	3,217	3,217	1,546	345	22%	31%	7%	14%	7%	3%	8%	6%	0%	108%	0%
SDO	2,975	2,975	1,561	298	19%	35%	2%	20%	4%	3%	7%	6%	3%	91%	0%
mean	1,807	1,258	499	32%	24%	4%	17%	4%	3%	6%	7%	4%	72%	70%	
median	1,226	856	305	25%	23%	3%	14%	4%	3%	6%	6%	3%	74%	70%	
std dev	1,449	1,098	728	15.1%	9.1%	1.9%	7.2%	1.8%	1.6%	2.3%	7.0%	5.8%	63.1%	1.3%	
min	143	117	26	18%	11%	1%	5%	1%	1%	1%	2%	0%	2%	69%	
max	5,698	4,699	2,918	62%	44%	7%	31%	7%	7%	8%	30%	23%	253%	71%	

* Balance and launch hardware

Table A-9.

Nam
ACE
MAP
SOHO
STEREO
Triana
GOES 12
TDRS 7
Spitzer
IBEX
GOES-N
SDO

Table A

Cassi
Cleme
Galile
LRO
Mage
Mars
NEAR
MESS
MRO
New
Starc

* Bal

Table A-9. High-Earth Spacecraft Used for Power Analysis (Percentage of Total Power).

Name	Power Total (W)	Payload Total	Payload (%)	Struc. & Mech. (%)	Thermal (%)	Power (inc. harness) (%)	TT&C (%)	On-Board Proc. (%)	ADCS (%)	Prop. (%)
ACE	305	108	35%	0%	29%	4%		27%	4%	0%
MAP	371	137	37%	0%	7%	6%	9%	13%	24%	0%
SOHO	1,186	649	55%	0%	11%	8%	—	11%	13%	3%
STEREO	514	116	23%	0%	10%	10%	25%	8%	16%	9%
Triana	394	150	38%	0%	16%	10%	9%	4%	23%	0%
GOES 12	1,046	462	44%	0%	13%	2%	22%	3%	15%	0%
TDRS 7	1,621	872	54%	0%	29%	1%	—	4%	12%	1%
Spitzer	434	61	14%	—	—	—	—	—	—	—
IBEX	70	22	31%	0%	0%	12%	17%	22%	17%	2%
GOES-N	—	386	—	—	—	—	—	—	—	—
SDO	969	344	35%	0%	11%	15%	13%	7%	18%	1%
mean	691	301	37%	0%	14%	7%	16%	11%	16%	2%
median	474	150	36%	0%	11%	8%	15%	8%	16%	1%
std dev	462	259	11.9%	0.0%	8.9%	4.4%	6.1%	7.9%	5.7%	2.7%
min	70	22	14%	0%	0%	1%	9%	3%	4%	0%
max	1,621	872	55%	0%	29%	15%	25%	27%	24%	9%

Table A-10. Planetary Spacecraft Used for Mass Analysis (Percentage of Dry Mass).

Name	Total Mass (kg)	Wet Mass (total less kick) (kg)	Dry Total Mass (kg)	Payload Mass (kg)	Payload (%)	Struc. & Mech. (% of wet)	Thermal (%)	Power (inc. harness) (%)	TT&C (%)	On-Board Proc. (%)	ADCS (%)	Prop. (%)	Other* (%)	Propellant (%)	Kick motor, incl prop (%of wet)
Cassini	5,565	5,565	2,558	688	27%	14%	3%	14%	6%	5%	6%	19%	5%	259%	0%
Clementine	1,646	491	259	14	5%	24%	9%	14%	5%	4%	6%	21%	13%	197%	70%
Galileo	1,816	1,816	1,782	531	30%	17%	4%	13%	4%	3%	8%	10%	10%	4%	0%
LRO	1,915	1,915	1,018	96	9%	31%	8%	24%	7%	2%	6%	12%	0%	88%	0%
Magellan	3,455	1,260	1,127	130	12%	29%	5%	18%	9%	4%	8%	7%	9%	26%	64%
Mars Odyssey	727	727	377	45	12%	28%	5%	23%	6%	3%	6%	13%	3%	204%	0%
NEAR	808	808	482	56	12%	19%	4%	23%	5%	6%	6%	25%	1%	149%	0%
MESSENGER	1,102	1,102	508	47	9%	25%	10%	24%	6%	2%	7%	16%	0%	117%	0%
MRO	2,114	2,114	953	130	14%	33%	3%	25%	7%	3%	6%	10%	0%	122%	0%
New Horizons	478	478	401	30	8%	32%	7%	27%	10%	6%	5%	7%	0%	19%	0%
Stardust	385	385	299	71	24%	20%	3%	33%	6%	3%	3%	6%	0%	28%	0%
mean	—	1,515	888	167	15%	25%	6%	21%	7%	4%	6%	13%	4%	110%	67%
median	—	1,102	508	71	12%	25%	5%	23%	6%	3%	6%	12%	1%	117%	67%
std dev	—	1,407	689	214	7.9%	6.1%	2.4%	5.9%	1.5%	1.4%	1.2%	6.0%	4.7%	81.8%	3.3%
min	—	385	259	14	5%	14%	3%	13%	4%	2%	3%	6%	0%	4%	64%
max	—	5,565	2,558	688	30%	33%	10%	33%	10%	6%	8%	25%	13%	259%	70%

* Balance and launch hardware

Table A-11. Planetary Spacecraft Used for Power Analysis (Percentage of Total Power)

Name	Power Total (W)	Payload Total	Payload (%)	Struc. & Mechs. (%)	Thermal (%)	Power (inc. harness) (%)	TT&C (%)	On-Board Proc. (%)	ADCS (%)	Prop. (%)
Cassini	1,338	745	56%	0%	11%	4%	—	11%	11%	7%
LRO	620	112	18%	0%	20%	10%	17%	15%	18%	2%
NEAR	852	94	11%	0%	18%	1%	7%	10%	6%	47%
MESSENGER	633	95	15%	0%	5%	17%	20%	5%	19%	19%
MRO	815	155	19%	5%	25%	5%	26%	11%	9%	0%
New Horizons	—	28	—	—	—	—	—	—	—	—
Stardust	234	28	12%	0%	15%	21%	22%	16%	11%	3%
mean	749	179	22%	1%	15%	10%	18%	11%	12%	13%
median	724	95	17%	0%	17%	7%	20%	11%	11%	5%
std dev	331	235	15.4%	1.9%	6.4%	7.1%	6.5%	3.5%	4.7%	16.3%
min	234	28	11%	0%	5%	1%	7%	5%	6%	0%
max	1,338	745	56%	5%	25%	21%	26%	16%	19%	47%

Appendix B
Physical and Orbit Properties of the Sun, Earth, Moon, and Planets

Nicola Sarzi-Amade, *Microcosm, Inc.*

B.1 Gravitational Constants of Major Solar System Bodies
B.2 Planetary Data
B.3 Physical Properties of the Sun
B.4 Physical and Orbit Properties of the Earth
Geocentric and Geodetic Coordinates on the Earth
B.5 Physical and Orbit Properties of the Moon

This appendix provides physical and orbit data for the Sun, Moon, and planets. Properties of orbits about these bodies are listed in tables throughout the text. Numerical properties of orbits about the Earth are given in App. I. For additional data on virtually all aspects of the Solar System, astrophysics, and astronomy, we highly recommend that the reader consult Cox [2000]. For a detailed explanation and high accuracy numerical methods and tables for computing planetary orbit ephemerides, see Seidelmann [2006]. For lower accuracy, but computationally convenient techniques, see Meeus [1998].

B.1 Gravitational Constants of Major Solar System Bodies



See website for other solar system bodies.

In Table B-1, values of $\mu = GM$ are given to their full available accuracy. Other values are rounded. Values for the smaller satellites are estimates based on a typical density and the object's size. Except for the Sun, the last 3 columns are evaluated at the object's surface (or at the largest dimension for irregular bodies). Data from Seidelmann [2006], IAU [2009], and Astronomical Almanac [2010].

Table B-1. Gravitational Parameters for Orbits About Major Solar System Bodies.

Object	$\mu = GM$ (m^3/s^2)	Orbital Velocity (km/s)	Orbit Period (min)	Escape Velocity (km/s)
Sun (at surface)	$1.327\,124\,400\,41 \times 10^{20}$	436.6678	166.9117	617.5416
Sun (at 1 AU)	$1.327\,124\,400\,41 \times 10^{20}$	29.7847	525,970	42.1219
PLANETS AND SATELLITES				
Mercury	$2.203\,21 \times 10^{13}$	3.0051	85.0169	4.2499
Venus	$3.248\,585\,917 \times 10^{14}$	7.3266	86.4984	10.3614
Earth	$3.986\,004\,356 \times 10^{14}$	7.9054	84.4891	11.1799
Moon	$4.902\,800\,15 \times 10^{12}$	1.6799	108.3069	2.3757
Mars	$4.282\,837\,522 \times 10^{13}$	3.5512	100.1500	5.0221
Phobos	7.161×10^{5}	0.0073	191.9561	0.0103
Deimos	1.041×10^{5}	0.0037	210.8393	0.0053
Jupiter	$1.267\,127\,626 \times 10^{17}$	42.0999	177.8298	59.5383
Io	5.961×10^{12}	1.8052	106.0975	2.5530
Europa	3.203×10^{12}	1.4321	114.2224	2.0252
Ganymede	9.890×10^{12}	1.9384	142.1873	2.7414
Callisto	7.181×10^{12}	1.7265	146.1158	2.4416

Table B-1. Gravitational Parameters for Orbits About Major Solar System Bodies. (Continued)

Object	$\mu = GM$ (m^3/s^2)	Orbital Velocity (km/s)	Orbit Period (min)	Escape Velocity (km/s)
Amalthea	1.39×10^8	0.0334	392.0009	0.0472
Himalia	2.8×10^8	0.0573	155.4302	0.0810
Elara	5.80×10^7	0.0367	122.5712	0.0520
Pasiphae	2.00×10^7	0.0258	121.6107	0.0365
Thebe	9.998×10^7	0.0415	146.2925	0.0587
Saturn	$3.794\,058\,49 \times 10^{16}$	25.0905	251.5398	35.4833
Mimas	2.508×10^9	0.1101	196.9388	0.1557
Enceladus	7.21×10^9	0.1675	160.6939	0.2369
Tethys	4.14×10^{10}	0.2767	204.3403	0.3914
Dione	7.32×10^{10}	0.3610	163.0431	0.5105
Rhea	1.540×10^{11}	0.4490	178.1778	0.6350
Titan	8.977×10^{12}	1.8671	144.4226	2.6405
Hyperion	3.79×10^8	0.0481	357.0610	0.0680
Iapetus	1.2054×10^{11}	0.4047	190.4516	0.5723
Janus	1.2608×10^8	0.0360	283.4979	0.0509
Uranus	$5.794\,5490 \times 10^{15}$	15.0570	177.7602	21.2938
Ariel	9.0×10^{10}	0.3944	154.2484	0.5578
Umbriel	7.8×10^{10}	0.3657	167.5273	0.5171
Titania	2.35×10^{11}	0.5461	151.3113	0.7722
Oberon	2.01×10^{11}	0.5140	155.0355	0.7269
Miranda	4.64×10^9	0.1390	180.8384	0.1965
Portia	1.11×10^8	0.0399	183.8718	0.0564
Puck	1.93×10^8	0.0488	173.7898	0.0690
Sycorax	3.59×10^8	0.0614	161.9045	0.0869
Neptune	$6.836\,527 \times 10^{15}$	16.6153	156.0781	23.4975
Triton	1.428×10^{12}	1.0274	137.9076	1.4530
Nereid	2.06×10^9	0.1100	161.8084	0.1556
Despina	1.40×10^8	0.0432	181.6878	0.0611
Galatea	2.50×10^8	0.0533	172.8201	0.0754
Larissa	3.30×10^8	0.0583	174.0984	0.0825
Proteus	3.359×10^9	0.1265	173.8690	0.1789
Pluto	9.7178×10^{11}	0.9018	138.7701	1.2753
Charon	1.132×10^{11}	0.4322	146.8216	0.6113
ASTEROIDS LARGER THAN 300 KM DIAMETER				
Ceres	6.264×10^{10}	0.3585	142.3300	0.5070
Pallas	1.367×10^{10}	0.1533	397.6845	0.2167
Vesta	1.792×10^{10}	0.1761	343.7935	0.2490

B.2 Planetary Data



See website for Natural Satellite Data.

Table B-2. Orbit Data for the Planets. Orbit elements are defined with respect to the mean ecliptic and equinox of J2000.0 (epoch JD 2,451,545.0). Data from Seidelmann [2006].

Planet	Orbit Eccentricity	Mean Distance (AU)	Tropical Period (Julian years)	Synodic Period (d)	Mean Daily Motion, n (deg)	Orbital Velocity (km/s)
Mercury	0.205 630 69	0.387 098 93	0.240 844 45	115.877 5	4.092 377 06	47.872 5
Venus	0.006 773 23	0.723 321 99	0.615 182 57	583.921 4	1.602 168 74	35.021 4
Earth	0.016 710 22	1.000 000 11	0.999 978 62		0.985 647 36	29.785 9
Mars	0.093 412 33	1.523 662 31	1.880 711 05	779.936 1	0.524 071 09	24.130 9
Jupiter	0.048 392 66	5.203 363 01	11.856 525 02	398.8840	0.083 129 44	13.069 7
Saturn	0.054 150 60	9.537 070 32	29.423 519 35	378.091 9	0.033 497 91	9.672 4
Uranus	0.047 167 71	19.191 263 93	83.747 406 82	369.6560	0.011 769 04	6.835 2
Neptune	0.008 585 87	30.068 963 48	163.723 204 5	367.486 7	0.006 020 076	5.477 8
Pluto	0.248 807 66	39.481 686 77	248.020 8	366.720 7	0.003 973 966	4.7490

Table B-3. Orbit Data for the Planets. Data from Seidelmann [2006] and Astronomical Almanac [1998].

Planet	Inclination to Ecliptic (deg)	Longitude of Ascending Node (deg)	Longitude of Perihelion (deg)	Planet Longitude on Jan. 1.5 2000 (deg)	Last Perihelion before 2000
Mercury	7.004 87	48.331 67	77.456 45	252.250 84	1999 Nov. 19
Venus	3.394 71	76.680 69	131.532 98	181.979 73	1999 Dec. 1
Earth	0.000 05	–11.260 64	102.947 19	100.464 35	1999 Jan. 3
Mars	1.850 61	49.578 54	336.040 84	355.453 32	1999 Nov. 25
Jupiter	1.305 30	100.556 15	14.753 85	34.404 38	1999 May 20
Saturn	2.484 46	113.715 04	92.431 94	49.944 32	1974 Jan. 8
Uranus	0.769 86	74.229 68	170.964 24	313.232 18	1966 May 20
Neptune	1.769 17	131.721 69	44.971 35	304.880 03	1876 Sept. 2
Pluto	17.141 75	110.303 47	224.066 76	238.928 81	1989 Sept. 5

Table B-4. Physical Data for the Planets. Data from Astronomical Almanac [2010] and Seidelmann [2006].

Planet	Mass[1] (10^{24} kg)	Radius[2] (km)	Flattening[3] (geometric)	Mean Density (g/cm^3)	J_2 ($\times 10^{-3}$)	Sidereal Rotation Period[4] (d)	Incl. of Equator to Orbit (deg)
Mercury	0.330 22	2,439.7	0	5.43		58.6462	0
Venus	4.8690	6,051.8	0	5.24	0.027	–243.0185	177.3
Earth	5.9722	6,378.1366	0.003 353 64	5.515	1.082 6359	0.997 269 63	23.45
Mars	0.641 91	3,396.19	0.006 772	3.94	1.964	1.02 595 676	25.19
Jupiter	1898.8	71,492	0.064 874	1.33	14.75	0.413 54	3.12
Saturn	568.52	60,268	0.097 962	0.69	16.45	0.444 01	26.73
Uranus	86.840	25,559	0.022 927	1.27	12	–0.718 33	97.86
Neptune	102.45	24,764	0.017 081	1.64	4	0.671 25	29.56
Pluto	0.013	1,195	0	1.8		–6.3872	118

Notes for Table B-4:

- The values for the masses include the atmospheres but exclude the satellites.
- The mean equatorial radii are given.
- The flattening is the ratio of the difference of the equatorial and polar radii to the equatorial radius.
- The sidereal rotation period refers to the rotation at the equator with respect to a fixed frame of reference: a negative sign indicates that the rotation is retrograde with respect to the pole that lies to the north of the invariable plane of the solar system. The period is given in days of 86,400 SI seconds.
- The data on the equator, flattening (ellipticity), and sidereal rotation period are based on Seidelmann et al. [2007].

B.3 Physical Properties of the Sun

Table B-5. Physical Properties of the Sun. Data from Cox [2000] and Astronomical Almanac [2010].

Radius of the photosphere	$6.960\ 00 \times 10^5$ km
Angular diameter of the photosphere at 1 AU	0.533 13 deg
Mass	1.9884×10^{30} kg
Mean density	1.409 g/cm^3
Gravity at surface	2.740×10^4 cm/sec^2
Moment of inertia	5.7×10^{53} g cm^2
Angular rotation velocity at equator	2.85×10^{-6} rad/sec
Angular momentum (based on surface rotation)	1.63×10^{48} g cm^2 /sec
Escape velocity at solar surface	6.177×10^7 g cm/sec
Total radiation emitted	3.845×10^{26} W
Total radiation per unit area at 1 AU (solar constant)*	1,366 W/m^2
Apparent visual magnitude at 1 AU	–26.75
Absolute visual magnitude (magnitude at distance of 10 parsecs)	+4.82
Color index, B–V	+0.65
Spectral type	G2 V
Effective temperature	5777 K
Inclination of the equator to the ecliptic	7.25 deg
Adopted period of sidereal rotation	25.38 days
Corresponding synodic rotation period (relative to Earth)	27.2753 days
Oblateness: semidiameter equator-pole difference	0."0086
Velocity relative to nearby stars	19.7 km sec

* The World Radiation Center in Switzerland has taken the data from a number of satellites and created a consensus on the solar constant. The World Radiation Center is the official international keeper of the solar constant. The value of the constant has a variation of –0.5 W/m^2 at solar min and +0.5 W/m^2 at solar max [World Radiation Center, 2006].

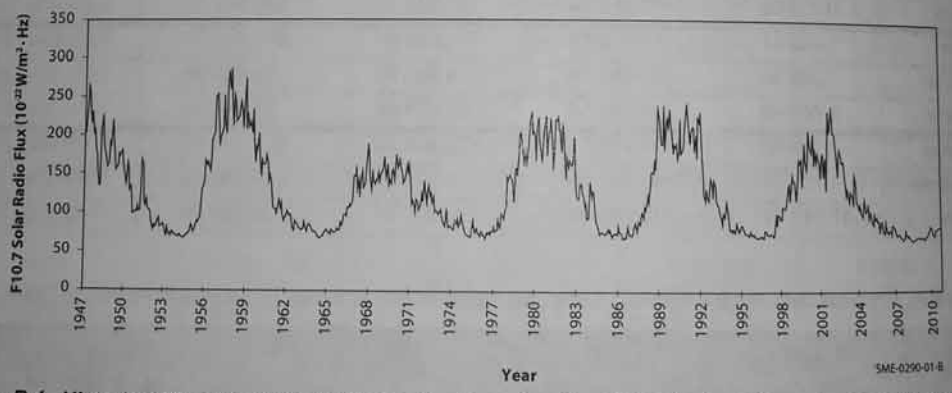


Fig. B-1. Historical Monthly 10.7 cm Radio Flux from the Sun (F10.7 Index) Since January 1947. For daily variations, see Fig. 9-16 in Sec. 9.4.4. (Plot courtesy Dave Bouwer, Space Environment Technologies)

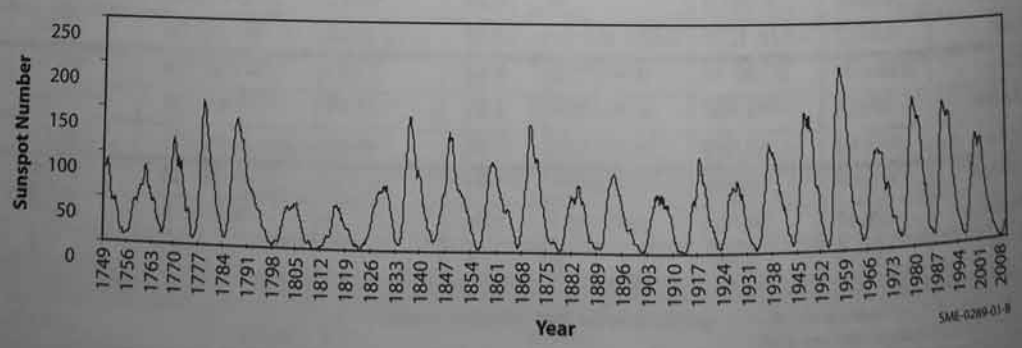


Fig. B-2. Historical Smoothed Sunspot Values from the 18th Century to Present. (Plot courtesy Dave Bouwer, Space Environment Technologies)

B.4 Physical and Orbit Properties of the Earth

Table B-6. Physical and Orbit Properties of the Earth.

		Reference
Equatorial radius, a	6,378.1366 km	IAU [2009]
Flattening factor (ellipticity), $f \equiv (a - c)/a$	1/298.256 42 ≈ 0.003 352 820	Astronomical Almanac [2010]
Polar radius,* c	6.356 752 × 10^6 m	
Mean radius,* $(a^2c)^{1/3}$	6,371.0004 km	
Eccentricity,* $(a^2 - c^2)^{1/2}/a$	0.081 819 301	
Surface area	5.100 657 × 10^8 km^2	
Volume	1.083 207 × 10^{12} km^3	Cox [2000]
Ellipticity of the equator $(a_{max} - a_{min})/a_{mean}$	~ 1.1 × 10^{-5}	Cox [2000]
Longitude of the maxima	15° W, 165°E	Groten [2000]
Ratio of the mass of the Sun to the mass of the Earth	332 946.0487	Groten [2000]
Geocentric gravitational constant, $GM_E \equiv \mu_E$	3.986 004 356 × 10^{14} m^3/ s^2	IAU [2009]
Mass of the Earth	5.9722 × 10^{24} kg	IAU [2009]
Mean density	5.515 g/cm^3	Astronomical Almanac [2010]
Gravitational field constants (See Eq. (9-23) in Sec. 9.4.1.)	J_2 = +1.082 63 × 10^{-3}	Astronomical Almanac [2010]
	J_3 = − 2.64 × 10^{-6}	Astronomical Almanac [2010]
	J_4 = − 1.61 × 10^{-6}	Astronomical Almanac [2010]
Mean distance of Earth center from Earth-Moon barycenter	4,671 km	Seidelmann [2006]
Average lengthening of the day (See Fig. B-3)	0.0015 sec/century	Seidelmann [2006]
General precession in longitude (i.e., precession of the equinoxes) per Julian century at epoch J2000	1.396 887 83 deg/century	Astronomical Almanac [2010]
Rate of change of precession at epoch J2000	+6.184 × 10^{-4} deg/century^2	Wertz [2009]
Rate of change of the obliquity at epoch J2000	−1.301 021 × 10^{-2} deg/Julian Century	Astronomical Almanac [2010]
Amplitude of the Earth's nutation	2.557 01 × 10^{-3} deg	Astronomical Almanac [2010]
Sidereal period of rotation, epoch J2000	0.997 269 68 d= 86 164.090 53 s = 23h56m4.09053 s	Astronomical Almanac [2010]
Length of tropical year (ref. = Υ), epoch J2000	3.155 692 522 × 10^7 s = 365.242 190 d	Astronomical Almanac [2010]
Length of sidereal year (ref. = fixed stars), epoch J2000	3.155 8150 × 10^7 s = 365.256 36 d	Cox [2000]
Length of anomalistic year (perihelion to perihelion), epoch J2000	3.155 843 255 × 10^7 s = 365.259 636 d	Astronomical Almanac [2010]

* Based on adopted values of f and a. "Eccentricity" here refers to the eccentricity of the Earth's surface, not the orbit eccentricity.

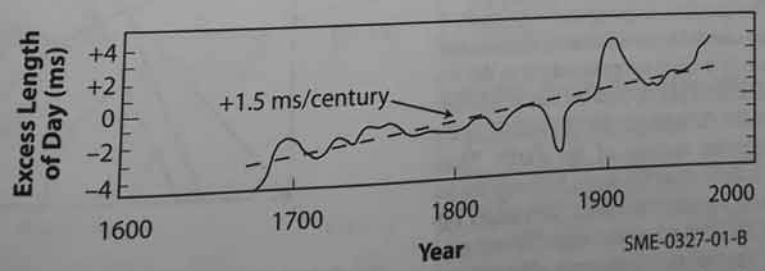


Fig. B-3. Variations in the Length of Day. On average, the Earth is slowing down by 1.5 ms/century, but local variations are large and very difficult to model.

Table B-7. Atmospheric Layers and Transitions. Data from Cox [2000].

Layer	Height, h (km)	Characteristics
Troposphere	0–11	Weather, T decreases with h, radiative-convective equilibrium
Tropopause	11	Temperature minimum, limit of upward mixing of heat
Stratosphere	11–48	T increases with h due to absorption of solar UV by O_3, dry
Stratopause	48	Maximum heating due to absorption of solar UV by O_3
Mesophere	48–85	T decreases with h
Mesopause	85	Coldest part of atmosphere, noctilucent clouds
Thermosphere	85–exobase	T increases with h, solar cycle and geomagnetic variations
Exobase	500–1,000 km	
Exosphere	> exobase	Region of Rayleigh-Jeans escape
Ozonosphere	15–35 km	Ozone layer (full width at e^{-1} of maximum)
Ionosphere	> 70 km	Ionized layers
Homosphere	< 85 km	Major constituents well-mixed
Heterosphere	> 85 km	Constituents diffusively separate

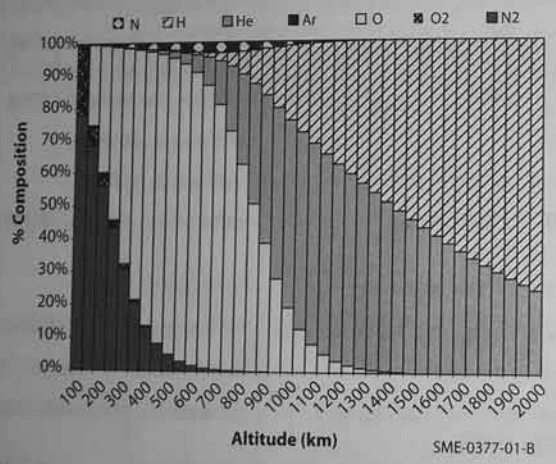


Fig. B-4. Mass Spectrometer Incoherent Scatter (MSIS) Atmospheric Species Percentage Composition vs. Altitude. [Hedin, 1987]. For atmospheric density, see Fig. 9-17 in Sec. 9.4.4.

B.4.1 Geocentric and Geodetic Coordinates on the Earth

The *geocentric latitude*, ϕ', of a point, P, on the surface of the Earth is the angle at the Earth's center between P and the equatorial plane. The *geodetic* or *geographic latitude*, ϕ, is the angle between the normal to an arbitrarily defined reference ellipsoid (chosen as a close approximation to the oblate Earth) and the equatorial plane. *Astronomical latitude* and longitude are defined relative to the *local vertical*, or the normal to the equipotential surface of the Earth. Thus, *astronomical latitude* is defined as the angle between the local vertical and the Earth's equatorial plane. Maximum values of the *deviation of the vertical*, or the angle between the local vertical and the normal to a reference ellipsoid, are about 1 minute of arc. Maximum variations in the height between the reference ellipsoid and *mean sea level* (also called the *equipotential surface*) are about 100 m.

Table B-7, Fig. B-5, Eq. B-3

The shape of the reference ellipsoid is most commonly defined by the *ellipticity* or *flattening factor*, $f \equiv (a-b)/a \approx 1/298.256\ 42 \approx 0.003\ 352\ 820$, where a is the equatorial radius of the Earth and b is the polar radius. Also used is the eccentricity of the reference ellipsoid, $e \equiv (a^2 - b^2)^{1/2}/a \approx 0.081\ 819\ 301$. These are related by:

$$e \equiv \sqrt{f(2-f)} \tag{B-1}$$

$$f = 1 - \sqrt{1-e^2} \tag{B-2}$$

On the surface of the Earth, the geodetic and geocentric latitude are related by:

$$\tan\phi = \tan\phi' / (1-f)^2 = 1.006\ 739\ 515\ \tan\phi' \tag{B-3}$$

where f is the flattening factor. At satellite altitudes the computation is more complex. As shown in Fig. B-5, the line normal to the oblate Earth through the satellite does not go through the Earth's center.

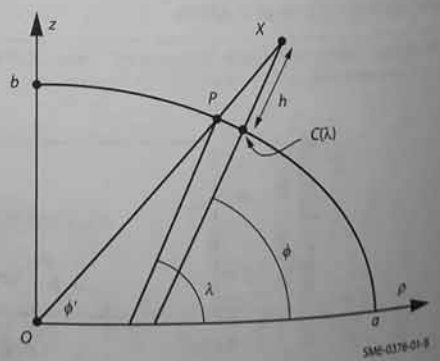


Fig. B-5. Relationship Between Geocentric Latitude, ϕ', and Geodetic Latitude, ϕ.

Geocentric coordinates are commonly expressed as Cartesian coordinates, (x, y, z). We then define the geocentric

latitude, ϕ', and the radius in the equatorial plane, ρ, by:

$$\rho \equiv \sqrt{x^2 + y^2} \tag{B-4}$$

$$\tan\phi' \equiv z / \rho \tag{B-5}$$

Given the geodetic coordinates, ϕ and h, as defined in Fig. B-5, we can immediately determine the geocentric coordinates from:

$$\rho = \left(N_\phi + h\right)\cos\phi \tag{B-6}$$

$$z = \left[\left(1 - e^2\right)N_\phi + h\right]\sin\phi \tag{B-7}$$

where the radius of curvature of the ellipse, N_ϕ, is given by:

$$N_\phi = \frac{a}{\sqrt{1 - e^2 \sin^2\phi}} \tag{B-8}$$

Determining the geodetic coordinates from the geocentric coordinates is more complex and requires an iterative technique. The approach used here is that of Nievergelt and Keeler [2000], which includes references to a number of earlier, less satisfactory methods. With a single iteration, this approach is good to 2×10^{-6} deg for the geodetic latitude and 1 mm in geodetic altitude. Successive iterations can improve this, although that would rarely be needed. The 4-step iterative approach is as follows:

Step 1

Set the iteration counter, $n = 0$. Compute $\rho \equiv \sqrt{x^2 + y^2}$ and the initial altitude estimate:

$$h_0 = \left[1 - 1/\sqrt{(\rho/a)^2 + (z/b)^2}\right]\sqrt{\rho^2 + z^2} \tag{B-9}$$

Step 2

Compute the cosine, $u_0 = \cos(\lambda)$, and sine, $v_0 = \sin(\lambda)$, of the initial latitude estimate $\phi_0 = \lambda$ and the initial value of the intermediate variable, w_0:

$$u_0 = \frac{\sigma^2\rho}{\sqrt{\sigma^4\rho^2 + z^2}}, \quad v_0 = \frac{z}{\sqrt{\sigma^4\rho^2 + z^2}}, \tag{B-10}$$

$$w_0 = \sqrt{1 - e^2 v_0^2}$$

where

$$\sigma \equiv b / a \tag{B-11}$$

Step 3

Compute the cosine, u_{n+1}, and sine, v_{n+1}, of the improved latitude estimate ϕ_{n+1}, the corresponding value w_{n+1}, and the improved altitude estimate h_{n+1}:

$$u_{n+1} = \frac{\left[a\sigma^2 + h_n w_n\right]\rho}{\sqrt{\left[a\sigma^2 + h_n w_n\right]^2 \rho^2 + \left[a + h_n w_n\right]^2 z^2}}$$

$$v_{n+1} = \frac{\left[a + h_n w_n\right] z}{\sqrt{\left[a\sigma^2 + h_n w_n\right]^2 \rho^2 + \left[a + h_n w_n\right]^2 z^2}} \tag{B-12}$$

$$w_{n+1} = \sqrt{1 - e^2 v_{n+1}^2}$$

$$h_{n+1} = \sqrt{\left[\rho - \frac{a u_{n+1}}{w_{n+1}}\right]^2 + \left[z - \frac{a\sigma^2 v_{n+1}}{w_{n+1}}\right]^2}$$

Step 4

Compute $\phi_{n+1} = \arctan(v_{n+1}/u_{n+1})$ with a standard algorithm that is stable near ±90 deg.

B.5 Physical and Orbit Properties of the Moon

Table B-8. Physical Parameters of the Moon. Data from Cox [2000] and Astronomical Almanac [2010]. For an extended discussion of lunar properties see Eckart [2006] and Heiken et al., [1991].

Radii: (a) Toward Earth, (b) Along orbit, (c) Toward pole	
Mean radius (b + c) / 2	1,738.2 km
a – c	1.09 km
a – b	0.31 km
b – c	0.78 km
Semi-diameter at mean distance	15′32″.6
Mass	7.3483×10^{22} kg
Mean density	3.341 g cm^{-3}
Surface gravity	162.2 cm/s^2 = 0.1654 g
Surface escape velocity	2.38 km/s
Extreme range	356,400 – 406,700 km
Inclination of orbit to ecliptic oscillating ±9′ with period of 173 d	5° 8′ 43″.42
Sidereal period (fixed stars)	27.321 661
Mean orbital speed	1.023 km/sec
Synodical month (new Moon to new Moon)	29. 530 588
Surface area of Moon at some time visible from Earth	59%
Inclination of lunar equator	
To ecliptic	1° 32′ 32″.7
To orbit	6° 41′

Table B-9. Gravity Field of the Moon. Data from Cox [2000].

$\alpha = (C - B) / A = 0.000\,400$		$C / MR^2 = 0.392$
$\beta = (C - A) / B = 0.000\,628$		$I = 5{,}552''.7 = 1°32'32$
$\gamma = (B - A) / C = 0.0002278$		
$C_{20} = -0.000\,202\,7$	$C_{30} = -0.000\,006$	$C_{32} = +0.000\,004\,8$
$C_{22} = +0.000\,022\,3$	$C_{31} = +0.000\,029$	$S_{32} = +0.000\,001\,7$
	$S_{31} = +0.000\,004$	$C_{33} = +0.000\,001\,8$
		$S_{33} = -0.000\,001$

Table B-9◀, Fig. B-5, Eq. B-12

Table B-10. Orbit of the Moon About the Earth. Data from Cox [2000] and Seidelmann [2006].

Sidereal mean motion of Moon	$2.661\,699\,489 \times 10^{-6}$ rad s^{-1}
Mean distance of Moon from Earth	384 400 km
	60.27 Earth radii
	0.002 570 AU
Equatorial horizontal parallax	57′02″.608
at mean distance	3,422″.608
Mean distance of center of Earth from Earth-Moon barycenter	4.671×10^3 km
Mean eccentricity	0.054 90
Mean inclination to ecliptic	5.145 396 deg
Mean inclination to lunar equator	6°41′
Limits of geocentric declination	± 29 deg
Period of revolution of node	18.612 Julian years
Period of revolution of perigee	8.849 Julian years
Mean orbital speed	1,023 ms^{-1} = 0.000 591 AU/d
Mean centripetal acceleration	0.00272 ms^{-2} = 0.0003 g
Optical libration in longitude (selenocentric displacement)	± 7.883 deg
Optical libration in latitude (selenocentric displacement)	± 6.85 deg
Saros = 223 lunations = 19 passages of Sun through node = 6,585 1/3 days	
Moment of inertia (about rotation axis)	0.396 $M_\oplus b^2$
Gravitational potential term	$J_2 = 2.027 \times 10^{-4}$
No. of strong mascons on the near side of the Moon	4 exceeding 80 milligals
Mean surface temperature	+107 C (day), 153 C (night)
Temperature extremes	−233 C, +123 C
Moon's atmospheric density	~10^4 molecules cm^{-3} (day); 2×10^5 molecules cm^{-3} (night)
No. of maria & craters on lunar surface w/ diam. > d	$5 \times 10^{10} d^{-2.0}$ per 10^6 km^2 (d in m)

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For annotated bibliography of Appendix B, see the book website.

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Appendix C
Summary of Keplerian Orbit and Coverage Equations

Nicola Sarzi-Amade, *Microcosm, Inc.*

C.1 Equations for Circular Orbits
C.2 Equations for Elliptical Orbits
C.3 Equations for Parabolic Orbits
C.4 Equations for Hyperbolic Orbits

This appendix provides summary formulas in addition to those given in Table 9-4 in Sec. 9.1. Formulas are provided for all four types of Keplerian orbits—circular, elliptical, parabolic, and hyperbolic. See Sec. 9.1 for definition of orbit variables, derivations, and conditions. See Table C-1 for values of the gravitational constant, μ, for the Earth, Moon, Sun, and Mars. See Table B-1 in App. B for values of μ for all of the major bodies in the Solar System. See Sec. 8.3 for definition of most of the geometric variables. See the website for the source of each of the equations. Finally, numerical evaluations of most of the characteristics of the motion are contained on the pages preceding the inside rear cover for orbits about the Earth, and on the website for orbits about other celestial bodies.

μ = *GM* is the gravitational constant of the central body and is much more accurately known than either the mass, *M*, or the universal gravitational constant, *G*, the least accurately known of the fundamental physical constants.

Table C-1. Values of the Gravitational Constant, μ, for the Earth, Sun, Moon, and Mars. Values are those given by the International Astronomical Union [2009]. See Table B-1 in App. B for other central bodies.

Central Body	μ (m^3/s^2)	$\sqrt{\mu}$ ($m^{1.5}/s$)
Earth	$3.986\,004\,356 \times 10^{14}$	19,964,980.230
Moon	$4.902\,800\,15 \times 10^{12}$	2,214,226.76
Mars	$4.282\,837\,522 \times 10^{13}$	6,544,339.17
Sun	$1.327\,124\,400\,41 \times 10^{20}$	11,520,088,543.11

For all of the orbit types, ν is the *true anomaly* = the angle from perifocus to the orbiting body and $M \equiv n\,(t - T)$ is the *mean anomaly*, where t is the time of observation, T is the time of perifocal passage, and n is the mean angular motion. Other parameters are defined in the table and discussed at the text reference cited in the table.

C.1 Equations for Circular Orbits

See Sec. 9.1 for definition of orbit variables, Sec. 8.3 for definition of geometric variables, and Sec. 8.3.3 in OCDM = Wertz [2009] for application of the Euler axis formulas to satellite ground tracks. See rear end-papers for properties of orbits about the Earth, and on the website for orbits about other celestial bodies.

Note: A bold **"C"** is put in front of each descriptor to denote formulas for circular orbits.

C. Defining parameter:

$$a = \text{semimajor axis} = r = \text{radius} \tag{C-1}$$

(For a discussion of Keplerian orbits, see Sec. 9.1; for transformations between orbital elements and position and velocity, see Sec. 2.7.1.1 and 2.7.1.2 in OCDM; for the apparent shape of a circular orbit viewed from nearby, see Sec. 8.4.1.)

C. Parametric equation:

$$x^2 + y^2 = a^2 \tag{C-2}$$

C. Distance from focus, *r*:

$$r = a \tag{C-3}$$

C. Specific energy, ε (= total energy per unit mass):

$$\varepsilon = -\frac{\mu}{2a} = \frac{V^2}{2} - \frac{\mu}{a} < 0 \tag{C-4}$$

C. Specific angular momentum, *h*:

$$h = |\mathbf{h}| = \sqrt{\mu a} \tag{C-5}$$

C. Angular momentum vector, h:

$$\mathbf{h} = \mathbf{r} \times \mathbf{V} \tag{C-6}$$

C. Nodal vector, N:

$$\mathbf{N} = \hat{\mathbf{z}} \times \hat{\mathbf{h}} = \hat{\mathbf{z}} \times \mathbf{h} / h \tag{C-7}$$

C. Inclination, *i*:

$$i = \text{acos}\,(h_z / h) \tag{C-8}$$

(For a discussion of apparent inclination, see the boxed example at the end of Sec. 9.3.1 in OCDM.)

C. Right ascension of the ascending node, Ω :

$$\Omega = \text{atan2}\,(N_x, N_y) \tag{C-9}$$

(For the definition of atan2, see Sec. D.4)

C. Flight path angle, ϕ_{fpa}:

$$\phi_{fpa} = 0 \qquad \text{(C-10)}$$

C. Perifocal distance, q:

$$q = a = r \qquad \text{(C-11)}$$

C. Semi-parameter, p:

$$p = a \qquad \text{(C-12)}$$

C. Semimajor axis, a:

$$a = r \qquad \text{(C-13)}$$

C. Eccentricity, e:

$$e = 0 \qquad \text{(C-14)}$$

C. Mean motion, n:

$$n = \sqrt{\mu / a^3} = 2\pi / P \qquad \text{(C-15)}$$

C. Mean anomaly, M:

$$M = M_0 + nt \qquad \text{(C-16)}$$

C. True anomaly, ν:

$$\nu = M \qquad \text{(C-17)}$$

C. Rate of change of true anomaly, $\dot{\nu}$:

$$\dot{\nu} = n \qquad \text{(C-18)}$$

C. Period, P:

$$P = 2\pi / n = 2\pi\sqrt{a^3 / \mu} \qquad \text{(C-19)}$$

$= 1.658\,669\,021\,3 \times 10^{-4}\ a^{3/2}$
for Earth; P in min, a in km)
$= 1.495\,569\,234 \times 10^{-3}\ a^{3/2}$
for Moon; P in min, a in km)
$= 5.060\,143\,330 \times 10^{-4}\ a^{3/2}$
for Mars; P in min, a in km)
$= 1.996\,228\,762\,54 \times 10^{-10}\ a^{3/2}$
(for Sun; P in calendar days, a in km)

C. Velocity, V:

$$V = \sqrt{\mu / a} = an \qquad \text{(C-20)}$$

C. Range • range rate, $r\dot{r}$:

$$r\dot{r} = 0 \qquad \text{(C-21)}$$

C. Areal velocity, $\dot{A}$:

$$\dot{A} = 0.5\sqrt{\mu a} = 0.5\, a^2 n \qquad \text{(C-22)}$$

C. Escape velocity, V_E:

$$V_E = \sqrt{2\mu / a} = \sqrt{2}\, V \qquad \text{(C-23)}$$

C. Euler axis co-latitude, δ'_{ea}:

$$\delta'_{ea} = \operatorname{atan}\left(\frac{n \sin i}{-\omega_E + n\cos i}\right) \qquad \text{(C-24)}$$

ω_E = inertial rotation rate of central body
$= 0.004\,178\,074\,1$ deg/sec (Earth)
$= 0.000\,152\,504\,145\,25$ deg/sec (Moon)
$= 0.004\,061\,249\,794\,68$ deg/sec (Mars)

(For a complete set of closed-form ground track equations, see Table 8-11 in Sec. 8.4.)

C. Euler rotation rate, ω_{ea}:

$$\omega_{ea} = n \sin i / \sin \delta'_{ea} = \sqrt{\omega_E^2 + n^2 + 2n\omega_E \cos i} \qquad \text{(C-25)}$$

C. Angular radius of the Earth, ρ:

$$\rho = \operatorname{asin}(R_E / a) \qquad \text{(C-26)}$$

R_E = radius of central body
$= 6378.1366$ km (Earth)
$= 1737.4$ km (Moon)
$= 3396.19$ km (Mars)

(Assumes a spherical Earth. For the effect of oblateness, see Sec. 9.4.1, and Sec. 9.1.5 in OCDM.)

C. Distance to the horizon, D:

$$D = a\cos\rho = \sqrt{a^2 - R_E^2} \qquad \text{(C-27)}$$

(For calculation of angles and distance for targets not on the horizon, see Sec. 8.3.1.)

C. Maximum Earth central angle, λ_{max}:

$$\lambda_{max} = 90 \text{ deg} - \rho = \operatorname{acos}(R_E / a) \qquad \text{(C-28)}$$

C. Instantaneous Access Area, IAA:

$$IAA = K_A(1 - \cos\lambda_{max}) \qquad \text{(C-29)}$$

$K_A = 2\pi \approx 6.283\,185\,311$ for area in steradians
$= 360^2 / (2\pi) \approx 20{,}626.480\,62$ for area in deg^2
$= 2.55\,603\,915 \times 10^8$ for area in km^2 (Earth)
$= 1.896\,62 \times 10^7$ for area in km^2 (Moon)
$= 7.247\,093 \times 10^7$ for area in km^2 (Mars)

(For alternative ways to calculate ground coverage, see Sec. 10.2.1; for other ground area formulas, see Table 10-3 in Sec. 10.2.1; for transformations between geocentric and access area coordinates, see the boxed example in Sec. 8.3; for projections of sensor fields of view onto the central body, see Sec. 8.3.4.)

C. Area Access Rate, AAR:

$$AAR = 2K_A \frac{\sin\lambda_{max}}{P} \qquad \text{(C-30)}$$

C. Maximum time in view, T_{max}:

$$T_{max} = P\lambda_{max} / 180 \text{ deg} \qquad \text{(C-31)}$$

(For a complete set of target or ground station coverage formulas, see Table 8-11 in Sec. 8.4.1.)

C. Maximum angular rate seen from ground, $\dot{\theta}_{max}$:

$$\dot{\theta}_{max} = \frac{a\ 360^\circ}{P(a - R_E)} \qquad \text{(C-32)}$$

C. Node precession rate due to J_2, $\Delta\Omega_{J2}$:

$$\Delta\Omega_{J2} = -1.5 J_2 n (R_E/a)^2 \cos i \equiv K_{J2} a^{-7/2} \cos i \quad \text{(C-33)}$$
$$= -2.064\,745\,722 \times 10^{14}\, a^{-7/2} \cos i \text{ (Earth, } a \text{ in km)}$$
$$= -3.193\,05 \times 10^{11}\, a^{-7/2} \cos i \quad \text{(Moon, } a \text{ in km)}$$
$$= -3.468\,016 \times 10^{13}\, a^{-7/2} \cos i \quad \text{(Mars, } a \text{ in km)}$$

(In the numerical forms, $\Delta\Omega_{J2}$ is in deg / calendar day*)

C. Sun-synchronous inclination, i_{SS}:

$$i_{ss} = \text{acos}\left(\frac{360 a^{7/2}}{K_{J2} SP}\right) \quad \text{(C-34)}$$
$$= \text{acos}(-4.773\,513\,28 \times 10^{-15}\, a^{7/2}) \text{ (Earth, } a \text{ in km)}$$
$$= \text{acos}(-3.086\,7 \times 10^{-12}\, a^{7/2}) \quad \text{(Moon, } a \text{ in km)}$$
$$= \text{acos}(-1.511\,0 \times 10^{-14}\, a^{7/2}) \quad \text{(Mars, } a \text{ in km)}$$

(Where SP is the sidereal period of the planet or the Moon about the Sun in calendar days. For a discussion of Sun synchronous orbits, see Sec. 9.5.3.)

C. Orbital revolutions per day, Rev/d:

$$Rev/d = \text{Day} / \text{P} \quad \text{(C-35)}$$
$$= 1.436\,068\,34 \times 10^3 / P \quad \text{(Earth, } P \text{ in min)}$$
$$= 3.934\,319\,286\,9 \times 10^4 / P \quad \text{(Moon, } P \text{ in min)}$$
$$= 1.477\,377\,729\,355 \times 10^3 / P \text{(Mars, } P \text{ in min)}$$

(This is the number of orbit revolutions for each rotation of the planet on its axis relative to the fixed stars (i.e., the inertial rotation period). "Day" is the length of the sidereal day (in minutes) for the given central body. For more accurate expressions, see the discussion of repeating ground tracks in Sec. 2.5.2.)

C. Node spacing, ΔN:

$$\Delta N = 360 \text{ deg } (P / \text{Day}) \quad \text{(C-36)}$$
$$= 0.250\,684\,448\, P \quad \text{(Earth, } P \text{ in min)}$$
$$= 0.009\,150\,248\,715\, P \quad \text{(Moon, } P \text{ in min)}$$
$$= 0.243\,674\,987\,681\, P \quad \text{(Mars, } P \text{ in min)}$$

(For the equations for repeating ground track orbits, see Sec. 9.5.2.)

C. Maximum eclipse, T_E:

$$T_E = P(\rho / 180 \text{ deg}) = P \,\text{asin}(R_E / a)/180 \text{ deg} \quad \text{(C-37)}$$

(For computation of eclipse duration for any Sun angle, see Sec. 8.2.4; for the formulas for eclipse conditions, see Sec. 11.4 in OCDM.)

C. Sun angle constraints for terminator visibility, β':

$$90° + \xi - \rho < \beta' < 90° - \xi + \rho \quad \text{(C-38)}$$

(ξ is the dark angle defined in Sec. 11.5.2 in OCDM. For general conditions of terminator visibility, see Table 11-3 in Sec. 11.5.2 in OCDM.)

* Calendar day = 24 h in contrast to sidereal day = 23 h 56 m 4.09053 s. See App. B.

C. Transit time for a spacecraft whose celestial coordinates are known, T:

$$T = \alpha + L - \text{GST} \quad \text{(C-39)}$$

(where α is the right ascension, L is the observer East longitude, and GST is the Greenwich Sidereal Time. For computation of observability times of spacecraft for which the celestial coordinates are known, see Table 9-6 in Sec. 9.4.4 in OCDM.)

C.2 Equations for Elliptical Orbits

See Sec. 9.1 for definition of orbit variables, Sec. 8.3 for definition of geometric variables, and Sec. 8.3.3 in OCDM for application of the Euler axis formulas to satellite ground tracks. See rear end-papers for properties of orbits about the Earth, and on the website for orbits about other celestial bodies.

Note: An **"E"** is put in front of each descriptor to denote formulas for Elliptical Orbits. Subscript A stands for evaluation of parameters at apogee; subscript P stands for evaluation of parameters at perigee.

E. Defining parameters:

$$a = \text{semimajor axis} \quad \text{(C-40)}$$
$$b = \text{semiminor axis}$$

(For a discussion of Keplerian orbits, see Sec. 9.1; for transformations between orbital elements and position and velocity, see Secs. 2.7.1.1 and 2.7.1.2 in OCDM.)

E. Parametric equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{(C-41)}$$

E. Distance from focus, r:

$$r = \frac{r_P(1+e)}{1+e\cos v} = a(1 - e\cos E) \quad \text{(C-42)}$$
$$= \frac{p}{1+e\cos v} = \frac{r_A(1-e)}{1+e\cos v}$$

E. Specific energy, ε (= total energy per unit mass):

$$\varepsilon = \frac{V^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} < 0 \quad \text{(C-43)}$$

E. Specific angular momentum, h:

$$h = |\mathbf{h}| = \sqrt{\mu p} = rV\cos\phi_{fpa} = r^2\dot{v} \quad \text{(C-44)}$$
$$= r_A V_A = r_P V_P$$

E. Angular momentum vector, h:

$$\mathbf{h} = \mathbf{r} \times \mathbf{V} \quad \text{(C-45)}$$

E. Nodal vector, N:

$$\mathbf{N} = \hat{\mathbf{z}} \times \hat{\mathbf{h}} = \hat{\mathbf{z}} \times \mathbf{h} / h \quad \text{(C-46)}$$

E. Eccentricity vector, e:

$$\mathbf{e} = \frac{\mathbf{V}\times\mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \qquad \text{(C-47)}$$

E. Inclination, *i*:

$$i = \text{acos}(h_z / h) \qquad \text{(C-48)}$$

(For a discussion of apparent inclination, see the boxed example at the end of Sec. 9.3.1 in OCDM.)

E. Right ascension of the ascending node, Ω:

$$\Omega = \text{atan2}(N_x, N_y) \qquad \text{(C-49)}$$

(For the definition of atan2, see Sec. D.4)

E. Flight path angle, ϕ_{fpa}:

$$\phi_{fpa} = \text{atan2}\left(\cos\phi_{fpa}, \sin\phi_{fpa}\right) \qquad \text{(C-50)}$$

$$= \text{acos}\left(\sqrt{\mu p}\,/\,rV\right)$$

$$\sin\phi_{fpa} = \frac{e\sin E}{\sqrt{1-e^2\cos^2 E}}$$

$$\cos\phi_{fpa} = \sqrt{\frac{1-e^2}{1-e^2\cos^2 E}}$$

E. Perifocal distance, q = radius of perigee, r_P :

$$q = r_P = a(1-e) = \frac{p}{1+e} = r_A\left(\frac{1-e}{1+e}\right) \qquad \text{(C-51)}$$

$$= \frac{b^2}{r_A} = r_A\left(V_A / V_P\right)$$

E. Semi-parameter, *p*:

$$p = a(1-e^2) = r(1 + e\cos\nu) = q(1+e) \qquad \text{(C-52)}$$

$$= b^2/a = h^2/\mu$$

E. Radius of apogee, r_A:

$$r_A = a(1+e) = 2a - r_P = b^2 / r_P = q\left(\frac{1+e}{1-e}\right) = \frac{b^2}{a-c} \qquad \text{(C-53)}$$

(Where the distance c is defined in Fig. 9-5, Sec. 9.1.5.)

E. Semimajor axis, *a*:

$$a = \frac{r_A + r_P}{2} = \frac{r_P}{1-e} = \frac{r_A}{1+e} = \frac{p}{1-e^2} = -\frac{\mu}{2\varepsilon} = \left(\frac{\mu}{n^2}\right)^{1/3} \qquad \text{(C-54)}$$

E. Semiminor axis, *b*:

$$b = a\sqrt{1-e^2} = \sqrt{ap} = \sqrt{r_A r_P} = \sqrt{\mu p / h} \qquad \text{(C-55)}$$

E. Eccentricity, *e*:

$$e = |\mathbf{e}| = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}} = \frac{\sqrt{a^2 - b^2}}{a} \qquad \text{(C-56)}$$

$$= \frac{r_A - r_P}{r_A + r_P} = \frac{r_A}{a} - 1 = 1 - \frac{r_P}{a} = \frac{c}{a}$$

$$0 < e < 1$$

E. Mean motion, *n*:

$$n = \sqrt{\mu / a^3} = \frac{2\pi}{P} \qquad \text{(C-57)}$$

E. Mean anomaly, *M*:

$$M = M_o + nt = E - e\sin E \qquad \text{(C-58)}$$

E. Eccentric anomaly, *E*:

$$E = \text{atan2}\left(\cos E, \sin E\right) = 2\text{atan}\left[\sqrt{\frac{1-e}{1+e}}\tan\left(\nu/2\right)\right] \qquad \text{(C-59)}$$

$$= \text{acos}\left(\frac{a-r}{ae}\right) = \text{asin}\left(\frac{r}{b}\sin\nu\right) = \text{acos}\left(\frac{a-r}{a-r_P}\right)$$

$$\cos E = \frac{e+\cos\nu}{1+e\cos\nu} \qquad \sin E = \frac{\sin\nu\sqrt{1-e^2}}{1+e\cos\nu}$$

as an iterative equation with successive estimates E_i:

$$E_{i+1} = E_i + \frac{M - E_i + e\sin E_i}{1 - e\cos E_i}$$

E. True anomaly, ν:

$$\nu = \text{atan2}\left(\cos\nu, \sin\nu\right) = 2\text{atan}\left[\sqrt{\frac{1+e}{1-e}}\tan\left(E/2\right)\right] \qquad \text{(C-60)}$$

$$= \text{acos}\left(\frac{p/r - 1}{e}\right) = \text{acos}\left(\frac{\cos E - e}{1 - e\cos E}\right)$$

$$\cos\nu = \frac{\cos E - e}{1 - e\cos E} \qquad \sin\nu = \frac{\sin E\sqrt{1-e^2}}{1-e\cos E}$$

as a power series in M:

$$\nu \approx M + \left(2e - \frac{e^3}{4} + \frac{5e^5}{96}\right)\sin(M) \qquad \text{(C-61)}$$

$$+ \left(\frac{5e^2}{4} - \frac{11e^4}{24}\right)\sin(2M) + \left(\frac{13e^3}{12} - \frac{43e^5}{64}\right)\sin(3M)$$

$$+ \left(\frac{103e^4}{96}\right)\sin(4M) + \mathcal{O}\left(e^6\right)$$

E. Rate of change of true anomaly, $\dot{\nu}$:

$$\dot{\nu} = \frac{\sqrt{\mu p}}{r^2} = \frac{V_P r_P}{r^2} = \frac{V_A r_A}{r^2} = \frac{na^2}{r^2}\sqrt{1-e^2} \qquad \text{(C-62)}$$

E. Period, *P*:

$$P = 2\pi / n = 2\pi\sqrt{a^3/\mu} \qquad \text{(C-63)}$$

$= 1.658\,669\,021\,3 \times 10^{-4}\, a^{3/2}$
(for Earth; P in min, a in km)

$= 1.495\,569\,234 \times 10^{-3}\, a^{3/2}$
(for Moon; P in min, a in km)

$= 5.060\,143\,330 \times 10^{-4}\, a^{3/2}$
(for Mars; P in min, a in km)

$= 1.996\,228\,762\,54 \times 10^{-10}\, a^{3/2}$
(for Sun; P in days, a in km)

E. Velocity, V:

$$V = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} = \sqrt{2\left(\frac{\mu}{r} + \varepsilon\right)} = \sqrt{\frac{\mu}{r}\left(2 - \frac{1-e^2}{1+e\cos\nu}\right)} = \sqrt{\mu p} \,/\, \left(r\cos\phi_{fpa}\right) \quad \text{(C-64)}$$

$$\frac{V_A}{V_P} = \frac{r_P}{r_A}$$

$$V_P = \sqrt{\frac{\mu}{r_P}(1+e)} = \sqrt{\frac{2\mu r_A}{r_P\left(r_A + r_P\right)}} = \sqrt{\frac{\mu}{r_P}\left(2 - \frac{r_P}{a}\right)} = V_A\left(\frac{r_A}{r_P}\right)$$

$$V_A = \sqrt{\frac{\mu}{r_A}(1-e)} = \sqrt{\frac{\mu r_P}{a r_A}} = \sqrt{\frac{\mu}{r_A}\left(2 - \frac{r_A}{a}\right)} = V_P\left(\frac{r_P}{r_A}\right)$$

E. Azimuthal velocity, V_{az}:

$$V_{az} = V\cos\phi_{fpa} = r\dot{\nu} \quad \text{(C-65)}$$

E. Radial velocity, V_r:

$$V_r = V\sin\phi_{fpa} = \dot{r} = V_{az}\frac{e\sin\nu}{1+e\cos\nu} = \frac{r\dot{\nu}e\sin\nu}{1+e\cos\nu} \quad \text{(C-66)}$$

E. Range • range rate, $r\dot{r}$:

$$r\dot{r} = e\sqrt{a\mu}\,\sin E \quad \text{(C-67)}$$

E. Areal velocity, $\dot{A}$:

$$\dot{A} = 0.5\sqrt{a\mu\left(1-e^2\right)} = 0.5r^2\dot{\nu} \quad \text{(C-68)}$$

E. Escape velocity, V_E:

$$V_E = \sqrt{2\mu / r} \quad \text{(C-69)}$$

E. Euler axis co-latitude, δ'_{ea}:

$$\delta'_{ea} = \operatorname{atan}\left(\frac{\dot{\nu}\sin i}{-\omega_E + \dot{\nu}\cos i}\right) \quad \text{(C-70)}$$

ω_E = inertial rotation of central body
= 0.004 178 074 1 deg/sec (Earth)
= 0.000 152 504 145 25 deg/sec (Moon)
= 0.004 061 249 794 68 deg/sec (Mars)

(For a complete set of ground track equations, see Table 8-11 in Sec. 8.4 and Sec. 8.4.1, which also includes a discussion of approximations for small eccentricities.)

E. Euler rotation rate, ω_{ea}:

$$\omega_{ea} = \dot{\nu}\sin i / \sin\delta'_{ea} = \sqrt{\omega_E^2 + \dot{\nu}^2 + 2\dot{\nu}\omega_E\cos i} \quad \text{(C-71)}$$

E. Angular radius of the Earth, ρ:

$$\rho = \operatorname{asin}\left(R_E / r\right) \quad \text{(C-72)}$$

R_E = radius of central body
= 6378.1366 km (Earth)
= 1737.4 km (Moon)
= 3396.19 km (Mars)

(Assumes a spherical Earth. For the effect of oblateness, see Sec. 9.4.1, and Sec. 9.1.5 in OCDM.)

E. Maximum angular radius of the Earth, ρ_{max}:

$$\rho_{max} = \operatorname{asin}(R_E / r_P) \quad \text{(C-73)}$$

E. Minimum angular radius of the Earth, ρ_{min}:

$$\rho_{min} = \operatorname{asin}(R_E / r_A) \quad \text{(C-74)}$$

E. Distance to the horizon, D:

$$D = r\cos\rho = \sqrt{r^2 - R_E^2} \quad \text{(C-75)}$$

(For calculation of angles and distance for targets not on the horizon, see Sec. 8.3.1.)

E. Maximum distance to the horizon, D_{max}:

$$D_{max} = r_A\cos\rho_{min} = \sqrt{r_A^2 - R_E^2} \quad \text{(C-76)}$$

E. Minimum distance to the horizon, D_{min}:

$$D_{min} = r_P\cos\rho_{max} = \sqrt{r_P^2 - R_E^2} \quad \text{(C-77)}$$

E. Maximum Earth central angle, λ_{max}:

$$\lambda_{max} = 90\ \text{deg} - \rho_{min} = \operatorname{acos}(R_E / r_A) \quad \text{(C-78)}$$

E. Instantaneous Access Area, IAA:

$$IAA = K_A(1 - \cos\lambda_{max}) \quad \text{(C-79)}$$

$K_A = 2\pi \approx 6.283\,185\,311$ for area in steradians
$= 360^2 / (2\pi) \approx 20{,}626.480\,62$ for area in deg^2
$= 2.55\,603\,915 \times 10^8$ for area in km^2 (Earth)
$= 1.896\,62 \times 10^7$ for area in km^2 (Moon)
$= 7.247\,093 \times 10^7$ for area in km^2 (Mars)

(For alternative ways to calculate ground coverage, see Sec. 10.2.1; for other ground area formulas, see Table 10-3 in Sec. 10.2.1; for transformations between geocentric and access area coordinates, see the boxed example in Sec. 8.3; for projections of sensor fields of view onto the central body, see Sec. 8.3.4.)

E. Area Access Rate, AAR:

$$AAR = \left(\frac{K_A\dot{\nu}}{\pi}\right)\sin\lambda_{max} \quad \text{(C-80)}$$

E. Maximum time in view, T_{max}:

$$T_{max} \approx 2\lambda_{max} / \dot{\nu} \quad \text{(C-81)}$$

(For a complete set of target or ground station coverage formulas for elliptical orbits, see Table 8-12 in Sec. 8.4.2.)

E. Maximum angular rate seen from ground, $\dot{\theta}_{max}$:

$$\dot{\theta}_{max} = \frac{\dot{v}r}{(r - R_E)} \quad \text{(C-82)}$$

E. Node precession rate due to J_2, $\Delta\Omega_{J2}$:

$$\Delta\Omega_{J2} = -1.5\, J_2 n (R_E/a)^2 (1-e^2)^{-2} \cos i \quad \text{(C-83)}$$
$$\approx K_{J2}\, a^{-7/2} (1-e^2)^{-2} \cos i$$
$$= -2.064\,745\,722 \times 10^{14}\, a^{-7/2} (1-e^2)^{-2} \cos i$$
(Earth, a in km)
$$= -3.193\,05 \times 10^{11}\, a^{-7/2} (1-e^2)^{-2} \cos i$$
(Moon, a in km)
$$= -3.468\,016 \times 10^{13}\, a^{-7/2} (1-e^2)^{-2} \cos i$$
(Mars, a in km)

(In the numerical forms, $\Delta\Omega_{J2}$ is in deg/calendar day.)

E. Sun-synchronous inclination, i_{SS}:

$$i_{SS} = \text{acos}\left(\frac{360\, a^{7/2}}{K_{J2}\left(1-e^2\right)^{-2} SP}\right) \quad \text{(C-84)}$$

$$= \text{acos}(-4.773\,513\,28 \times 10^{-15}\, a^{7/2}\ (1-e^2)^{-2})$$
(Earth, a in km)
$$= \text{acos}(-3.086\,7 \times 10^{-12}\, a^{7/2}\ (1-e^2)^{-2})$$
(Moon, a in km)
$$= \text{acos}(-1.511\,0 \times 10^{-14}\, a^{7/2}\ (1-e^2)^{-2})$$
(Mars, a in km)

(Where SP is the sidereal period of the planet or the Moon about the Sun in calendar days. For a discussion of Sun synchronous orbits, see Sec. 9.5.3.)

E. Perigee rotation rate due to J_2, $\Delta\omega_{J2}$:

$$\Delta\omega_{J2} = 1.5 J_2 n \left(R_E/a\right)^2 \left(1-e^2\right)^{-2} \left(2 - \frac{5}{2}\sin^2 i\right) \quad \text{(C-85)}$$
$$= K_{J2} a^{-7/2} \left(1-e^2\right)^{-2} \left(2 - \frac{5}{2}\sin^2 i\right)$$

where K_{J2} is the same as for $\Delta\Omega_{J2}$

(In the numerical forms, $\Delta\omega_{J2}$ is in deg/calendar day. Note that the perigee rotation rate depends on the central body. However, the fact that perigee rotation goes to 0 at the critical inclination of 63.4 deg is independent of the central body.)

E. Orbital revolutions per day, *Rev/d*:

$$Rev/d = \text{Day}/P \quad \text{(C-86)}$$
$$= 1.436\,068\,34 \times 10^3 / P \quad \text{(Earth, } P \text{ in min)}$$
$$= 3.934\,319\,286\,9 \times 10^4 / P \quad \text{(Moon, } P \text{ in min)}$$
$$= 1.477\,377\,729\,355 \times 10^3 / P \quad \text{(Mars, } P \text{ in min)}$$

(This is the number of orbit revolutions for each rotation of the planet on its axis relative to the fixed stars (i.e., the inertial rotation period). "Day" is the length of the sidereal day (in minutes) for the given central body. For more accurate expressions, see the discussion of repeating ground tracks in Sec. 9.5.2.)

E. Node spacing, ΔN:

$$\Delta N = 360 \text{ deg}(P/\text{Day}) \quad \text{(C-87)}$$
$$= 0.250\,684\,448\, P \quad \text{(Earth, } P \text{ in min)}$$
$$= 0.009\,150\,248\,715\, P \quad \text{(Moon, } P \text{ in min)}$$
$$= 0.243\,674\,987\,681\, P \quad \text{(Mars, } P \text{ in min)}$$

(For the equations for repeating ground track orbits, see Sec. 9.5.2.)

E. Maximum eclipse at a specified true anomaly, T_E:

$$T_E \approx 2\rho/\dot{v} = (2/\dot{v})\text{asin}\left(R_E/r\right) \quad \text{(C-88)}$$

(For computation of eclipse duration for any Sun angle, see Sec. 8.2.4; for the formulas for eclipse conditions, see Sec. 11.4 in OCDM.)

E. Sun angle constraints for terminator visibility, β':

$$90° + \xi - \rho < \beta' < 90° - \xi + \rho \quad \text{(C-89)}$$

(ξ is the dark angle defined in Sec. 11.5.2 in OCDM. For general conditions of terminator visibility, see Table 11-3 in Sec. 11.5.2 in OCDM.)

E. Transit time for a spacecraft whose celestial coordinates are known, *T*:

$$T = \alpha + L - \text{GST} \quad \text{(C-90)}$$

(where α is the right ascension, L is the observer East longitude, and GST is the Greenwich Sidereal Time. For computation of observability times of spacecraft for which the celestial coordinates are known, see Table 9-6 in Sec. 9.4.4 in OCDM.)

C.3 Equations for Parabolic Orbits

As discussed in Sec. 9.1 which defines the orbit variables, parabolic orbits represent the boundary between closed, negative total energy elliptical orbits and unbounded, positive total energy hyperbolic orbits. Sec. 8.3 defines the geometric variables involved.

Note: A "**P**" is put in front of each descriptor to denote formulas for Parabolic orbits. Subscript P stands for the value of the parameter at perigee.

P. Defining parameters:

p = semilatus rectum = semiparameter (C-91)

q = perifocal distance

(For a discussion of Keplerian orbits, see Sec. 9.1; for transformations between orbital elements and position and velocity, see Sec. 2.7.1.1 and 2.7.1.2 in OCDM.)

P. Parametric equation:

$$x^2 = 4qy \quad \text{(C-92)}$$

P. Distance from focus, *r*:

$$r = \frac{p}{1+\cos\nu} = \frac{2q}{1+\cos\nu} = q\left(1 + D_P^2\right) \quad \text{(C-93)}$$

P. Specific energy, ε (= total energy per unit mass):

$$\varepsilon = 0 \quad \text{(C-94)}$$

P. Specific angular momentum, h:

$$h = |\mathbf{h}| = \sqrt{\mu p} = rV\cos\phi_{fpa} = r^2\dot{\nu} = r_P V_P \quad \text{(C-95)}$$

P. Angular momentum vector, h:

$$\mathbf{h} = \mathbf{r} \times \mathbf{V} \quad \text{(C-96)}$$

P. Nodal vector, N:

$$\mathbf{N} = \hat{\mathbf{z}} \times \hat{\mathbf{h}} = \hat{\mathbf{z}} \times \mathbf{h} / h \quad \text{(C-97)}$$

P. Eccentricity vector, e:

$$\mathbf{e} = \frac{\mathbf{V} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \quad \text{(C-98)}$$

P. Inclination, *i*:

$$i = \text{acos}\,(h_z / h) \quad \text{(C-99)}$$

(For a discussion of apparent inclination, see the boxed example at the end of Sec. 9.3.1 in OCDM.)

P. Right ascension of the ascending node, Ω:

$$\Omega = \text{atan2}(N_x, N_y) \quad \text{(C-100)}$$

(For the definition of atan2, see Sec. D.4)

P. Flight path angle, ϕ_{fpa}:

$$\phi_{fpa} = \nu / 2 = \text{acos}\left(\sqrt{\mu p} / rV\right) \quad \text{(C-101)}$$

P. Perifocal distance, q = radius of perigee, r_P :

$$q = r_P = p/2 \quad \text{(C-102)}$$

P. Semi-parameter, *p*:

$$p = 2q = h^2/\mu = r(1 + \cos\nu) \quad \text{(C-103)}$$

P. Semimajor axis, *a*:

$$a = \infty \quad \text{(C-104)}$$

P. Semiminor axis, *b*:

$$b = \infty \quad \text{(C-105)}$$

P. Eccentricity, *e*:

$$e = 1 \quad \text{(C-106)}$$

P. Mean motion, *n*:

$$n = 2\sqrt{\mu / p^3} = \left(\frac{1}{t}\right)\left[\tan\left(\frac{\nu}{2}\right) + \frac{1}{3}\tan^3\left(\frac{\nu}{2}\right)\right] \quad \text{(C-107)}$$

where t = time from perigee passage.

P. Mean anomaly, *M*:

$$M = M_0 + nt = D_P + D_P^3 / 3 \quad \text{(C-108)}$$

P. Parabolic anomaly, D_P:

$$D_P = \tan\left(\nu / 2\right) \quad \text{(C-109)}$$

P. True anomaly, ν:

$$\nu = \text{atan2}(\cos\nu, \sin\nu)$$
$$\cos\nu = (p - r)/r$$
$$\sin\nu = 2qD_p / r \quad \text{(C-110)}$$

P. Rate of change of true anomaly, $\dot{\nu}$:

$$\dot{\nu} = \sqrt{\mu p} / r^2 = V_P r_P / r^2 \quad \text{(C-111)}$$

P. Period, *P*:

$$P = \infty \quad \text{(C-112)}$$

P. Velocity, *V*:

$$V = V_E = \sqrt{2\mu / r} = \sqrt{\mu p} / \left(r\cos\phi_{fpa}\right) \quad \text{(C-113)}$$

P. Velocity at perigee, V_P:

$$V_P = \sqrt{\frac{2\mu}{r_P}} \quad \text{(C-114)}$$

P. Parabolic velocity = Escape Velocity, V_E:

$$V_E = \sqrt{2\mu / r} \quad \text{(C-115)}$$

P. Azimuthal velocity, V_{az}:

$$V_{az} = V\cos\phi_{fpa} = r\dot{\nu} \quad \text{(C-116)}$$

P. Radial velocity, V_r:

$$V_r = V\sin\phi_{fpa} = V_{az}\frac{\sin\nu}{q} = \frac{r\dot{\nu}\sin\nu}{1 + \cos\nu} \quad \text{(C-117)}$$

P. Range • range rate, $r\dot{r}$:

$$r\dot{r} = \sqrt{\mu} D_P \quad \text{(C-118)}$$

P. Areal velocity, $\dot{A}$:

$$\dot{A} = \sqrt{\mu q / 2} = 0.5 r^2 \dot{\nu} \quad \text{(C-119)}$$

(For a complete set of ground track equations, see Table 8-11 in Sec. 8.4, and Sec. 8.4.1.)

All Euler Axis and coverage formulas for Parabolic Orbits are the same as for Hyperbolic Orbits

C.4 Equations for Hyperbolic Orbits

As discussed in Sec. 9.1 which defines the orbit variables, hyperbolic orbits are unbounded with positive total energy. As shown in Figs. 9-1 and 9-3 in Sec. 9.1.1, the hyperbola has two discrete segments, only one of which represents the orbit. Section 8.3 defines the geometric variables involved. Sections 12.7.3 to 12.7.5 in OCDM discuss hyperbolic orbit parameters with respect to planetary arrival and departure.

Note: An "**H**" is put in front of each descriptor to denote formulas for Hyperbolic orbits. Subscript P stands for the value of the parameter at perigee.

H. Defining parameters:

a = semi-transverse axis ($a < 0$) (C-120)

b = semi-conjugate axis

(For a discussion of Keplerian orbits, see Sec. 9.1; for transformations between orbital elements and position and velocity, see Secs. 2.7.1.1 and 2.7.1.2 in OCDM; for planetary arrival and departure, see Secs. 12.7.3 to 12.7.5 in OCDM.)

H. Parametric equation:

$$x^2/a^2 - y^2/b^2 = 1 \qquad \text{(C-121)}$$

H. Distance from focus, r:

$$r = \frac{q(1+e)}{1+e\cos\nu} = a(1-e\cosh F) = -a\left(\frac{e}{\cos H} - 1\right) \qquad \text{(C-122)}$$

$$= -a\sqrt{e^2-1}\frac{\tan H}{\sin\nu} = \frac{p}{1+e\cos\nu} = \frac{a\left(1-e^2\right)}{1+e\cos\nu}$$

H. Specific energy, ε (= total energy per unit mass):

$$\varepsilon = -\frac{\mu}{2a} = \frac{V^2}{2} - \frac{\mu}{r} > 0 \qquad \text{(C-123)}$$

H. Specific angular momentum, h:

$$h = |\mathbf{h}| = \sqrt{\mu p} = rV\cos\phi_{fpa} = r^2\dot{\nu} = r_P V_P \qquad \text{(C-124)}$$

H. Angular momentum vector, h:

$$\mathbf{h} = \mathbf{r} \times \mathbf{V} \qquad \text{(C-125)}$$

H. Nodal vector, N:

$$\mathbf{N} = \hat{\mathbf{z}} \times \hat{\mathbf{h}} = \hat{\mathbf{z}} \times \mathbf{h}/h \qquad \text{(C-126)}$$

H. Eccentricity vector, e:

$$\mathbf{e} = \frac{\mathbf{V} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \qquad \text{(C-127)}$$

H. Inclination, i:

$$i = \text{acos}\,(h_z/h) \qquad \text{(C-128)}$$

(For a discussion of apparent inclination, see the boxed example at the end of Sec. 9.3.1 in OCDM.)

H. Right ascension of the ascending node, Ω:

$$\Omega = \text{atan2}\,(N_x, N_y) \qquad \text{(C-129)}$$

(For the definition of atan2, see Sec. D.4)

H. Flight path angle, ϕ_{fpa}:

$$\phi_{fpa} = \text{acos}\left(\sqrt{\mu p}/rV\right) \qquad \text{(C-130)}$$

H. Auxiliary angle of the hyperbola, H^*:

$$H = \text{acos}\left(\frac{ae}{a-r}\right) = \text{acos}\left(\frac{e}{1 - rV_\infty^2/\mu}\right) \qquad \text{(C-131)}$$

Table C-1◀, Fig. C-0, Eq. C-139

H. Turn angle, ψ:

$$\psi = 2\text{asin}\left(\frac{1}{e}\right) = 2\text{asin}\left(\frac{1}{1+V_\infty^2 q/\mu}\right) = 2\text{atan}\left(\frac{-a}{b}\right) \qquad \text{(C-132)}$$

$$= 2\text{atan}\left(\frac{\mu}{V_P V_\infty r_P}\right) = 2\text{atan}\left(\frac{\mu}{\rho V^2\sqrt{\eta}\cos\phi_{fpa}}\right)$$

(For a discussion of planetary fly-bys, see Sec. 9.6.3, and Secs. 12.7.3 to 12.7.5 in OCDM.)

H. Perifocal distance, q = radius of perigee, r_p:

$$q = r_P = a(1-e) = \frac{p}{1+e} = \frac{2\mu}{V_P^2 - V_\infty^2} = \frac{\mu}{V_\infty^2}(e-1) \qquad \text{(C-133)}$$

H. Semi-parameter, p:

$$p = a\left(1-e^2\right) = r(1+e\cos\nu) = q(1+e) \qquad \text{(C-134)}$$

$$= \frac{b^2}{a} = \frac{h^2}{\mu}$$

H. Semimajor axis, a:

$$a = -\frac{\mu}{2\varepsilon} = \left(\frac{\mu}{n^2}\right)^{1/3} = -\frac{\mu}{V_\infty^2} = \frac{q}{1-e} = \frac{p}{1-e^2} \qquad \text{(C-135)}$$

$a < 0$

H. Semiminor axis = semi-conjugate axis = impact parameter, b:

$$b = -a\sqrt{e^2-1} = \sqrt{-ap} = \sqrt{\mu p}/V_\infty = \frac{V_P}{V_\infty} r_P \qquad \text{(C-136)}$$

$$= \frac{-a}{\tan(\psi/2)} = r_P\sqrt{\frac{e+1}{e-1}}$$

H. Eccentricity, e:

$$e = |\mathbf{e}| = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}} = \sqrt{\frac{a^2+b^2}{-a}} = 1 - \frac{r_P}{a} \qquad \text{(C-137)}$$

$$= \sqrt{1 - \frac{p}{a}} = 1 + \frac{r_P}{\mu} V_\infty^2 \qquad e > 1$$

H. Mean motion, n:

$$n = \sqrt{-\mu/a^3} \qquad \text{(C-138)}$$

H. Mean anomaly, M:

$$M = M_0 + nt = e\sinh F - F \qquad \text{(C-139)}$$

---

* Also known as the Gudermannian function of the hyperbolic anomaly: H = gd F = 2atan(e^F) − π/2. See Battin [1999] for more details.

H. Hyperbolic anomaly, F:

$$F = 2\text{atanh}\left[\sqrt{\frac{e-1}{e+1}}\tan\frac{\nu}{2}\right] \qquad \text{(C-140)}$$

$$\sinh F = \frac{\sin\nu\left(\sqrt{e^2-1}\right)}{1+e\cos\nu}$$

$$\cosh F = \frac{e+\cos\nu}{1+e\cos\nu}$$

as an iterative equation with successive estimates F_i:

$$F_{i+1} = F_i + \frac{M - e\sinh F_i + F_i}{e\cosh F_i - 1}$$

H. True anomaly, ν:

$$\nu = \text{atan2}(\cos\nu, \sin\nu) = 2\text{atan}\left[\sqrt{\frac{e+1}{e-1}}\tanh\frac{F}{2}\right]$$

$$= \text{acos}\left(\frac{p/r-1}{e}\right) \qquad \text{(C-141)}$$

$$= \text{asin}\left(\frac{a}{r}\sqrt{e^2-1}\,\tan H\right)$$

$$= 2\text{atan}\left[\sqrt{\frac{e+1}{e-1}}\tan\frac{H}{2}\right]$$

$$\cos\nu = \frac{\cosh F - e}{1 - e\cosh F}$$

$$\sin\nu = \frac{-\sinh F\sqrt{e^2-1}}{1-e\cosh F}$$

H. Rate of change of true anomaly, $\dot{\nu}$:

$$\dot{\nu} = \sqrt{\mu p}\,/\,r^2 = V_P r_P / r^2 \qquad \text{(C-142)}$$

H. Period, P:

$$P = \infty \qquad \text{(C-143)}$$

H. Velocity, V:

$$V = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} = \frac{\sqrt{\mu p}}{\left(r\cos\phi_{fpa}\right)} = \sqrt{\frac{2\mu}{r} + V_\infty^2} \qquad \text{(C-144)}$$

$$= \sqrt{\frac{2\mu}{r} + 2\varepsilon}$$

H. Velocity at perigee, V_P:

$$V_P = \sqrt{\frac{2\mu}{r_P} + V_\infty^2} = \sqrt{\frac{\mu}{r_P}(1+e)} \qquad \text{(C-145)}$$

H. Hyperbolic excess velocity, V_∞:

$$V_\infty = \sqrt{V^2 - \frac{2\mu}{r}} = \sqrt{V_P^2 - \frac{2\mu}{r_P}} \qquad \text{(C-146)}$$

$$= \frac{r_P V_P}{b} = \sqrt{\frac{\mu}{r_P}(e-1)}$$

(The role of V_∞, C_3, and related parameters in interplanetary mission analysis is discussed in Sec. 9.6.3 and Sec. 12.7.1 in OCDM.)

H. Reference launch energy, C_3:

$$C_3 = V_\infty^2 \qquad \text{(C-147)}$$

H. Azimuthal velocity, V_{az}:

$$V_{az} = r\dot{\nu} = V\cos\phi_{fpa} \qquad \text{(C-148)}$$

H. Radial velocity, V_r:

$$V_r = V\sin\phi_{fpa} = \left(\frac{V_{az}e}{q}\right)\sin\nu = \frac{r\dot{\nu}e\sin\nu}{1+e\cos\nu} \qquad \text{(C-149)}$$

H. Range • range rate, $r\dot{r}$:

$$r\dot{r} = e\sqrt{-a\mu}\sinh F \qquad \text{(C-150)}$$

H. Areal velocity, $\dot{A}$:

$$\dot{A} = \frac{1}{2}\sqrt{a\mu\left(1-e^2\right)} \qquad \text{(C-151)}$$

H. Escape velocity, V_E:

$$V_E = \sqrt{2\mu/r} \qquad \text{(C-152)}$$

H. Euler axis co-latitude, δ'_{ea}:

$$\delta'_{ea} = \text{atan}\left(\frac{\dot{\nu}\sin i}{-\omega_E + \dot{\nu}\cos i}\right) \qquad \text{(C-153)}$$

ω_E = inertial rotation of central body
= 0.004 178 074 1 deg/sec (Earth)
= 0.000 152 504 145 25 deg/sec (Moon)
= 0.004 061 249 794 68 deg/sec (Mars)

(For a complete set of ground track equations, see Table 8-11 in Sec. 8.4, and Sec. 8.4.1.)

H. Euler rotation rate, ω_{ea}:

$$\omega_{ea} = \dot{\nu}\sin i / \sin\delta'_{ea} \qquad \text{(C-154)}$$

$$= \sqrt{\omega_E^2 + \dot{\nu}^2 + 2\dot{\nu}\omega_E\cos i}$$

H. Angular radius of the Earth, ρ:

$$\rho = \text{asin}\left(R_E / r\right) \qquad \text{(C-155)}$$

R_E = radius of central body
= 6378.1366 km (Earth)
= 1737.4 km (Moon)
= 3396.19 km (Mars)

(Assumes a spherical Earth. For the effect of oblateness, see Sec. 9.4.1, and Sec. 9.1.5 in OCDM.)

H. Maximum angular radius of the Earth, ρ_{max}:

$$\rho_{max} = \text{asin}\left(R_E / r_P\right) \quad \text{(C-156)}$$

H. Distance to the horizon, D:

$$D = r\cos\rho = \sqrt{r^2 - R_E^2} \quad \text{(C-157)}$$

(For calculation of angles and distance for targets not on the horizon, see Sec. 8.3.1.)

H. Minimum distance to the horizon, D_{min}:

$$D_{min} = r_P \cos\rho_{max} = \sqrt{r_P^2 - R_E^2} \quad \text{(C-158)}$$

H. Maximum Earth central angle, λ_{max}:

$$\lambda_{max} = 90 \text{ deg} - \rho = \text{acos}\,(R_E / r) \quad \text{(C-159)}$$

H. Instantaneous Access Area, *IAA*:

$$IAA = K_A(1 - \cos\lambda_{max}) \quad \text{(C-160)}$$

$K_A = 2\pi \approx 6.283\ 185\ 311$ for area in steradians
$= 360^2 / (2\pi) \approx 20{,}626.480\ 62$ for area in deg^2
$= 2.55\ 603\ 915 \times 10^8$ for area in km^2 (Earth)
$= 1.896\ 62 \times 10^7$ for area in km^2 (Moon)
$= 7.247\ 093 \times 10^7$ for area in km^2 (Mars)

(For alternative ways to calculate ground coverage, see Sec. 10.2.1; for other ground area formulas, see Table 10-3 in Sec. 10.2.1; for transformations between geocentric and access area coordinates, see the boxed example in Sec. 8.3; for projections of sensor fields of view onto the central body, see Sec. 8.3.4.)

H. Area Access Rate, *AAR*:

$$AAR = \left(\frac{K_A \dot{\nu}}{\pi}\right)\sin\lambda_{max} \quad \text{(C-161)}$$

H. Maximum time in view, T_{max}:

$$T_{max} \approx 2\lambda_{max} / \dot{\nu} \quad \text{(C-162)}$$

(This approximation holds only when the rate of change of true anomaly and the distance to the Earth do not change strongly over the time in view. For more precise calculations the start time and stop time of a viewing event should be calculated independently.)

H. Maximum angular rate seen from ground, $\dot{\theta}_{max}$:

$$\dot{\theta}_{max} = \frac{\dot{\nu} r}{(r - R_E)} \quad \text{(C-163)}$$

H. Maximum eclipse at a specified true anomaly, T_E:

$$T_E \approx 2\rho / \dot{\nu} = \left(\frac{2}{\dot{\nu}}\right)\text{asin}\left(\frac{R_E}{r}\right) \quad \text{(C-164)}$$

(This approximation holds only when the rate of change of true anomaly and the distance to the Earth do not change strongly over the eclipse duration. For more precise calculations, the start time and stop time of the eclipse should be calculated independently. For computation of eclipse duration for any Sun angle, see Sec. 8.2.4; for the formulas for eclipse conditions, see Sec. 11.4 in OCDM.)

H. Sun angle constraints for terminator visibility, β':

$$90° + \xi - \rho < \beta' < 90° - \xi + \rho \quad \text{(C-165)}$$

(ξ is the dark angle defined in Sec. 11.5.2 in OCDM. For general conditions of terminator visibility, see Table 11-3 in Sec. 11.5.2 in OCDM.)

H. Transit time for a spacecraft whose celestial coordinates are known, T:

$$T = \alpha + L - \text{GST} \quad \text{(C-166)}$$

(where α is the right ascension, L is the observer East longitude, and GST is the Greenwich Sidereal Time. For computation of observability times of spacecraft for which the celestial coordinates are known, see Table 9-6 in Sec. 9.4.4 in OCDM.)

References



For annotated bibliography of Appendix C, see the book website.

Battin, Richard H. 1999. *Introduction to the Mathematics and Methods of Astrodynamics.* Reston, VA: AIAA.

International Astronomical Union Working Group on Numerical Standards for Fundamental Astronomy. 2009. *Current Best Estimates.*

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Appendix D
Mission Geometry Formulas

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D.1 Arc Length and Rotation Angle Formulas
D.2 Area Formulas
D.3 Spherical Triangles
D.4 The atan2 and acos2 functions

This appendix provides a summary of basic rules for computing angles and areas on either the celestial sphere, the surface of the Earth, other planets and moons, or for the solution of most spherical triangles. As discussed in Chap. 8 and Sec. 10.5, spherical geometry is the natural choice for problems of what the spacecraft sees, and geometry on the surface of the Earth or other planets. Many, though not all, of the computations can also be done using unit vectors, although this approach does not usually provide the physical insight that we would like to have for mission analysis and design. Where possible, both spherical geometry and vector formulations are given. In the past, a principal limitation of spherical geometry solutions has been the quadrant ambiguities associated with working with angles that cover the full range of 0 to 360 deg because most spherical geometry solutions only cover the range of 0 to 180 deg.* This is no longer a problem. *Orbit and Constellation Design and Management* [= OCDM = Wertz, 2009] provides a complete set of "*full sky geometry*" solutions for all spherical triangles with all values covering the full range of 0 to 360 deg that eliminate all quadrant ambiguities not inherent in the solution itself. Thus, the only remaining anomalies are those that are inherent to the problem, such as when two reference vectors are exactly 180 deg apart and, therefore, are effectively the same as two that are on top of each other.

Note that full sky geometry solutions are particularly useful in providing closed form solutions for the motion of a satellite in circular orbit over the surface of the Earth. These provide significantly more accurate solutions than traditional methods that assume motion on a great circle arc and are summarized in Table 8-11. Developing these equations requires more background than appropriate here, so the solution development is left to OCDM. The most common mission geometry problems are listed in Table D-1, along with where they are addressed both in this volume and in OCDM. Other basic references on spherical geometry are provided in the web bibliography.

* Spherical geometry has been used for several centuries for solving basic navigation problems. However, for most of this time, these solutions were implemented by hand using trig tables and it was a trivial matter when any of the angles got larger than 180 deg to "turn the triangles around," such that all of the angles were in the range of 0 to 180 deg. With today's automated methods, having solutions covering the full range of input and output angles from 0 to 360 deg is substantially more important.

Table D-1. Principal Geometrically-Oriented Mission Design Problems and Where Discussed. "Live Calc" means that the SME-SMAD table or figure is available for download from the website. FoV is the Field of View. See also Table 8-2 in Sec. 8.1 for where the application formulas are derived.

Topic	SME-SMAD	OCDM
Basic Geometry on the Celestial Sphere (or on the Earth or Planetary Surface)		
Introduction to Mission Geometry — Formula summaries	Secs. 8.0, 8.1 App. D	Secs. 6.1, 6.2 App. A
Coordinate Systems	Sec. 8.1.2, Table 8-3, 8-4	Sec. 6.1.2, Table 6-2, 6-3
Transformation Between Spherical Coordinates		Sec. 8.3.4
Angular Measurements on the Celestial Sphere	Sec. 8.1.3, Fig. 8-11	Table 6-4, Sec. 7.1, Sec. A.1
Rotation Angle Measurement	Table 8-5, Fig. 8-11, 8-19	Table 6-4, Sec. 7.4, Sec. A.1
Rhumb Line		Sec. 6.3 boxed note
Measurement Uncertainty on the Celestial Sphere		Sec. 7.2

Table D-1. Principal Geometrically-Oriented Mission Design Problems and Where Discussed. (Continued) "Live Calc" means that the SME-SMAD table or figure is available for download from the website. FoV is the Field of View. See also Table 8-2 in Sec. 8.1 for where the application formulas are derived.

Topic	SME-SMAD	OCDM
Good and Bad Measurement Sets		Sec. 7.6, Sec. 7.4 boxed example
Area Formulas	Sec. D.2	Table 8.3, Sec. A.3
Effect of Oblateness		Sec. 9.1.5
Full Sky Geometry	App. D*	Chap. 8
Full Sky Spherical Trig Solutions		Sec. 8.1, Sec. A.4, A.7
Dual Axis Spiral (see also satellite ground track)		Sec. 8.2
— Equations for		Table 8-8
Spherical Geometry Formulas		App. A
— Arc length and rotation angles	Sec. D.1	Sec. A.1
— Equations of great and small circles		Sec. A.2
— Area formulas	Sec. D.2	Sec. A.3
— Napier's rules		Sec. A.5
— Right and quadrantal spherical triangles	Sec. D.3	Sec. A.6
— Oblique full sky spherical triangles		Sec. A.4, A.7
— Differential spherical trig		Sec. A.8
— Relations between trig functions		Sec. A.9
Applications		
Live Calc ***Array Sensor FoV***	Secs. 8.2.2, 8.2.3, Table 8-8	Secs. 6.3.1, 7.2
Earth Surface Seen from Space	Sec. 8.3.2	Sec. 9.1
— Effect of oblateness	Fig. 9-14*	Sec. 9.1.5
Directions, Shapes, and Areas on the Earth as Seen from Space	Sec. 8.3.3	Sec. 9.1.3
Transformation Between Earth-based and Space-based Spherical Coordinates	Sec. 8.3.1 boxed note	Sec 9.1.1 boxed note
Live Calc ***Projection of Sensor FoVs onto the Earth***	Sec 8.3.4,Table 8-10	Sec. 9.1.4
Apparent Motion of Ground Targets seen from Space		Sec. 9.2
Live Calc ***Mapping and Pointing budgets***	Sec. 8.6, Table 8-16, 8-17, Fig. 8-50, 8-51	Sec. 5.5
Earth Coverage		Chap. 9
— Coverage areas	Sec. 10.2	Secs. 9.1, 9.5.1
Live Calc ***— Coverage patterns***	Sec. 10.2.1.2	Secs. 9.5.1.2
— Histograms for	Sec. 10.2.1.2	Fig. 9-42
Live Calc ***— Coverage vs. latitude***	Fig. 10-10	Fig. 9-44, 9-45
— Formulas for	Table 10-3, 10-4	App. D
— Example computations	App. C	Sec. 9.6
Eclipse Duration	Sec. 8.1.1	Sec. 6.3.2
Sun Angle on an Arbitrary Spacecraft Face		Sec. 6.3.3
Magnetometer Measurement Accuracy		Sec. 7.3.2
Live Calc ***Satellite Ground Track***		Secs. 8.3.3, 9.3
— Equations for		Table 8-9
Live Calc ***Ground Station Passes***	Sec. 8.4	Sec. 9.4
— Equations for	Table 8-11, 8-12	Table 9-4, 9-5
Motion of the Satellite as Seen from Earth	Sec. 8.4	Sec. 9.4
Satellite Relative Motion	Sec. 8.5	Chap. 10
Viewing and Lighting Conditions		Chap. 11, Table 11-1
Observability of Interplanetary Spacecraft		Sec. 9.4.4

* See OCDM for computations.

D.1 Arc Length and Rotation Angle Formulas

Let P_i, $i = 1, 2, 3$, be three points on the unit sphere with coordinates (α_i, δ_i). The **arc-length** distance, θ_{12}, between P_1 and P_2 is given by:

$$\cos\theta_{12} = \cos\theta_{21} \qquad 0 \le \theta \le 180°$$
$$= \sin\delta_1 \sin\delta_2 + \cos\delta_1 \cos\delta_2 \cos(\alpha_1 - \alpha_2) \quad \text{(D-1a)}$$

If $\hat{\mathbf{P}}_i$ are unit vectors corresponding to P_i, then

$$\cos\theta = \hat{\mathbf{P}}_1 \cdot \hat{\mathbf{P}}_2 \quad 0 \le \theta \le 180° \quad \text{(D-1b)}$$

The **rotation angle**, $\Phi(P_1, P_2; P_3)$, from P_1 to P_2 about P_3, is cumbersome to calculate and is most easily obtained from spherical triangles (see Sec. D.3 and OCDM Secs. A.4, A.5, and A.7) if any of the triangle components are already known. To calculate directly from coordinates, obtain as intermediaries the arc-length distances θ_{ij}, between pairs of points P_i, P_j. Then

$$\cos\Phi(\hat{P}_1, \hat{P}_2; \hat{P}_3) = \frac{\cos\theta_{12} - \cos\theta_{13}\cos\theta_{23}}{\sin\theta_{13}\sin\theta_{23}} \quad \text{(D-2a)}$$
$$0 \le \Phi \le 360°$$

with the quadrant determined by inspection. For automated quadrant resolution use the side-side-side triangle solution in OCDM Sec. A.7.1.

If the three points are expressed as unit vectors, then

$$\tan\Phi = \frac{\hat{\mathbf{P}}_3 \cdot (\hat{\mathbf{P}}_1 \times \hat{\mathbf{P}}_2)}{(\hat{\mathbf{P}}_1 \cdot \hat{\mathbf{P}}_2) - (\hat{\mathbf{P}}_3 \cdot \hat{\mathbf{P}}_1)(\hat{\mathbf{P}}_3 \cdot \hat{\mathbf{P}}_2)} \quad \text{(D-2b)}$$

with the quadrant of Φ determined by the signs of the numerator and denominator, as done by the ATAN2 computer function. (See Sec. D.4 and OCDM Sec. A.7)

D.2 Area Formulas

All areas are measured on the curved surface of the unit sphere. See OCDM Sec. 8.1 and Table 8-3 for a discussion of angular area. For a sphere of radius R, multiply each area by R^2. **In the area formulas, all arc lengths are in radians and all angular areas are in steradians (sr)**, where

1 sr ≡ solid angle enclosing an area equal to the square of the radius

$= (180/\pi)^2 \text{ deg}^2 \approx 3{,}282.806\,350\,011\,743\,794 \text{ deg}^2$

The **surface area of the sphere** is

$$\Omega_s = 4\pi \quad \text{(D-3)}$$

The **area of a lune** bounded by two great circles which intersect at an angle of Θ radians is

$$\Omega_l = 2\Theta \quad \text{(D-4)}$$

The **area of a spherical triangle** whose three rotation angles are θ_1, θ_2, and θ_3, is

$$\Omega_t = \theta_1 + \theta_2 + \theta_3 - \pi \quad \text{(D-5)}$$

The **area of a spherical polygon** of n sides, where S is the sum of its rotation angles in radians, is

$$\Omega_p = S - (n-2)\pi \quad \text{(D-6)}$$

The **area of a small circle** of angular radius ρ is

$$\Omega_c = 2\pi(1 - \cos\rho) \quad \text{(D-7)}$$

The **area of a segment of rotation angle, ϕ,** in a small circle of angular radius ρ (i.e., a "pie piece") is

$$\Omega_{sc} = \phi(1 - \cos\rho) \quad \text{(D-8)}$$

The **area of a ring or annulus** of inner radius ρ_i and outer radius ρ_0 is

$$\Omega_r = 2\pi(\cos\rho_i - \cos\rho_0) \quad \text{(D-9)}$$

The **area of a segment of rotation angle, ϕ,** in an annulus of inner radius ρ_i and outer radius ρ_0 is

$$\Omega_{sr} = \phi(\cos\rho_i - \cos\rho_0) \quad \text{(D-10)}$$

The **overlap area between two small circles** of angular radii ρ and ε, separated by a center-to-center distance, α, is

$$\Omega_0 = 2\pi - 2\cos\rho \operatorname{acos}\left[\frac{\cos\varepsilon - \cos\rho\cos\alpha}{\sin\rho\sin\alpha}\right] \quad \text{(D-11)}$$
$$-2\cos\varepsilon \operatorname{acos}\left[\frac{\cos\rho - \cos\varepsilon\cos\alpha}{\sin\varepsilon\sin\alpha}\right]$$
$$-2\operatorname{acos}\left[\frac{\cos\alpha - \cos\varepsilon\cos\rho}{\sin\varepsilon\sin\rho}\right]$$
$$(|\rho - \varepsilon| \le \alpha < \rho + \varepsilon)$$

If ε goes to 90 deg, then the above simpliefies to the formula for the **overlap area between a great circle and a small circle** of angular radius ρ, separated by a center-to-center distance, α:

$$\Omega_0 = 2\pi - 2\cos\rho \operatorname{acos}(\cot\alpha \cot\rho) \quad \text{(D-12)}$$
$$-2 \operatorname{acos}(\cos\alpha/\sin\rho)$$
$$|90° - \rho| \le \alpha < 90° + \rho$$

Recall that area is measured on the curved surface.

D.3 Spherical Triangles

A *spherical triangle* consists of any three non-collinear points on the sphere connected by great circle arcs. A *right spherical triangle* is one with at least one right angle. (Unlike plane triangles, spherical triangles can have 1, 2, or 3 right angles.) Any two of the remain-

ing components, including the two remaining angles, serve to completely define the triangle. *Napier's Rules*, given in any of the books in the bibliography, provide a concise formulation for all possible right spherical triangles. However, experience has shown that it is substantially more convenient to write out explicitly the rules for the relatively small number of possible combinations of known and unknown sides and angles. These are listed in Table D-2.

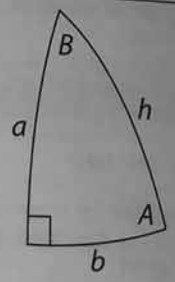
A *quadrantal spherical triangle* is one with at least one side that is 90 deg in length. As with right spherical triangles, any two of the remaining five components completely define the triangle. These are given by a corresponding set of Napier's Rules. Again, it is more practical to write out explicitly all possible relationships. These are given in Table D-4.

An *oblique spherical triangle* has arbitrary sides and angles. Sides and angles are generally defined over the range of 0 to 180 deg, although most of the spherical geometry relations continue to hold in the angular range up to 360 deg. A set of basic rules which can be applied to any spherical triangle are given in Table D-3. Finally, these general rules can be used to write explicit expressions for any of the unknown components in any oblique spherical triangle with any three components known. These are given in full in OCDM, including the rules for all possible full sky triangles, i.e., those in which any of the sides or angles can range from 0 to 360 deg.

Table D-2. Right Spherical Triangles.

Right Spherical Triangle

The line below each formula indicates the quadrant of the answer. $Q(A) = Q(a)$ means that the quadrant of angle A is the same as that of side a. "2 possible solutions" means that either quadrant provides a correct solution to the defined triangle.



Given	Find		
a, b	$\cos h = \cos a \cos b$ $Q(h) = \{Q(a)\,Q(b)\}$*	$\tan A = \tan a / \sin b$ $Q(A) = Q(a)$	$\tan B = \tan b / \sin a$ $Q(B) = Q(b)$
a, h	$\cos b = \cos h / \cos a$ $Q(b) = \{Q(a) / Q(h)\}$**	$\sin A = \sin a / \sin h$ $Q(A) = Q(a)$	$\cos B = \tan a / \tan h$ $Q(B) = \{Q(a) / Q(h)\}$**
b, h	$\cos a = \cos h / \cos b$ $Q(a) = \{Q(b) / Q(h)\}$**	$\cos A = \tan b / \tan h$ $Q(A) = \{Q(b) / Q(h)\}$**	$\sin B = \sin b / \sin h$ $Q(B) = Q(b)$
a, A	$\sin b = \tan a / \tan A$ 2 possible solutions	$\sin h = \sin a / \sin A$ 2 possible solutions	$\sin B = \cos A / \cos a$ 2 possible solutions
a, B	$\tan b = \sin a \tan B$ $Q(b) = Q(B)$	$\tan h = \tan a / \cos B$ $Q(h) = \{Q(a)\,Q(B)\}$*	$\cos A = \cos a \sin B$ $Q(A) = Q(a)$
b, A	$\tan a = \sin b \tan A$ $Q(a) = Q(A)$	$\tan h = \tan b / \cos A$ $Q(h) = \{Q(b)\,Q(A)\}$*	$\cos B = \cos b \sin A$ $Q(B) = Q(b)$
b, B	$\sin a = \tan b / \tan B$ 2 possible solutions	$\sin h = \sin b / \sin B$ 2 possible solutions	$\sin A = \cos B / \cos b$ 2 possible solutions
h, A	$\sin a = \sin h \sin A$ $Q(a) = Q(A)$	$\tan b = \tan h \cos A$ $Q(b) = \{Q(A) / Q(h)\}$**	$\tan B = 1 / \cos h \tan A$ $Q(B) = \{Q(A) / Q(h)\}$**
h, B	$\sin b = \sin h \sin B$ $Q(b) = Q(B)$	$\tan a = \tan h \cos B$ $Q(a) = \{Q(B) / Q(h)\}$**	$\tan A = 1 / \cos h \tan B$ $Q(A) = \{Q(B) / Q(h)\}$**
A, B	$\cos a = \cos A / \sin B$ $Q(a) = Q(A)$	$\cos b = \cos B / \sin A$ $Q(b) = Q(B)$	$\cos h = 1 / \tan A \tan B$ $Q(h) = \{Q(A)\,Q(B)\}$*

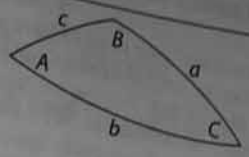
* $\{Q(x)\,Q(y)\}$ ≡ 1st quadrant if $Q(x) = Q(y)$, 2nd quadrant if $Q(x) \neq Q(y)$

** $\{Q(x) / Q(h)\}$ ≡ quadrant of x if $h \leq 90$ deg, quadrant opposite x if $h > 90$ deg.

Table D-3. Oblique Spherical Triangles.

Oblique Spherical Triangle

The following rules hold for any spherical triangle:



The Law of Sines: $\dfrac{\sin a}{\sin A} = \dfrac{\sin b}{\sin B} = \dfrac{\sin c}{\sin C}$

The Law of Cosines for Sides:

$\cos a = \cos b \cos c + \sin b \sin c \cos A$
$\cos b = \cos c \cos a + \sin c \sin a \cos B$
$\cos c = \cos a \cos b + \sin a \sin b \cos C$

The Law of Cosines for Angles:

$\cos A = -\cos B \cos C + \sin B \sin C \cos a$
$\cos B = -\cos C \cos A + \sin C \sin A \cos b$
$\cos C = -\cos A \cos B + \sin A \sin B \cos c$

Gauss's Formula:

$$\sin\left[\frac{1}{2}(A-B)\right] = \frac{\sin\left[\frac{1}{2}(a-b)\right]}{\sin(c/2)}\cos(C/2)$$

Useful Derived Formulas:

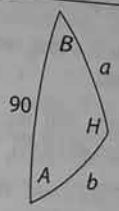
$c = \text{atan}(\tan b \cos A) \pm \text{atan}(\tan a \cos B)$

$$C = \text{atan}\left(\frac{1}{\tan A \cos b}\right) \pm \text{atan}\left(\frac{1}{\tan B \cos a}\right)$$

Table D-4. Quadrantal Spherical Triangles.

Quadrantal Spherical Triangle

The line below each formula indicates the quadrant of the answer. $Q(A) = Q(a)$ means that the quadrant of angle A is the same as that of side a. "2 possible solutions" means that either quadrant provides a correct solution to the defined triangle.



Given	Find		
A, B	$\cos H = -\cos A \cos B$ $Q(H) = \{Q(A)\ Q(B)\}$*	$\tan a = \tan A / \sin B$ $Q(a) = Q(A)$	$\tan b = \tan B / \sin A$ $Q(b) = Q(B)$
A, H	$\cos B = -\cos H / \cos A$ $Q(B) = \{Q(A) \setminus Q(H)\}$**	$\sin a = \sin A / \sin H$ $Q(a) = Q(A)$	$\cos b = -\tan A / \tan H$ $Q(b) = \{Q(A) \setminus Q(H)\}$**
B, H	$\cos A = -\cos H / \cos B$ $Q(A) = \{Q(B) \setminus Q(H)\}$**	$\cos a = -\tan B / \tan H$ $Q(a) = \{Q(B) \setminus Q(H)\}$**	$\sin b = \sin B / \sin H$ $Q(b) = Q(B)$
A, a	$\sin B = \tan A / \tan a$ 2 possible solutions	$\sin H = \sin A / \sin a$ 2 possible solutions	$\sin b = \cos a / \cos A$ 2 possible solutions
A, b	$\tan B = \sin A \tan b$ $Q(B) = Q(b)$	$\tan H = -\tan A / \cos b$ $Q(H) = \{Q(A)\ Q(b)\}$*	$\cos a = \cos A \sin b$ $Q(a) = Q(A)$
B, a	$\tan A = \sin B \tan a$ $Q(A) = Q(a)$	$\tan H = -\tan B / \cos a$ $Q(H) = \{Q(B)\ Q(a)\}$*	$\cos b = \cos B \sin a$ $Q(b) = Q(B)$
B, b	$\sin A = \tan B / \tan b$ 2 possible solutions	$\sin H = \sin B / \sin b$ 2 possible solutions	$\sin a = \cos b / \cos B$ 2 possible solutions
H, a	$\sin A = \sin H \sin a$ $Q(A) = Q(a)$	$\tan B = -\tan H \cos a$ $Q(B) = \{Q(a) \setminus Q(H)\}$**	$\tan b = -1/\cos H \tan a$ $Q(b) = \{Q(a) \setminus Q(H)\}$**
H, b	$\sin B = \sin H \sin b$ $Q(B) = Q(b)$	$\tan A = -\tan H \cos b$ $Q(A) = \{Q(b) \setminus Q(H)\}$**	$\tan a = -1/\cos H \tan b$ $Q(a) = \{Q(b) \setminus Q(H)\}$**
a, b	$\cos A = \cos a / \sin b$ $Q(A) = Q(a)$	$\cos B = \cos b / \sin a$ $Q(B) = Q(b)$	$\cos H = -1 / \tan a \tan b$ $Q(H) = \{Q(a)\ Q(b)\}$*

*$\{Q(x)\ Q(y)\}$ ≡ 1st quadrant if $Q(x) = Q(y)$, 2nd quadrant if $Q(x) \neq Q(y)$

**$\{Q(x) \setminus Q(H)\}$ ≡ quadrant of x if $H > 90$ deg, quadrant opposite x if $H \leq 90$ deg.

D.4 The atan2 and acos2 functions

In working with full-sky triangles*, we need to have a mechanism for evaluating inverse trigonometric functions over the full range of 0 to 360 deg. For the arc tangent, the computer function atan2 is convenient. With the angle's sine and cosine specified, atan2 uniquely specifies the angle. In general, any inverse trig function covering the full range of 0 to 360 deg must be a function of two arguments, because all of the trig functions have two solutions over this range. For many triangles, it is convenient to define an acos2 function that plays a role similar to that of atan2. To do this, we first define the hemisphere function, $H(\phi)$:

$$H(\phi) \equiv +1 \text{ if } 0 \text{ deg} \le \phi \bmod^{\dagger}_{360 \text{ deg}} < 180 \text{ deg}$$
$$H(\phi) \equiv -1 \text{ if } 180 \text{ deg} \le \phi \bmod_{360 \text{ deg}} < 360 \text{ deg} \quad \text{(D-13)}$$

The acos2 function is then defined by:

$$\text{acos2}[\cos(\phi), H(\phi)] \equiv [H(\phi)\ \text{acos}(\cos(\phi))] \bmod_{360 \text{ deg}}$$
$$\text{where } 0 \text{ deg} \le \text{acos}(\cos(\phi)) < 180 \text{ deg} \quad \text{(D-14)}$$

Thus, over the range of 0 to 360 deg, acos2 and the set (cos, H) are inverse functions:

$$\text{If } 0 \text{ deg} \le \phi < 360 \text{ deg},$$
$$\text{then } \phi = \text{acos2}[\cos(\phi), H(\phi)] \quad \text{(D-15)}$$

Additional properties of the acos2 function are summarized in Table D-5.

Table D-5. Properties of the acos2 Function. The acos2 function serves the same purpose as atan2—i.e., it is a function of two variables rather than one so that it can return an angle over the range 0 to 360 deg.

- First define the hemisphere function, $H(\phi)$:
  $H(\phi) \equiv +1$ if $0 \text{ deg} \le \phi \bmod_{360 \text{ deg}} < 180$ deg
  $H(\phi) \equiv -1$ if $180 \text{ deg} \le \phi \bmod_{360 \text{ deg}} < 360$ deg
- Then the acos2 function is defined by:
  $\text{acos2}[\cos(\phi), H(\phi)] \equiv [H(\phi)\ \text{acos}(\cos\phi)] \bmod_{360 \text{ deg}}$
  where $0 \text{ deg} \le \text{acos}(\cos(\phi)) < 180$ deg
- Thus:
  If $0 \text{ deg} \le \phi < 360$ deg, then $\phi = \text{acos2}[\cos(\phi), H(\phi)]$
- Also:
  $\text{acos2}[\cos(\phi), -H(\phi)] = 360 \text{ deg} - \phi$
  $\text{acos2}[-\cos(\phi), H(\phi)] = (180 \text{ deg} - \phi) \bmod_{360 \text{ deg}}$
  $= 180 \text{ deg} - \phi$, if $\phi < 180$ deg, or $360 \text{ deg} - (\phi - 180 \text{ deg})$, if $\phi > 180$ deg
  $\text{acos2}[-\cos(\phi), -H(\phi)] = (180 \text{ deg} + \phi) \bmod_{360 \text{ deg}}$
  $= 180 \text{ deg} + \phi$, if $\phi < 180$ deg, or $\phi - 180$ deg, if $\phi > 180$ deg

* For a complete treatment of full-sky spherical geometry see OCDM Chap. 8 and App. A.

† The modulo function adds or subtracts multiples of 360 deg in order to express the result as a quantity from 0 to 360 deg.

The atan2 function has been used in computer programming languages for many years and is intended to resolve the problem of quadrant ambiguities in inverse trig functions. Specifically, if ϕ is an angle defined over the range 0 to 360 deg and $\sin\phi$ and $\cos\phi$ are known, then the atan2 function is defined by:

$$\text{atan2}(\cos\phi, \sin\phi) \equiv \phi$$
$$= [\text{atan}(\sin\phi/\cos\phi) + 90^{\circ}\ (1 - \text{sign}(\cos\phi))] \bmod_{360 \text{ deg}} \quad \text{(D-16b)}$$

where atan is defined over the usual range of –90 deg to +90 deg and the atan2 function is defined over the range 0 to 360 deg.

While the above expression is correct, it is not well-defined when $\cos\phi$ goes to 0 and can result in mathematical singularities when used in practice. An alternative definition which avoids these singularities and is well-defined over the entire range is as follows:

$$\text{atan2}(x, y) \equiv \begin{cases} \text{atan}(y/x) & \text{if } x>0,\ y \ge 0 \\ 90^{\circ} & \text{if } x=0,\ y>0 \\ 180^{\circ} + \text{atan}(y/x) & \text{if } x<0 \\ 270^{\circ} & \text{if } x=0,\ y<0 \\ 360^{\circ} + \text{atan}(y/x) & \text{if } x>0,\ y<0 \\ \text{Undefined} & \text{if } x=0,\ y=0 \end{cases} \quad \text{(D-17)}$$

where, as above, atan is defined over the usual range of –90 to +90 deg and atan2 is defined over the range 0 to 360 deg.

Unfortunately, the definition of the atan2 function is implemented differently in common system engineering tools. Specifically,

$$\text{atan2}(x, y) = \text{atan}(y/x) \quad \text{(D-18)}$$

in Excel, Visual Basic, Google Spreadsheets, and in OCDM and this volume. Since the website tables and plots are implemented in Excel, the definition on the website is consistent with one given in the text. In contrast,

$$\text{atan2}(y, x) = \text{atan}(y/x) \quad \text{(D-19)}$$

in programming languages FORTRAN, MatLab, C++, Java, and Pascal. In addition, all of the languages and spreadsheets define atan2 over the range –180 deg to +180 deg, whereas it is more convenient for mission geometry to define all angles over the range of 0 to 360 deg and this is done in the text. Fortunately, this change is trivial and easily handled by all of the relevant tools.

References



For annotated bibliography of Appendix D, see the book website.

Wertz, James R. 2009. *Orbit & Constellation Design and Management: Spacecraft Orbit and Attitude Systems.* Hawthorne, CA: Microcosm Press.

Appendix E
Time and Date Systems

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Time is something we intuitively understand extremely well and, consequently, understand poorly in detail. For example, any phenomenon which repeats on a regular basis, such as a pendulum or the motion of electrons in an atom, can be used as the basis for timekeeping. However, "regularly repeating" implies that we have some standard to compare with. The only way to determine a clock's accuracy is to compare it with another clock, which of course leads to the question of which clock we should assume to be correct. Historically, the fundamental "clocks" chosen for maintaining absolute time have been the rotation of the Earth on its axis and the revolution of the Earth about the Sun. However, modern time measurements have shown these systems to be both nonuniform and extremely difficult to model with precision. For example, Fig. B-3 in App. B shows the variations in the length of the day over the past several centuries. On average, the Earth is slowing down by 1.5 ms/century but local variations are very substantial.*

Unfortunately, clock errors are cumulative, and therefore, high accuracy can prove important. For example, assume a spacecraft in low Earth orbit has a clock that is slowing down by 1 ms/day, which is small enough to be difficult to measure by many processes. This would result in a cumulative error of 0.365 sec over a year which corresponds to an error in position of the satellite of 3 km. Thus, if we wish to know the positions of satellites to tens of meters, we will need reasonably good clocks to do so.

Two basic types of time measurement are used in spacecraft systems: (1) *time intervals* between 2 events such as the spacecraft's spin period, orbital period, or the length of time a sensor sees the Earth; and (2) *absolute times* or *calendar times* of specific events, such as the time associated with some particular spacecraft observation. Of course, calendar time is simply a time interval for which the beginning event is an agreed standard.

E.1 Calendar Time and Long Duration Intervals

Calendar time in the usual form of date and time is used only for input and output because generally, arithmetic is cumbersome in months, days, hours, minutes, and seconds (for computations, absolute time is used instead, as explained next). Nonetheless, this is used for most human interaction with space systems because it's the system with which we are most familiar. Even with date and time systems, problems can arise, because time zones are different throughout the world and spacecraft operations typically involve a worldwide network. The uniformly adopted solution to this problem is to use the local standard time corresponding to 0 deg longitude (i.e., the Greenwich meridian) as the assigned time for events anywhere in the world or in space. This is referred to as *Universal Time* (UT), *Greenwich Mean Time* (GMT), or *Zulu* (Z), all of which are equivalent for most practical spacecraft operations. For precise computations, UT has replaced GMT, since the term GMT can be ambiguous [Seidelmann, 2006]. The name Greenwich Mean Time was chosen because 0 deg longitude is defined by the site of the former Royal Greenwich Observatory southeast of central London.

Civil time, T_{civil}, as measured by a standard wall clock or time signals, differs from Universal Time by an integral number of hours, corresponding approximately to the longitude of the observer. The approximate relation is:

$$T_{civil} \approx UT \pm L / 15 \qquad \text{(E-1)}$$

where T_{civil} and UT are in hours, and L is the longitude in degrees with the plus sign corresponding to East longitude and the minus sign corresponding to West longitude.

Calendar time is remarkably inconvenient for computation, particularly over long time intervals of months or years. We need an absolute time that is a continuous count of time units from some arbitrary reference. The time interval between any two events is then found by simply subtracting the absolute time of the second event from that of the first. The universally adopted solution for astronomical problems is the *Julian Date*, JD, a continuous count of the number of days since Greenwich noon (12:00 UT) on January 1, 4713 BC,† or, as astron-

* The overall slowing of the Earth's rotation and lengthening of the day are caused primarily by tidal friction with the Moon. Local variations are caused principally by the growth and decline in the polar ice caps which shifts large quantities of water from the pole (smaller moment of inertia and higher spin rate) to the equator (larger moment of inertia and lower spin rate).

† This strange starting point was suggested by an Italian scholar of Greek and Hebrew, Joseph Scaliger, in 1582 as the beginning of the current *Julian period* of 7,980 years. This period is the product of three numbers: the *solar cycle*, or the interval at which all dates recur on the same days of the week (28 years); the *lunar cycle*, containing an integral number of lunar months (19 years); and the *indiction* or the tax period introduced by the Emperor Constantine in 313 AD (15 years). The last time that these started together was 4713 BC and the next time will be 3267 AD. Scaliger was interested in reducing the astronomical dating problems associated with calendar reforms of his time and his proposal had the convenient selling point that it pre-dated the ecclesiastically approved date of creation, October 4, 4004 BC.

Table E-1. Julian Date at the Beginning of Each Year from 2000 to 2041. See text for explanation of use. The day number for the beginning of the year is called "Jan. 0.0" (actually Dec. 31st of the preceding year) so that day numbers can be found by simply using dates. Thus, Jan. 1 is day number 1 and has a JD 1 greater than that for Jan. 0. * = leap year.

Year	*JD* 2,400,000+ for Jan 0.0 *UT*	Year	*JD* 2,400,000+ for Jan 0.0 *UT*	Year	*JD* 2,400,000+ for Jan 0.0 *UT*
2000*	51,543.5	2014	56,657.5	2028*	61,770.5
2001	51,909.5	2015	57,022.5	2029	62,136.5
2002	52,274.5	2016*	57,387.5	2030	62,501.5
2003	52,639.5	2017	57,753.5	2031	62,866.5
2004*	53,004.5	2018	58,118.5	2032	63,231.5
2005	53,370.5	2019	58,483.5	2033	63,597.5
2006	53,735.5	2020*	58,848.5	2034	63,962.5
2007	54,100.5	2021	59,214.5	2035	64,327.5
2008*	54,465.5	2022	59,579.5	2036	64,692.5
2009	54,831.5	2023	59,944.5	2037	65,058.5
2010	55,196.5	2024*	60,309.5	2038	65,423.5
2011	55,561.5	2025	60,675.5	2039	65,788.5
2012*	55,926.5	2026	61,040.5	2040	66,153.5
2013	56,292.5	2027	61,405.5	2041	66,519.5

omers now say, –4712. Because Julian Days start at noon UT, they will be a half day off with respect to civil dates. While this is inconvenient for transforming from civil dates to Julian Dates, it was useful for astronomers because this way the date didn't change in the middle of the night (for European observers).

As described below, there are 4 general approaches for converting between calendar dates and Julian Dates.

Table Look-Up

Tabulations of the current Julian Date are in most astronomical ephemerides and almanacs. Table E-1 lists the Julian Dates at the beginning of each year from 2000 through 2041. To find the Julian Date for any given calendar date, simply add the *day number* within the year (and fractional day number, if appropriate) to the Julian Date for Jan 0.0 of that year from Table E-2. Day numbers for each day of the year are on many calendars or can be found by adding the date to the day number for day 0 of the month from Table E-2. Thus 18:00 UT on April 15, 2014 = day number 15.75 + 90 = 105.75 in 2014 = JD 105.75 + 2,456,657.5 = JD 2,456,763.25.

Table E-2. Day Numbers for Day 0.0 of Each Month. Leap years (in which February has 29 days) are those evenly divisible by 4. However, years evenly divisible by 100 are not leap years, except that those evenly divisible by 400 are. Leap years are indicated by * in Table E-2.

Month	Non-Leap Years	Leap Years
January	0	0
February	31	31
March	59	60
April	90	91
May	120	121
June	151	152
July	181	182
August	212	213
September	243	244
October	273	274
November	304	305
December	334	335

Table E-2, Fig. E-0, Eq. E-1◀

To convert from Julian Days to dates, determine the year in which the Julian Date falls from Table E-1. Subtract the Julian Date from the JD for January 0.0 of that year to determine the day number within the year. This can be converted to a date (and time, if appropriate) by using day numbers on a calendar or subtracting from the day number for the beginning of the appropriate month from Table E-2. Thus, from Table E-1, JD 2,456,073.25 is in the year 2012. The day number is 2,456,073.25 – 2,455,926.5 = 146.75. From Table E-2, this is 18:00 UT, May 25, 2012.

E.2 Modern Time Measurement —Short Duration Intervals

As one would expect, modern technology has lead to ever increasing precision in the measurement of time. However, in addition, new processes for measuring time have been introduced, as well as new and fundamentally different definitions of the meaning of time in both science and engineering. In the 1950s, *Ephemeris Time, ET*, was introduced, based on the dynamic equations of motion of the Earth. For many years, this was used as the basis of astronomical and astrophysical ephemerides, i.e., the most precise orbit calculations. In 1967, the second was redefined as having an atomic standard but ephemeris time remained in use for the motion of planets and satellites. In 1984, ephemeris time was unceremoniously abandoned in favor of *Terrestrial Dynamic Time, TDT*, in which the unit of measure was the atomic second and the scale was chosen to agree with ephemeris time in 1984. In 1991, the general theory of relativity was explicitly adopted as the theoretical background for defining both space and time reference frames, *TDT* was renamed *Terrestrial Time, TT*, and the definition of the second was "adjusted" to correspond to atomic measurements at a specific location (i.e., at mean sea level on the surface of the Earth).*

* An excellent discussion of the history of time systems is provided by Seidelmann [2006].

Table E-3. Common Time Systems.

Kind of Time	Defined By	Fundamental Unit	Regularity	Application
Sidereal (ST)	Earth's rotation relative to stars	Sidereal day, 1 rotation of Earth with respect to stars	Irregular	Astronomical observations; determining *UT* and rotational orientation of Earth
Solar				
Apparent	Earth's rotation relative to true Sun	Successive transits of Sun	Irregular and annual variations	Sundials*
Mean	Earth's rotation relative to fictitious mean Sun	Mean solar day	Uniform and annual variations	Confuse students and engineers; use uniform time
Universal				
UT0	Observed *UT*	Mean solar day	Irregular	Study of Earth's wandering pole
UT1	Corrected *UT0*	Mean solar day	Irregular	Shows seasonal variation of Earth's rotation
UT2	Corrected *UT1*	Mean solar day	Irregular	Basic rotation of Earth
UTC=GMT=Z	Atomic sec and leap sec to approximate *UT1*	Mean solar day	Uniform except for leap sec	Civil timekeeping; terrestrial navigation and surveying; broadcast time signals
Ephemeris (ET)	Fraction of tropical year 1900	Ephemeris sec	Uniform	Ephemerides prior to 1994; no longer in use
Atomic (TAI)	Frequency of Ce-133 radiation	Atomic sec = Ephemeris sec	Uniform	Basis of *ET* and *UTC*
GPS	Atomic sec without leap sec	Atomic sec	Uniform	Time component of GPS signals
Relativistic				
Terrestrial (TT)	Atomic sec at mean sea level on Earth	Atomic sec at Earth's surface	Uniform	Ephemerides
Barycentric Dynamic (TDB)	Orbital equations of motion with respect to barycenter of the Solar System	Atomic sec adjusted for relativistic effects	Uniform	Transforms Earth-based time to time kept by the motions of the planets

*Devices showing the time of day by the shadow of a pointer (called style or gnomon) cast by the Sun on a plate or surface marked with lines that indicate the hours of the day.

Currently, there are 4 main types of time systems in use:

- *Atomic Time, TAI*, for which the unit of duration corresponds to a defined number of wavelengths for a specific atomic transition of a specific isotope
- *Universal Time, UT*, for which the unit of duration is the rotation of the Earth with respect to the Sun, defined to be as uniform as possible despite variations in the physical rotation rate of the Earth
- *Sidereal Time, ST*, for which the unit of duration is the rotation of the Earth with respect to the vernal equinox which, in turn, is nearly fixed with respect to the mean positions of the stars
- *Dynamical Time, DT*, for which the unit of duration is based on the orbital motion of the Earth, Moon, and planets. Terrestrial Time, *TT*, belongs to this family of timescales

In addition, rapidly rotating pulsars may provide an even more accurate standard for future time systems. While the differences between the various time systems are subtle, they can have important implications for spacecraft systems and applications. Commonly used modern time systems are defined in Table E-3.

Fortunately, the basis for all of the modern time systems is the Système International, SI, *second*. This is defined as the duration of 9,192,631,770 periods of the radiation corresponding to the transition between 2 hyperfine levels of the ground state of the Cesium-133 atom measured at mean sea level on the Earth. This definition of the second corresponds more-or-less to 1 / 86,400 (= 1 / [60 × 60 × 24]) of the rotation period of the Earth, relative to the mean Sun. It is, of course, the "more-or-less" part which ultimately causes most of the problems in time measurement systems. Some of the time systems are described in more detail below.

Atomic Time (TAI)

International Atomic Time, TAI (*Temps Atomique International*), is a practical implementation of a time standard based on the SI second. An excellent approximation to *TAI* can be maintained by laboratory Cesium clocks. A large number of such clocks are compared from time to time and a weighted average is prepared which provides a fine adjustment for each of the individual clocks. The use of these Cesium-based clocks provides a readily available basis for timekeeping for all types of physical, astronomical, and space-related observation.

Table E-3, Fig. E-0, Eq. E-1◀

Universal Time (UT)

Universal Time, UT, follows the irregular rotation of the Earth, and is often referred to as a type of solar time because the objective in universal time is to remain synchronized with the orientation of the Earth relative to the Sun. The most important subcategory of universal time is *Coordinated Universal Time (UTC),* which is the basis for civil timekeeping and broadcast time signals worldwide. *UTC* uses the SI second as the basic unit of time and then adds (or, in principle, subtracts) a leap second at the end of the last day of June and the last day of December as needed to maintain close agreement with the rotational orientation of the Earth. Thus, *UTC* lags behind *TAI* by an integral number of seconds. For example, for January 1, 1996, *TAI* minus *UTC* equaled 30 sec exactly. There is a long-term continuous drift between *UTC* and *TAI* that cannot be predicted in advance, because *UTC* takes into account irregularities in the motion of the Earth.

In applications where precision is not critical, *UTC* frequently goes by the name Greenwich Mean Time (*GMT*), Zulu (*Z*), or simply Universal Time (*UT*). The latter definition is ambiguous since *UT* is also used for *UT1*, another subcategory of universal time which even more closely follows the real motion of the Earth. Fig. E-1 shows historical differences between *TAI*, *UTC*, and *UT1*.

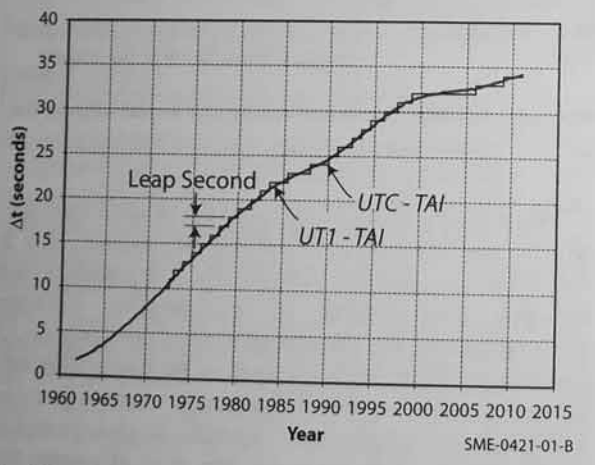


Fig. E-1. Historical Differences Between Universal Time and Atomic Time.

Terrestrial Time (TT)

Terrestrial Time, TT, is the current substitute for what was previously identified as terrestrial dynamic time, TDT, which in turn replaced the earlier ephemeris time, *ET*. Terrestrial time also uses the SI second as the unit of measure but is more precisely defined in terms of the dynamic equations of motion for the Earth. For most practical purposes, *TT* can be defined in terms of *TAI* by a simple offset, i.e., $TT - TAI = 32.184$ sec. (This offset is due to the historical origins of the time system. *TT* matched *ET* in 1984, when the use of *ET* was discontinued.) Consequently, for example, in January 1996, $TT - UTC = 32.184 + 30 \text{ sec} = 62.184 \text{ sec}$. Because atomic clocks have a small drift rate, changes between *TT* and *TAI* on the order of microseconds can accumulate over a period of years. (A more precise definition of terrestrial time based on the theory of relativity is given on the website.)

Table E-4, Fig. E-1, Eq. E-1◀

GPS Time

The GPS satellite system uses its own unique time called *GPS time*. Although presumably it could be accommodated with modern computer systems, the addition of a leap second is certainly inconvenient for GPS processing algorithms. Consequently, the GPS clock uses the SI second but does not introduce leap seconds. Like *TT* and *TAI*, *GPS time* maintains a fixed offset from *UTC* that changes by 1 sec whenever a leap second is introduced in *UTC*. The difference between *GPS time* and *TAI* is constant: $GPS\ time - TAI = 19$ sec., except for a quantity of the order of tens of nanoseconds, varying with time. In order to allow *UTC* to be recovered from the time signals broadcast by GPS, the integral number of seconds by which the two differ is included in subframe 4 of the GPS navigation message. (See Fig. E-9 in Sec. E.2.) Consequently, a GPS receiver may provide time output which is either *GPS time* or *UTC*. The GPS system is becoming a common mechanism for time transfer between 2 locations on the Earth or between 2 satellites in low-Earth orbit. An error budget for *GPS time* transfer when both locations have a common GPS satellite in view is given in Table E-4.

Table E-4. Error Budgets for GPS Time Transfer (with Two Satellites in View). [Seidelmann, 2006].

Type of Delay	Best Case RMS (ns)	Worst Case RMS (ns)
Satellite Ephemeris	3	10
Ionospheric	2	100
Tropospheric	1	20
User-Position	1	1
Multipath	1	2
Receiver	1	1
Signal-to-Noise	7	1
Total RMS (for single 13-minute track)	4.2	103

NOTE: Some of the errors depend on the distance (for example, the satellite ephemeris). The best case applies to distances of 2,000–3,000 km; the worst case is for distances of 6,000–8,000 km.

Get More — See website for discontinuous time, Solar time, sidereal time, and relativisitic time.

References

Get Ref — **For annotated bibliography of Appendix E, see the book website.**

Seidelmann, P. Kenneth, ed., USNO. 2006. *Explanatory Supplement to the Astronomical Almanac.* Mill Valley, CA: University Science Books.

Appendix F
Coordinate Transformations; Vector, Matrix, and Quaternion Algebra

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F.1 Coordinate Transformations

Transformation Between Cartesian and Spherical Coordinates

The components of a vector, **R**, in cartesian (x, y, z) and spherical (r, θ, ϕ) coordinates are shown in Fig. F-1 and listed below. Note that θ is measured from the positive z-axis. The *declination*, δ, used in celestial coordinates is measured from the equatorial (x–y) plane and is related to θ by

$$\delta \equiv 90° - \theta \tag{F-1}$$

The components in cartesian and spherical coordinates are related by the following equations:

Arbitrary Vector	Unit Vector ($\|\mathbf{R}\| \equiv 1$)	
$x = r \sin\theta \cos\phi$	$x = \sin\theta \cos\phi$	(F-2a)
$y = r \sin\theta \sin\phi$	$y = \sin\theta \sin\phi$	(F-2b)
$z = r \cos\theta$	$z = \cos\theta$	(F-2c)
$r = (x^2 + y^2 + z^2)^{1/2}$	$r = 1$	(F-2d)
$\theta = \text{acos}\{z/(x^2 + y^2 + z^2)^{1/2}\}$	$\theta = \text{acos}\, z$	(F-2e)
$\phi = \text{atan2}(x, y)$	$\phi = \text{atan2}(x, y)$	(F-2f)

The correct quadrant for ϕ in Eq. (F-2f) is obtained from the signs of x and y by, for example, using the atan2 function defined in App. D.4.

Transformation Between Cartesian Coordinates

Given a cartesian rectangular coordinate frame (x, y, z) as defined in Fig. F-1, we can define a second cartesian coordinate frame (u, v, w) by defining 3 mutually perpendicular unit vectors along the axes of the new coordinate frame. (Both reference frames are assumed to be orthogonal, and right-handed.) The unit vectors defining the new frame in terms of the original coordinates are:

$$\hat{\mathbf{u}} \equiv \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} \quad \hat{\mathbf{v}} \equiv \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \quad \hat{\mathbf{w}} \equiv \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \tag{F-3}$$

Given these vectors, we can form the *direction cosine matrix*[*], or *attitude matrix*, A, as follows:

$$A \equiv \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} \tag{F-4}$$

[*] The matrix A is called the *direction cosine matrix* because each of the elements is the cosine of the angle between the two relevant vectors. For example, $u_x = \hat{\mathbf{u}} \cdot \hat{\mathbf{x}} = \cos\theta$, where θ is the angle between $\hat{\mathbf{u}}$ and $\hat{\mathbf{x}}$.

So long as $\hat{\mathbf{u}}$, $\hat{\mathbf{v}}$, $\hat{\mathbf{w}}$ and form a right-handed triad of mutually perpendicular unit vectors, A will be a proper real orthogonal matrix. Only three of the nine elements are independent and $\det A = 1$. The attitude matrix is a coordinate transformation that maps vectors from the original frame to the new frame. Thus, if $\mathbf{b}$ is an arbitrary vector with components b_x, b_y, and b_z, then the components in the new frame are given by:

$$\begin{bmatrix} b_u \\ b_v \\ b_w \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{u}} \cdot \mathbf{b} \\ \hat{\mathbf{v}} \cdot \mathbf{b} \\ \hat{\mathbf{w}} \cdot \mathbf{b} \end{bmatrix} = A\mathbf{b} \equiv \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} \tag{F-5}$$

The inverse matrix, A^{-1}, is the coordinate transformation that maps vectors from the new frame (u, v, w) back into the original frame (x, y, z). For the attitude matrix, the inverse is simply the transpose. That is:

$$A^{-1} = A^T \equiv \begin{bmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{bmatrix}, \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = A^{-1} \begin{bmatrix} b_u \\ b_v \\ b_w \end{bmatrix} \tag{F-6}$$

Transformation Between Spherical Coordinates

We can use the formalism developed for full sky geometry to significantly reduce the complexity of transforming between two spherical coordinate systems. Specifically, assume that two spherical coordinate systems are related as shown in Fig. F-2, with the variables defined as follows:

ρ ≡ arc length between the two positive poles ($0 \le \rho \le 180$ deg)

ψ_1 ≡ azimuth of the second pole in the first coordinate system ($0 \le \psi_1 < 360$ deg)

ψ_2 ≡ azimuth of the first pole in the second coordinate system ($0 \le \psi_2 < 360$ deg)

Given the coordinates (ϕ_2, δ_2') of a point, **P**, in the second coordinate system, we wish to find the coordinates (ϕ_1, δ_1') of **P** in the first system, where δ' is the co-elevation angle = 90 deg − δ. We first define the triangle shown in Fig. F-3, with:

$$\Delta\phi_1 = \psi_1 - \phi_1 \tag{F-7}$$

$$\Delta\phi_2 = \psi_2 - \phi_2 \tag{F-8}$$

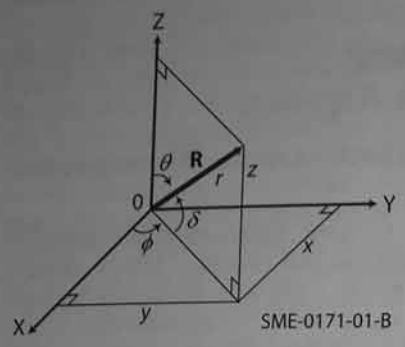


Fig. F-1. Components of a Vector, R, of length *r* in Cartesian (*x, y, z*), and Spherical (*r, θ, φ*) Coordinates.

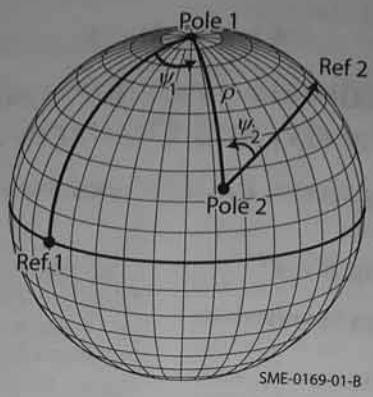


Fig. F-2. Relationship Between the Coordinate Systems.

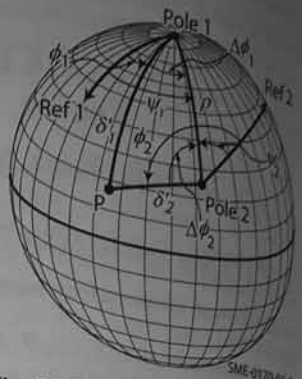


Fig. F-3. Definition of the Spherical Triangle for Coordinate Transformations. (Note that $\Delta\phi_2 < 0$)

From the general solution for side-angle-side spherical triangles (see A.7.3 and Eqs. (8-4) to (8-6) in Wertz [2009]), we have two solutions:

$$\delta_1' = \text{acos2}\Big[\cos\rho\cos\delta_2' + \sin\rho\sin\delta_2'\cos\Delta\phi_2, \text{H}(\Delta\phi_2)\Big] \quad \text{(F-9)}$$

$$\Delta\phi_1 = \text{acos2}\left[\frac{\cos\delta_2' - \cos\rho\cos\delta_1'}{\sin\rho\sin\delta_1'}, \text{H}\left(\delta_2'\right)\right] \quad \text{(F-10)}$$

and

$$\delta_1'^{*} = 360\text{ deg} - \delta_1' \quad \text{(F-11)}$$

$$\Delta\phi_1^{*} = \Delta\phi_1 + 180\text{ deg} \quad \text{(F-12)}$$

where the acos2 function is defined in App. D.4. If we allow the angles to range from 0 to 360 deg, but restrict the sides to the range of 0 to 180 deg, then we have a unique solution (see Eq. (8-6) in Wertz [2009]) (except in the singular case in which the poles are identical or 180 deg apart):

$$\delta_1' = \text{acos}\Big[\cos\rho\cos\delta_2' + \sin\rho\sin\delta_2'\cos\Delta\phi_2\Big] \quad \text{(F-13)}$$

$$\Delta\phi_1 = \left[\text{acos}\left[\frac{\cos\delta_2' - \cos\rho\cos\delta_1'}{\sin\rho\sin\delta_1'}\right]\cdot\text{H}\left(\Delta\phi_2\right)\right]_{\text{MOD360}}$$

This can be further reduced to the normal coordinate expressions (α, δ), where α is azimuth and δ is elevation:

$$\delta_1 = 90\text{ deg} - \text{acos}[\cos\rho\ \sin\delta_2 + \sin\rho\ \cos\delta_2\cos\Delta\phi_2] \quad \text{(F-14)}$$

$$\Delta\phi_1 = \left[\text{acos}\left[\frac{\sin\delta_2 - \cos\rho\sin\delta_1}{\sin\rho\cos\delta_1}\right]\cdot\text{H}\left(\Delta\phi_2\right)\right]_{\text{MOD360}}$$

Table F-0, Fig. F-3, Eq. F-16

By symmetry, the reverse transformation will be:

$$\delta_2 = 90\text{ deg} - \text{acos}\big[\cos\rho\sin\delta_1 + \sin\rho\cos\delta_1\cos\Delta\phi_1\big] \quad \text{(F-15)}$$

$$\Delta\phi_2 = \left[\text{acos}\left[\frac{\sin\delta_1 - \cos\rho\sin\delta_2}{\sin\rho\cos\delta_2}\right]\cdot\text{H}\left(\Delta\phi_1\right)\right]_{\text{MOD360}}$$

where acos is the normal inverse cosine function defined over the range 0 to 180 deg and H is the hemisphere function defined in App. D.4. Finally,

$$\phi_1 = \psi_1 - \Delta\phi_1 \quad \text{(F-16)}$$

and δ_1' is defined by Eq. (F-13).

F.2 Vector, Matrix, & Quaternion Algebra

Get Info

The details of vector, matrix, and quaternion algebra are beyond the scope of this book, but are nonetheless relevant to many of the orbit, attitude, and mission computations. Therefore, we have reproduced, with permission, the relevant appendices on these topics from the book *Spacecraft Attitude Determination and Control* [Wertz, 1978] and included them on the book website in the App. F material. The appendix in Wertz [1978] on spherical geometry calculations has been superceded by the Full Sky Spherical Geometry approach described in Chap. 8 and App. A of Wertz [2009].

References

Get Ref

For annotated bibliography of Appendix F, see the book website.

Wertz, James R., ed. 1978. *Spacecraft Attitude Determination and Control.* Dordrecht, The Netherlands: D. Reidel Publishing Company.

Wertz, James R. 2009. *Orbit & Constellation Design and Management: Spacecraft Orbit and Attitude Systems.* Hawthorne, CA: Microcosm Press.

Appendix H
Units and Conversion Factors

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The metric system of units, officially known as the *International System of Units*, or *SI*, is used throughout this book, with the exception that angles are expressed in degrees rather than the SI unit of radians. By international agreement, the fundamental SI units of length, mass, and time are defined by NIST [2008]:

The *meter* is the length of the path traveled by light in vacuum during a time interval of 1/299,792,458 of a second.

The *kilogram* is equal to the mass of the international prototype of the kilogram.

The *second* is the duration of 9,192,631,770 periods of the radiation corresponding to the transition between two hyperfine levels of the ground state of the cesium-133 atom.

Additional base units in the SI system are the *ampere* for electric current, the *kelvin* for thermodynamic temperature, the *mole* for amount of substance, and the *candela* for luminous intensity. Thompson and Taylor [2008] provide an excellent summary of SI units.

The names of multiples and submultiples of SI units are formed by application of the following prefixes:

Multiplier	Prefix	Symbol	Multiplier	Prefix	Symbol
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	μ
10^{12}	tera	T	10^{-9}	nano	n
10^{9}	giga	G	10^{-12}	pico	p
10^{6}	mega	M	10^{-15}	femto	f
10^{3}	kilo	k	10^{-18}	atto	a
10^{2}	hecto	h	10^{-21}	zepto	z
10^{1}	deka	da	10^{-24}	yocto	y

For each quantity, the SI unit and its abbreviation are given in brackets. For convenience in computer use, most conversion factors are given to the greatest available accuracy. Note that some conversions are exact definitions and some (speed of light, astronomical unit) depend on the value of physical constants. ". . ." indicates a repeating decimal. All notes are on the last page.

To Convert from	*To*	*Multiply by*	*Notes*
Acceleration [meter/second², m/s²]			
Gal (galileo)	m/s^2	0.01	E
Inch/second², in/s²	m/s^2	0.025 4	E
Foot/second², ft/s²	m/s^2	0.304 8	E
Free fall (standard), g	m/s^2	9.806 65	E
Angle (see angular measure)			
Angular Acceleration [radian/second², rad/s²]			
Degrees/sec², deg/s²	rad/s^2	$\pi/180$	E
Revolutions/sec², rev/s²	rad/s^2	$\approx 0.017\,453\,292\,519\,943\,295$	E
Revolutions/min², rev/min²	rad/s^2	2π	E
Revolutions/min²	deg/s^2	$\approx 6.283\,185\,307\,179\,586\,477$	E
Radians/sec², rad/s²	deg/s^2	$\pi/1{,}800$	E
Revolutions/sec², rev/s²	deg/s^2	$\approx 1.745\,329\,251\,994 \times 10^{-3}$	E
		0.1	E
		$180/\pi$	E
		$\approx 57.29\,577\,951\,308\,232$	E
		360	E
Angular Area [sr], book also uses deg²			
Degree², deg²	sr	$(\pi/180)^2$	E
Minute², min²	sr	$\approx 3.046\,174\,197\,867 \times 10^{-4}$	E
Second², s²	sr	$(\pi/10{,}800)^2$	E
Steradian, sr	deg^2	$\approx 8.461\,594\,994\,075 \times 10^{-8}$	E
		$(\pi/648{,}000)^2$	E
		$\approx 2.350\,443\,053\,909 \times 10^{-11}$	E
		$(180/\pi)^2$	E
Minute², min²	deg^2	$\approx 3.282\,806\,350\,011 \times 10^{3}$	E
Second², s²	deg^2	$1/3{,}600 = 2.777\ldots \times 10^{-4}$	E
		$(1/3{,}600)^2$	E
Steradian, sr	rad^2	$\approx 7.716\,049\,382\,716 \times 10^{-8}$	E
		1	E
Angular Measure [radian, rad]. ***This book uses degree (abbreviated "deg") as the basic unit.***			
Degree, deg	rad	$\pi/180$	E
Minute (of arc), min (of arc), min	rad	$\approx 0.017\,453\,292\,519\,943$	E
Second (of arc), s (of arc), s	rad	$\pi/10{,}800$	E
Radian, rad	deg	$\approx 2.908\,882\,086\,657 \times 10^{-4}$	E
		$\pi/648\,000$	E
		$\approx 4.848\,136\,811\,095 \times 10^{-6}$	E
		$180/\pi$	E
		$\approx 57.295\,779\,513\,082\,320$	E
Minute (of arc), min	deg	$1/60 = 0.01666\ldots$	E
Second (of arc), s	deg	$1/3{,}600 = 2.777\ldots \times 10^{-4}$	E
Angular Momentum [kilogram·meter²/second, kg·m²/s]			
Gram · cm²/sec, g · cm²/s	$kg \cdot m^2/s$	1×10^{-7}	E
lbm · inch²/sec, lbm · in²/s	$kg \cdot m^2/s$	$2.926\,396\,534\,292 \times 10^{-4}$	E
Slug · inch²/sec, slug · in²/s	$kg \cdot m^2/s$	$9.415\,402\,418\,968 \times 10^{-3}$	D
lbm · foot²/sec, lbm · ft²/s	$kg \cdot m^2/s$	0.042 140 110 093 80	D
Inch · lbf · sec, in · lbf · s	$kg \cdot m^2/s$	0.112 984 829 027 6	D
Slug · foot²/sec, slug · ft²/s = foot · lbf · sec, ft · lbf · s	$kg \cdot m^2/s$	1.355 817 948 331	D
Angular Velocity [radian/second, rad/s]. ***This book uses degrees/second as the basic unit.***			
Degrees/sec, deg/s	rad/s	$\pi/180$	E
		$\approx 0.017\,453\,292\,519\,943\,295$	E

To Convert from	To	Multiply by	Notes
Revolutions/min, rpm	rad/s	$\pi/30$ $\approx 0.104\ 719\ 755\ 119\ 659\ 774$	E
Revolutions/sec, rev/s	rad/s	2π $\approx 6.283\ 185\ 307\ 179\ 586\ 477$	E
Revolutions/min, rpm	deg/s	6	E
Radians/sec, rad/s	deg/s	$180/\pi$ $\approx 57.295\ 779\ 513\ 082\ 320\ 88$	E
Revolutions/sec , rev/s	deg/s	360	E
Area [meter^2, m^2]			
Acre	m^2	$4.046\ 856\ 422 \times 10^3$	E
Foot^2, ft^2	m^2	0.092 903 04	E
Hectare	m^2	1×10^4	E
Inch^2, in^2	m^2	$6.451\ 6 \times 10^{-4}$	E
Mile^2 (U.S. statute)	m^2	$2.589\ 110\ 336 \times 10^6$	E
Yard^2, yd^2	m^2	0.836 127 36	E
(Nautical mile)^2	m^2	$3.429\ 904 \times 10^6$	E
Capacity (see Volume)			
Conductance (see Electric Conductance)			
Current (see Electric Current)			
Density [kilogram/meter^3, kg/m^3]			
Gram/centimeter^3, g/cm^3	kg/m^3	1.0×10^3	E
Pound mass/inch^3, lbm/in^3	kg/m^3	$2.767\ 990\ 471\ 020 \times 10^4$	D
Pound mass/foot^3, lbm/ft^3	kg/m^3	16.018 463 373 96	D
Slug/ft^3	kg/m^3	515.378 818 393 2	D
Distance (see Length)			
Electric Charge [coulomb, C]			
Abcoulomb	C	10	E
Faraday (based on carbon–12)	C	$9.648\ 534 \times 10^4$	NIST
Statcoulomb	C	$3.335\ 641 \times 10^{-10}$	NIST
Electric Conductance [siemens, S]			
Abmho	S	1×10^9	E
Mho (Ω^{-1})	S	1	E
Electric Current [ampere, A]			
Abampere	A	10	E
Gilbert	A	$10/4\pi$ $\approx 0.795\ 774\ 715\ 459\ 5$	E
Statampere	A	$3.335\ 641 \times 10^{-10}$	NIST
Electric Field Intensity [volt/meter ≡ kilogram · meter · ampere^{-1} · second^{-3}, V/m ≡ kg · m · A^{-1} · s^{-3}]			
Electric Potential Difference [volt ≡ watt/ampere ≡ kilogram · meter^2 · ampere^{-1} · second^{-3}, V ≡ W/A ≡ kg · m^2 · A^{-1} · s^{-3}]			
Abvolt	V	1×10^{-8}	E
Statvolt	V	299.792 5	NIST
Electric Resistance [ohm ≡ volt/ampere ≡ kilogram · meter^2 · ampere^{-2} · second^{-3}, Ω ≡ V/A ≡ kg · m^2 · A^{-2} · s^{-3}]			
Abohm	Ω	1×10^{-9}	E
Statohm	Ω	$8.987\ 552 \times 10^{11}$	NIST
Energy (= torque = work; torque is a vector, energy is a scalar, but both have same units) [joule ≡ newton · meter ≡ kilogram · meter^2/s^2 , J ≡ N · m ≡ kg · m^2/s^2]			
British thermal unit, Btu (mean)	J	$1.055\ 055\ 852\ 62 \times 10^3$	E
Calorie (IT), cal	J	4.186 8	E
Kilocalorie (IT), kcal	J	$4.186\ 8 \times 10^3$	E
Electron volt, eV	J	$1.602\ 176\ 487 \times 10^{-19}$	
Erg ≡ gram · cm^2/s^2 = pole · cm · oersted	J	1×10^{-7}	E
Foot poundal	J	0.042 140 110 093 80	E
Foot lbf = slug · foot^2/s^2	J	1.355 817 948 331 4	E
Kilowatt hour, kW · hr	J	3.6×10^6	D
Ton equivalent of TNT	J	4.184×10^9	E
Force [newton ≡ kilogram · meter/second^2, N ≡ kg · m/s^2]			
Dyne	N	1×10^{-5}	E
Kilogram-force (kgf)	N	9.806 65	E
Ounce force (avoirdupois)	N	0.278 013 850 953 8	E
Poundal	N	0.138 254 954 376	E
Pound force (avoirdupois), lbf ≡ slug · foot/s^2	N	4.448 221 615 260 5	D
Illuminance [lux ≡ candela · steradian/meter^2, lx ≡ cd · sr/m^2]			
Footcandle	cd · sr/m^2	10.763 910 416 709 70	E
Phot	cd · sr/m^2	1×10^4	E
Inductance, L [henry ≡ weber/ampere ≡ kilogram · meter^2 · ampere^{-2} · second^{-2}, H ≡ Wb/A ≡ kg · m^2 · A^{-2} · s^{-2}]			
Abhenry	H	1×10^{-9}	E
Stathenry	Hd	8.987552×10^{11}	E
Length [meter, m]			
Angstrom, Å	m	1×10^{-10}	E
Astronomical unit (SI)	m	$1.495\ 978\ 707\ 00 \times 10^{11}$	IAU
Earth equatorial radius, R_E	m	$6.378\ 136\ 6 \times 10^6$	IERS
Fermi (1 fermi = 1 fm)	m	1×10^{-15}	E
Foot, ft	m	0.304 8	E
Inch, in	m	0.025 4	E
Light year	m	$9.460\ 730\ 472\ 580\ 8 \times 10^{15}$	D
Micron, μm	m	1×10^{-6}	E
Mil (10^{-3} inch)	m	2.54×10^{-5}	E
Mile (U.S. statute), mi	m	$1.609\ 344 \times 10^3$	E
Nautical mile (U.S.), NM	m	1.852×10^3	E
Parsec (IAU)	m	$3.085\ 677\ 597\ 49 \times 10^{16}$	D
Solar radius	m	$6.960\ 00 \times 10^8$	AA
Yard, yd	m	0.914 4	E
Luminance [candela/meter^2, cd/m^2]			
Footlambert	cd/m^2	3.426 259 099 635 39	D
Lambert	cd/m^2	$(1/\pi) \times 10^4$ $\approx 3.183\ 098\ 862 \times 10^3$	E
Stilb	cd/m^2	1×10^4	E
Magnetic Field Strength, H [ampere turn/meter, A/m]			
Oersted (EMU)	A/m	$(1/4\pi) \times 10^3$ $\approx 79.577\ 471\ 545\ 947\ 667$	E,I
Magnetic Flux [weber ≡ volt · s ≡ kilogram · meter^2 · ampere^{-1} · second^{-2}, Wb ≡ V · s ≡ kg · m^2 · A^{-1} · s^{-2}]			
Maxwell (EMU)	Wb	1×10^{-8}	E
Unit pole	Wb	$1.256\ 637 \times 10^{-7}$	NIST
Magnetic Flux Density (see Magnetic Induction)			
Magnetic Induction, B [tesla ≡ weber/meter^2 ≡ kilogram · ampere^{-1} · second^{-2}, T ≡ Wb/m^2 ≡ kg · A^{-1} · s^{-2}]			E,I
Gamma (EMU) (γ)	T	1×10^{-9}	E,I
Gauss (EMU)	T	1×10^{-4}	
Magnetic Dipole Moment [weber · meter ≡ kilogram · meter^3 · ampere^{-1} · second^{-2}, Wb · m ≡ kg · m^3 · A^{-1} · s^{-2}]			E,I
Pole · centimeter (EMU)	Wb · m	$4\pi \times 10^{-10}$ $\approx 1.256\ 637\ 061\ 435 \times 10^{-9}$	E,I
Gauss · centimeter^3 (Practical)	Wb · m	1×10^{-10}	

To Convert from	To	Multiply by	Notes
Magnetic Moment [ampere turn · meter² ≡ joule/tesla, A · m² ≡ J/T]			
Abampere · centimeter² (EMU)	A · m²	1×10^{-3}	E, I
Ampere · centimeter²	A · m²	1×10^{-4}	E, I
Mass [kilogram, kg]			
γ (= 1 μg)	kg	1×10^{-9}	
Atomic unit (electron mass)	kg	$9.109\,382\,15 \times 10^{-31}$	E
Atomic mass unit (unified), amu	kg	$1.660\,538\,782 \times 10^{-27}$	C
Metric carat	kg	2.0×10^{-4}	C
Metric ton	kg	1×10^{3}	
Ounce mass (avoirdupois), oz	kg	0.028 349 231 25	E
Pound mass, lbm (avoirdupois)	kg	0.453 592 37	E
Slug	kg	14.593 902 937 21	E
Short ton (2,000 lbm)	kg	907.184 74	D
Solar mass	kg	$1.988\,4 \times 10^{30}$	E
Mass Density (see Density)			AA
Moment of Inertia [kilogram · meter², kg · m²]			
Gram · centimeter², g · cm²	kg · m²	1×10^{-7}	E
Pound mass · inch², lbm · in²	kg · m²	$2.926\,396\,534\,292 \times 10^{-4}$	E
Pound mass · foot², lbm · ft²	kg · m²	$4.214\,011\,009\,380 \times 10^{-2}$	D
Slug · inch², slug · in²	kg · m²	$9.415\,402\,418\,968 \times 10^{-3}$	D
Inch · pound force · s², in · lbf · s²	kg · m²	0.112 984 829 027 6	D
Slug · foot² = ft · lbf · s²	kg · m²	1.355 817 948 331 4	D
Plane Angle (see Angular Measure)			
Potential Difference (see Electric Potential Difference)			
Power [watt ≡ joule/second ≡ kilogram · meter²/second³, W ≡ J/s ≡ kg · m²/s³]			
Foot · pound force/second, ft lbf/s	W	1.355 817 948 331	D
Horsepower (550 ft · lbf/s), hp	W	745.699 871 582 3	D
Horsepower (electrical), hp	W	746.0	E
Solar luminosity	W	3.845×10^{26}	AQ
Pressure or Stress [pascal ≡ newton/meter² ≡ kilogram · meter⁻¹ · second⁻², Pa ≡ N/m² ≡ kg · m⁻¹ · s⁻²]			
Atmosphere, atm	Pa	$1.013\,25 \times 10^{5}$	E
Bar	Pa	1×10^{5}	E
Centimeter of mercury (0° C)	Pa	$\approx 1.333\,223\,874\,145 \times 10^{3}$	E
Dyne/centimeter², dyne/cm²	Pa	0.1	E
Inch of mercury (32° F)	Pa	$3.386\,388\,640\,341 \times 10^{3}$	D
Pound force/foot², lbf/ft², psf	Pa	47.880 258 980 34	D
Pound force/inch², lbf/in², psi	Pa	$6.894\,757\,293\,168 \times 10^{3}$	D
Torr (0° C)	Pa	(101325/760) ≈ 133.322 368 421 052 631	E
Resistance (see Electric Resistance)			
Solid Angle (see Angular Area)			
Specific Heat Capacity [joule · kilogram⁻¹ · kelvin⁻¹ ≡ meter² · second² · kelvin⁻¹, J · kg⁻¹ · K⁻¹ ≡ m² · s² · K⁻¹]			
cal · g⁻¹ · K⁻¹ (mean)	J · kg⁻¹ · K⁻¹	$4.186\,80 \times 10^{3}$	E
Btu · lbm⁻¹ · °F⁻¹ (mean)	J · kg⁻¹ · K⁻¹	$4.186\,80 \times 10^{3}$	E
Stress (see Pressure)			
Temperature [kelvin, K]			
Celsius, °C	K	$t_K = t_C + 273.15$	
Fahrenheit, °F	K	$t_K = (5/9)(t_F + 459.67)$	E
Rankine °R	K	$t_K = (5/9)\,t_R$	E
Fahrenheit, °F	C	$t_C = (5/9)(t_F - 32.0)$	E
Rankine °R	C	$t_C = (5/9)(t_R - 491.67)$	E
Thermal Conductivity [watt · meter⁻¹ · kelvin⁻¹ ≡ kilogram · meter · second⁻³ · kelvin⁻¹, W · m⁻¹ · K⁻¹ ≡ kg · m · s⁻³ · K⁻¹]			
cal · cm⁻¹ · s⁻¹ · K⁻¹	W · m⁻¹ · K⁻¹	418.68	E
Btu · ft⁻¹ · hr⁻¹ · °F⁻¹	W · m⁻¹ · K⁻¹	1.730 734 666 371 39	D
Thrust (see Force)			
Time [second, s]			
Sidereal day, d_* (ref. = fixed stars)	s	$8.616\,409\,053 \times 10^{4}$ = 23h 56m 4.09 053s	AA
Ephemeris day (= calendar day), d_e	s	8.64×10^{4}	AQ
Ephemeris day (= calendar day), d_e	d_*	1.002 737 909 35	AA
Keplerian period of a satellite in low-Earth orbit	min	$1.658\,669\,010\,080 \times 10^{-4} \times a^{3/2}$ (a in km)	Table 8–3
Keplerian period of a satellite of the Sun	d_e	$3.652\,568\,983\,276 \times 10^{2} \times a^{3/2}$ (a in AU)	IAU
Tropical year (ref. = ϒ)	s	$3.155\,692\,522 \times 10^{7}$	AA
Tropical year (ref. = ϒ)	d_e	365.242 190	AA
Sidereal year (ref. = fixed stars)	s	$3.155\,814\,976 \times 10^{7}$	AA
Sidereal year (ref. = fixed stars)	d_e	365.256 363	AA
Calendar year (365 days), yr	s	$3.153\,6 \times 10^{7}$	E
Julian century	d_e	36,525	E
Gregorian calendar century	d_e	36,524.25	E
Torque (see Energy)			
Velocity [meter/second, m/s]			
Foot/minute, ft/min	m/s	5.08×10^{-3}	E
Inch/second, ips	m/s	0.025 4	E
Kilometer/hour, km/hr	m/s	$(3.6)^{-1} = 0.277777\ldots$	E
Foot/second, fps or ft/s	m/s	0.304 8	E
Miles/hour, mph	m/s	0.447 04	E
Knot (international)	m/s	$(1852/3600) = 0.514444\ldots$	E
Miles/minute, mi/min	m/s	26.822 4	E
Miles/second, mi/s	m/s	$1.609\,344 \times 10^{3}$	E
Velocity of Light	m/s	$2.997\,924\,58 \times 10^{8}$	E
Viscosity [pascal · second ≡ kilogram · meter⁻¹ · second⁻¹, Pa · s ≡ kg · m⁻¹ · s⁻¹]			
Stoke	m²/s	1.0×10^{-4}	E
Foot² · second, ft² · s	m²/s	0.092 903 04	E
Pound mass · foot⁻¹ · second⁻¹, lbm · ft⁻¹ · s⁻¹	Pa · s	1.488 163 943 570	D
Pound force · second/foot², lbf · s/ft²	Pa · s	47.880 258 980 34	D
Poise	Pa · s	0.1	E
Poundal second/foot², poundal s/ft²	Pa · s	1.488 163 943 570	D
Slug · foot⁻¹ · second⁻¹, slug · ft⁻¹ · s⁻¹	Pa · s	47.880 258 980 34	D
Rhe	(Pa · s)⁻¹	10	E